Ответы и официальные разборы

Том 8. Официальные оригиналы

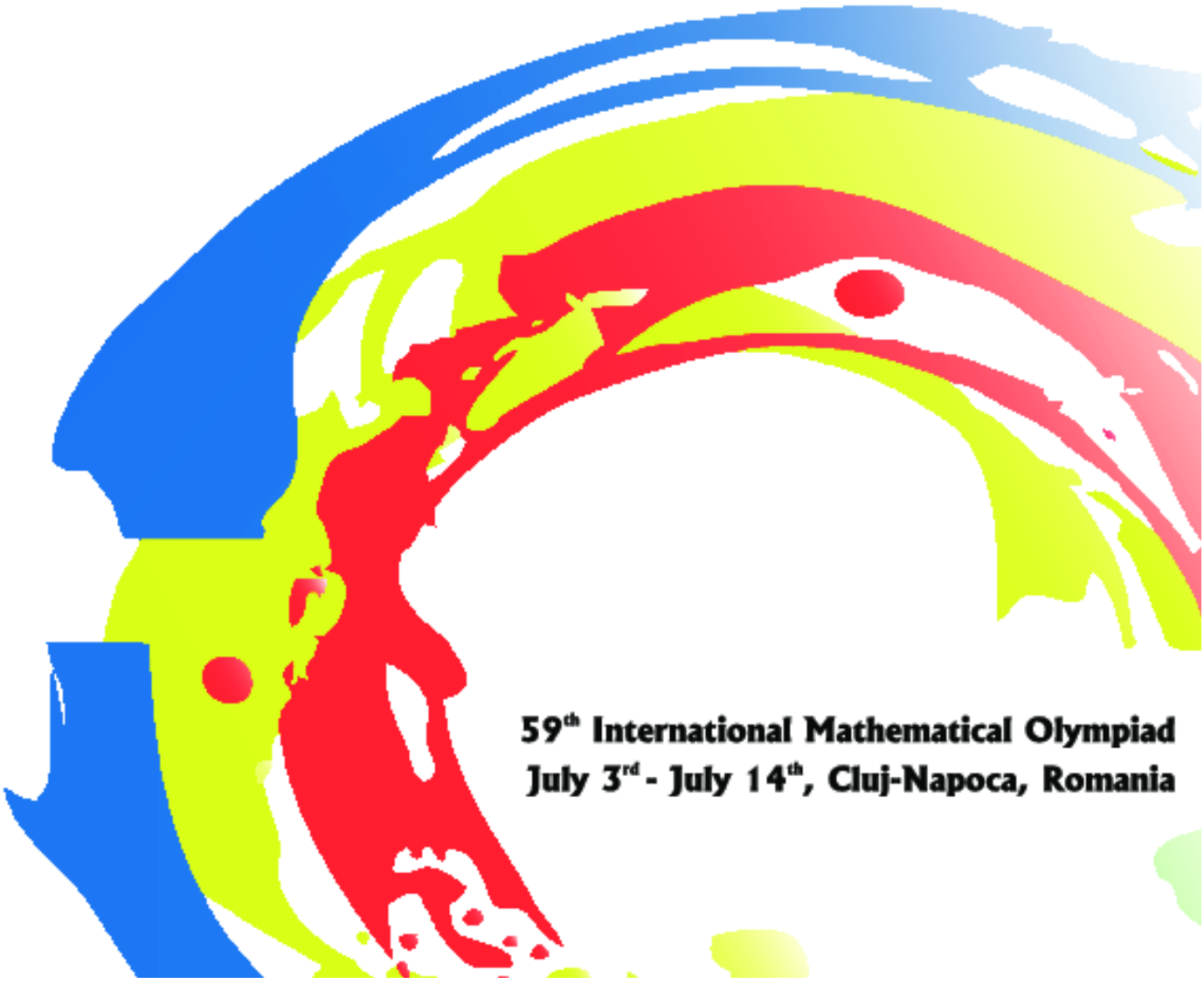
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SHORTLISTED PROBLEMS

WITH SOLUTIONS







Shortlisted Problems

(with solutions)

59th International Mathematical Olympiad

Cluj-Napoca — Romania, 3–14 July 2018

Note of Confidentiality

The Shortlist has to be kept strictly confidential until the conclusion of the following International Mathematical Olympiad.

IMO General Regulations §6.6

Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2018 thank the following 49 countries for contributing 168 problem proposals:

Armenia, Australia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Brazil, Bulgaria, Canada, China, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Germany, Greece, Hong Kong, Iceland, India, Indonesia, Iran, Ireland, Israel, Japan, Kosovo, Luxembourg, Mexico, Moldova, Mongolia, Netherlands, Nicaragua, Poland, Russia, Serbia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Switzerland, Taiwan, Tanzania, Thailand, Turkey, Ukraine, United Kingdom, U.S.A.

Problem Selection Committee



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Problems

Algebra

A1. Let $\mathbb{Q}_{>0}$ denote the set of all positive rational numbers. Determine all functions $f\colon \mathbb{Q}_{>0} \to \mathbb{Q}_{>0}$ satisfying

$$f\left(x^2 f(y)^2\right) = f(x)^2 f(y)$$

for all $x, y \in \mathbb{Q}_{>0}$.

(Switzerland)

A2. Find all positive integers $n \geqslant 3$ for which there exist real numbers $a_1, a_2, \ldots, a_n$, $a_{n+1} = a_1$, $a_{n+2} = a_2$ such that

$$a_i a_{i+1} + 1 = a_{i+2}$$

for all $i = 1, 2, \ldots, n$.

(Slovakia)

A3. Given any set S of positive integers, show that at least one of the following two assertions holds:

(1) There exist distinct finite subsets F and G of S such that $\sum_{x\in F} 1/x = \sum_{x\in G} 1/x$;

(2) There exists a positive rational number $r < 1$ such that $\sum_{x\in F} 1/x \neq r$ for all finite subsets F of S.

(Luxembourg)

A4. Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers such that $a_0 = 0$, $a_1 = 1$, and for every $n \geqslant 2$ there exists $1 \leqslant k \leqslant n$ satisfying

$$a_n = \frac{a_{n-1} + \cdots + a_{n-k}}{k}.$$

Find the maximal possible value of $a_{2018} - a_{2017}$.

(Belgium)

A5. Determine all functions $f : (0, \infty) \to \mathbb{R}$ satisfying

$$\left(x + \frac{1}{x}\right) f(y) = f(xy) + f\left(\frac{y}{x}\right)$$

for all $x, y > 0$.

(South Korea)

A6. Let $m, n \geqslant 2$ be integers. Let $f(x_1, \ldots, x_n)$ be a polynomial with real coefficients such that

$$f(x_1, \ldots, x_n) = \left\lfloor \frac{x_1 + \ldots + x_n}{m} \right\rfloor \quad \text{for every } x_1, \ldots, x_n \in \{0, 1, \ldots, m-1\}.$$

Prove that the total degree of f is at least n.

(Brazil)

A7. Find the maximal value of

$$S = \sqrt[3]{\frac{a}{b+7}} + \sqrt[3]{\frac{b}{c+7}} + \sqrt[3]{\frac{c}{d+7}} + \sqrt[3]{\frac{d}{a+7}},$$

where a, b, c, d are nonnegative real numbers which satisfy $a + b + c + d = 100$.

(Taiwan)

Combinatorics

C1. Let $n \geqslant 3$ be an integer. Prove that there exists a set S of $2n$ positive integers satisfying the following property: For every $m = 2, 3, \ldots, n$ the set S can be partitioned into two subsets with equal sums of elements, with one of subsets of cardinality m.

(Iceland)

C2. Queenie and Horst play a game on a 20×20 chessboard. In the beginning the board is empty. In every turn, Horst places a black knight on an empty square in such a way that his new knight does not attack any previous knights. Then Queenie places a white queen on an empty square. The game gets finished when somebody cannot move.

Find the maximal positive K such that, regardless of the strategy of Queenie, Horst can put at least K knights on the board.

(Armenia)

C3. Let n be a given positive integer. Sisyphus performs a sequence of turns on a board consisting of $n + 1$ squares in a row, numbered 0 to n from left to right. Initially, n stones are put into square 0, and the other squares are empty. At every turn, Sisyphus chooses any nonempty square, say with k stones, takes one of those stones and moves it to the right by at most k squares (the stone should stay within the board). Sisyphus' aim is to move all n stones to square n.

Prove that Sisyphus cannot reach the aim in less than

$$\left\lceil \frac{n}{1} \right\rceil + \left\lceil \frac{n}{2} \right\rceil + \left\lceil \frac{n}{3} \right\rceil + \cdots + \left\lceil \frac{n}{n} \right\rceil$$

turns. (As usual, $\lceil x \rceil$ stands for the least integer not smaller than x.)

(Netherlands)

C4. An *anti-Pascal pyramid* is a finite set of numbers, placed in a triangle-shaped array so that the first row of the array contains one number, the second row contains two numbers, the third row contains three numbers and so on; and, except for the numbers in the bottom row, each number equals the absolute value of the difference of the two numbers below it. For instance, the triangle below is an anti-Pascal pyramid with four rows, in which every integer from 1 to $1 + 2 + 3 + 4 = 10$ occurs exactly once:

$$\begin{array}{ccccccc} & & & 4 & & & \\ & & 2 & & 6 & & \\ & 5 & & 7 & & 1 & \\ 8 & & 3 & & 10 & & 9 \end{array}.$$

Is it possible to form an anti-Pascal pyramid with 2018 rows, using every integer from 1 to $1 + 2 + \cdots + 2018$ exactly once?

(Iran)

C5. Let k be a positive integer. The organising committee of a tennis tournament is to schedule the matches for $2k$ players so that every two players play once, each day exactly one match is played, and each player arrives to the tournament site the day of his first match, and departs the day of his last match. For every day a player is present on the tournament, the committee has to pay 1 coin to the hotel. The organisers want to design the schedule so as to minimise the total cost of all players' stays. Determine this minimum cost.

(Russia)

C6. Let a and b be distinct positive integers. The following infinite process takes place on an initially empty board.

(i) If there is at least a pair of equal numbers on the board, we choose such a pair and increase one of its components by a and the other by b.

(ii) If no such pair exists, we write down two times the number 0.

Prove that, no matter how we make the choices in (i), operation (ii) will be performed only finitely many times.

(Serbia)

C7. Consider 2018 pairwise crossing circles no three of which are concurrent. These circles subdivide the plane into regions bounded by circular *edges* that meet at *vertices*. Notice that there are an even number of vertices on each circle. Given the circle, alternately colour the vertices on that circle red and blue. In doing so for each circle, every vertex is coloured twice — once for each of the two circles that cross at that point. If the two colourings agree at a vertex, then it is assigned that colour; otherwise, it becomes yellow. Show that, if some circle contains at least 2061 yellow points, then the vertices of some region are all yellow.

(India)

Geometry

G1. Let ABC be an acute-angled triangle with circumcircle Γ. Let D and E be points on the segments AB and AC, respectively, such that $AD = AE$. The perpendicular bisectors of the segments BD and CE intersect the small arcs $\overset{\frown}{AB}$ and $\overset{\frown}{AC}$ at points F and G respectively. Prove that $DE \parallel FG$.

(Greece)

G2. Let ABC be a triangle with $AB = AC$, and let M be the midpoint of BC. Let P be a point such that $PB < PC$ and PA is parallel to BC. Let X and Y be points on the lines PB and PC, respectively, so that B lies on the segment PX, C lies on the segment PY, and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic.

(Australia)

G3. A circle ω of radius 1 is given. A collection T of triangles is called *good*, if the following conditions hold:

(i) each triangle from T is inscribed in ω;

(ii) no two triangles from T have a common interior point.

Determine all positive real numbers t such that, for each positive integer n, there exists a good collection of n triangles, each of perimeter greater than t.

(South Africa)

G4. A point T is chosen inside a triangle ABC. Let A_1, B_1, and C_1 be the reflections of T in BC, CA, and AB, respectively. Let Ω be the circumcircle of the triangle $A_1B_1C_1$. The lines A_1T, B_1T, and C_1T meet Ω again at A_2, B_2, and C_2, respectively. Prove that the lines AA_2, BB_2, and CC_2 are concurrent on Ω.

(Mongolia)

G5. Let ABC be a triangle with circumcircle ω and incentre I. A line ℓ intersects the lines AI, BI, and CI at points D, E, and F, respectively, distinct from the points A, B, C, and I. The perpendicular bisectors x, y, and z of the segments AD, BE, and CF, respectively determine a triangle Θ. Show that the circumcircle of the triangle Θ is tangent to ω.

(Denmark)

G6. A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. A point X is chosen inside the quadrilateral so that $\angle XAB = \angle XCD$ and $\angle XBC = \angle XDA$. Prove that $\angle AXB + \angle CXD = 180^\circ$.

(Poland)

G7. Let O be the circumcentre, and Ω be the circumcircle of an acute-angled triangle ABC. Let P be an arbitrary point on Ω, distinct from A, B, C, and their antipodes in Ω. Denote the circumcentres of the triangles AOP, BOP, and COP by O_A, O_B, and O_C, respectively. The lines ℓ_A, ℓ_B, and ℓ_C perpendicular to BC, CA, and AB pass through O_A, O_B, and O_C, respectively. Prove that the circumcircle of the triangle formed by ℓ_A, ℓ_B, and ℓ_C is tangent to the line OP.

(Russia)

Number Theory

N1. Determine all pairs (n, k) of distinct positive integers such that there exists a positive integer s for which the numbers of divisors of sn and of sk are equal.

(Ukraine)

N2. Let $n > 1$ be a positive integer. Each cell of an $n \times n$ table contains an integer. Suppose that the following conditions are satisfied:

(*i*) Each number in the table is congruent to 1 modulo n;

(*ii*) The sum of numbers in any row, as well as the sum of numbers in any column, is congruent to n modulo n^2.

Let R_i be the product of the numbers in the ith row, and C_j be the product of the numbers in the jth column. Prove that the sums $R_1 + \cdots + R_n$ and $C_1 + \cdots + C_n$ are congruent modulo n^4.

(Indonesia)

N3. Define the sequence $a_0, a_1, a_2, \ldots$ by $a_n = 2^n + 2^{\lfloor n/2 \rfloor}$. Prove that there are infinitely many terms of the sequence which can be expressed as a sum of (two or more) distinct terms of the sequence, as well as infinitely many of those which cannot be expressed in such a way.

(Serbia)

N4. Let a_1, a_2, …, a_n, … be a sequence of positive integers such that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}$$

is an integer for all $n \geqslant k$, where k is some positive integer. Prove that there exists a positive integer m such that $a_n = a_{n+1}$ for all $n \geqslant m$.

(Mongolia)

N5. Four positive integers x, y, z, and t satisfy the relations

$$xy - zt = x + y = z + t.$$

Is it possible that both xy and zt are perfect squares?

(Russia)

N6. Let $f\colon \{1, 2, 3, \ldots\} \to \{2, 3, \ldots\}$ be a function such that $f(m + n) \mid f(m) + f(n)$ for all pairs m, n of positive integers. Prove that there exists a positive integer $c > 1$ which divides all values of f.

(Mexico)

N7. Let $n \geqslant 2018$ be an integer, and let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be pairwise distinct positive integers not exceeding $5n$. Suppose that the sequence

$$\frac{a_1}{b_1}, \frac{a_2}{b_2}, \ldots, \frac{a_n}{b_n}$$

forms an arithmetic progression. Prove that the terms of the sequence are equal.

(Thailand)

Solutions

Algebra

A1. Let $\mathbb{Q}_{>0}$ denote the set of all positive rational numbers. Determine all functions $f\colon \mathbb{Q}_{>0} \to \mathbb{Q}_{>0}$ satisfying

$$f\left(x^2 f(y)^2\right) = f(x)^2 f(y) \tag{$*$}$$

for all $x, y \in \mathbb{Q}_{>0}$.

(Switzerland)

Answer: $f(x) = 1$ for all $x \in \mathbb{Q}_{>0}$.

Solution. Take any $a, b \in \mathbb{Q}_{>0}$. By substituting $x = f(a)$, $y = b$ and $x = f(b)$, $y = a$ into $(*)$ we get

$$f\big(f(a)\big)^2 f(b) = f\big(f(a)^2 f(b)^2\big) = f\big(f(b)\big)^2 f(a),$$

which yields

$$\frac{f\big(f(a)\big)^2}{f(a)} = \frac{f\big(f(b)\big)^2}{f(b)} \qquad \text{for all } a, b \in \mathbb{Q}_{>0}.$$

In other words, this shows that there exists a constant $C \in \mathbb{Q}_{>0}$ such that $f\big(f(a)\big)^2 = C f(a)$, or

$$\left(\frac{f\big(f(a)\big)}{C}\right)^2 = \frac{f(a)}{C} \qquad \text{for all } a \in \mathbb{Q}_{>0}. \tag{1}$$

Denote by $f^n(x) = \underbrace{f(f(\ldots(f}_{n}(x))\ldots))$ the nth iteration of f. Equality (1) yields

$$\frac{f(a)}{C} = \left(\frac{f^2(a)}{C}\right)^2 = \left(\frac{f^3(a)}{C}\right)^4 = \cdots = \left(\frac{f^{n+1}(a)}{C}\right)^{2^n}$$

for all positive integer n. So, $f(a)/C$ is the 2^n-th power of a rational number for all positive integer n. This is impossible unless $f(a)/C = 1$, since otherwise the exponent of some prime in the prime decomposition of $f(a)/C$ is not divisible by sufficiently large powers of 2. Therefore, $f(a) = C$ for all $a \in \mathbb{Q}_{>0}$.

Finally, after substituting $f \equiv C$ into $(*)$ we get $C = C^3$, whence $C = 1$. So $f(x) \equiv 1$ is the unique function satisfying $(*)$.

Comment 1. There are several variations of the solution above. For instance, one may start with finding $f(1) = 1$. To do this, let $d = f(1)$. By substituting $x = y = 1$ and $x = d^2$, $y = 1$ into $(*)$ we get $f(d^2) = d^3$ and $f(d^6) = f(d^2)^2 \cdot d = d^7$. By substituting now $x = 1$, $y = d^2$ we obtain $f(d^6) = d^2 \cdot d^3 = d^5$. Therefore, $d^7 = f(d^6) = d^5$, whence $d = 1$.

After that, the rest of the solution simplifies a bit, since we already know that $C = \frac{f(f(1))^2}{f(1)} = 1$. Hence equation (1) becomes merely $f(f(a))^2 = f(a)$, which yields $f(a) = 1$ in a similar manner.

Comment 2. There exist nonconstant functions $f\colon \mathbb{R}^+ \to \mathbb{R}^+$ satisfying $(*)$ for all real $x, y > 0$ — e.g., $f(x) = \sqrt{x}$.

A2. Find all positive integers $n \geqslant 3$ for which there exist real numbers $a_1, a_2, \ldots, a_n$, $a_{n+1} = a_1$, $a_{n+2} = a_2$ such that

$$a_i a_{i+1} + 1 = a_{i+2}$$

for all $i = 1, 2, \ldots, n$.

(Slovakia)

Answer: n can be any multiple of 3.

Solution 1. For the sake of convenience, extend the sequence $a_1, \ldots, a_{n+2}$ to an infinite periodic sequence with period n. (n is not necessarily the shortest period.)

If n is divisible by 3, then $(a_1, a_2, \ldots) = (-1, -1, 2, -1, -1, 2, \ldots)$ is an obvious solution.

We will show that in every periodic sequence satisfying the recurrence, each positive term is followed by two negative values, and after them the next number is positive again. From this, it follows that n is divisible by 3.

If the sequence contains two consecutive positive numbers a_i, a_{i+1}, then $a_{i+2} = a_i a_{i+1} + 1 > 1$, so the next value is positive as well; by induction, all numbers are positive and greater than 1. But then $a_{i+2} = a_i a_{i+1} + 1 \geqslant 1 \cdot a_{i+1} + 1 > a_{i+1}$ for every index i, which is impossible: our sequence is periodic, so it cannot increase everywhere.

If the number 0 occurs in the sequence, $a_i = 0$ for some index i, then it follows that $a_{i+1} = a_{i-1} a_i + 1$ and $a_{i+2} = a_i a_{i+1} + 1$ are two consecutive positive elements in the sequences and we get the same contradiction again.

Notice that after any two consecutive negative numbers the next one must be positive: if $a_i < 0$ and $a_{i+1} < 0$, then $a_{i+2} = a_1 a_{i+1} + 1 > 1 > 0$. Hence, the positive and negative numbers follow each other in such a way that each positive term is followed by one or two negative values and then comes the next positive term.

Consider the case when the positive and negative values alternate. So, if a_i is a negative value then a_{i+1} is positive, a_{i+2} is negative and a_{i+3} is positive again.

Notice that $a_i a_{i+1} + 1 = a_{i+2} < 0 < a_{i+3} = a_{i+1} a_{i+2} + 1$; by $a_{i+1} > 0$ we conclude $a_i < a_{i+2}$. Hence, the negative values form an infinite increasing subsequence, $a_i < a_{i+2} < a_{i+4} < \ldots$, which is not possible, because the sequence is periodic.

The only case left is when there are consecutive negative numbers in the sequence. Suppose that a_i and a_{i+1} are negative; then $a_{i+2} = a_i a_{i+1} + 1 > 1$. The number a_{i+3} must be negative. We show that a_{i+4} also must be negative.

Notice that a_{i+3} is negative and $a_{i+4} = a_{i+2} a_{i+3} + 1 < 1 < a_i a_{i+1} + 1 = a_{i+2}$, so

$$a_{i+5} - a_{i+4} = (a_{i+3} a_{i+4} + 1) - (a_{i+2} a_{i+3} + 1) = a_{i+3}(a_{i+4} - a_{i+2}) > 0,$$

therefore $a_{i+5} > a_{i+4}$. Since at most one of a_{i+4} and a_{i+5} can be positive, that means that a_{i+4} must be negative.

Now a_{i+3} and a_{i+4} are negative and a_{i+5} is positive; so after two negative and a positive terms, the next three terms repeat the same pattern. That completes the solution.

Solution 2. We prove that the shortest period of the sequence must be 3. Then it follows that n must be divisible by 3.

Notice that the equation $x^2 + 1 = x$ has no real root, so the numbers $a_1, \ldots, a_n$ cannot be all equal, hence the shortest period of the sequence cannot be 1.

By applying the recurrence relation for i and $i + 1$,

$$(a_{i+2} - 1) a_{i+2} = a_i a_{i+1} a_{i+2} = a_i (a_{i+3} - 1), \quad \text{so}$$
$$a_{i+2}^2 - a_i a_{i+3} = a_{i+2} - a_i.$$

By summing over $i = 1, 2, \ldots, n$, we get

$$\sum_{i=1}^{n} (a_i - a_{i+3})^2 = 0.$$

That proves that $a_i = a_{i+3}$ for every index i, so the sequence $a_1, a_2, \ldots$ is indeed periodic with period 3. The shortest period cannot be 1, so it must be 3; therefore, n is divisible by 3.

Comment. By solving the system of equations $ab + 1 = c, \quad bc + 1 = a, \quad ca + 1 = b$, it can be seen that the pattern $(-1, -1, 2)$ is repeated in all sequences satisfying the problem conditions.

A3. Given any set S of positive integers, show that at least one of the following two assertions holds:

(1) There exist distinct finite subsets F and G of S such that $\sum_{x\in F} 1/x = \sum_{x\in G} 1/x$;

(2) There exists a positive rational number $r < 1$ such that $\sum_{x\in F} 1/x \neq r$ for all finite subsets F of S.

(Luxembourg)

Solution 1. Argue indirectly. Agree, as usual, that the empty sum is 0 to consider rationals in $[0, 1)$; adjoining 0 causes no harm, since $\sum_{x\in F} 1/x = 0$ for no nonempty finite subset F of S. For every rational r in $[0, 1)$, let F_r be the unique finite subset of S such that $\sum_{x\in F_r} 1/x = r$. The argument hinges on the lemma below.

Lemma. If x is a member of S and q and r are rationals in $[0, 1)$ such that $q - r = 1/x$, then x is a member of F_q if and only if it is not one of F_r.

Proof. If x is a member of F_q, then

$$\sum_{y\in F_q\smallsetminus\{x\}} \frac{1}{y} = \sum_{y\in F_q} \frac{1}{y} - \frac{1}{x} = q - \frac{1}{x} = r = \sum_{y\in F_r} \frac{1}{y},$$

so $F_r = F_q \smallsetminus \{x\}$, and x is not a member of F_r. Conversely, if x is not a member of F_r, then

$$\sum_{y\in F_r\cup\{x\}} \frac{1}{y} = \sum_{y\in F_r} \frac{1}{y} + \frac{1}{x} = r + \frac{1}{x} = q = \sum_{y\in F_q} \frac{1}{y},$$

so $F_q = F_r \cup \{x\}$, and x is a member of F_q. □

Consider now an element x of S and a positive rational $r < 1$. Let $n = \lfloor rx\rfloor$ and consider the sets $F_{r-k/x}$, $k = 0, \ldots, n$. Since $0 \leqslant r - n/x < 1/x$, the set $F_{r-n/x}$ does not contain x, and a repeated application of the lemma shows that the $F_{r-(n-2k)/x}$ do not contain x, whereas the $F_{r-(n-2k-1)/x}$ do. Consequently, x is a member of F_r if and only if n is odd.

Finally, consider $F_{2/3}$. By the preceding, $\lfloor 2x/3\rfloor$ is odd for each x in $F_{2/3}$, so $2x/3$ is not integral. Since $F_{2/3}$ is finite, there exists a positive rational ε such that $\lfloor(2/3 - \varepsilon)x\rfloor = \lfloor 2x/3\rfloor$ for all x in $F_{2/3}$. This implies that $F_{2/3}$ is a subset of $F_{2/3-\varepsilon}$ which is impossible.

Comment. The solution above can be adapted to show that the problem statement still holds, if the condition $r < 1$ in **(2)** is replaced with $r < \delta$, for an arbitrary positive δ. This yields that, if S does not satisfy **(1)**, then there exist *infinitely many* positive rational numbers $r < 1$ such that $\sum_{x\in F} 1/x \neq r$ for all finite subsets F of S.

Solution 2. A finite S clearly satisfies **(2)**, so let S be infinite. If S fails both conditions, so does $S \smallsetminus \{1\}$. We may and will therefore assume that S consists of integers greater than 1. Label the elements of S increasingly $x_1 < x_2 < \cdots$, where $x_1 \geqslant 2$.

We first show that S satisfies **(2)** if $x_{n+1} \geqslant 2x_n$ for all n. In this case, $x_n \geqslant 2^{n-1}x_1$ for all n, so

$$s = \sum_{n\geqslant 1} \frac{1}{x_n} \leqslant \sum_{n\geqslant 1} \frac{1}{2^{n-1}x_1} = \frac{2}{x_1}.$$

If $x_1 \geqslant 3$, or $x_1 = 2$ and $x_{n+1} > 2x_n$ for some n, then $\sum_{x\in F} 1/x < s < 1$ for every finite subset F of S, so S satisfies **(2)**; and if $x_1 = 2$ and $x_{n+1} = 2x_n$ for all n, that is, $x_n = 2^n$ for all n, then every finite subset F of S consists of powers of 2, so $\sum_{x\in F} 1/x \neq 1/3$ and again S satisfies **(2)**.

Finally, we deal with the case where $x_{n+1} < 2x_n$ for some n. Consider the positive rational $r = 1/x_n - 1/x_{n+1} < 1/x_{n+1}$. If $r = \sum_{x\in F} 1/x$ for no finite subset F of S, then S satisfies **(2)**.

We now assume that $r = \sum_{x \in F_0} 1/x$ for some finite subset F_0 of S, and show that S satisfies **(1)**. Since $\sum_{x \in F_0} 1/x = r < 1/x_{n+1}$, it follows that x_{n+1} is not a member of F_0, so

$$\sum_{x \in F_0 \cup \{x_{n+1}\}} \frac{1}{x} = \sum_{x \in F_0} \frac{1}{x} + \frac{1}{x_{n+1}} = r + \frac{1}{x_{n+1}} = \frac{1}{x_n}.$$

Consequently, $F = F_0 \cup \{x_{n+1}\}$ and $G = \{x_n\}$ are distinct finite subsets of S such that $\sum_{x \in F} 1/x = \sum_{x \in G} 1/x$, and S satisfies **(1)**.

A4. Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers such that $a_0 = 0$, $a_1 = 1$, and for every $n \geqslant 2$ there exists $1 \leqslant k \leqslant n$ satisfying

$$a_n = \frac{a_{n-1} + \cdots + a_{n-k}}{k}.$$

Find the maximal possible value of $a_{2018} - a_{2017}$.

(Belgium)

Answer: The maximal value is $\frac{2016}{2017^2}$.

Solution 1. The claimed maximal value is achieved at

$$a_1 = a_2 = \cdots = a_{2016} = 1, \quad a_{2017} = \frac{a_{2016} + \cdots + a_0}{2017} = 1 - \frac{1}{2017},$$
$$a_{2018} = \frac{a_{2017} + \cdots + a_1}{2017} = 1 - \frac{1}{2017^2}.$$

Now we need to show that this value is optimal. For brevity, we use the notation

$$S(n,k) = a_{n-1} + a_{n-2} + \cdots + a_{n-k} \quad \text{for nonnegative integers } k \leqslant n.$$

In particular, $S(n,0) = 0$ and $S(n,1) = a_{n-1}$. In these terms, for every integer $n \geqslant 2$ there exists a positive integer $k \leqslant n$ such that $a_n = S(n,k)/k$.

For every integer $n \geqslant 1$ we define

$$M_n = \max_{1 \leqslant k \leqslant n} \frac{S(n,k)}{k}, \qquad m_n = \min_{1 \leqslant k \leqslant n} \frac{S(n,k)}{k}, \quad \text{and} \quad \Delta_n = M_n - m_n \geqslant 0.$$

By definition, $a_n \in [m_n, M_n]$ for all $n \geqslant 2$; on the other hand, $a_{n-1} = S(n,1)/1 \in [m_n, M_n]$. Therefore,

$$a_{2018} - a_{2017} \leqslant M_{2018} - m_{2018} = \Delta_{2018},$$

and we are interested in an upper bound for Δ_{2018}.

Also by definition, for any $0 < k \leqslant n$ we have $km_n \leqslant S(n,k) \leqslant kM_n$; notice that these inequalities are also valid for $k = 0$.

Claim 1. For every $n > 2$, we have $\Delta_n \leqslant \frac{n-1}{n}\Delta_{n-1}$.

Proof. Choose positive integers $k, \ell \leqslant n$ such that $M_n = S(n,k)/k$ and $m_n = S(n,\ell)/\ell$. We have $S(n,k) = a_{n-1} + S(n-1,k-1)$, so

$$k(M_n - a_{n-1}) = S(n,k) - ka_{n-1} = S(n-1,k-1) - (k-1)a_{n-1} \leqslant (k-1)(M_{n-1} - a_{n-1}),$$

since $S(n-1,k-1) \leqslant (k-1)M_{n-1}$. Similarly, we get

$$\ell(a_{n-1} - m_n) = (\ell-1)a_{n-1} - S(n-1,\ell-1) \leqslant (\ell-1)(a_{n-1} - m_{n-1}).$$

Since $m_{n-1} \leqslant a_{n-1} \leqslant M_{n-1}$ and $k, \ell \leqslant n$, the obtained inequalities yield

$$M_n - a_{n-1} \leqslant \frac{k-1}{k}(M_{n-1} - a_{n-1}) \leqslant \frac{n-1}{n}(M_{n-1} - a_{n-1}) \quad \text{and}$$
$$a_{n-1} - m_n \leqslant \frac{\ell-1}{\ell}(a_{n-1} - m_{n-1}) \leqslant \frac{n-1}{n}(a_{n-1} - m_{n-1}).$$

Therefore,

$$\Delta_n = (M_n - a_{n-1}) + (a_{n-1} - m_n) \leqslant \frac{n-1}{n}\big((M_{n-1} - a_{n-1}) + (a_{n-1} - m_{n-1})\big) = \frac{n-1}{n}\Delta_{n-1}. \quad \square$$

Back to the problem, if $a_n = 1$ for all $n \leqslant 2017$, then $a_{2018} \leqslant 1$ and hence $a_{2018} - a_{2017} \leqslant 0$. Otherwise, let $2 \leqslant q \leqslant 2017$ be the minimal index with $a_q < 1$. We have $S(q, i) = i$ for all $i = 1, 2, \ldots, q-1$, while $S(q, q) = q - 1$. Therefore, $a_q < 1$ yields $a_q = S(q, q)/q = 1 - \frac{1}{q}$.

Now we have $S(q+1, i) = i - \frac{1}{q}$ for $i = 1, 2, \ldots, q$, and $S(q+1, q+1) = q - \frac{1}{q}$. This gives us

$$m_{q+1} = \frac{S(q+1,1)}{1} = \frac{S(q+1,q+1)}{q+1} = \frac{q-1}{q} \quad \text{and} \quad M_{q+1} = \frac{S(q+1,q)}{q} = \frac{q^2-1}{q^2},$$

so $\Delta_{q+1} = M_{q+1} - m_{q+1} = (q-1)/q^2$. Denoting $N = 2017 \geqslant q$ and using Claim 1 for $n = q+2, q+3, \ldots, N+1$ we finally obtain

$$\Delta_{N+1} \leqslant \frac{q-1}{q^2} \cdot \frac{q+1}{q+2} \cdot \frac{q+2}{q+3} \cdots \frac{N}{N+1} = \frac{1}{N+1}\left(1 - \frac{1}{q^2}\right) \leqslant \frac{1}{N+1}\left(1 - \frac{1}{N^2}\right) = \frac{N-1}{N^2},$$

as required.

Comment 1. One may check that the maximal value of $a_{2018} - a_{2017}$ is attained at the unique sequence, which is presented in the solution above.

Comment 2. An easier question would be to determine the maximal value of $|a_{2018} - a_{2017}|$. In this version, the answer $\frac{1}{2018}$ is achieved at

$$a_1 = a_2 = \cdots = a_{2017} = 1, \quad a_{2018} = \frac{a_{2017} + \cdots + a_0}{2018} = 1 - \frac{1}{2018}.$$

To prove that this value is optimal, it suffices to notice that $\Delta_2 = \frac{1}{2}$ and to apply Claim 1 obtaining

$$|a_{2018} - a_{2017}| \leqslant \Delta_{2018} \leqslant \frac{1}{2} \cdot \frac{2}{3} \cdots \frac{2017}{2018} = \frac{1}{2018}.$$

Solution 2. We present a different proof of the estimate $a_{2018} - a_{2017} \leqslant \frac{2016}{2017^2}$. We keep the same notations of $S(n, k)$, m_n and M_n from the previous solution.

Notice that $S(n, n) = S(n, n-1)$, as $a_0 = 0$. Also notice that for $0 \leqslant k \leqslant \ell \leqslant n$ we have $S(n, \ell) = S(n, k) + S(n-k, \ell-k)$.

Claim 2. For every positive integer n, we have $m_n \leqslant m_{n+1}$ and $M_{n+1} \leqslant M_n$, so the segment $[m_{n+1}, M_{n+1}]$ is contained in $[m_n, M_n]$.

Proof. Choose a positive integer $k \leqslant n+1$ such that $m_{n+1} = S(n+1, k)/k$. Then we have

$$km_{n+1} = S(n+1, k) = a_n + S(n, k-1) \geqslant m_n + (k-1)m_n = km_n,$$

which establishes the first inequality in the Claim. The proof of the second inequality is similar. □

Claim 3. For every positive integers $k \geqslant n$, we have $m_n \leqslant a_k \leqslant M_n$.

Proof. By Claim 2, we have $[m_k, M_k] \subseteq [m_{k-1}, M_{k-1}] \subseteq \cdots \subseteq [m_n, M_n]$. Since $a_k \in [m_k, M_k]$, the claim follows. □

Claim 4. For every integer $n \geqslant 2$, we have $M_n = S(n, n-1)/(n-1)$ and $m_n = S(n,n)/n$.

Proof. We use induction on n. The base case $n = 2$ is routine. To perform the induction step, we need to prove the inequalities

$$\frac{S(n,n)}{n} \leqslant \frac{S(n,k)}{k} \quad \text{and} \quad \frac{S(n,k)}{k} \leqslant \frac{S(n,n-1)}{n-1} \tag{1}$$

for every positive integer $k \leqslant n$. Clearly, these inequalities hold for $k = n$ and $k = n-1$, as $S(n,n) = S(n,n-1) > 0$. In the sequel, we assume that $k < n-1$.

Now the first inequality in (1) rewrites as $nS(n,k) \geqslant kS(n,n) = k\big(S(n,k)+S(n-k,n-k)\big)$, or, cancelling the terms occurring on both parts, as

$$(n-k)S(n,k) \geqslant kS(n-k,n-k) \iff S(n,k) \geqslant k \cdot \frac{S(n-k,n-k)}{n-k}.$$

By the induction hypothesis, we have $S(n-k,n-k)/(n-k) = m_{n-k}$. By Claim 3, we get $a_{n-i} \geqslant m_{n-k}$ for all $i = 1, 2, \ldots, k$. Summing these k inequalities we obtain

$$S(n,k) \geqslant km_{n-k} = k \cdot \frac{S(n-k,n-k)}{n-k},$$

as required.

The second inequality in (1) is proved similarly. Indeed, this inequality is equivalent to

$$\begin{aligned}(n-1)S(n,k) \leqslant kS(n,n-1) &\iff (n-k-1)S(n,k) \leqslant kS(n-k,n-k-1)\\ &\iff S(n,k) \leqslant k \cdot \frac{S(n-k,n-k-1)}{n-k-1} = kM_{n-k};\end{aligned}$$

the last inequality follows again from Claim 3, as each term in $S(n,k)$ is at most M_{n-k}. □

Now we can prove the required estimate for $a_{2018} - a_{2017}$. Set $N = 2017$. By Claim 4,

$$\begin{aligned}a_{N+1} - a_N \leqslant M_{N+1} - a_N = \frac{S(N+1,N)}{N} - a_N &= \frac{a_N + S(N,N-1)}{N} - a_N\\ &= \frac{S(N,N-1)}{N} - \frac{N-1}{N} \cdot a_N.\end{aligned}$$

On the other hand, the same Claim yields

$$a_N \geqslant m_N = \frac{S(N,N)}{N} = \frac{S(N,N-1)}{N}.$$

Noticing that each term in $S(N,N-1)$ is at most 1, so $S(N,N-1) \leqslant N-1$, we finally obtain

$$a_{N+1} - a_N \leqslant \frac{S(N,N-1)}{N} - \frac{N-1}{N} \cdot \frac{S(N,N-1)}{N} = \frac{S(N,N-1)}{N^2} \leqslant \frac{N-1}{N^2}.$$

Comment 1. Claim 1 in Solution 1 can be deduced from Claims 2 and 4 in Solution 2.

By Claim 4 we have $M_n = \frac{S(n,n-1)}{n-1}$ and $m_n = \frac{S(n,n)}{n} = \frac{S(n,n-1)}{n}$. It follows that $\Delta_n = M_n - m_n = \frac{S(n,n-1)}{(n-1)n}$ and so $M_n = n\Delta_n$ and $m_n = (n-1)\Delta_n$

Similarly, $M_{n-1} = (n-1)\Delta_{n-1}$ and $m_{n-1} = (n-2)\Delta_{n-1}$. Then the inequalities $m_{n-1} \leqslant m_n$ and $M_n \leqslant M_{n-1}$ from Claim 2 write as $(n-2)\Delta_{n-1} \leqslant (n-1)\Delta_n$ and $n\Delta_n \leqslant (n-1)\Delta_{n-1}$. Hence we have the double inequality

$$\frac{n-2}{n-1}\Delta_{n-1} \leqslant \Delta_n \leqslant \frac{n-1}{n}\Delta_{n-1}.$$

Comment 2. Both solutions above discuss the properties of an arbitrary sequence satisfying the problem conditions. Instead, one may investigate only an *optimal* sequence which maximises the value of $a_{2018} - a_{2017}$. Here we present an observation which allows to simplify such investigation — for instance, the proofs of Claim 1 in Solution 1 and Claim 4 in Solution 2.

The sequence (a_n) is uniquely determined by choosing, for every $n \geqslant 2$, a positive integer $k(n) \leqslant n$ such that $a_n = S(n, k(n))/k(n)$. Take an arbitrary $2 \leqslant n_0 \leqslant 2018$, and assume that all such integers $k(n)$, for $n \neq n_0$, are fixed. Then, for every n, the value of a_n is a linear function in a_{n_0} (whose possible values constitute some discrete subset of $[m_{n_0}, M_{n_0}]$ containing both endpoints). Hence, $a_{2018} - a_{2017}$ is also a linear function in a_{n_0}, so it attains its maximal value at one of the endpoints of the segment $[m_{n_0}, M_{n_0}]$.

This shows that, while dealing with an optimal sequence, we may assume $a_n \in \{m_n, M_n\}$ for all $2 \leqslant n \leqslant 2018$. Now one can easily see that, if $a_n = m_n$, then $m_{n+1} = m_n$ and $M_{n+1} \leqslant \frac{m_n + nM_n}{n+1}$; similar estimates hold in the case $a_n = M_n$. This already establishes Claim 1, and simplifies the inductive proof of Claim 4, both applied to an optimal sequence.

A5. Determine all functions $f : (0, \infty) \to \mathbb{R}$ satisfying

$$\left(x + \frac{1}{x}\right) f(y) = f(xy) + f\left(\frac{y}{x}\right) \tag{1}$$

for all $x, y > 0$.

(South Korea)

Answer: $f(x) = C_1 x + \dfrac{C_2}{x}$ with arbitrary constants C_1 and C_2.

Solution 1. Fix a real number $a > 1$, and take a new variable t. For the values $f(t)$, $f(t^2)$, $f(at)$ and $f(a^2t^2)$, the relation (1) provides a system of linear equations:

$$\begin{aligned}
x = y = t\colon &\quad \left(t + \frac{1}{t}\right) f(t) &&= f(t^2) + f(1) && \text{(2a)}\\
x = \frac{t}{a},\ y = at\colon &\quad \left(\frac{t}{a} + \frac{a}{t}\right) f(at) &&= f(t^2) + f(a^2) && \text{(2b)}\\
x = a^2t,\ y = t\colon &\quad \left(a^2t + \frac{1}{a^2t}\right) f(t) &&= f(a^2t^2) + f\left(\frac{1}{a^2}\right) && \text{(2c)}\\
x = y = at\colon &\quad \left(at + \frac{1}{at}\right) f(at) &&= f(a^2t^2) + f(1) && \text{(2d)}
\end{aligned}$$

In order to eliminate $f(t^2)$, take the difference of (2a) and (2b); from (2c) and (2d) eliminate $f(a^2t^2)$; then by taking a linear combination, eliminate $f(at)$ as well:

$$\left(t + \frac{1}{t}\right) f(t) - \left(\frac{t}{a} + \frac{a}{t}\right) f(at) = f(1) - f(a^2) \quad \text{and}$$

$$\left(a^2t + \frac{1}{a^2t}\right) f(t) - \left(at + \frac{1}{at}\right) f(at) = f(1/a^2) - f(1), \quad \text{so}$$

$$\left(\left(at + \frac{1}{at}\right)\left(t + \frac{1}{t}\right) - \left(\frac{t}{a} + \frac{a}{t}\right)\left(a^2t + \frac{1}{a^2t}\right)\right) f(t)$$

$$= \left(at + \frac{1}{at}\right)\big(f(1) - f(a^2)\big) - \left(\frac{t}{a} + \frac{a}{t}\right)\big(f(1/a^2) - f(1)\big).$$

Notice that on the left-hand side, the coefficient of $f(t)$ is nonzero and does not depend on t:

$$\left(at + \frac{1}{at}\right)\left(t + \frac{1}{t}\right) - \left(\frac{t}{a} + \frac{a}{t}\right)\left(a^2t + \frac{1}{a^2t}\right) = a + \frac{1}{a} - \left(a^3 + \frac{1}{a^3}\right) < 0.$$

After dividing by this fixed number, we get

$$f(t) = C_1 t + \frac{C_2}{t} \tag{3}$$

where the numbers C_1 and C_2 are expressed in terms of a, $f(1)$, $f(a^2)$ and $f(1/a^2)$, and they do not depend on t.

The functions of the form (3) satisfy the equation:

$$\left(x + \frac{1}{x}\right) f(y) = \left(x + \frac{1}{x}\right)\left(C_1 y + \frac{C_2}{y}\right) = \left(C_1 xy + \frac{C_2}{xy}\right) + \left(C_1 \frac{y}{x} + C_2 \frac{x}{y}\right) = f(xy) + f\left(\frac{y}{x}\right).$$

Solution 2. We start with an observation. If we substitute $x = a \neq 1$ and $y = a^n$ in (1), we obtain

$$f(a^{n+1}) - \left(a + \frac{1}{a}\right) f(a^n) + f(a^{n-1}) = 0.$$

For the sequence $z_n = a^n$, this is a homogeneous linear recurrence of the second order, and its characteristic polynomial is $t^2 - \left(a + \frac{1}{a}\right) t + 1 = (t - a)(t - \frac{1}{a})$ with two distinct nonzero roots, namely a and $1/a$. As is well-known, the general solution is $z_n = C_1 a^n + C_2 (1/a)^n$ where the index n can be as well positive as negative. Of course, the numbers C_1 and C_2 may depend of the choice of a, so in fact we have two functions, C_1 and C_2, such that

$$f(a^n) = C_1(a) \cdot a^n + \frac{C_2(a)}{a^n} \quad \text{for every } a \neq 1 \text{ and every integer } n. \tag{4}$$

The relation (4) can be easily extended to rational values of n, so we may conjecture that C_1 and C_2 are constants, and whence $f(t) = C_1 t + \frac{C_2}{t}$. As it was seen in the previous solution, such functions indeed satisfy (1).

The equation (1) is linear in f; so if some functions f_1 and f_2 satisfy (1) and c_1, c_2 are real numbers, then $c_1 f_1(x) + c_2 f_2(x)$ is also a solution of (1). In order to make our formulas simpler, define

$$f_0(x) = f(x) - f(1) \cdot x.$$

This function is another one satisfying (1) and the extra constraint $f_0(1) = 0$. Repeating the same argument on linear recurrences, we can write $f_0(a) = K(a)a^n + \frac{L(a)}{a^n}$ with some functions K and L. By substituting $n = 0$, we can see that $K(a) + L(a) = f_0(1) = 0$ for every a. Hence,

$$f_0(a^n) = K(a) \left(a^n - \frac{1}{a^n}\right).$$

Now take two numbers $a > b > 1$ arbitrarily and substitute $x = (a/b)^n$ and $y = (ab)^n$ in (1):

$$\left(\frac{a^n}{b^n} + \frac{b^n}{a^n}\right) f_0\big((ab)^n\big) = f_0\big(a^{2n}\big) + f_0\big(b^{2n}\big), \quad \text{so}$$

$$\left(\frac{a^n}{b^n} + \frac{b^n}{a^n}\right) K(ab) \left((ab)^n - \frac{1}{(ab)^n}\right) = K(a)\left(a^{2n} - \frac{1}{a^{2n}}\right) + K(b)\left(b^{2n} - \frac{1}{b^{2n}}\right), \quad \text{or equivalently}$$

$$K(ab)\left(a^{2n} - \frac{1}{a^{2n}} + b^{2n} - \frac{1}{b^{2n}}\right) = K(a)\left(a^{2n} - \frac{1}{a^{2n}}\right) + K(b)\left(b^{2n} - \frac{1}{b^{2n}}\right). \tag{5}$$

By dividing (5) by a^{2n} and then taking limit with $n \to +\infty$ we get $K(ab) = K(a)$. Then (5) reduces to $K(a) = K(b)$. Hence, $K(a) = K(b)$ for all $a > b > 1$.

Fix $a > 1$. For every $x > 0$ there is some b and an integer n such that $1 < b < a$ and $x = b^n$. Then

$$f_0(x) = f_0(b^n) = K(b) \left(b^n - \frac{1}{b^n}\right) = K(a) \left(x - \frac{1}{x}\right).$$

Hence, we have $f(x) = f_0(x) + f(1)x = C_1 x + \frac{C_2}{x}$ with $C_1 = K(a) + f(1)$ and $C_2 = -K(a)$.

Comment. After establishing (5), there are several variants of finishing the solution. For example, instead of taking a limit, we can obtain a system of linear equations for $K(a)$, $K(b)$ and $K(ab)$ by substituting two positive integers n in (5), say $n = 1$ and $n = 2$. This approach leads to a similar ending as in the first solution.

Optionally, we define another function $f_1(x) = f_0(x) - C\left(x - \frac{1}{x}\right)$ and prescribe $K(c) = 0$ for another fixed c. Then we can choose $ab = c$ and decrease the number of terms in (5).

A6. Let $m, n \geqslant 2$ be integers. Let $f(x_1, \ldots, x_n)$ be a polynomial with real coefficients such that

$$f(x_1, \ldots, x_n) = \left\lfloor \frac{x_1 + \ldots + x_n}{m} \right\rfloor \quad \text{for every } x_1, \ldots, x_n \in \{0, 1, \ldots, m-1\}.$$

Prove that the total degree of f is at least n.

(Brazil)

Solution. We transform the problem to a single variable question by the following

Lemma. Let $a_1, \ldots, a_n$ be nonnegative integers and let $G(x)$ be a nonzero polynomial with $\deg G \leqslant a_1 + \ldots + a_n$. Suppose that some polynomial $F(x_1, \ldots, x_n)$ satisfies

$$F(x_1, \ldots, x_n) = G(x_1 + \ldots + x_n) \quad \text{for } (x_1, \ldots, x_n) \in \{0, 1, \ldots, a_1\} \times \ldots \times \{0, 1, \ldots, a_n\}.$$

Then F cannot be the zero polynomial, and $\deg F \geqslant \deg G$.

For proving the lemma, we will use *forward differences* of polynomials. If $p(x)$ is a polynomial with a single variable, then define $(\Delta p)(x) = p(x+1) - p(x)$. It is well-known that if p is a nonconstant polynomial then $\deg \Delta p = \deg p - 1$.

If $p(x_1, \ldots, x_n)$ is a polynomial with n variables and $1 \leqslant k \leqslant n$ then let

$$(\Delta_k p)(x_1, \ldots, x_n) = p(x_1, \ldots, x_{k-1}, x_k + 1, x_{k+1}, \ldots, x_n) - p(x_1, \ldots, x_n).$$

It is also well-known that either $\Delta_k p$ is the zero polynomial or $\deg(\Delta_k p) \leqslant \deg p - 1$.

Proof of the lemma. We apply induction on the degree of G. If G is a constant polynomial then we have $F(0, \ldots, 0) = G(0) \neq 0$, so F cannot be the zero polynomial.

Suppose that $\deg G \geqslant 1$ and the lemma holds true for lower degrees. Since $a_1 + \ldots + a_n \geqslant \deg G > 0$, at least one of $a_1, \ldots, a_n$ is positive; without loss of generality suppose $a_1 \geqslant 1$.

Consider the polynomials $F_1 = \Delta_1 F$ and $G_1 = \Delta G$. On the grid $\{0, \ldots, a_1 - 1\} \times \{0, \ldots, a_2\} \times \ldots \times \{0, \ldots, a_n\}$ we have

$$\begin{aligned} F_1(x_1, \ldots, x_n) &= F(x_1 + 1, x_2, \ldots, x_n) - F(x_1, x_2, \ldots, x_n) = \\ &= G(x_1 + \ldots + x_n + 1) - G(x_1 + \ldots + x_n) = G_1(x_1 + \ldots + x_n). \end{aligned}$$

Since G is nonconstant, we have $\deg G_1 = \deg G - 1 \leqslant (a_1 - 1) + a_2 + \ldots + a_n$. Therefore we can apply the induction hypothesis to F_1 and G_1 and conclude that F_1 is not the zero polynomial and $\deg F_1 \geqslant \deg G_1$. Hence, $\deg F \geqslant \deg F_1 + 1 \geqslant \deg G_1 + 1 = \deg G$. That finishes the proof. □

To prove the problem statement, take the unique polynomial $g(x)$ so that $g(x) = \left\lfloor \frac{x}{m} \right\rfloor$ for $x \in \{0, 1, \ldots, n(m-1)\}$ and $\deg g \leqslant n(m-1)$. Notice that precisely $n(m-1) + 1$ values of g are prescribed, so $g(x)$ indeed exists and is unique. Notice further that the constraints $g(0) = g(1) = 0$ and $g(m) = 1$ together enforce $\deg g \geqslant 2$.

By applying the lemma to $a_1 = \ldots = a_n = m - 1$ and the polynomials f and g, we achieve $\deg f \geqslant \deg g$. Hence we just need a suitable lower bound on $\deg g$.

Consider the polynomial $h(x) = g(x + m) - g(x) - 1$. The degree of $g(x + m) - g(x)$ is $\deg g - 1 \geqslant 1$, so $\deg h = \deg g - 1 \geqslant 1$, and therefore h cannot be the zero polynomial. On the other hand, h vanishes at the points $0, 1, \ldots, n(m-1) - m$, so h has at least $(n-1)(m-1)$ roots. Hence,

$$\deg f \geqslant \deg g = \deg h + 1 \geqslant (n-1)(m-1) + 1 \geqslant n.$$

Comment 1. In the lemma we have equality for the choice $F(x_1,\ldots,x_n)=G(x_1+\ldots+x_n)$, so it indeed transforms the problem to an equivalent single-variable question.

Comment 2. If $m\geqslant 3$, the polynomial $h(x)$ can be replaced by Δg. Notice that

$$(\Delta g)(x)=\begin{cases}1 & \text{if } x\equiv -1 \pmod m\\ 0 & \text{otherwise}\end{cases}\qquad \text{for } x=0,1,\ldots,n(m-1)-1.$$

Hence, Δg vanishes at all integers x with $0\leqslant x<n(m-1)$ and $x\not\equiv -1 \pmod m$. This leads to $\deg g\geqslant \frac{(m-1)^2 n}{m}+1$.

If m is even then this lower bound can be improved to $n(m-1)$. For $0\leqslant N<n(m-1)$, the $(N+1)^{\text{st}}$ forward difference at $x=0$ is

$$\big(\Delta^{N+1}\big)g(0)=\sum_{k=0}^{N}(-1)^{N-k}\binom{N}{k}(\Delta g)(k)=\sum_{\substack{0\leqslant k\leqslant N\\ k\equiv -1 \pmod m}}(-1)^{N-k}\binom{N}{k}. \qquad (*)$$

Since m is even, all signs in the last sum are equal; with $N=n(m-1)-1$ this proves $\Delta^{n(m-1)}g(0)\neq 0$, indicating that $\deg g\geqslant n(m-1)$.

However, there are infinitely many cases when all terms in $(*)$ cancel out, for example if m is an odd divisor of $n+1$. In such cases, $\deg f$ can be less than $n(m-1)$.

Comment 3. The lemma is closely related to the so-called

> **Alon–Füredi bound.** *Let $S_1,\ldots,S_n$ be nonempty finite sets in a field and suppose that the polynomial $P(x_1,\ldots,x_n)$ vanishes at the points of the grid $S_1\times\ldots\times S_n$, except for a single point. Then* $\deg P\geqslant \sum_{i=1}^{n}\big(|S_i|-1\big)$.
>
> (A well-known application of the Alon–Füredi bound was the former IMO problem 2007/6. Since then, this result became popular among the students and is part of the IMO training for many IMO teams.)

The proof of the lemma can be replaced by an application of the Alon–Füredi bound as follows. Let $d=\deg G$, and let G_0 be the unique polynomial such that $G_0(x)=G(x)$ for $x\in\{0,1,\ldots,d-1\}$ but $\deg G_0<d$. The polynomials G_0 and G are different because they have different degrees, and they attain the same values at $0,1,\ldots,d-1$; that enforces $G_0(d)\neq G(d)$.

Choose some nonnegative integers $b_1,\ldots,b_n$ so that $b_1\leqslant a_1$, …, $b_n\leqslant a_n$, and $b_1+\ldots+b_n=d$, and consider the polynomial

$$H(x_1,\ldots,x_n)=F(x_1,\ldots,x_n)-G_0(x_1+\ldots+x_n)$$

on the grid $\{0,1,\ldots,b_1\}\times\ldots\times\{0,1,\ldots,b_n\}$.

At the point $(b_1,\ldots,b_n)$ we have $H(b_1,\ldots,b_n)=G(d)-G_0(d)\neq 0$. At all other points of the grid we have $F=G$ and therefore $H=G-G_0=0$. So, by the Alon–Füredi bound, $\deg H\geqslant b_1+\ldots+b_n=d$. Since $\deg G_0<d$, this implies $\deg F=\deg(H+G_0)=\deg H\geqslant d=\deg G$. □

A7. Find the maximal value of

$$S = \sqrt[3]{\frac{a}{b+7}} + \sqrt[3]{\frac{b}{c+7}} + \sqrt[3]{\frac{c}{d+7}} + \sqrt[3]{\frac{d}{a+7}},$$

where a, b, c, d are nonnegative real numbers which satisfy $a + b + c + d = 100$.

(Taiwan)

Answer: $\frac{8}{\sqrt[3]{7}}$, reached when (a, b, c, d) is a cyclic permutation of $(1, 49, 1, 49)$.

Solution 1. Since the value $8/\sqrt[3]{7}$ is reached, it suffices to prove that $S \leqslant 8/\sqrt[3]{7}$.

Assume that x, y, z, t is a permutation of the variables, with $x \leqslant y \leqslant z \leqslant t$. Then, by the rearrangement inequality,

$$S \leqslant \left(\sqrt[3]{\frac{x}{t+7}} + \sqrt[3]{\frac{t}{x+7}}\right) + \left(\sqrt[3]{\frac{y}{z+7}} + \sqrt[3]{\frac{z}{y+7}}\right).$$

Claim. The first bracket above does not exceed $\sqrt[3]{\frac{x+t+14}{7}}$.

Proof. Since

$$X^3 + Y^3 + 3XYZ - Z^3 = \frac{1}{2}(X + Y - Z)\left((X - Y)^2 + (X + Z)^2 + (Y + Z)^2\right),$$

the inequality $X + Y \leqslant Z$ is equivalent (when $X, Y, Z \geqslant 0$) to $X^3 + Y^3 + 3XYZ \leqslant Z^3$. Therefore, the claim is equivalent to

$$\frac{x}{t+7} + \frac{t}{x+7} + 3\sqrt[3]{\frac{xt(x+t+14)}{7(x+7)(t+7)}} \leqslant \frac{x+t+14}{7}.$$

Notice that

$$3\sqrt[3]{\frac{xt(x+t+14)}{7(x+7)(t+7)}} = 3\sqrt[3]{\frac{t(x+7)}{7(t+7)} \cdot \frac{x(t+7)}{7(x+7)} \cdot \frac{7(x+t+14)}{(t+7)(x+7)}}$$
$$\leqslant \frac{t(x+7)}{7(t+7)} + \frac{x(t+7)}{7(x+7)} + \frac{7(x+t+14)}{(t+7)(x+7)}$$

by the AM–GM inequality, so it suffices to prove

$$\frac{x}{t+7} + \frac{t}{x+7} + \frac{t(x+7)}{7(t+7)} + \frac{x(t+7)}{7(x+7)} + \frac{7(x+t+14)}{(t+7)(x+7)} \leqslant \frac{x+t+14}{7}.$$

A straightforward check verifies that the last inequality is in fact an equality. □

The claim leads now to

$$S \leqslant \sqrt[3]{\frac{x+t+14}{7}} + \sqrt[3]{\frac{y+z+14}{7}} \leqslant 2\sqrt[3]{\frac{x+y+z+t+28}{14}} = \frac{8}{\sqrt[3]{7}},$$

the last inequality being due to the AM–CM inequality (or to the fact that $\sqrt[3]{\cdot}$ is concave on $[0, \infty)$).

Solution 2. We present a different proof for the estimate $S \leqslant 8/\sqrt[3]{7}$.

Start by using Hölder's inequality:

$$S^3 = \left(\sum_{\text{cyc}} \frac{\sqrt[6]{a}\cdot\sqrt[6]{a}}{\sqrt[3]{b+7}}\right)^3 \leqslant \sum_{\text{cyc}}\left(\sqrt[6]{a}\right)^3 \cdot \sum_{\text{cyc}}\left(\sqrt[6]{a}\right)^3 \cdot \sum_{\text{cyc}}\left(\frac{1}{\sqrt[3]{b+7}}\right)^3 = \left(\sum_{\text{cyc}}\sqrt{a}\right)^2 \sum_{\text{cyc}}\frac{1}{b+7}.$$

Notice that

$$\frac{(x-1)^2(x-7)^2}{x^2+7} \geqslant 0 \iff x^2 - 16x + 71 \geqslant \frac{448}{x^2+7}$$

yields

$$\sum \frac{1}{b+7} \leqslant \frac{1}{448}\sum\left(b - 16\sqrt{b} + 71\right) = \frac{1}{448}\left(384 - 16\sum\sqrt{b}\right) = \frac{48 - 2\sum\sqrt{b}}{56}.$$

Finally,

$$S^3 \leqslant \frac{1}{56}\left(\sum\sqrt{a}\right)^2\left(48 - 2\sum\sqrt{a}\right) \leqslant \frac{1}{56}\left(\frac{\sum\sqrt{a} + \sum\sqrt{a} + \left(48 - 2\sum\sqrt{a}\right)}{3}\right)^3 = \frac{512}{7}$$

by the AM–GM inequality. The conclusion follows.

Comment. All the above works if we replace 7 and 100 with $k > 0$ and $2(k^2+1)$, respectively; in this case, the answer becomes

$$2\sqrt[3]{\frac{(k+1)^2}{k}}.$$

Even further, a linear substitution allows to extend the solutions to a version with 7 and 100 being replaced with arbitrary positive real numbers p and q satisfying $q \geqslant 4p$.

Combinatorics

C1. Let $n \geqslant 3$ be an integer. Prove that there exists a set S of $2n$ positive integers satisfying the following property: For every $m = 2, 3, \ldots, n$ the set S can be partitioned into two subsets with equal sums of elements, with one of subsets of cardinality m.

(Iceland)

Solution. We show that one of possible examples is the set

$$S = \{1 \cdot 3^k,\ 2 \cdot 3^k \colon k = 1, 2, \ldots, n-1\} \cup \left\{1,\ \frac{3^n + 9}{2} - 1\right\}.$$

It is readily verified that all the numbers listed above are distinct (notice that the last two are not divisible by 3).

The sum of elements in S is

$$\Sigma = 1 + \left(\frac{3^n + 9}{2} - 1\right) + \sum_{k=1}^{n-1}(1 \cdot 3^k + 2 \cdot 3^k) = \frac{3^n + 9}{2} + \sum_{k=1}^{n-1} 3^{k+1} = \frac{3^n + 9}{2} + \frac{3^{n+1} - 9}{2} = 2 \cdot 3^n.$$

Hence, in order to show that this set satisfies the problem requirements, it suffices to present, for every $m = 2, 3, \ldots, n$, an m-element subset $A_m \subset S$ whose sum of elements equals 3^n.

Such a subset is

$$A_m = \{2 \cdot 3^k \colon k = n - m + 1, n - m + 2, \ldots, n - 1\} \cup \{1 \cdot 3^{n-m+1}\}.$$

Clearly, $|A_m| = m$. The sum of elements in A_m is

$$3^{n-m+1} + \sum_{k=n-m+1}^{n-1} 2 \cdot 3^k = 3^{n-m+1} + \frac{2 \cdot 3^n - 2 \cdot 3^{n-m+1}}{2} = 3^n,$$

as required.

Comment. Let us present a more general construction. Let $s_1, s_2, \ldots, s_{2n-1}$ be a sequence of pairwise distinct positive integers satisfying $s_{2i+1} = s_{2i} + s_{2i-1}$ for all $i = 2, 3, \ldots, n-1$. Set $s_{2n} = s_1 + s_2 + \cdots + s_{2n-4}$.

Assume that s_{2n} is distinct from the other terms of the sequence. Then the set $S = \{s_1, s_2, \ldots, s_{2n}\}$ satisfies the problem requirements. Indeed, the sum of its elements is

$$\Sigma = \sum_{i=1}^{2n-4} s_i + (s_{2n-3} + s_{2n-2}) + s_{2n-1} + s_{2n} = s_{2n} + s_{2n-1} + s_{2n-1} + s_{2n} = 2s_{2n} + 2s_{2n-1}.$$

Therefore, we have

$$\frac{\Sigma}{2} = s_{2n} + s_{2n-1} = s_{2n} + s_{2n-2} + s_{2n-3} = s_{2n} + s_{2n-2} + s_{2n-4} + s_{2n-5} = \ldots,$$

which shows that the required sets A_m can be chosen as

$$A_m = \{s_{2n}, s_{2n-2}, \ldots, s_{2n-2m+4}, s_{2n-2m+3}\}.$$

So, the only condition to be satisfied is $s_{2n} \notin \{s_1, s_2, \ldots, s_{2n-1}\}$, which can be achieved in many different ways (e.g., by choosing properly the number s_1 after specifying $s_2, s_3, \ldots, s_{2n-1}$).

The solution above is an instance of this general construction. Another instance, for $n > 3$, is the set

$$\{F_1, F_2, \ldots, F_{2n-1}, F_1 + \cdots + F_{2n-4}\},$$

where $F_1 = 1$, $F_2 = 2$, $F_{n+1} = F_n + F_{n-1}$ is the usual Fibonacci sequence.

C2. Queenie and Horst play a game on a 20×20 chessboard. In the beginning the board is empty. In every turn, Horst places a black knight on an empty square in such a way that his new knight does not attack any previous knights. Then Queenie places a white queen on an empty square. The game gets finished when somebody cannot move.

Find the maximal positive K such that, regardless of the strategy of Queenie, Horst can put at least K knights on the board.

(Armenia)

Answer: $K = 20^2/4 = 100$. In case of a $4N \times 4M$ board, the answer is $K = 4NM$.

Solution. We show two strategies, one for Horst to place at least 100 knights, and another strategy for Queenie that prevents Horst from putting more than 100 knights on the board.

A strategy for Horst: *Put knights only on black squares, until all black squares get occupied.*

Colour the squares of the board black and white in the usual way, such that the white and black squares alternate, and let Horst put his knights on black squares as long as it is possible. Two knights on squares of the same colour never attack each other. The number of black squares is $20^2/2 = 200$. The two players occupy the squares in turn, so Horst will surely find empty black squares in his first 100 steps.

A strategy for Queenie: *Group the squares into cycles of length* 4*, and after each step of Horst, occupy the opposite square in the same cycle.*

Consider the squares of the board as vertices of a graph; let two squares be connected if two knights on those squares would attack each other. Notice that in a 4×4 board the squares can be grouped into 4 cycles of length 4, as shown in Figure 1. Divide the board into parts of size 4×4, and perform the same grouping in every part; this way we arrange the 400 squares of the board into 100 cycles (Figure 2).

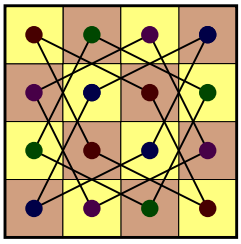


Figure 1

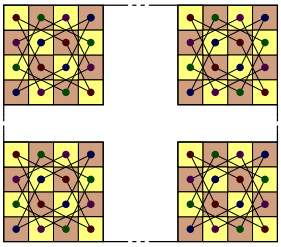


Figure 2

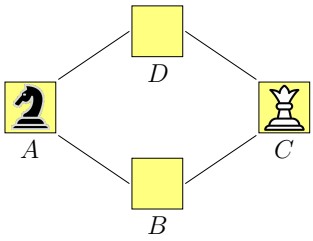


Figure 3

The strategy of Queenie can be as follows: Whenever Horst puts a new knight to a certain square A, which is part of some cycle $A - B - C - D - A$, let Queenie put her queen on the opposite square C in that cycle (Figure 3). From this point, Horst cannot put any knight on A or C because those squares are already occupied, neither on B or D because those squares are attacked by the knight standing on A. Hence, Horst can put at most one knight on each cycle, that is at most 100 knights in total.

Comment 1. Queenie's strategy can be prescribed by a simple rule: divide the board into 4×4 parts; whenever Horst puts a knight in a part P, Queenie reflects that square about the centre of P and puts her queen on the reflected square.

Comment 2. The result remains the same if Queenie moves first. In the first turn, she may put her first queen arbitrarily. Later, if she has to put her next queen on a square that already contains a queen, she may move arbitrarily again.

C3. Let n be a given positive integer. Sisyphus performs a sequence of turns on a board consisting of $n+1$ squares in a row, numbered 0 to n from left to right. Initially, n stones are put into square 0, and the other squares are empty. At every turn, Sisyphus chooses any nonempty square, say with k stones, takes one of those stones and moves it to the right by at most k squares (the stone should stay within the board). Sisyphus' aim is to move all n stones to square n.

Prove that Sisyphus cannot reach the aim in less than

$$\left\lceil \frac{n}{1} \right\rceil + \left\lceil \frac{n}{2} \right\rceil + \left\lceil \frac{n}{3} \right\rceil + \cdots + \left\lceil \frac{n}{n} \right\rceil$$

turns. (As usual, $\lceil x \rceil$ stands for the least integer not smaller than x.)

(Netherlands)

Solution. The stones are indistinguishable, and all have the same origin and the same final position. So, at any turn we can prescribe which stone from the chosen square to move. We do it in the following manner. Number the stones from 1 to n. At any turn, after choosing a square, Sisyphus moves the stone with the largest number from this square.

This way, when stone k is moved from some square, that square contains not more than k stones (since all their numbers are at most k). Therefore, stone k is moved by at most k squares at each turn. Since the total shift of the stone is exactly n, at least $\lceil n/k \rceil$ moves of stone k should have been made, for every $k = 1, 2, \ldots, n$.

By summing up over all $k = 1, 2, \ldots, n$, we get the required estimate.

Comment. The original submission contained the second part, asking for which values of n the equality can be achieved. The answer is $n = 1, 2, 3, 4, 5, 7$. The Problem Selection Committee considered this part to be less suitable for the competition, due to technicalities.

C4. An *anti-Pascal pyramid* is a finite set of numbers, placed in a triangle-shaped array so that the first row of the array contains one number, the second row contains two numbers, the third row contains three numbers and so on; and, except for the numbers in the bottom row, each number equals the absolute value of the difference of the two numbers below it. For instance, the triangle below is an anti-Pascal pyramid with four rows, in which every integer from 1 to $1+2+3+4=10$ occurs exactly once:

$$\begin{array}{cccccccc} & & & 4 & & & & \\ & & 2 & & 6 & & & \\ & 5 & & 7 & & 1 & & \\ 8 & & 3 & & 10 & & 9 & . \end{array}$$

Is it possible to form an anti-Pascal pyramid with 2018 rows, using every integer from 1 to $1+2+\cdots+2018$ exactly once?

(Iran)

Answer: No, it is not possible.

Solution. Let T be an anti-Pascal pyramid with n rows, containing every integer from 1 to $1+2+\cdots+n$, and let a_1 be the topmost number in T (Figure 1). The two numbers below a_1 are some a_2 and $b_2=a_1+a_2$, the two numbers below b_2 are some a_3 and $b_3=a_1+a_2+a_3$, and so on and so forth all the way down to the bottom row, where some a_n and $b_n=a_1+a_2+\cdots+a_n$ are the two neighbours below $b_{n-1}=a_1+a_2+\cdots+a_{n-1}$. Since the a_k are n pairwise distinct positive integers whose sum does not exceed the largest number in T, which is $1+2+\cdots+n$, it follows that they form a permutation of $1,2,\ldots,n$.

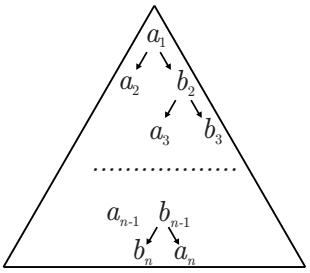


Figure 1

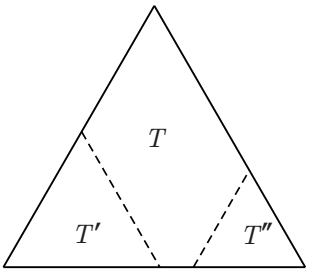


Figure 2

Consider now (Figure 2) the two 'equilateral' subtriangles of T whose bottom rows contain the numbers to the left, respectively right, of the pair a_n, b_n. (One of these subtriangles may very well be empty.) At least one of these subtriangles, say T', has side length $\ell \geqslant \lceil (n-2)/2 \rceil$. Since T' obeys the anti-Pascal rule, it contains ℓ pairwise distinct positive integers $a'_1, a'_2, \ldots, a'_\ell$, where a'_1 is at the apex, and a'_k and $b'_k = a'_1 + a'_2 + \cdots + a'_k$ are the two neighbours below b'_{k-1} for each $k = 2, 3 \ldots, \ell$. Since the a_k all lie outside T', and they form a permutation of $1, 2, \ldots, n$, the a'_k are all greater than n. Consequently,

$$\begin{aligned} b'_\ell &\geqslant (n+1) + (n+2) + \cdots + (n+\ell) = \frac{\ell(2n+\ell+1)}{2} \\ &\geqslant \frac{1}{2} \cdot \frac{n-2}{2} \left(2n + \frac{n-2}{2} + 1 \right) = \frac{5n(n-2)}{8}, \end{aligned}$$

which is greater than $1+2+\cdots+n = n(n+1)/2$ for $n = 2018$. A contradiction.

Comment. The above estimate may be slightly improved by noticing that $b'_\ell \neq b_n$. This implies $n(n+1)/2 = b_n > b'_\ell \geqslant \lceil (n-2)/2 \rceil \left(2n + \lceil (n-2)/2 \rceil + 1\right)/2$, so $n \leqslant 7$ if n is odd, and $n \leqslant 12$ if n is even. It seems that the largest anti-Pascal pyramid whose entries are a permutation of the integers from 1 to $1+2+\cdots+n$ has 5 rows.

C5. Let k be a positive integer. The organising committee of a tennis tournament is to schedule the matches for $2k$ players so that every two players play once, each day exactly one match is played, and each player arrives to the tournament site the day of his first match, and departs the day of his last match. For every day a player is present on the tournament, the committee has to pay 1 coin to the hotel. The organisers want to design the schedule so as to minimise the total cost of all players' stays. Determine this minimum cost.

(Russia)

Answer: The required minimum is $k(4k^2+k-1)/2$.

Solution 1. Enumerate the days of the tournament $1, 2, \ldots, \binom{2k}{2}$. Let $b_1 \leqslant b_2 \leqslant \cdots \leqslant b_{2k}$ be the days the players arrive to the tournament, arranged in *nondecreasing* order; similarly, let $e_1 \geqslant \cdots \geqslant e_{2k}$ be the days they depart arranged in *nonincreasing* order (it may happen that a player arrives on day b_i and departs on day e_j, where $i \neq j$). If a player arrives on day b and departs on day e, then his stay cost is $e-b+1$. Therefore, the total stay cost is

$$\Sigma = \sum_{i=1}^{2k} e_i - \sum_{i=1}^{2k} b_i + n = \sum_{i=1}^{2k} (e_i - b_i + 1).$$

Bounding the total cost from below. To this end, estimate $e_{i+1} - b_{i+1} + 1$. Before day b_{i+1}, only i players were present, so at most $\binom{i}{2}$ matches could be played. Therefore, $b_{i+1} \leqslant \binom{i}{2} + 1$. Similarly, at most $\binom{i}{2}$ matches could be played after day e_{i+1}, so $e_i \geqslant \binom{2k}{2} - \binom{i}{2}$. Thus,

$$e_{i+1} - b_{i+1} + 1 \geqslant \binom{2k}{2} - 2\binom{i}{2} = k(2k-1) - i(i-1).$$

This lower bound can be improved for $i > k$: List the i players who arrived first, and the i players who departed last; at least $2i-2k$ players appear in both lists. The matches between these players were counted twice, though the players in each pair have played only once. Therefore, if $i > k$, then

$$e_{i+1} - b_{i+1} + 1 \geqslant \binom{2k}{2} - 2\binom{i}{2} + \binom{2i-2k}{2} = (2k-i)^2.$$

An optimal tournament, We now describe a schedule in which the lower bounds above are all achieved simultaneously. Split players into two groups X and Y, each of cardinality k. Next, partition the schedule into three parts. During the first part, the players from X arrive one by one, and each newly arrived player immediately plays with everyone already present. During the third part (after all players from X have already departed) the players from Y depart one by one, each playing with everyone still present just before departing.

In the middle part, everyone from X should play with everyone from Y. Let $S_1, S_2, \ldots, S_k$ be the players in X, and let $T_1, T_2, \ldots, T_k$ be the players in Y. Let $T_1, T_2, \ldots, T_k$ arrive in this order; after T_j arrives, he immediately plays with all the S_i, $i > j$. Afterwards, players S_k, $S_{k-1}, \ldots, S_1$ depart in this order; each S_i plays with all the T_j, $i \leqslant j$, just before his departure, and S_k departs the day T_k arrives. For $0 \leqslant s \leqslant k-1$, the number of matches played between T_{k-s}'s arrival and S_{k-s}'s departure is

$$\sum_{j=k-s}^{k-1} (k-j) + 1 + \sum_{j=k-s}^{k-1} (k-j+1) = \frac{1}{2}\, s(s+1) + 1 + \frac{1}{2}\, s(s+3) = (s+1)^2.$$

Thus, if $i > k$, then the number of matches that have been played between T_{i-k+1}'s arrival, which is b_{i+1}, and S_{i-k+1}'s departure, which is e_{i+1}, is $(2k-i)^2$; that is, $e_{i+1} - b_{i+1} + 1 = (2k-i)^2$, showing the second lower bound achieved for all $i > k$.

If $i \leqslant k$, then the matches between the i players present before b_{i+1} all fall in the first part of the schedule, so there are $\binom{i}{2}$ such, and $b_{i+1} = \binom{i}{2} + 1$. Similarly, after e_{i+1}, there are i players left, all $\binom{i}{2}$ matches now fall in the third part of the schedule, and $e_{i+1} = \binom{2k}{2} - \binom{i}{2}$. The first lower bound is therefore also achieved for all $i \leqslant k$.

Consequently, all lower bounds are achieved simultaneously, and the schedule is indeed optimal.

Evaluation. Finally, evaluate the total cost for the optimal schedule:

$$\Sigma = \sum_{i=0}^{k} \big(k(2k-1) - i(i-1)\big) + \sum_{i=k+1}^{2k-1} (2k-i)^2 = (k+1)k(2k-1) - \sum_{i=0}^{k} i(i-1) + \sum_{j=1}^{k-1} j^2$$
$$= k(k+1)(2k-1) - k^2 + \frac{1}{2}\,k(k+1) = \frac{1}{2}\,k(4k^2+k-1).$$

Solution 2. Consider any tournament schedule. Label players P_1, P_2, …, P_{2k} in order of their arrival, and label them again Q_{2k}, Q_{2k-1}, …, Q_1 in order of their departure, to define a permutation a_1, a_2, …, a_{2k} of 1, 2, …, $2k$ by $P_i = Q_{a_i}$.

We first describe an optimal tournament for any given permutation a_1, a_2, …, a_{2k} of the indices 1, 2, …, $2k$. Next, we find an optimal permutation and an optimal tournament.

Optimisation for a fixed a_1, …, a_{2k}. We say that the *cost* of the match between P_i and P_j is the number of players present at the tournament when this match is played. Clearly, the Committee pays for each day the cost of the match of that day. Hence, we are to minimise the total cost of all matches.

Notice that Q_{2k}'s departure does not precede P_{2k}'s arrival. Hence, the number of players at the tournament monotonically increases (non-strictly) until it reaches $2k$, and then monotonically decreases (non-strictly). So, the best time to schedule the match between P_i and P_j is either when $P_{\max(i,j)}$ arrives, or when $Q_{\max(a_i,a_j)}$ departs, in which case the cost is $\min\big(\max(i,j), \max(a_i,a_j)\big)$.

Conversely, assuming that $i > j$, if this match is scheduled between the arrivals of P_i and P_{i+1}, then its cost will be exactly $i = \max(i,j)$. Similarly, one can make it cost $\max(a_i,a_j)$. Obviously, these conditions can all be simultaneously satisfied, so the minimal cost for a fixed sequence a_1, a_2, …, a_{2k} is

$$\Sigma(a_1,\dots,a_{2k}) = \sum_{1\leqslant i<j\leqslant 2k} \min\big(\max(i,j), \max(a_i,a_j)\big). \tag{1}$$

Optimising the sequence (a_i). Optimisation hinges on the lemma below.

Lemma. If $a \leqslant b$ and $c \leqslant d$, then

$$\min\big(\max(a,x), \max(c,y)\big) + \min\big(\max(b,x), \max(d,y)\big)$$
$$\geqslant \min\big(\max(a,x), \max(d,y)\big) + \min\big(\max(b,x), \max(c,y)\big).$$

Proof. Write $a' = \max(a,x) \leqslant \max(b,x) = b'$ and $c' = \max(c,y) \leqslant \max(d,y) = d'$ and check that $\min(a',c') + \min(b',d') \geqslant \min(a',d') + \min(b',c')$. □

Consider a permutation a_1, a_2, …, a_{2k} such that $a_i < a_j$ for some $i < j$. Swapping a_i and a_j does not change the (i,j)th summand in (1), and for $\ell \notin \{i,j\}$ the sum of the (i,ℓ)th and the (j,ℓ)th summands does not increase by the Lemma. Hence the optimal value does not increase, but the number of disorders in the permutation increases. This process stops when $a_i = 2k+1-i$ for all i, so the required minimum is

$$S(2k, 2k-1, \dots, 1) = \sum_{1\leqslant i<j\leqslant 2k} \min\big(\max(i,j), \max(2k+1-i, 2k+1-j)\big)$$
$$= \sum_{1\leqslant i<j\leqslant 2k} \min(j, 2k+1-i).$$

The latter sum is fairly tractable and yields the stated result; we omit the details.

Comment. If the number of players is odd, say, $2k-1$, the required minimum is $k(k-1)(4k-1)/2$. In this case, $|X| = k$, $|Y| = k-1$, the argument goes along the same lines, but some additional technicalities are to be taken care of.

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C6. Let a and b be distinct positive integers. The following infinite process takes place on an initially empty board.

(i) If there is at least a pair of equal numbers on the board, we choose such a pair and increase one of its components by a and the other by b.

(ii) If no such pair exists, we write down two times the number 0.

Prove that, no matter how we make the choices in (i), operation (ii) will be performed only finitely many times.

(Serbia)

Solution 1. We may assume $\gcd(a,b)=1$; otherwise we work in the same way with multiples of $d=\gcd(a,b)$.

Suppose that after N moves of type (ii) and some moves of type (i) we have to add two new zeros. For each integer k, denote by $f(k)$ the number of times that the number k appeared on the board up to this moment. Then $f(0)=2N$ and $f(k)=0$ for $k<0$. Since the board contains at most one $k-a$, every second occurrence of $k-a$ on the board produced, at some moment, an occurrence of k; the same stands for $k-b$. Therefore,

$$f(k)=\left\lfloor\frac{f(k-a)}{2}\right\rfloor+\left\lfloor\frac{f(k-b)}{2}\right\rfloor, \tag{1}$$

yielding

$$f(k)\geqslant\frac{f(k-a)+f(k-b)}{2}-1. \tag{2}$$

Since $\gcd(a,b)=1$, every integer $x>ab-a-b$ is expressible in the form $x=sa+tb$, with integer $s,t\geqslant 0$.

We will prove by induction on $s+t$ that if $x=sa+bt$, with s,t nonnegative integers, then

$$f(x)>\frac{f(0)}{2^{s+t}}-2. \tag{3}$$

The base case $s+t=0$ is trivial. Assume now that (3) is true for $s+t=v$. Then, if $s+t=v+1$ and $x=sa+tb$, at least one of the numbers s and t – say s – is positive, hence by (2),

$$f(x)=f(sa+tb)\geqslant\frac{f\big((s-1)a+tb\big)}{2}-1>\frac{1}{2}\left(\frac{f(0)}{2^{s+t-1}}-2\right)-1=\frac{f(0)}{2^{s+t}}-2.$$

Assume now that we must perform moves of type (ii) *ad infinitum*. Take $n=ab-a-b$ and suppose $b>a$. Since each of the numbers $n+1,n+2,\ldots,n+b$ can be expressed in the form $sa+tb$, with $0\leqslant s\leqslant b$ and $0\leqslant t\leqslant a$, after moves of type (ii) have been performed 2^{a+b+1} times and we have to add a new pair of zeros, each $f(n+k)$, $k=1,2,\ldots,b$, is at least 2. In this case (1) yields inductively $f(n+k)\geqslant 2$ for all $k\geqslant 1$. But this is absurd: after a finite number of moves, f cannot attain nonzero values at infinitely many points.

Solution 2. We start by showing that the result of the process in the problem does not depend on the way the operations are performed. For that purpose, it is convenient to modify the process a bit.

Claim 1. Suppose that the board initially contains a finite number of nonnegative integers, and one starts performing type (i) moves only. Assume that one had applied k moves which led to a *final* arrangement where no more type (i) moves are possible. Then, if one starts from the same initial arrangement, performing type (i) moves in an arbitrary fashion, then the process will necessarily stop at the same final arrangement

Proof. Throughout this proof, all moves are supposed to be of type (i).

Induct on k; the base case $k = 0$ is trivial, since no moves are possible. Assume now that $k \geqslant 1$. Fix some *canonical* process, consisting of k moves $M_1, M_2, \ldots, M_k$, and reaching the final arrangement A. Consider any *sample* process $m_1, m_2, \ldots$ starting with the same initial arrangement and proceeding as long as possible; clearly, it contains at least one move. We need to show that this process stops at A.

Let move m_1 consist in replacing two copies of x with $x + a$ and $x + b$. If move M_1 does the same, we may apply the induction hypothesis to the arrangement appearing after m_1. Otherwise, the canonical process should still contain at least one move consisting in replacing $(x, x) \mapsto (x + a, x + b)$, because the initial arrangement contains at least two copies of x, while the final one contains at most one such.

Let M_i be the first such move. Since the copies of x are indistinguishable and no other copy of x disappeared before M_i in the canonical process, the moves in this process can be permuted as $M_i, M_1, \ldots, M_{i-1}, M_{i+1}, \ldots, M_k$, without affecting the final arrangement. Now it suffices to perform the move $m_1 = M_i$ and apply the induction hypothesis as above. □

Claim 2. Consider any process starting from the empty board, which involved exactly n moves of type (ii) and led to a final arrangement where all the numbers are distinct. Assume that one starts with the board containing $2n$ zeroes (as if n moves of type (ii) were made in the beginning), applying type (i) moves in an arbitrary way. Then this process will reach the same final arrangement.

Proof. Starting with the board with $2n$ zeros, one may indeed model the first process mentioned in the statement of the claim, omitting the type (ii) moves. This way, one reaches the same final arrangement. Now, Claim 1 yields that this final arrangement will be obtained when type (i) moves are applied arbitrarily. □

Claim 2 allows now to reformulate the problem statement as follows: *There exists an integer n such that, starting from $2n$ zeroes, one may apply type (i) moves indefinitely.*

In order to prove this, we start with an obvious induction on $s + t = k \geqslant 1$ to show that if we start with 2^{s+t} zeros, then we can get simultaneously on the board, at some point, each of the numbers $sa + tb$, with $s + t = k$.

Suppose now that $a < b$. Then, an appropriate use of separate groups of zeros allows us to get two copies of each of the numbers $sa + tb$, with $1 \leqslant s, t \leqslant b$.

Define $N = ab - a - b$, and notice that after representing each of numbers $N + k$, $1 \leqslant k \leqslant b$, in the form $sa + tb$, $1 \leqslant s, t \leqslant b$ we can get, using enough zeros, the numbers $N + 1, N + 2, \ldots, N + a$ and the numbers $N + 1, N + 2, \ldots, N + b$.

From now on we can perform only moves of type (i). Indeed, if $n \geqslant N$, the occurrence of the numbers $n + 1, n + 2, \ldots, n + a$ and $n + 1, n + 2, \ldots, n + b$ and the replacement $(n + 1, n + 1) \mapsto (n + b + 1, n + a + 1)$ leads to the occurrence of the numbers $n + 2, n + 3, \ldots, n + a + 1$ and $n + 2, n + 3, \ldots, n + b + 1$.

Comment. The proofs of Claims 1 and 2 may be extended in order to show that in fact the number of moves in the canonical process is the same as in an arbitrary sample one.

C7. Consider 2018 pairwise crossing circles no three of which are concurrent. These circles subdivide the plane into regions bounded by circular *edges* that meet at *vertices*. Notice that there are an even number of vertices on each circle. Given the circle, alternately colour the vertices on that circle red and blue. In doing so for each circle, every vertex is coloured twice — once for each of the two circles that cross at that point. If the two colourings agree at a vertex, then it is assigned that colour; otherwise, it becomes yellow. Show that, if some circle contains at least 2061 yellow points, then the vertices of some region are all yellow.

(India)

Solution 1. Letting $n = 2018$, we will show that, if every region has at least one non-yellow vertex, then every circle contains at most $n + \lfloor\sqrt{n-2}\rfloor - 2$ yellow points. In the case at hand, the latter equals $2018 + 44 - 2 = 2060$, contradicting the hypothesis.

Consider the natural geometric graph G associated with the configuration of n circles. Fix any circle C in the configuration, let k be the number of yellow points on C, and find a suitable lower bound for the total number of yellow vertices of G in terms of k and n. It turns out that k is even, and G has at least

$$k + 2\binom{k/2}{2} + 2\binom{n - k/2 - 1}{2} = \frac{k^2}{2} - (n-2)k + (n-2)(n-1) \qquad (*)$$

yellow vertices. The proof hinges on the two lemmata below.

Lemma 1. Let two circles in the configuration cross at x and y. Then x and y are either both yellow or both non-yellow.

Proof. This is because the numbers of interior vertices on the four arcs x and y determine on the two circles have like parities. □

In particular, each circle in the configuration contains an even number of yellow vertices.

Lemma 2. If $\overset{\frown}{xy}$, $\overset{\frown}{yz}$, and $\overset{\frown}{zx}$ are circular arcs of three pairwise distinct circles in the configuration, then the number of yellow vertices in the set $\{x, y, z\}$ is odd.

Proof. Let C_1, C_2, C_3 be the three circles under consideration. Assume, without loss of generality, that C_2 and C_3 cross at x, C_3 and C_1 cross at y, and C_1 and C_2 cross at z. Let k_1, k_2, k_3 be the numbers of interior vertices on the three circular arcs under consideration. Since each circle in the configuration, different from the C_i, crosses the cycle $\overset{\frown}{xy} \cup \overset{\frown}{yz} \cup \overset{\frown}{zx}$ at an even number of points (recall that no three circles are concurrent), and self-crossings are counted twice, the sum $k_1 + k_2 + k_3$ is even.

Let Z_1 be the colour z gets from C_1 and define the other colours similarly. By the preceding, the number of bichromatic pairs in the list (Z_1, Y_1), (X_2, Z_2), (Y_3, X_3) is odd. Since the total number of colour changes in a cycle Z_1–Y_1–Y_3–X_3–X_2–Z_2–Z_1 is even, the number of bichromatic pairs in the list (X_2, X_3), (Y_1, Y_3), (Z_1, Z_2) is odd, and the lemma follows. □

We are now in a position to prove that $(*)$ bounds the total number of yellow vertices from below. Refer to Lemma 1 to infer that the k yellow vertices on C pair off to form the pairs of points where C is crossed by $k/2$ circles in the configuration. By Lemma 2, these circles cross pairwise to account for another $2\binom{k/2}{2}$ yellow vertices. Finally, the remaining $n - k/2 - 1$ circles in the configuration cross C at non-yellow vertices, by Lemma 1, and Lemma 2 applies again to show that these circles cross pairwise to account for yet another $2\binom{n-k/2-1}{2}$ yellow vertices. Consequently, there are at least $(*)$ yellow vertices.

Next, notice that G is a plane graph on $n(n-1)$ degree 4 vertices, having exactly $2n(n-1)$ edges and exactly $n(n-1) + 2$ faces (regions), the outer face inclusive (by Euler's formula for planar graphs).

Lemma 3. Each face of G has equally many red and blue vertices. In particular, each face has an even number of non-yellow vertices.

Proof. Trace the boundary of a face once in circular order, and consider the colours each vertex is assigned in the colouring of the two circles that cross at that vertex, to infer that colours of non-yellow vertices alternate. □

Consequently, if each region has at least one non-yellow vertex, then it has at least two such. Since each vertex of G has degree 4, consideration of vertex-face incidences shows that G has at least $n(n-1)/2+1$ non-yellow vertices, and hence at most $n(n-1)/2-1$ yellow vertices. (In fact, Lemma 3 shows that there are at least $n(n-1)/4+1/2$ red, respectively blue, vertices.)

Finally, recall the lower bound $(*)$ for the total number of yellow vertices in G, to write $n(n-1)/2-1 \geqslant k^2/2-(n-2)k+(n-2)(n-1)$, and conclude that $k \leqslant n+\lfloor\sqrt{n-2}\rfloor-2$, as claimed in the first paragraph.

Solution 2. The first two lemmata in Solution 1 show that the circles in the configuration split into two classes: Consider any circle C along with all circles that cross C at yellow points to form one class; the remaining circles then form the other class. Lemma 2 shows that any pair of circles in the same class cross at yellow points; otherwise, they cross at non-yellow points.

Call the circles from the two classes *white* and *black*, respectively. Call a region *yellow* if its vertices are all yellow. Let w and b be the numbers of white and black circles, respectively; clearly, $w+b=n$. Assume that $w \geqslant b$, and that there is no yellow region. Clearly, $b \geqslant 1$, otherwise each region is yellow. The white circles subdivide the plane into $w(w-1)+2$ larger regions — call them *white*. The white regions (or rather their boundaries) subdivide each black circle into *black arcs*. Since there are no yellow regions, each white region contains at least one black arc.

Consider any white region; let it contain $t \geqslant 1$ black arcs. We claim that the number of points at which these t arcs cross does not exceed $t-1$. To prove this, consider a multigraph whose vertices are these black arcs, two vertices being joined by an edge for each point at which the corresponding arcs cross. If this graph had more than $t-1$ edges, it would contain a cycle, since it has t vertices; this cycle would correspond to a closed contour formed by black sub-arcs, lying inside the region under consideration. This contour would, in turn, define at least one yellow region, which is impossible.

Let t_i be the number of black arcs inside the ith white region. The total number of black arcs is $\sum_i t_i = 2wb$, and they cross at $2\binom{b}{2} = b(b-1)$ points. By the preceding,

$$b(b-1) \leqslant \sum_{i=1}^{w^2-w+2} (t_i-1) = \sum_{i=1}^{w^2-w+2} t_i - (w^2-w+2) = 2wb-(w^2-w+2),$$

or, equivalently, $(w-b)^2 \leqslant w+b-2 = n-2$, which is the case if and only if $w-b \leqslant \lfloor\sqrt{n-2}\rfloor$. Consequently, $b \leqslant w \leqslant \left(n+\lfloor\sqrt{n-2}\rfloor\right)/2$, so there are at most $2(w-1) \leqslant n+\lfloor\sqrt{n-2}\rfloor-2$ yellow vertices on each circle — a contradiction.

Geometry

G1. Let ABC be an acute-angled triangle with circumcircle Γ. Let D and E be points on the segments AB and AC, respectively, such that $AD = AE$. The perpendicular bisectors of the segments BD and CE intersect the small arcs $\overset{\frown}{AB}$ and $\overset{\frown}{AC}$ at points F and G respectively. Prove that $DE \parallel FG$.

(Greece)

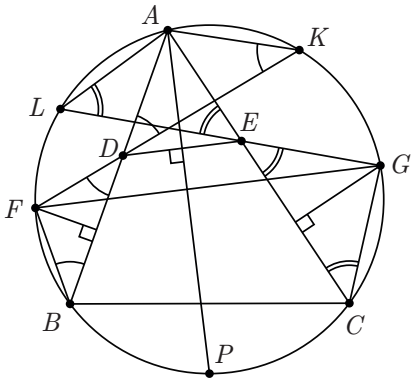
Solution 1. In the sequel, all the considered arcs are small arcs.

Let P be the midpoint of the arc $\overset{\frown}{BC}$. Then AP is the bisector of $\angle BAC$, hence, in the isosceles triangle ADE, $AP \perp DE$. So, the statement of the problem is equivalent to $AP \perp FG$.

In order to prove this, let K be the second intersection of Γ with FD. Then the triangle FBD is isosceles, therefore

$$\angle AKF = \angle ABF = \angle FDB = \angle ADK,$$

yielding $AK = AD$. In the same way, denoting by L the second intersection of Γ with GE, we get $AL = AE$. This shows that $AK = AL$.



Now $\angle FBD = \angle FDB$ gives $\overset{\frown}{AF} = \overset{\frown}{BF} + \overset{\frown}{AK} = \overset{\frown}{BF} + \overset{\frown}{AL}$, hence $\overset{\frown}{BF} = \overset{\frown}{LF}$. In a similar way, we get $\overset{\frown}{CG} = \overset{\frown}{GK}$. This yields

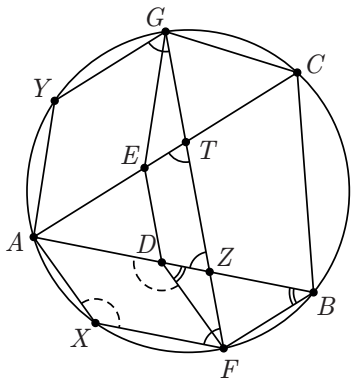
$$\angle(AP, FG) = \frac{\overset{\frown}{AF} + \overset{\frown}{PG}}{2} = \frac{\overset{\frown}{AL} + \overset{\frown}{LF} + \overset{\frown}{PC} + \overset{\frown}{CG}}{2} = \frac{\overset{\frown}{KL} + \overset{\frown}{LB} + \overset{\frown}{BC} + \overset{\frown}{CK}}{4} = 90^\circ.$$

Solution 2. Let $Z = AB \cap FG$, $T = AC \cap FG$. It suffices to prove that $\angle ATZ = \angle AZT$.

Let X be the point for which $FXAD$ is a parallelogram. Then

$$\angle FXA = \angle FDA = 180^\circ - \angle FDB = 180^\circ - \angle FBD,$$

where in the last equality we used that $FD = FB$. It follows that the quadrilateral $BFXA$ is cyclic, so X lies on Γ.

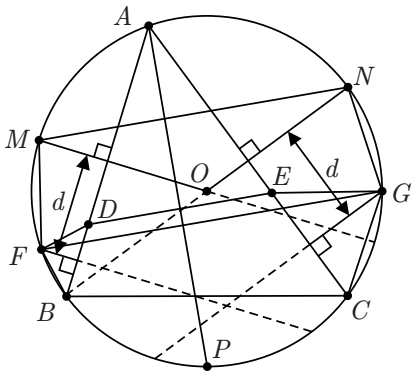


Analogously, if Y is the point for which $GYAE$ is a parallelogram, then Y lies on Γ. So the quadrilateral $XFGY$ is cyclic and $FX = AD = AE = GY$, hence $XFGY$ is an isosceles trapezoid.

Now, by $XF \parallel AZ$ and $YG \parallel AT$, it follows that $\angle ATZ = \angle YGF = \angle XFG = \angle AZT$.

Solution 3. As in the first solution, we prove that $FG \perp AP$, where P is the midpoint of the small arc $\widehat{BC}$.

Let O be the circumcentre of the triangle ABC, and let M and N be the midpoints of the small arcs $\widehat{AB}$ and $\widehat{AC}$, respectively. Then OM and ON are the perpendicular bisectors of AB and AC, respectively.



The distance d between OM and the perpendicular bisector of BD is $\frac{1}{2}AB - \frac{1}{2}BD = \frac{1}{2}AD$, hence it is equal to the distance between ON and the perpendicular bisector of CE.

This shows that the isosceles trapezoid determined by the diameter δ of Γ through M and the chord parallel to δ through F is congruent to the isosceles trapezoid determined by the diameter δ' of Γ through N and the chord parallel to δ' through G. Therefore $MF = NG$, yielding $MN \parallel FG$.

Now

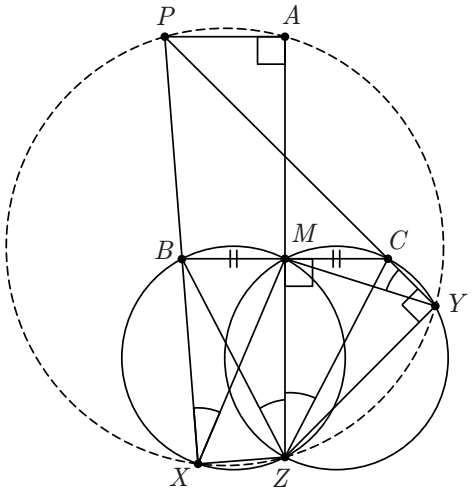
$$\angle(MN, AP) = \frac{1}{2}\left(\widehat{AM} + \widehat{PC} + \widehat{CN}\right) = \frac{1}{4}\left(\widehat{AB} + \widehat{BC} + \widehat{CA}\right) = 90^\circ,$$

hence $MN \perp AP$, and the conclusion follows.

G2. Let ABC be a triangle with $AB = AC$, and let M be the midpoint of BC. Let P be a point such that $PB < PC$ and PA is parallel to BC. Let X and Y be points on the lines PB and PC, respectively, so that B lies on the segment PX, C lies on the segment PY, and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic.

(Australia)

Solution. Since $AB = AC$, AM is the perpendicular bisector of BC, hence $\angle PAM = \angle AMC = 90^\circ$.



Now let Z be the common point of AM and the perpendicular through Y to PC (notice that Z lies on to the ray AM beyond M). We have $\angle PAZ = \angle PYZ = 90^\circ$. Thus the points P, A, Y, and Z are concyclic.

Since $\angle CMZ = \angle CYZ = 90^\circ$, the quadrilateral $CYZM$ is cyclic, hence $\angle CZM = \angle CYM$. By the condition in the statement, $\angle CYM = \angle BXM$, and, by symmetry in ZM, $\angle CZM = \angle BZM$. Therefore, $\angle BXM = \angle BZM$. It follows that the points B, X, Z, and M are concyclic, hence $\angle BXZ = 180^\circ - \angle BMZ = 90^\circ$.

Finally, we have $\angle PXZ = \angle PYZ = \angle PAZ = 90^\circ$, hence the five points P, A, X, Y, Z are concyclic. In particular, the quadrilateral $APXY$ is cyclic, as required.

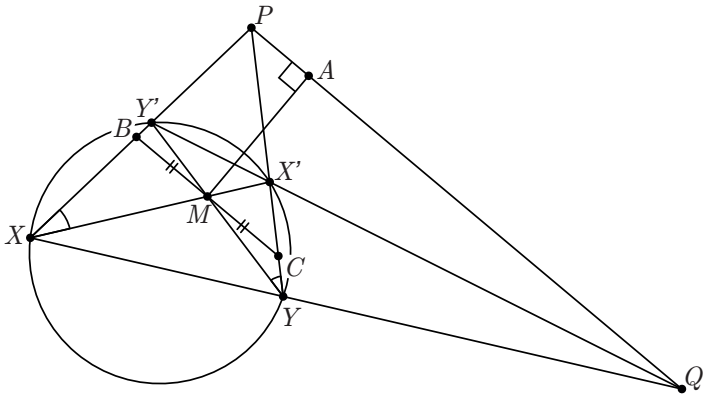
Comment 1. Clearly, the key point Z from the solution above can be introduced in several different ways, e.g., as the second meeting point of the circle CMY and the line AM, or as the second meeting point of the circles CMY and BMX, etc.

For some of definitions of Z its location is not obvious. For instance, if Z is defined as a common point of AM and the perpendicular through X to PX, it is not clear that Z lies on the ray AM beyond M. To avoid such slippery details some more restrictions on the construction may be required.

Comment 2. Let us discuss a connection to the Miquel point of a cyclic quadrilateral. Set $X' = MX \cap PC$, $Y' = MY \cap PB$, and $Q = XY \cap X'Y'$ (see the figure below).

We claim that $BC \parallel PQ$. (One way of proving this is the following. Notice that the quadruple of lines PX, PM, PY, PQ is harmonic, hence the quadruple B, M, C, $PQ \cap BC$ of their intersection points with BC is harmonic. Since M is the midpoint of BC, $PQ \cap BC$ is an ideal point, i.e., $PQ \parallel BC$.)

It follows from the given equality $\angle PXM = \angle PYM$ that the quadrilateral $XYX'Y'$ is cyclic. Note that A is the projection of M onto PQ. By a known description, A is the Miquel point for the sidelines $XY, XY', X'Y, X'Y'$. In particular, the circle PXY passes through A.



Comment 3. An alternative approach is the following. One can note that the (oriented) lengths of the segments CY and BX are both linear functions of a parameter $t = \cot \angle PXM$. As t varies, the intersection point S of the perpendicular bisectors of PX and PY traces a fixed line, thus the family of circles PXY has a fixed common point (other than P). By checking particular cases, one can show that this fixed point is A.

Comment 4. The problem states that $\angle PXM = \angle PYM$ implies that $APXY$ is cyclic. The original submission claims that these two conditions are in fact equivalent. The Problem Selection Committee omitted the converse part, since it follows easily from the direct one, by reversing arguments.

G3. A circle ω of radius 1 is given. A collection T of triangles is called *good*, if the following conditions hold:

(i) each triangle from T is inscribed in ω;

(ii) no two triangles from T have a common interior point.

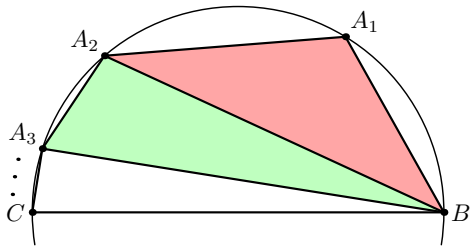
Determine all positive real numbers t such that, for each positive integer n, there exists a good collection of n triangles, each of perimeter greater than t.

(South Africa)

Answer: $t \in (0, 4]$.

Solution. First, we show how to construct a good collection of n triangles, each of perimeter greater than 4. This will show that all $t \leqslant 4$ satisfy the required conditions.

Construct inductively an $(n+2)$-gon $BA_1A_2 \ldots A_nC$ inscribed in ω such that BC is a diameter, and BA_1A_2, BA_2A_3, $\ldots$, $BA_{n-1}A_n$, BA_nC is a good collection of n triangles. For $n = 1$, take any triangle BA_1C inscribed in ω such that BC is a diameter; its perimeter is greater than $2BC = 4$. To perform the inductive step, assume that the $(n+2)$-gon $BA_1A_2 \ldots A_nC$ is already constructed. Since $A_nB + A_nC + BC > 4$, one can choose a point A_{n+1} on the small arc $\overset{\frown}{CA_n}$, close enough to C, so that $A_nB + A_nA_{n+1} + BA_{n+1}$ is still greater than 4. Thus each of these new triangles BA_nA_{n+1} and $BA_{n+1}C$ has perimeter greater than 4, which completes the induction step.



We proceed by showing that no $t > 4$ satisfies the conditions of the problem. To this end, we assume that there exists a good collection T of n triangles, each of perimeter greater than t, and then bound n from above.

Take $\varepsilon > 0$ such that $t = 4 + 2\varepsilon$.

Claim. There exists a positive constant $\sigma = \sigma(\varepsilon)$ such that any triangle Δ with perimeter $2s \geqslant 4 + 2\varepsilon$, inscribed in ω, has area $S(\Delta)$ at least σ.

Proof. Let a, b, c be the side lengths of Δ. Since Δ is inscribed in ω, each side has length at most 2. Therefore, $s - a \geqslant (2 + \varepsilon) - 2 = \varepsilon$. Similarly, $s - b \geqslant \varepsilon$ and $s - c \geqslant \varepsilon$. By Heron's formula, $S(\Delta) = \sqrt{s(s-a)(s-b)(s-c)} \geqslant \sqrt{(2+\varepsilon)\varepsilon^3}$. Thus we can set $\sigma(\varepsilon) = \sqrt{(2+\varepsilon)\varepsilon^3}$. □

Now we see that the total area S of all triangles from T is at least $n\sigma(\varepsilon)$. On the other hand, S does not exceed the area of the disk bounded by ω. Thus $n\sigma(\varepsilon) \leqslant \pi$, which means that n is bounded from above.

Comment 1. One may prove the Claim using the formula $S = \dfrac{abc}{4R}$ instead of Heron's formula.

Comment 2. In the statement of the problem condition (i) could be replaced by a weaker one: each triangle from T lies within ω. This does not affect the solution above, but reduces the number of ways to prove the Claim.

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G4. A point T is chosen inside a triangle ABC. Let A_1, B_1, and C_1 be the reflections of T in BC, CA, and AB, respectively. Let Ω be the circumcircle of the triangle $A_1B_1C_1$. The lines A_1T, B_1T, and C_1T meet Ω again at A_2, B_2, and C_2, respectively. Prove that the lines AA_2, BB_2, and CC_2 are concurrent on Ω.

(Mongolia)

Solution. By $\measuredangle(\ell, n)$ we always mean the directed angle of the lines ℓ and n, taken modulo 180°.

Let CC_2 meet Ω again at K (as usual, if CC_2 is tangent to Ω, we set $T = C_2$). We show that the line BB_2 contains K; similarly, AA_2 will also pass through K. For this purpose, it suffices to prove that

$$\measuredangle(C_2C, C_2A_1) = \measuredangle(B_2B, B_2A_1). \tag{1}$$

By the problem condition, CB and CA are the perpendicular bisectors of TA_1 and TB_1, respectively. Hence, C is the circumcentre of the triangle A_1TB_1. Therefore,

$$\measuredangle(CA_1, CB) = \measuredangle(CB, CT) = \measuredangle(B_1A_1, B_1T) = \measuredangle(B_1A_1, B_1B_2).$$

In circle Ω we have $\measuredangle(B_1A_1, B_1B_2) = \measuredangle(C_2A_1, C_2B_2)$. Thus,

$$\measuredangle(CA_1, CB) = \measuredangle(B_1A_1, B_1B_2) = \measuredangle(C_2A_1, C_2B_2). \tag{2}$$

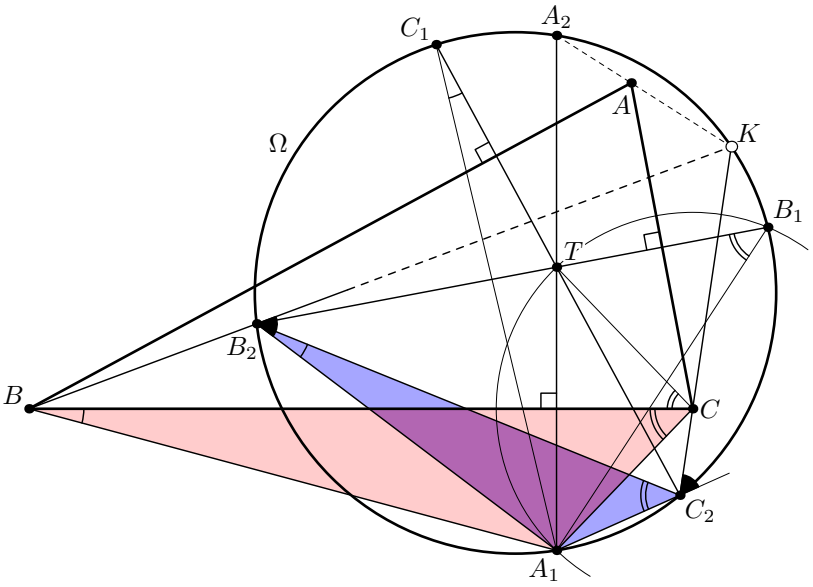
Similarly, we get

$$\measuredangle(BA_1, BC) = \measuredangle(C_1A_1, C_1C_2) = \measuredangle(B_2A_1, B_2C_2). \tag{3}$$

The two obtained relations yield that the triangles A_1BC and $A_1B_2C_2$ are similar and equioriented, hence

$$\frac{A_1B_2}{A_1B} = \frac{A_1C_2}{A_1C} \quad \text{and} \quad \measuredangle(A_1B, A_1C) = \measuredangle(A_1B_2, A_1C_2).$$

The second equality may be rewritten as $\measuredangle(A_1B, A_1B_2) = \measuredangle(A_1C, A_1C_2)$, so the triangles A_1BB_2 and A_1CC_2 are also similar and equioriented. This establishes (1).



Comment 1. In fact, the triangle A_1BC is an image of $A_1B_2C_2$ under a spiral similarity centred at A_1; in this case, the triangles ABB_2 and ACC_2 are also spirally similar with the same centre.

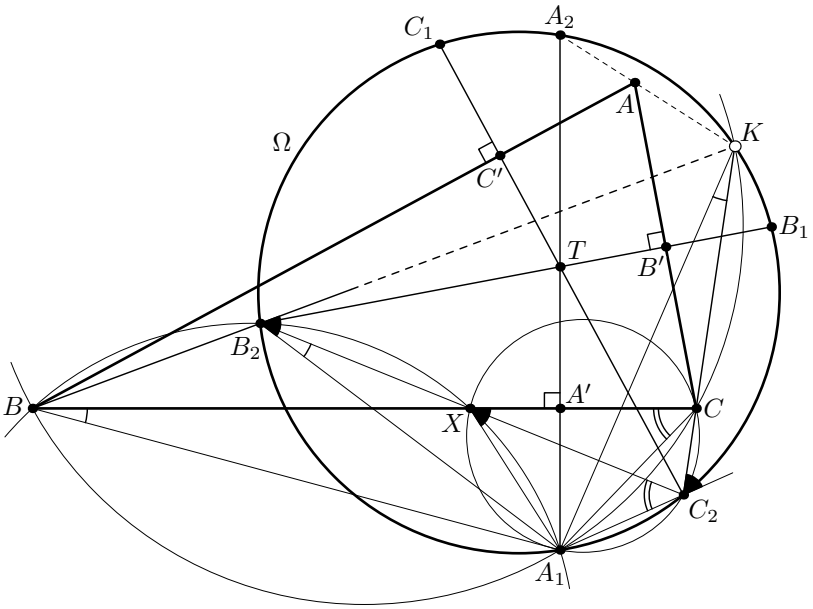
Comment 2. After obtaining (2) and (3), one can finish the solution in different ways.

For instance, introducing the point $X = BC \cap B_2C_2$, one gets from these relations that the 4-tuples (A_1, B, B_2, X) and (A_1, C, C_2, X) are both cyclic. Therefore, K is the Miquel point of the lines BB_2, CC_2, BC, and B_2C_2; this yields that the meeting point of BB_2 and CC_2 lies on Ω.

Yet another way is to show that the points A_1, B, C, and K are concyclic, as

$$\measuredangle(KC, KA_1) = \measuredangle(B_2C_2, B_2A_1) = \measuredangle(BC, BA_1).$$

By symmetry, the second point K' of intersection of BB_2 with Ω is also concyclic to A_1, B, and C, hence $K' = K$.



Comment 3. The requirement that the common point of the lines AA_2, BB_2, and CC_2 should lie on Ω may seem to make the problem easier, since it suggests some approaches. On the other hand, there are also different ways of showing that the lines AA_2, BB_2, and CC_2 are just concurrent.

In particular, the problem conditions yield that the lines A_2T, B_2T, and C_2T are perpendicular to the corresponding sides of the triangle ABC. One may show that the lines AT, BT, and CT are also perpendicular to the corresponding sides of the triangle $A_2B_2C_2$, i.e., the triangles ABC and $A_2B_2C_2$ are *orthologic*, and their orthology centres coincide. It is known that such triangles are also *perspective*, i.e. the lines AA_2, BB_2, and CC_2 are concurrent (in projective sense).

To show this mutual orthology, one may again apply angle chasing, but there are also other methods. Let A', B', and C' be the projections of T onto the sides of the triangle ABC. Then $A_2T \cdot TA' = B_2T \cdot TB' = C_2T \cdot TC'$, since all three products equal (minus) half the power of T with respect to Ω. This means that A_2, B_2, and C_2 are the poles of the sidelines of the triangle ABC with respect to some circle centred at T and having *pure imaginary* radius (in other words, the reflections of A_2, B_2, and C_2 in T are the poles of those sidelines with respect to some regular circle centred at T). Hence, dually, the vertices of the triangle ABC are also the poles of the sidelines of the triangle $A_2B_2C_2$.

G5. Let ABC be a triangle with circumcircle ω and incentre I. A line ℓ intersects the lines AI, BI, and CI at points D, E, and F, respectively, distinct from the points A, B, C, and I. The perpendicular bisectors x, y, and z of the segments AD, BE, and CF, respectively determine a triangle Θ. Show that the circumcircle of the triangle Θ is tangent to ω.

(Denmark)

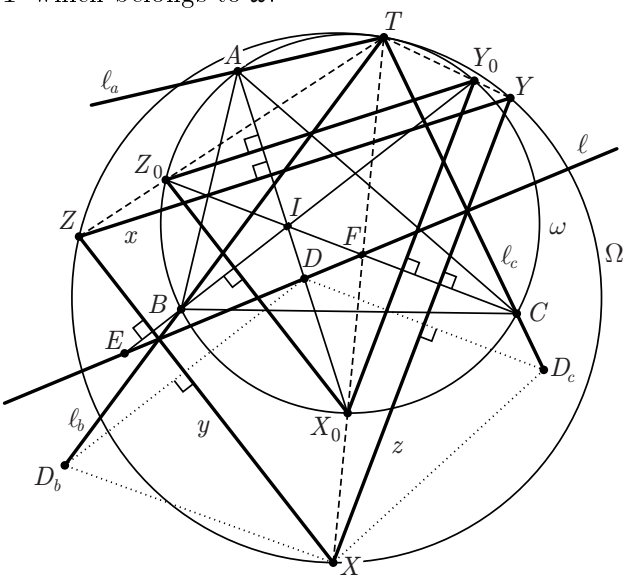
Preamble. Let $X = y \cap z$, $Y = x \cap z$, $Z = x \cap y$ and let Ω denote the circumcircle of the triangle XYZ. Denote by X_0, Y_0, and Z_0 the second intersection points of AI, BI and CI, respectively, with ω. It is known that Y_0Z_0 is the perpendicular bisector of AI, Z_0X_0 is the perpendicular bisector of BI, and X_0Y_0 is the perpendicular bisector of CI. In particular, the triangles XYZ and $X_0Y_0Z_0$ are homothetic, because their corresponding sides are parallel.

The solutions below mostly exploit the following approach. Consider the triangles XYZ and $X_0Y_0Z_0$, or some other pair of homothetic triangles Δ and δ inscribed into Ω and ω, respectively. In order to prove that Ω and ω are tangent, it suffices to show that the centre T of the homothety taking Δ to δ lies on ω (or Ω), or, in other words, to show that Δ and δ are perspective (i.e., the lines joining corresponding vertices are concurrent), with their perspector lying on ω (or Ω).

We use directed angles throughout all the solutions.

Solution 1.

Claim 1. The reflections ℓ_a, ℓ_b and ℓ_c of the line ℓ in the lines x, y, and z, respectively, are concurrent at a point T which belongs to ω.



Proof. Notice that $\measuredangle(\ell_b, \ell_c) = \measuredangle(\ell_b, \ell) + \measuredangle(\ell, \ell_c) = 2\measuredangle(y, \ell) + 2\measuredangle(\ell, z) = 2\measuredangle(y, z)$. But $y \perp BI$ and $z \perp CI$ implies $\measuredangle(y, z) = \measuredangle(BI, IC)$, so, since $2\measuredangle(BI, IC) = \measuredangle(BA, AC)$, we obtain

$$\measuredangle(\ell_b, \ell_c) = \measuredangle(BA, AC). \tag{1}$$

Since A is the reflection of D in x, A belongs to ℓ_a; similarly, B belongs to ℓ_b. Then (1) shows that the common point T' of ℓ_a and ℓ_b lies on ω; similarly, the common point T'' of ℓ_c and ℓ_b lies on ω.

If $B \notin \ell_a$ and $B \notin \ell_c$, then T' and T'' are the second point of intersection of ℓ_b and ω, hence they coincide. Otherwise, if, say, $B \in \ell_c$, then $\ell_c = BC$, so $\measuredangle(BA, AC) = \measuredangle(\ell_b, \ell_c) = \measuredangle(\ell_b, BC)$, which shows that ℓ_b is tangent at B to ω and $T' = T'' = B$. So T' and T'' coincide in all the cases, and the conclusion of the claim follows. □

Now we prove that X, X_0, T are collinear. Denote by D_b and D_c the reflections of the point D in the lines y and z, respectively. Then D_b lies on ℓ_b, D_c lies on ℓ_c, and

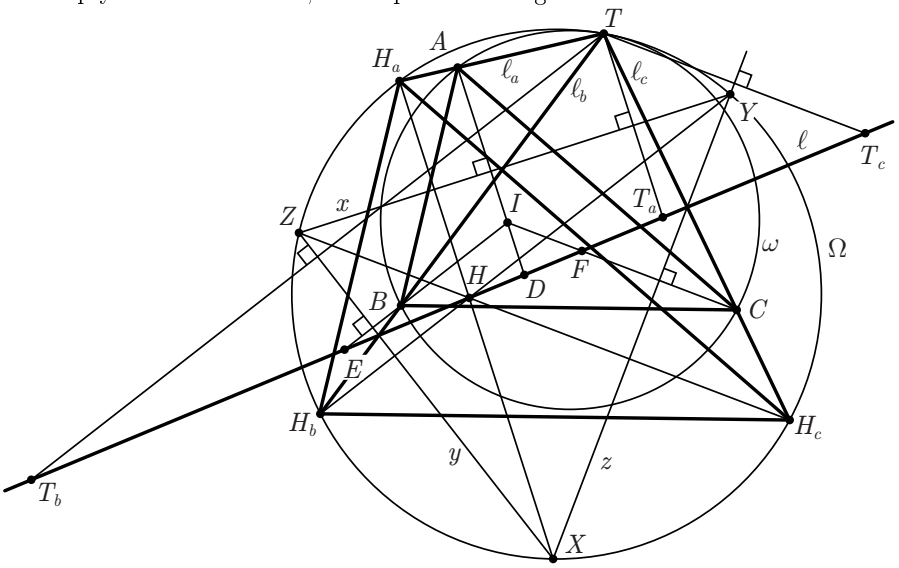
$$\begin{aligned}\measuredangle(D_bX, XD_c) &= \measuredangle(D_bX, DX) + \measuredangle(DX, XD_c) = 2\measuredangle(y, DX) + 2\measuredangle(DX, z) = 2\measuredangle(y, z)\\ &= \measuredangle(BA, AC) = \measuredangle(BT, TC),\end{aligned}$$

hence the quadrilateral XD_bTD_c is cyclic. Notice also that since $XD_b = XD = XD_c$, the points D, D_b, D_c lie on a circle with centre X. Using in this circle the diameter D_cD_c' yields $\measuredangle(D_bD_c, D_cX) = 90^\circ + \measuredangle(D_bD_c', D_c'X) = 90^\circ + \measuredangle(D_bD, DD_c)$. Therefore,

$$\begin{aligned}\measuredangle(\ell_b, XT) = \measuredangle(D_bT, XT) = \measuredangle(D_bD_c, D_cX) = 90^\circ + \measuredangle(D_bD, DD_c)\\ = 90^\circ + \measuredangle(BI, IC) = \measuredangle(BA, AI) = \measuredangle(BA, AX_0) = \measuredangle(BT, TX_0) = \measuredangle(\ell_b, X_0T),\end{aligned}$$

so the points X, X_0, T are collinear. By a similar argument, Y, Y_0, T and Z, Z_0, T are collinear. As mentioned in the preamble, the statement of the problem follows.

Comment 1. After proving Claim 1 one may proceed in another way. As it was shown, the reflections of ℓ in the sidelines of XYZ are concurrent at T. Thus ℓ is the Steiner line of T with respect to ΔXYZ (that is the line containing the reflections T_a, T_b, T_c of T in the sidelines of XYZ). The properties of the Steiner line imply that T lies on Ω, and ℓ passes through the orthocentre H of the triangle XYZ.



Let H_a, H_b, and H_c be the reflections of the point H in the lines x, y, and z, respectively. Then the triangle $H_aH_bH_c$ is inscribed in Ω and homothetic to ABC (by an easy angle chasing). Since $H_a \in \ell_a$, $H_b \in \ell_b$, and $H_c \in \ell_c$, the triangles $H_aH_bH_c$ and ABC form a required pair of triangles Δ and δ mentioned in the preamble.

Comment 2. The following observation shows how one may guess the description of the tangency point T from Solution 1.

Let us fix a direction and move the line ℓ parallel to this direction with constant speed.

Then the points D, E, and F are moving with constant speeds along the lines AI, BI, and CI, respectively. In this case x, y, and z are moving with constant speeds, defining a family of homothetic triangles XYZ with a common centre of homothety T. Notice that the triangle $X_0Y_0Z_0$ belongs to this family (for ℓ passing through I). We may specify the location of T considering the degenerate case when x, y, and z are concurrent. In this degenerate case all the lines x, y, z, ℓ, ℓ_a, ℓ_b, ℓ_c have a common point. Note that the lines ℓ_a, ℓ_b, ℓ_c remain constant as ℓ is moving (keeping its direction). Thus T should be the common point of ℓ_a, ℓ_b, and ℓ_c, lying on ω.

Solution 2. As mentioned in the preamble, it is sufficient to prove that the centre T of the homothety taking XYZ to $X_0Y_0Z_0$ belongs to ω. Thus, it suffices to prove that $\sphericalangle(TX_0, TY_0) = \sphericalangle(Z_0X_0, Z_0Y_0)$, or, equivalently, $\sphericalangle(XX_0, YY_0) = \sphericalangle(Z_0X_0, Z_0Y_0)$.

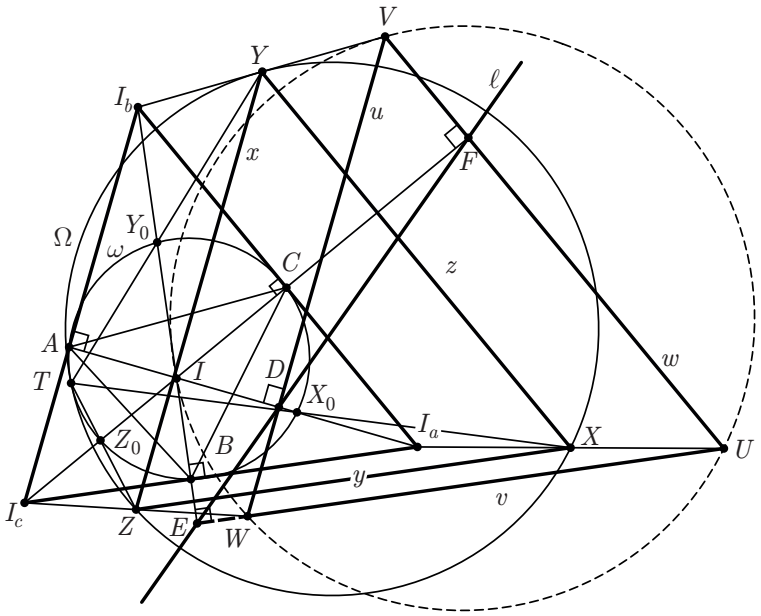
Recall that YZ and Y_0Z_0 are the perpendicular bisectors of AD and AI, respectively. Then, the vector $\overrightarrow{x}$ perpendicular to YZ and shifting the line Y_0Z_0 to YZ is equal to $\frac{1}{2}\overrightarrow{ID}$. Define the shifting vectors $\overrightarrow{y} = \frac{1}{2}\overrightarrow{IE}$, $\overrightarrow{z} = \frac{1}{2}\overrightarrow{IF}$ similarly. Consider now the triangle UVW formed by the perpendiculars to AI, BI, and CI through D, E, and F, respectively (see figure below). This is another triangle whose sides are parallel to the corresponding sides of XYZ.

Claim 2. $\overrightarrow{IU} = 2\overrightarrow{X_0X}$, $\overrightarrow{IV} = 2\overrightarrow{Y_0Y}$, $\overrightarrow{IW} = 2\overrightarrow{Z_0Z}$.

Proof. We prove one of the relations, the other proofs being similar. To prove the equality of two vectors it suffices to project them onto two non-parallel axes and check that their projections are equal.

The projection of $\overrightarrow{X_0X}$ onto IB equals $\vec{y}$, while the projection of $\overrightarrow{IU}$ onto IB is $\overrightarrow{IE} = 2\vec{y}$. The projections onto the other axis IC are $\vec{z}$ and $\overrightarrow{IF} = 2\vec{z}$. Then $\overrightarrow{IU} = 2\overrightarrow{X_0X}$ follows. □

Notice that the line ℓ is the Simson line of the point I with respect to the triangle UVW; thus U, V, W, and I are concyclic. It follows from Claim 2 that $\sphericalangle(XX_0, YY_0) = \sphericalangle(IU, IV) = \sphericalangle(WU, WV) = \sphericalangle(Z_0X_0, Z_0Y_0)$, and we are done.



Solution 3. Let I_a, I_b, and I_c be the excentres of triangle ABC corresponding to A, B, and C, respectively. Also, let u, v, and w be the lines through D, E, and F which are perpendicular to AI, BI, and CI, respectively, and let UVW be the triangle determined by these lines, where $u = VW$, $v = UW$ and $w = UV$ (see figure above).

Notice that the line u is the reflection of I_bI_c in the line x, because u, x, and I_bI_c are perpendicular to AD and x is the perpendicular bisector of AD. Likewise, v and I_aI_c are reflections of each other in y, while w and I_aI_b are reflections of each other in z. It follows that X, Y, and Z are the midpoints of UI_a, VI_b and WI_c, respectively, and that the triangles UVW, XYZ and $I_aI_bI_c$ are either translates of each other or homothetic with a common homothety centre.

Construct the points T and S such that the quadrilaterals $UVIW$, $XYTZ$ and $I_aI_bSI_c$ are homothetic. Then T is the midpoint of IS. Moreover, note that ℓ is the Simson line of the point I with respect to the triangle UVW, hence I belongs to the circumcircle of the triangle UVW, therefore T belongs to Ω.

Consider now the homothety or translation h_1 that maps $XYZT$ to $I_aI_bI_cS$ and the homothety h_2 with centre I and factor $\frac{1}{2}$. Furthermore, let $h = h_2 \circ h_1$. The transform h can be a homothety or a translation, and

$$h(T) = h_2(h_1(T)) = h_2(S) = T,$$

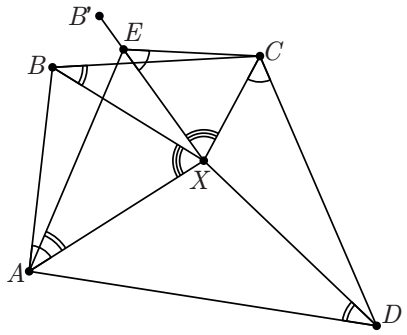
hence T is a fixed point of h. So, h is a homothety with centre T. Note that h_2 maps the excentres I_a, I_b, I_c to X_0, Y_0, Z_0 defined in the preamble. Thus the centre T of the homothety taking XYZ to $X_0Y_0Z_0$ belongs to Ω, and this completes the proof.

G6. A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. A point X is chosen inside the quadrilateral so that $\angle XAB = \angle XCD$ and $\angle XBC = \angle XDA$. Prove that $\angle AXB + \angle CXD = 180^\circ$.

(Poland)

Solution 1. Let B' be the reflection of B in the internal angle bisector of $\angle AXC$, so that $\angle AXB' = \angle CXB$ and $\angle CXB' = \angle AXB$. If X, D, and B' are collinear, then we are done. Now assume the contrary.

On the ray XB' take a point E such that $XE \cdot XB = XA \cdot XC$, so that $\triangle AXE \sim \triangle BXC$ and $\triangle CXE \sim \triangle BXA$. We have $\angle XCE + \angle XCD = \angle XBA + \angle XAB < 180^\circ$ and $\angle XAE + \angle XAD = \angle XDA + \angle XAD < 180^\circ$, which proves that X lies inside the angles $\angle ECD$ and $\angle EAD$ of the quadrilateral $EADC$. Moreover, X lies in the interior of exactly one of the two triangles EAD, ECD (and in the exterior of the other).



The similarities mentioned above imply $XA \cdot BC = XB \cdot AE$ and $XB \cdot CE = XC \cdot AB$. Multiplying these equalities with the given equality $AB \cdot CD = BC \cdot DA$, we obtain $XA \cdot CD \cdot CE = XC \cdot AD \cdot AE$, or, equivalently,

$$\frac{XA \cdot DE}{AD \cdot AE} = \frac{XC \cdot DE}{CD \cdot CE}. \qquad (*)$$

Lemma. Let PQR be a triangle, and let X be a point in the interior of the angle QPR such that $\angle QPX = \angle PRX$. Then $\dfrac{PX \cdot QR}{PQ \cdot PR} < 1$ if and only if X lies in the interior of the triangle PQR.

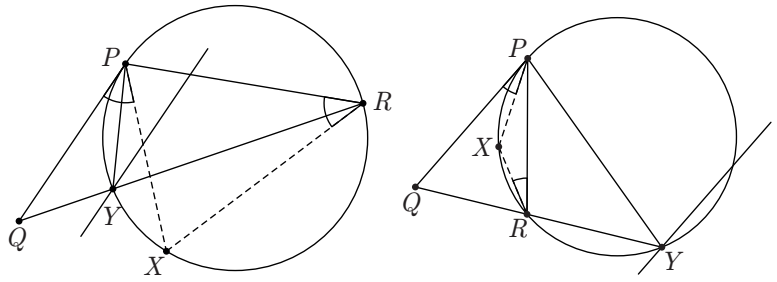
Proof. The locus of points X with $\angle QPX = \angle PRX$ lying inside the angle QPR is an arc α of the circle γ through R tangent to PQ at P. Let γ intersect the line QR again at Y (if γ is tangent to QR, then set $Y = R$). The similarity $\triangle QPY \sim \triangle QRP$ yields $PY = \dfrac{PQ \cdot PR}{QR}$. Now it suffices to show that $PX < PY$ if and only if X lies in the interior of the triangle PQR. Let m be a line through Y parallel to PQ. Notice that the points Z of γ satisfying $PZ < PY$ are exactly those between the lines m and PQ.

Case 1: Y lies in the segment QR (see the left figure below).

In this case Y splits α into two arcs $\overset{\frown}{PY}$ and $\overset{\frown}{YR}$. The arc $\overset{\frown}{PY}$ lies inside the triangle PQR, and $\overset{\frown}{PY}$ lies between m and PQ, hence $PX < PY$ for points $X \in \overset{\frown}{PY}$. The other arc $\overset{\frown}{YR}$ lies outside triangle PQR, and $\overset{\frown}{YR}$ is on the opposite side of m than P, hence $PX > PY$ for $X \in \overset{\frown}{YR}$.

Case 2: Y lies on the ray QR beyond R (see the right figure below).

In this case the whole arc α lies inside triangle PQR, and between m and PQ, thus $PX < PY$ for all $X \in \alpha$. □



Applying the Lemma (to $\triangle EAD$ with the point X, and to $\triangle ECD$ with the point X), we obtain that exactly one of two expressions $\dfrac{XA \cdot DE}{AD \cdot AE}$ and $\dfrac{XC \cdot DE}{CD \cdot CE}$ is less than 1, which contradicts $(*)$.

Comment 1. One may show that $AB \cdot CD = XA \cdot XC + XB \cdot XD$. We know that D, X, E are collinear and $\angle DCE = \angle CXD = 180^\circ - \angle AXB$. Therefore,

$$AB \cdot CD = XB \cdot \frac{\sin \angle AXB}{\sin \angle BAX} \cdot DE \cdot \frac{\sin \angle CED}{\sin \angle DCE} = XB \cdot DE.$$

Furthermore, $XB \cdot DE = XB \cdot (XD + XE) = XB \cdot XD + XB \cdot XE = XB \cdot XD + XA \cdot XC$.

Comment 2. For a convex quadrilateral $ABCD$ with $AB \cdot CD = BC \cdot DA$, it is known that $\angle DAC + \angle ABD + \angle BCA + \angle CDB = 180^\circ$ (among other, it was used as a problem on the Regional round of All-Russian olympiad in 2012), but it seems that there is no essential connection between this fact and the original problem.

Solution 2. The solution consists of two parts. In Part 1 we show that it suffices to prove that

$$\frac{XB}{XD} = \frac{AB}{CD} \tag{1}$$

and

$$\frac{XA}{XC} = \frac{DA}{BC}. \tag{2}$$

In Part 2 we establish these equalities.

Part 1. Using the sine law and applying (1) we obtain

$$\frac{\sin \angle AXB}{\sin \angle XAB} = \frac{AB}{XB} = \frac{CD}{XD} = \frac{\sin \angle CXD}{\sin \angle XCD},$$

so $\sin \angle AXB = \sin \angle CXD$ by the problem conditions. Similarly, (2) yields $\sin \angle DXA = \sin \angle BXC$. If at least one of the pairs $(\angle AXB, \angle CXD)$ and $(\angle BXC, \angle DXA)$ consists of supplementary angles, then we are done. Otherwise, $\angle AXB = \angle CXD$ and $\angle DXA = \angle BXC$. In this case $X = AC \cap BD$, and the problem conditions yield that $ABCD$ is a parallelogram and hence a rhombus. In this last case the claim also holds.

Part 2. To prove the desired equality (1), invert $ABCD$ at centre X with unit radius; the images of points are denoted by primes.

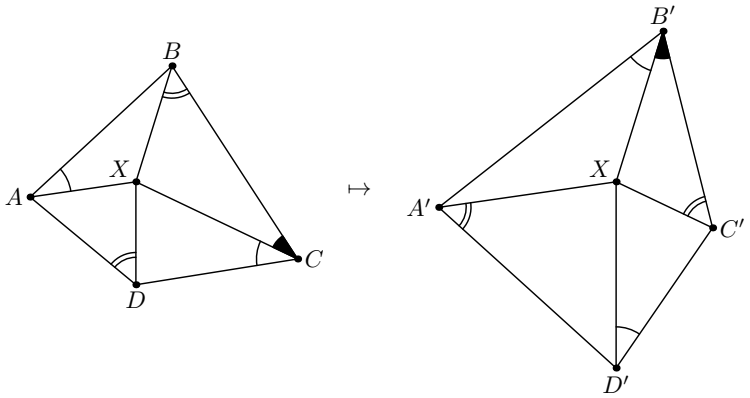
We have

$$\angle A'B'C' = \angle XB'A' + \angle XB'C' = \angle XAB + \angle XCB = \angle XCD + \angle XCB = \angle BCD.$$

Similarly, the corresponding angles of quadrilaterals $ABCD$ and $D'A'B'C'$ are equal.

Moreover, we have

$$A'B' \cdot C'D' = \frac{AB}{XA \cdot XB} \cdot \frac{CD}{XC \cdot XD} = \frac{BC}{XB \cdot XC} \cdot \frac{DA}{XD \cdot DA} = B'C' \cdot D'A'.$$



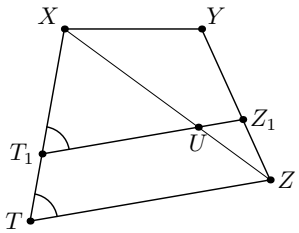
Now we need the following Lemma.

Lemma. Assume that the corresponding angles of convex quadrilaterals $XYZT$ and $X'Y'Z'T'$ are equal, and that $XY \cdot ZT = YZ \cdot TX$ and $X'Y' \cdot Z'T' = Y'Z' \cdot T'X'$. Then the two quadrilaterals are similar.

Proof. Take the quadrilateral XYZ_1T_1 similar to $X'Y'Z'T'$ and sharing the side XY with $XYZT$, such that Z_1 and T_1 lie on the rays YZ and XT, respectively, and $Z_1T_1 \parallel ZT$. We need to prove that $Z_1 = Z$ and $T_1 = T$. Assume the contrary. Without loss of generality, $TX > XT_1$. Let segments XZ and Z_1T_1 intersect at U. We have

$$\frac{T_1X}{T_1Z_1} < \frac{T_1X}{T_1U} = \frac{TX}{ZT} = \frac{XY}{YZ} < \frac{XY}{YZ_1},$$

thus $T_1X \cdot YZ_1 < T_1Z_1 \cdot XY$. A contradiction. □



It follows from the Lemma that the quadrilaterals $ABCD$ and $D'A'B'C'$ are similar, hence

$$\frac{BC}{AB} = \frac{A'B'}{D'A'} = \frac{AB}{XA \cdot XB} \cdot \frac{XD \cdot XA}{DA} = \frac{AB}{AD} \cdot \frac{XD}{XB},$$

and therefore

$$\frac{XB}{XD} = \frac{AB^2}{BC \cdot AD} = \frac{AB^2}{AB \cdot CD} = \frac{AB}{CD}.$$

We obtain (1), as desired; (2) is proved similarly.

Comment. Part 1 is an easy one, while part 2 seems to be crucial. On the other hand, after the proof of the similarity $D'A'B'C' \sim ABCD$ one may finish the solution in different ways, e.g., as follows. The similarity taking $D'A'B'C'$ to $ABCD$ maps X to the point X' *isogonally conjugate* of X with respect to $ABCD$ (i.e. to the point X' inside $ABCD$ such that $\angle BAX = \angle DAX'$, $\angle CBX = \angle ABX'$, $\angle DCX = \angle BCX'$, $\angle ADX = \angle CDX'$). It is known that the required equality $\angle AXB + \angle CXD = 180^\circ$ is one of known conditions on a point X inside $ABCD$ equivalent to the existence of its isogonal conjugate.

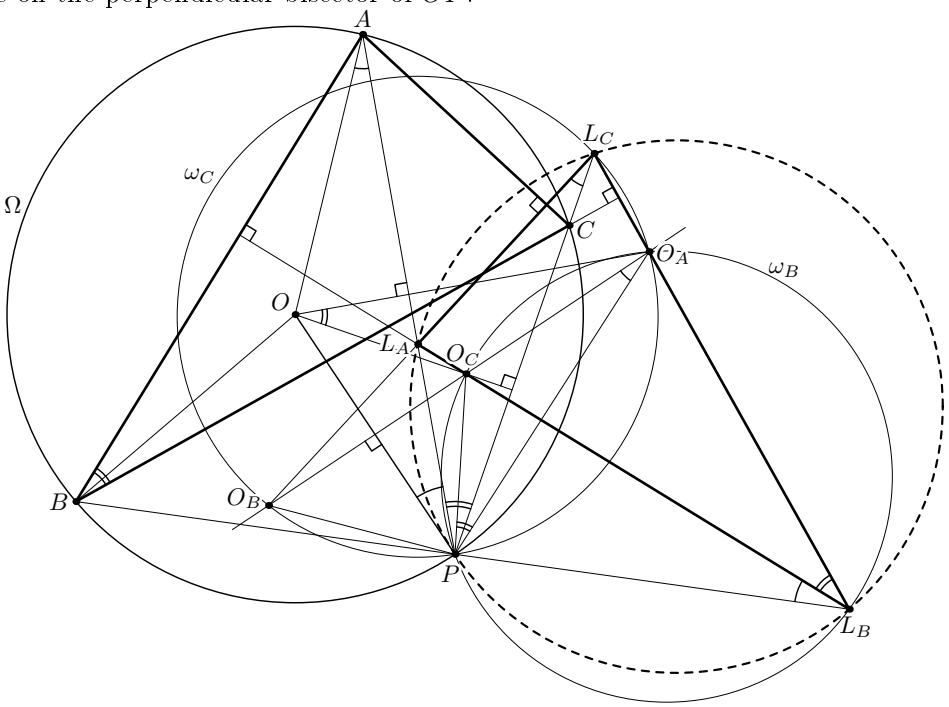
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G7. Let O be the circumcentre, and Ω be the circumcircle of an acute-angled triangle ABC. Let P be an arbitrary point on Ω, distinct from A, B, C, and their antipodes in Ω. Denote the circumcentres of the triangles AOP, BOP, and COP by O_A, O_B, and O_C, respectively. The lines ℓ_A, ℓ_B, and ℓ_C perpendicular to BC, CA, and AB pass through O_A, O_B, and O_C, respectively. Prove that the circumcircle of the triangle formed by ℓ_A, ℓ_B, and ℓ_C is tangent to the line OP.

(Russia)

Solution. As usual, we denote the directed angle between the lines a and b by $\measuredangle(a,b)$. We frequently use the fact that $a_1 \perp a_2$ and $b_1 \perp b_2$ yield $\measuredangle(a_1,b_1) = \measuredangle(a_2,b_2)$.

Let the lines ℓ_B and ℓ_C meet at L_A; define the points L_B and L_C similarly. Note that the sidelines of the triangle $L_AL_BL_C$ are perpendicular to the corresponding sidelines of ABC. Points O_A, O_B, O_C are located on the corresponding sidelines of $L_AL_BL_C$; moreover, O_A, O_B, O_C all lie on the perpendicular bisector of OP.



Claim 1. The points L_B, P, O_A, and O_C are concyclic.

Proof. Since O is symmetric to P in O_AO_C, we have

$$\measuredangle(O_AP, O_CP) = \measuredangle(O_CO, O_AO) = \measuredangle(CP, AP) = \measuredangle(CB, AB) = \measuredangle(O_AL_B, O_CL_B). \qquad \square$$

Denote the circle through L_B, P, O_A, and O_C by ω_B. Define the circles ω_A and ω_C similarly.

Claim 2. The circumcircle of the triangle $L_AL_BL_C$ passes through P.

Proof. From cyclic quadruples of points in the circles ω_B and ω_C, we have

$$\begin{aligned}\measuredangle(L_CL_A, L_CP) &= \measuredangle(L_CO_B, L_CP) = \measuredangle(O_AO_B, O_AP)\\ &= \measuredangle(O_AO_C, O_AP) = \measuredangle(L_BO_C, L_BP) = \measuredangle(L_BL_A, L_BP).\end{aligned} \qquad \square$$

Claim 3. The points P, L_C, and C are collinear.

Proof. We have $\measuredangle(PL_C, L_CL_A) = \measuredangle(PL_C, L_CO_B) = \measuredangle(PO_A, O_AO_B)$. Further, since O_A is the centre of the circle AOP, $\measuredangle(PO_A, O_AO_B) = \measuredangle(PA, AO)$. As O is the circumcentre of the triangle PCA, $\measuredangle(PA, AO) = \pi/2 - \measuredangle(CA, CP) = \measuredangle(CP, L_CL_A)$. We obtain $\measuredangle(PL_C, L_CL_A) = \measuredangle(CP, L_CL_A)$, which shows that $P \in CL_C$. $\square$

Similarly, the points P, L_A, A are collinear, and the points P, L_B, B are also collinear. Finally, the computation above also shows that

$$\measuredangle(OP, PL_A) = \measuredangle(PA, AO) = \measuredangle(PL_C, L_CL_A),$$

which means that OP is tangent to the circle $PL_AL_BL_C$.

Comment 1. The proof of Claim 2 may be replaced by the following remark: since P belongs to the circles ω_A and ω_C, P is the Miquel point of the four lines ℓ_A, ℓ_B, ℓ_C, and $O_AO_BO_C$.

Comment 2. Claims 2 and 3 can be proved in several different ways and, in particular, in the reverse order.

Claim 3 implies that the triangles ABC and $L_AL_BL_C$ are perspective with perspector P. Claim 2 can be derived from this observation using spiral similarity. Consider the centre Q of the spiral similarity that maps ABC to $L_AL_BL_C$. From known spiral similarity properties, the points L_A, L_B, P, Q are concyclic, and so are L_A, L_C, P, Q.

Comment 3. The final conclusion can also be proved it terms of spiral similarity: the spiral similarity with centre Q located on the circle ABC maps the circle ABC to the circle $PL_AL_BL_C$. Thus these circles are orthogonal.

Comment 4. Notice that the homothety with centre O and ratio 2 takes O_A to A' that is the common point of tangents to Ω at A and P. Similarly, let this homothety take O_B to B' and O_C to C'. Let the tangents to Ω at B and C meet at A'', and define the points B'' and C'' similarly. Now, replacing labels O with I, Ω with ω, and swapping labels $A \leftrightarrow A''$, $B \leftrightarrow B''$, $C \leftrightarrow C''$ we obtain the following

Reformulation. Let ω be the incircle, and let I be the incentre of a triangle ABC. Let P be a point of ω (other than the points of contact of ω with the sides of ABC). The tangent to ω at P meets the lines AB, BC, and CA at A', B', and C', respectively. Line ℓ_A parallel to the internal angle bisector of $\angle BAC$ passes through A'; define lines ℓ_B and ℓ_C similarly. Prove that the line IP is tangent to the circumcircle of the triangle formed by ℓ_A, ℓ_B, and ℓ_C.

Though this formulation is equivalent to the original one, it seems more challenging, since the point of contact is now "hidden".

Number Theory

N1. Determine all pairs (n, k) of distinct positive integers such that there exists a positive integer s for which the numbers of divisors of sn and of sk are equal.

(Ukraine)

Answer: All pairs (n, k) such that $n \nmid k$ and $k \nmid n$.

Solution. As usual, the number of divisors of a positive integer n is denoted by $d(n)$. If $n = \prod_i p_i^{\alpha_i}$ is the prime factorisation of n, then $d(n) = \prod_i (\alpha_i + 1)$.

We start by showing that one cannot find any suitable number s if $k \mid n$ or $n \mid k$ (and $k \neq n$). Suppose that $n \mid k$, and choose any positive integer s. Then the set of divisors of sn is a proper subset of that of sk, hence $d(sn) < d(sk)$. Therefore, the pair (n, k) does not satisfy the problem requirements. The case $k \mid n$ is similar.

Now assume that $n \nmid k$ and $k \nmid n$. Let $p_1, \ldots, p_t$ be all primes dividing nk, and consider the prime factorisations

$$n = \prod_{i=1}^{t} p_i^{\alpha_i} \quad \text{and} \quad k = \prod_{i=1}^{t} p_i^{\beta_i}.$$

It is reasonable to search for the number s having the form

$$s = \prod_{i=1}^{t} p_i^{\gamma_i}.$$

The (nonnegative integer) exponents γ_i should be chosen so as to satisfy

$$\frac{d(sn)}{d(sk)} = \prod_{i=1}^{t} \frac{\alpha_i + \gamma_i + 1}{\beta_i + \gamma_i + 1} = 1. \tag{1}$$

First of all, if $\alpha_i = \beta_i$ for some i, then, regardless of the value of γ_i, the corresponding factor in (1) equals 1 and does not affect the product. So we may assume that there is no such index i. For the other factors in (1), the following lemma is useful.

Lemma. Let $\alpha > \beta$ be nonnegative integers. Then, for every integer $M \geqslant \beta + 1$, there exists a nonnegative integer γ such that

$$\frac{\alpha + \gamma + 1}{\beta + \gamma + 1} = 1 + \frac{1}{M} = \frac{M + 1}{M}.$$

Proof.

$$\frac{\alpha + \gamma + 1}{\beta + \gamma + 1} = 1 + \frac{1}{M} \iff \frac{\alpha - \beta}{\beta + \gamma + 1} = \frac{1}{M} \iff \gamma = M(\alpha - \beta) - (\beta + 1) \geqslant 0. \qquad \square$$

Now we can finish the solution. Without loss of generality, there exists an index u such that $\alpha_i > \beta_i$ for $i = 1, 2, \ldots, u$, and $\alpha_i < \beta_i$ for $i = u + 1, \ldots, t$. The conditions $n \nmid k$ and $k \nmid n$ mean that $1 \leqslant u \leqslant t - 1$.

Choose an integer X greater than all the α_i and β_i. By the lemma, we can define the numbers γ_i so as to satisfy

$$\frac{\alpha_i + \gamma_i + 1}{\beta_i + \gamma_i + 1} = \frac{uX + i}{uX + i - 1} \qquad \text{for } i = 1, 2, \ldots, u, \text{ and}$$

$$\frac{\beta_{u+i} + \gamma_{u+i} + 1}{\alpha_{u+i} + \gamma_{u+i} + 1} = \frac{(t - u)X + i}{(t - u)X + i - 1} \qquad \text{for } i = 1, 2, \ldots, t - u.$$

Then we will have

$$\frac{d(sn)}{d(sk)} = \prod_{i=1}^{u} \frac{uX+i}{uX+i-1} \cdot \prod_{i=1}^{t-u} \frac{(t-u)X+i-1}{(t-u)X+i} = \frac{u(X+1)}{uX} \cdot \frac{(t-u)X}{(t-u)(X+1)} = 1,$$

as required.

Comment. The lemma can be used in various ways, in order to provide a suitable value of s. In particular, one may apply induction on the number t of prime factors, using identities like

$$\frac{n}{n-1} = \frac{n^2}{n^2-1} \cdot \frac{n+1}{n}.$$

N2. Let $n > 1$ be a positive integer. Each cell of an $n \times n$ table contains an integer. Suppose that the following conditions are satisfied:

(*i*) Each number in the table is congruent to 1 modulo n;

(*ii*) The sum of numbers in any row, as well as the sum of numbers in any column, is congruent to n modulo n^2.

Let R_i be the product of the numbers in the ith row, and C_j be the product of the numbers in the jth column. Prove that the sums $R_1 + \cdots + R_n$ and $C_1 + \cdots + C_n$ are congruent modulo n^4.

(Indonesia)

Solution 1. Let $A_{i,j}$ be the entry in the ith row and the jth column; let P be the product of all n^2 entries. For convenience, denote $a_{i,j} = A_{i,j} - 1$ and $r_i = R_i - 1$. We show that

$$\sum_{i=1}^{n} R_i \equiv (n-1) + P \pmod{n^4}. \tag{1}$$

Due to symmetry of the problem conditions, the sum of all the C_j is also congruent to $(n-1)+P$ modulo n^4, whence the conclusion.

By condition (*i*), the number n divides $a_{i,j}$ for all i and j. So, every product of at least two of the $a_{i,j}$ is divisible by n^2, hence

$$R_i = \prod_{j=1}^{n}(1+a_{i,j}) = 1+\sum_{j=1}^{n} a_{i,j} + \sum_{1\leqslant j_1<j_2\leqslant n} a_{i,j_1}a_{i,j_2} + \cdots \equiv 1+\sum_{j=1}^{n} a_{i,j} \equiv 1-n+\sum_{j=1}^{n} A_{i,j} \pmod{n^2}$$

for every index i. Using condition (*ii*), we obtain $R_i \equiv 1 \pmod{n^2}$, and so $n^2 \mid r_i$.

Therefore, every product of at least two of the r_i is divisible by n^4. Repeating the same argument, we obtain

$$P = \prod_{i=1}^{n} R_i = \prod_{i=1}^{n}(1+r_i) \equiv 1 + \sum_{i=1}^{n} r_i \pmod{n^4},$$

whence

$$\sum_{i=1}^{n} R_i = n + \sum_{i=1}^{n} r_i \equiv n + (P-1) \pmod{n^4},$$

as desired.

Comment. The original version of the problem statement contained also the condition

(*iii*) The product of all the numbers in the table is congruent to 1 modulo n^4.

This condition appears to be superfluous, so it was omitted.

Solution 2. We present a more straightforward (though lengthier) way to establish (1). We also use the notation of $a_{i,j}$.

By condition (*i*), all the $a_{i,j}$ are divisible by n. Therefore, we have

$$P = \prod_{i=1}^{n}\prod_{j=1}^{n}(1+a_{i,j}) \equiv 1 + \sum_{(i,j)} a_{i,j} + \sum_{(i_1,j_1),\,(i_2,j_2)} a_{i_1,j_1}a_{i_2,j_2} + \sum_{(i_1,j_1),\,(i_2,j_2),\,(i_3,j_3)} a_{i_1,j_1}a_{i_2,j_2}a_{i_3,j_3} \pmod{n^4},$$

where the last two sums are taken over all unordered pairs/triples of *pairwise different* pairs (i, j); such conventions are applied throughout the solution.

Similarly,

$$\sum_{i=1}^{n} R_i = \sum_{i=1}^{n} \prod_{j=1}^{n} (1 + a_{i,j}) \equiv n + \sum_i \sum_j a_{i,j} + \sum_i \sum_{j_1, j_2} a_{i,j_1} a_{i,j_2} + \sum_i \sum_{j_1, j_2, j_3} a_{i,j_1} a_{i,j_2} a_{i,j_3} \pmod{n^4}.$$

Therefore,

$$P + (n-1) - \sum_i R_i \equiv \sum_{\substack{(i_1,j_1),(i_2,j_2) \\ i_1 \neq i_2}} a_{i_1,j_1} a_{i_2,j_2} + \sum_{\substack{(i_1,j_1),(i_2,j_2),(i_3,j_3) \\ i_1 \neq i_2 \neq i_3 \neq i_1}} a_{i_1,j_1} a_{i_2,j_2} a_{i_3,j_3} + \sum_{\substack{(i_1,j_1),(i_2,j_2),(i_3,j_3) \\ i_1 \neq i_2 = i_3}} a_{i_1,j_1} a_{i_2,j_2} a_{i_3,j_3} \pmod{n^4}.$$

We show that in fact each of the three sums appearing in the right-hand part of this congruence is divisible by n^4; this yields (1). Denote those three sums by Σ_1, Σ_2, and Σ_3 in order of appearance. Recall that by condition (ii) we have

$$\sum_j a_{i,j} \equiv 0 \pmod{n^2} \qquad \text{for all indices } i.$$

For every two indices $i_1 < i_2$ we have

$$\sum_{j_1} \sum_{j_2} a_{i_1,j_1} a_{i_2,j_2} = \left(\sum_{j_1} a_{i_1,j_1} \right) \cdot \left(\sum_{j_2} a_{i_2,j_2} \right) \equiv 0 \pmod{n^4},$$

since each of the two factors is divisible by n^2. Summing over all pairs (i_1, i_2) we obtain $n^4 \mid \Sigma_1$.

Similarly, for every three indices $i_1 < i_2 < i_3$ we have

$$\sum_{j_1} \sum_{j_2} \sum_{j_3} a_{i_1,j_1} a_{i_2,j_2} a_{i_3,j_3} = \left(\sum_{j_1} a_{i_1,j_1} \right) \cdot \left(\sum_{j_2} a_{i_2,j_2} \right) \cdot \left(\sum_{j_3} a_{i_3,j_3} \right)$$

which is divisible even by n^6. Hence $n^4 \mid \Sigma_2$.

Finally, for every indices $i_1 \neq i_2 = i_3$ and $j_2 < j_3$ we have

$$a_{i_2,j_2} \cdot a_{i_2,j_3} \cdot \sum_{j_1} a_{i_1,j_1} \equiv 0 \pmod{n^4},$$

since the three factors are divisible by n, n, and n^2, respectively. Summing over all 4-tuples of indices (i_1, i_2, j_2, j_3) we get $n^4 \mid \Sigma_3$.

N3. Define the sequence $a_0, a_1, a_2, \ldots$ by $a_n = 2^n + 2^{\lfloor n/2 \rfloor}$. Prove that there are infinitely many terms of the sequence which can be expressed as a sum of (two or more) distinct terms of the sequence, as well as infinitely many of those which cannot be expressed in such a way.

(Serbia)

Solution 1. Call a nonnegative integer *representable* if it equals the sum of several (possibly 0 or 1) distinct terms of the sequence. We say that two nonnegative integers b and c are *equivalent* (written as $b \sim c$) if they are either both representable or both non-representable.

One can easily compute

$$S_{n-1} := a_0 + \cdots + a_{n-1} = 2^n + 2^{\lfloor n/2 \rfloor} + 2^{\lfloor n/2 \rfloor} - 3.$$

Indeed, we have $S_n - S_{n-1} = 2^n + 2^{\lfloor n/2 \rfloor} = a_n$ so we can use the induction. In particular, $S_{2k-1} = 2^{2k} + 2^{k+1} - 3$.

Note that, if $n \geqslant 3$, then $2^{\lfloor n/2 \rfloor} \geqslant 2^2 > 3$, so

$$S_{n-1} = 2^n + 2^{\lfloor n/2 \rfloor} + 2^{\lfloor n/2 \rfloor} - 3 > 2^n + 2^{\lfloor n/2 \rfloor} = a_n.$$

Also notice that $S_{n-1} - a_n = 2^{\lfloor n/2 \rfloor} - 3 < a_n$.

The main tool of the solution is the following claim.

Claim 1. Assume that b is a positive integer such that $S_{n-1} - a_n < b < a_n$ for some $n \geqslant 3$. Then $b \sim S_{n-1} - b$.

Proof. As seen above, we have $S_{n-1} > a_n$. Denote $c = S_{n-1} - b$; then $S_{n-1} - a_n < c < a_n$, so the roles of b and c are symmetrical.

Assume that b is representable. The representation cannot contain a_i with $i \geqslant n$, since $b < a_n$. So b is the sum of some subset of $\{a_0, a_1, \ldots, a_{n-1}\}$; then c is the sum of the complement. The converse is obtained by swapping b and c. □

We also need the following version of this claim.

Claim 2. For any $n \geqslant 3$, the number a_n can be represented as a sum of two or more distinct terms of the sequence if and only if $S_{n-1} - a_n = 2^{\lfloor n/2 \rfloor} - 3$ is representable.

Proof. Denote $c = S_{n-1} - a_n < a_n$. If a_n satisfies the required condition, then it is the sum of some subset of $\{a_0, a_1, \ldots, a_{n-1}\}$; then c is the sum of the complement. Conversely, if c is representable, then its representation consists only of the numbers from $\{a_0, \ldots, a_{n-1}\}$, so a_n is the sum of the complement. □

By Claim 2, in order to prove the problem statement, it suffices to find infinitely many representable numbers of the form $2^t - 3$, as well as infinitely many non-representable ones.

Claim 3. For every $t \geqslant 3$, we have $2^t - 3 \sim 2^{4t-6} - 3$, and $2^{4t-6} - 3 > 2^t - 3$.

Proof. The inequality follows from $t \geqslant 3$. In order to prove the equivalence, we apply Claim 1 twice in the following manner.

First, since $S_{2t-3} - a_{2t-2} = 2^{t-1} - 3 < 2^t - 3 < 2^{2t-2} + 2^{t-1} = a_{2t-2}$, by Claim 1 we have $2^t - 3 \sim S_{2t-3} - (2^t - 3) = 2^{2t-2}$.

Second, since $S_{4t-7} - a_{4t-6} = 2^{2t-3} - 3 < 2^{2t-2} < 2^{4t-6} + 2^{2t-3} = a_{4t-6}$, by Claim 1 we have $2^{2t-2} \sim S_{4t-7} - 2^{2t-2} = 2^{4t-6} - 3$.

Therefore, $2^t - 3 \sim 2^{2t-2} \sim 2^{4t-6} - 3$, as required. □

Now it is easy to find the required numbers. Indeed, the number $2^3 - 3 = 5 = a_0 + a_1$ is representable, so Claim 3 provides an infinite sequence of representable numbers

$$2^3 - 3 \sim 2^6 - 3 \sim 2^{18} - 3 \sim \cdots \sim 2^t - 3 \sim 2^{4t-6} - 3 \sim \cdots.$$

On the other hand, the number $2^7 - 3 = 125$ is non-representable (since by Claim 1 we have $125 \sim S_6 - 125 = 24 \sim S_4 - 24 = 17 \sim S_3 - 17 = 4$ which is clearly non-representable). So Claim 3 provides an infinite sequence of non-representable numbers

$$2^7 - 3 \sim 2^{22} - 3 \sim 2^{82} - 3 \sim \cdots \sim 2^t - 3 \sim 2^{4t-6} - 3 \sim \cdots.$$

Solution 2. We keep the notion of representability and the notation S_n from the previous solution. We say that an index n is *good* if a_n writes as a sum of smaller terms from the sequence $a_0, a_1, \ldots$. Otherwise we say it is *bad*. We must prove that there are infinitely many good indices, as well as infinitely many bad ones.

Lemma 1. If $m \geqslant 0$ is an integer, then 4^m is representable if and only if either of $2m+1$ and $2m+2$ is good.

Proof. The case $m = 0$ is obvious, so we may assume that $m \geqslant 1$. Let $n = 2m+1$ or $2m+2$. Then $n \geqslant 3$. We notice that

$$S_{n-1} < a_{n-2} + a_n.$$

The inequality writes as $2^n + 2^{\lceil n/2 \rceil} + 2^{\lfloor n/2 \rfloor} - 3 < 2^n + 2^{\lfloor n/2 \rfloor} + 2^{n-2} + 2^{\lfloor n/2 \rfloor - 1}$, i.e. as $2^{\lceil n/2 \rceil} < 2^{n-2} + 2^{\lfloor n/2 \rfloor - 1} + 3$. If $n \geqslant 4$, then $n/2 \leqslant n-2$, so $\lceil n/2 \rceil \leqslant n-2$ and $2^{\lceil n/2 \rceil} \leqslant 2^{n-2}$. For $n = 3$ the inequality verifies separately.

If n is good, then a_n writes as $a_n = a_{i_1} + \cdots + a_{i_r}$, where $r \geqslant 2$ and $i_1 < \cdots < i_r < n$. Then $i_r = n-1$ and $i_{r-1} = n-2$, for if $n-1$ or $n-2$ is missing from the sequence $i_1, \ldots, i_r$, then $a_{i_1} + \cdots + a_{i_r} \leqslant a_0 + \cdots + a_{n-3} + a_{n-1} = S_{n-1} - a_{n-2} < a_n$. Thus, if n is good, then both $a_n - a_{n-1}$ and $a_n - a_{n-1} - a_{n-2}$ are representable.

We now consider the cases $n = 2m+1$ and $n = 2m+2$ separately.

If $n = 2m+1$, then $a_n - a_{n-1} = a_{2m+1} - a_{2m} = (2^{2m+1} + 2^m) - (2^{2m} + 2^m) = 2^{2m}$. So we proved that, if $2m+1$ is good, then 2^{2m} is representable. Conversely, if 2^{2m} is representable, then $2^{2m} < a_{2m}$, so 2^{2m} is a sum of some distinct terms a_i with $i < 2m$. It follows that $a_{2m+1} = a_{2m} + 2^{2m}$ writes as a_{2m} plus a sum of some distinct terms a_i with $i < 2m$. Hence $2m+1$ is good.

If $n = 2m+2$, then $a_n - a_{n-1} - a_{n-2} = a_{2m+2} - a_{2m+1} - a_{2m} = (2^{2m+2} + 2^{m+1}) - (2^{2m+1} + 2^m) - (2^{2m} + 2^m) = 2^{2m}$. So we proved that, if $2m+2$ is good, then 2^{2m} is representable. Conversely, if 2^{2m} is representable, then, as seen in the previous case, it writes as a sum of some distinct terms a_i with $i < 2m$. Hence $a_{2m+2} = a_{2m+1} + a_{2m} + 2^{2m}$ writes as $a_{2m+1} + a_{2m}$ plus a sum of some distinct terms a_i with $i < 2m$. Thus $2m+2$ is good. □

Lemma 2. If $k \geqslant 2$, then 2^{4k-2} is representable if and only if 2^{k+1} is representable.

In particular, if $s \geqslant 2$, then 4^s is representable if and only if 4^{4s-3} is representable. Also, $4^{4s-3} > 4^s$.

Proof. We have $2^{4k-2} < a_{4k-2}$, so in a representation of 2^{4k-2} we can have only terms a_i with $i \leqslant 4k-3$. Notice that

$$a_0 + \cdots + a_{4k-3} = 2^{4k-2} + 2^{2k} - 3 < 2^{4k-2} + 2^{2k} + 2^k = 2^{4k-2} + a_{2k}.$$

Hence, any representation of 2^{4k-2} must contain all terms from a_{2k} to a_{4k-3}. (If any of these terms is missing, then the sum of the remaining ones is $\leqslant (a_0 + \cdots + a_{4k-3}) - a_{2k} < 2^{4k-2}$.) Hence, if 2^{4k-2} is representable, then $2^{4k-2} - \sum_{i=2k}^{4k-3} a_i$ is representable. But

$$2^{4k-2} - \sum_{i=2k}^{4k-3} a_i = 2^{4k-2} - (S_{4k-3} - S_{2k-1}) = 2^{4k-2} - (2^{4k-2} + 2^{2k} - 3) + (2^{2k} + 2^{k+1} - 3) = 2^{k+1}.$$

So, if 2^{4k-2} is representable, then 2^{k+1} is representable. Conversely, if 2^{k+1} is representable, then $2^{k+1} < 2^{2k} + 2^k = a_{2k}$, so 2^{k+1} writes as a sum of some distinct terms a_i with $i < 2k$. It follows that $2^{4k-2} = \sum_{i=2k}^{4k-3} a_i + 2^{k+1}$ writes as $a_{4k-3} + a_{4k-4} + \cdots + a_{2k}$ plus the sum of some distinct terms a_i with $i < 2k$. Hence 2^{4k-2} is representable.

For the second statement, if $s \geqslant 2$, then we just take $k = 2s-1$ and we notice that $2^{k+1} = 4^s$ and $2^{4k-2} = 4^{4s-3}$. Also, $s \geqslant 2$ implies that $4s - 3 > s$. □

Now $4^2 = a_2+a_3$ is representable, whereas $4^6 = 4096$ is not. Indeed, note that $4^6 = 2^{12} < a_{12}$, so the only available terms for a representation are $a_0, \ldots, a_{11}$, i.e., 2, 3, 6, 10, 20, 36, 72, 136, 272, 528, 1056, 2080. Their sum is $S_{11} = 4221$, which exceeds 4096 by 125. Then any representation of 4096 must contain all the terms from $a_0, \ldots, a_{11}$ that are greater that 125, i.e., 136, 272, 528, 1056, 2080. Their sum is 4072. Since $4096 - 4072 = 24$ and 24 is clearly not representable, 4096 is non-representable as well.

Starting with these values of m, by using Lemma 2, we can obtain infinitely many representable powers of 4, as well as infinitely many non-representable ones. By Lemma 1, this solves our problem.

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N4. Let a_1, a_2, …, a_n, … be a sequence of positive integers such that

$$\frac{a_1}{a_2}+\frac{a_2}{a_3}+\cdots+\frac{a_{n-1}}{a_n}+\frac{a_n}{a_1}$$

is an integer for all $n \geqslant k$, where k is some positive integer. Prove that there exists a positive integer m such that $a_n = a_{n+1}$ for all $n \geqslant m$.

(Mongolia)

Solution 1. The argument hinges on the following two facts: Let a, b, c be positive integers such that $N = b/c + (c-b)/a$ is an integer.

(1) If $\gcd(a,c) = 1$, then c divides b; and
(2) If $\gcd(a,b,c) = 1$, then $\gcd(a,b) = 1$.

To prove **(1)**, write $ab = c(aN + b - c)$. Since $\gcd(a,c) = 1$, it follows that c divides b. To prove **(2)**, write $c^2 - bc = a(cN - b)$ to infer that a divides $c^2 - bc$. Letting $d = \gcd(a,b)$, it follows that d divides c^2, and since the two are relatively prime by hypothesis, $d = 1$.

Now, let $s_n = a_1/a_2 + a_2/a_3 + \cdots + a_{n-1}/a_n + a_n/a_1$, let $\delta_n = \gcd(a_1, a_n, a_{n+1})$ and write

$$s_{n+1} - s_n = \frac{a_n}{a_{n+1}} + \frac{a_{n+1}-a_n}{a_1} = \frac{a_n/\delta_n}{a_{n+1}/\delta_n} + \frac{a_{n+1}/\delta_n - a_n/\delta_n}{a_1/\delta_n}.$$

Let $n \geqslant k$. Since $\gcd(a_1/\delta_n, a_n/\delta_n, a_{n+1}/\delta_n) = 1$, it follows by **(2)** that $\gcd(a_1/\delta_n, a_n/\delta_n) = 1$. Let $d_n = \gcd(a_1, a_n)$. Then $d_n = \delta_n \cdot \gcd(a_1/\delta_n, a_n/\delta_n) = \delta_n$, so d_n divides a_{n+1}, and therefore d_n divides d_{n+1}.

Consequently, from some rank on, the d_n form a nondecreasing sequence of integers not exceeding a_1, so $d_n = d$ for all $n \geqslant \ell$, where ℓ is some positive integer.

Finally, since $\gcd(a_1/d, a_{n+1}/d) = 1$, it follows by **(1)** that a_{n+1}/d divides a_n/d, so $a_n \geqslant a_{n+1}$ for all $n \geqslant \ell$. The conclusion follows.

Solution 2. We use the same notation s_n. This time, we explore the exponents of primes in the prime factorizations of the a_n for $n \geqslant k$.

To start, for every $n \geqslant k$, we know that the number

$$s_{n+1} - s_n = \frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} - \frac{a_n}{a_1} \qquad (*)$$

is integer. Multiplying it by a_1 we obtain that $a_1 a_n / a_{n+1}$ is integer as well, so that $a_{n+1} \mid a_1 a_n$. This means that $a_n \mid a_1^{n-k} a_k$, so all prime divisors of a_n are among those of $a_1 a_k$. There are finitely many such primes; therefore, it suffices to prove that the exponent of each of them in the prime factorization of a_n is eventually constant.

Choose any prime $p \mid a_1 a_k$. Recall that $v_p(q)$ is the standard notation for the exponent of p in the prime factorization of a nonzero rational number q. Say that an index $n \geqslant k$ is *large* if $v_p(a_n) \geqslant v_p(a_1)$. We separate two cases.

Case 1: There exists a large index n.

If $v_p(a_{n+1}) < v_p(a_1)$, then $v_p(a_n/a_{n+1})$ and $v_p(a_n/a_1)$ are nonnegative, while $v_p(a_{n+1}/a_1) < 0$; hence $(*)$ cannot be an integer. This contradiction shows that index $n+1$ is also large.

On the other hand, if $v_p(a_{n+1}) > v_p(a_n)$, then $v_p(a_n/a_{n+1}) < 0$, while $v_p\big((a_{n+1}-a_n)/a_1\big) \geqslant 0$, so $(*)$ is not integer again. Thus, $v_p(a_1) \leqslant v_p(a_{n+1}) \leqslant v_p(a_n)$.

The above arguments can now be applied successively to indices $n+1$, $n+2$, …, showing that all the indices greater than n are large, and the sequence $v_p(a_n), v_p(a_{n+1}), v_p(a_{n+2}), \ldots$ is nonincreasing — hence eventually constant.

Case 2: There is no large index.

We have $v_p(a_1) > v_p(a_n)$ for all $n \geqslant k$. If we had $v_p(a_{n+1}) < v_p(a_n)$ for some $n \geqslant k$, then $v_p(a_{n+1}/a_1) < v_p(a_n/a_1) < 0 < v_p(a_n/a_{n+1})$ which would also yield that $(*)$ is not integer. Therefore, in this case the sequence $v_p(a_k), v_p(a_{k+1}), v_p(a_{k+2}), \ldots$ is nondecreasing and bounded by $v_p(a_1)$ from above; hence it is also eventually constant.

Comment. Given any positive odd integer m, consider the m-tuple $(2, 2^2, \ldots, 2^{m-1}, 2^m)$. Appending an infinite string of 1's to this m-tuple yields an eventually constant sequence of integers satisfying the condition in the statement, and shows that the rank from which the sequence stabilises may be arbitrarily large.

There are more sophisticated examples. The solution to part **(b)** of **10532**, *Amer. Math. Monthly*, Vol. 105 No. 8 (Oct. 1998), 775–777 (available at https://www.jstor.org/stable/2589009), shows that, for every integer $m \geqslant 5$, there exists an m-tuple $(a_1, a_2, \ldots, a_m)$ of pairwise distinct positive integers such that $\gcd(a_1, a_2) = \gcd(a_2, a_3) = \cdots = \gcd(a_{m-1}, a_m) = \gcd(a_m, a_1) = 1$, and the sum $a_1/a_2 + a_2/a_3 + \cdots + a_{m-1}/a_m + a_m/a_1$ is an integer. Letting $a_{m+k} = a_1$, $k = 1, 2, \ldots$, extends such an m-tuple to an eventually constant sequence of positive integers satisfying the condition in the statement of the problem at hand.

Here is the example given by the proposers of **10532**. Let $b_1 = 2$, let $b_{k+1} = 1 + b_1 \cdots b_k = 1 + b_k(b_k - 1)$, $k \geqslant 1$, and set $B_m = b_1 \cdots b_{m-4} = b_{m-3} - 1$. The m-tuple $(a_1, a_2, \ldots, a_m)$ defined below satisfies the required conditions:

$$a_1 = 1, \quad a_2 = (8B_m + 1)B_m + 8, \quad a_3 = 8B_m + 1, \quad a_k = b_{m-k} \quad \text{for } 4 \leqslant k \leqslant m-1,$$
$$a_m = \frac{a_2}{2} \cdot a_3 \cdot \frac{B_m}{2} = \left(\frac{1}{2}(8B_m + 1)B_m + 4\right) \cdot (8B_m + 1) \cdot \frac{B_m}{2}.$$

It is readily checked that $a_1 < a_{m-1} < a_{m-2} < \cdots < a_3 < a_2 < a_m$. For further details we refer to the solution mentioned above. Acquaintance with this example (or more elaborated examples derived from) offers no advantage in tackling the problem.

N5. Four positive integers x, y, z, and t satisfy the relations

$$xy - zt = x + y = z + t. \qquad (*)$$

Is it possible that both xy and zt are perfect squares?

(Russia)

Answer: No.

Solution 1. Arguing indirectly, assume that $xy = a^2$ and $zt = c^2$ with $a, c > 0$.

Suppose that the number $x + y = z + t$ is odd. Then x and y have opposite parity, as well as z and t. This means that both xy and zt are even, as well as $xy - zt = x + y$; a contradiction. Thus, $x + y$ is even, so the number $s = \frac{x+y}{2} = \frac{z+t}{2}$ is a positive integer.

Next, we set $b = \frac{|x-y|}{2}$, $d = \frac{|z-t|}{2}$. Now the problem conditions yield

$$s^2 = a^2 + b^2 = c^2 + d^2 \qquad (1)$$

and

$$2s = a^2 - c^2 = d^2 - b^2 \qquad (2)$$

(the last equality in (2) follows from (1)). We readily get from (2) that $a, d > 0$.

In the sequel we will use only the relations (1) and (2), along with the fact that a, d, s are positive integers, while b and c are nonnegative integers, at most one of which may be zero. Since both relations are symmetric with respect to the simultaneous swappings $a \leftrightarrow d$ and $b \leftrightarrow c$, we assume, without loss of generality, that $b \geqslant c$ (and hence $b > 0$). Therefore, $d^2 = 2s + b^2 > c^2$, whence

$$d^2 > \frac{c^2 + d^2}{2} = \frac{s^2}{2}. \qquad (3)$$

On the other hand, since $d^2 - b^2$ is even by (2), the numbers b and d have the same parity, so $0 < b \leqslant d - 2$. Therefore,

$$2s = d^2 - b^2 \geqslant d^2 - (d-2)^2 = 4(d-1), \qquad \text{i.e.,} \qquad d \leqslant \frac{s}{2} + 1. \qquad (4)$$

Combining (3) and (4) we obtain

$$2s^2 < 4d^2 \leqslant 4\left(\frac{s}{2} + 1\right)^2, \qquad \text{or} \qquad (s-2)^2 < 8,$$

which yields $s \leqslant 4$.

Finally, an easy check shows that each number of the form s^2 with $1 \leqslant s \leqslant 4$ has a unique representation as a sum of two squares, namely $s^2 = s^2 + 0^2$. Thus, (1) along with $a, d > 0$ imply $b = c = 0$, which is impossible.

Solution 2. We start with a complete description of all 4-tuples (x, y, z, t) of positive integers satisfying $(*)$. As in the solution above, we notice that the numbers

$$s = \frac{x+y}{2} = \frac{z+t}{2}, \quad p = \frac{x-y}{2}, \quad \text{and} \quad q = \frac{z-t}{2}$$

are integers (we may, and will, assume that $p, q \geqslant 0$). We have

$$2s = xy - zt = (s+p)(s-p) - (s+q)(s-q) = q^2 - p^2,$$

so p and q have the same parity, and $q > p$.

Set now $k=\frac{q-p}{2}$, $\ell=\frac{q+p}{2}$. Then we have $s=\frac{q^2-p^2}{2}=2k\ell$ and hence

$$\begin{aligned} x&=s+p=2k\ell-k+\ell, \qquad & y&=s-p=2k\ell+k-\ell,\\ z&=s+q=2k\ell+k+\ell, \qquad & t&=s-q=2k\ell-k-\ell. \end{aligned} \tag{5}$$

Recall here that $\ell\geqslant k>0$ and, moreover, $(k,\ell)\neq(1,1)$, since otherwise $t=0$.

Assume now that both xy and zt are squares. Then $xyzt$ is also a square. On the other hand, we have

$$\begin{aligned} xyzt&=(2k\ell-k+\ell)(2k\ell+k-\ell)(2k\ell+k+\ell)(2k\ell-k-\ell)\\ &=\big(4k^2\ell^2-(k-\ell)^2\big)\big(4k^2\ell^2-(k+\ell)^2\big)=(4k^2\ell^2-k^2-\ell^2)^2-4k^2\ell^2. \end{aligned} \tag{6}$$

Denote $D=4k^2\ell^2-k^2-\ell^2>0$. From (6) we get $D^2>xyzt$. On the other hand,

$$\begin{aligned} (D-1)^2&=D^2-2(4k^2\ell^2-k^2-\ell^2)+1=(D^2-4k^2\ell^2)-(2k^2-1)(2\ell^2-1)+2\\ &=xyzt-(2k^2-1)(2\ell^2-1)+2<xyzt, \end{aligned}$$

since $\ell\geqslant 2$ and $k\geqslant 1$. Thus $(D-1)^2<xyzt<D^2$, and $xyzt$ cannot be a perfect square; a contradiction.

Comment. The first part of Solution 2 shows that all 4-tuples of positive integers $x\geqslant y$, $z\geqslant t$ satisfying $(*)$ have the form (5), where $\ell\geqslant k>0$ and $\ell\geqslant 2$. The converse is also true: every pair of positive integers $\ell\geqslant k>0$, except for the pair $k=\ell=1$, generates via (5) a 4-tuple of positive integers satisfying $(*)$.

N6. Let $f\colon \{1,2,3,\ldots\} \to \{2,3,\ldots\}$ be a function such that $f(m+n) \mid f(m)+f(n)$ for all pairs m, n of positive integers. Prove that there exists a positive integer $c > 1$ which divides all values of f.

(Mexico)

Solution 1. For every positive integer m, define $S_m = \{n\colon m \mid f(n)\}$.

Lemma. If the set S_m is infinite, then $S_m = \{d, 2d, 3d, \ldots\} = d \cdot \mathbb{Z}_{>0}$ for some positive integer d.

Proof. Let $d = \min S_m$; the definition of S_m yields $m \mid f(d)$.

Whenever $n \in S_m$ and $n > d$, we have $m \mid f(n) \mid f(n-d) + f(d)$, so $m \mid f(n-d)$ and therefore $n - d \in S_m$. Let $r \leqslant d$ be the least positive integer with $n \equiv r \pmod{d}$; repeating the same step, we can see that $n - d, n - 2d, \ldots, r \in S_m$. By the minimality of d, this shows $r = d$ and therefore $d \mid n$.

Starting from an arbitrarily large element of S_m, the process above reaches all multiples of d; so they all are elements of S_m. □

The solution for the problem will be split into two cases.

Case 1: The function f is bounded.

Call a prime p *frequent* if the set S_p is infinite, i.e., if p divides $f(n)$ for infinitely many positive integers n; otherwise call p *sporadic*. Since the function f is bounded, there are only a finite number of primes that divide at least one $f(n)$; so altogether there are finitely many numbers n such that $f(n)$ has a sporadic prime divisor. Let N be a positive integer, greater than all those numbers n.

Let $p_1, \ldots, p_k$ be the frequent primes. By the lemma we have $S_{p_i} = d_i \cdot \mathbb{Z}_{>0}$ for some d_i. Consider the number

$$n = N d_1 d_2 \cdots d_k + 1.$$

Due to $n > N$, all prime divisors of $f(n)$ are frequent primes. Let p_i be any frequent prime divisor of $f(n)$. Then $n \in S_{p_i}$, and therefore $d_i \mid n$. But $n \equiv 1 \pmod{d_i}$, which means $d_i = 1$. Hence $S_{p_i} = 1 \cdot \mathbb{Z}_{>0} = \mathbb{Z}_{>0}$ and therefore p_i is a common divisor of all values $f(n)$.

Case 2: f is unbounded.

We prove that $f(1)$ divides all $f(n)$.

Let $a = f(1)$. Since $1 \in S_a$, by the lemma it suffices to prove that S_a is an infinite set.

Call a positive integer p a *peak* if $f(p) > \max\big(f(1), \ldots, f(p-1)\big)$. Since f is not bounded, there are infinitely many peaks. Let $1 = p_1 < p_2 < \ldots$ be the sequence of all peaks, and let $h_k = f(p_k)$. Notice that for any peak p_i and for any $k < p_i$, we have $f(p_i) \mid f(k) + f(p_i - k) < 2f(p_i)$, hence

$$f(k) + f(p_i - k) = f(p_i) = h_i. \tag{1}$$

By the pigeonhole principle, among the numbers $h_1, h_2, \ldots$ there are infinitely many that are congruent modulo a. Let $k_0 < k_1 < k_2 < \ldots$ be an infinite sequence of positive integers such that $h_{k_0} \equiv h_{k_1} \equiv \ldots \pmod{a}$. Notice that

$$f(p_{k_i} - p_{k_0}) = f(p_{k_i}) - f(p_{k_0}) = h_{k_i} - h_{k_0} \equiv 0 \pmod{a},$$

so $p_{k_i} - p_{k_0} \in S_a$ for all $i = 1, 2, \ldots$. This provides infinitely many elements in S_a.

Hence, S_a is an infinite set, and therefore $f(1) = a$ divides $f(n)$ for every n.

Comment. As an extension of the solution above, it can be proven that if f is not bounded then $f(n) = an$ with $a = f(1)$.

Take an arbitrary positive integer n; we will show that $f(n+1) = f(n) + a$. Then it follows by induction that $f(n) = an$.

Take a peak p such that $p > n+2$ and $h = f(p) > f(n) + 2a$. By (1) we have $f(p-1) = f(p) - f(1) = h - a$ and $f(n+1) = f(p) - f(p-n-1) = h - f(p-n-1)$. From $h - a = f(p-1) \mid f(n) + f(p-n-1) < f(n) + h < 2(h-a)$ we get $f(n) + f(p-n-1) = h - a$. Then

$$f(n+1) - f(n) = \big(h - f(p-n-1)\big) - \big(h - a - f(p-n-1)\big) = a.$$

On the other hand, there exists a wide family of bounded functions satisfying the required properties. Here we present a few examples:

$$f(n) = c; \quad f(n) = \begin{cases} 2c & \text{if } n \text{ is even} \\ c & \text{if } n \text{ is odd;} \end{cases} \quad f(n) = \begin{cases} 2018c & \text{if } n \leqslant 2018 \\ c & \text{if } n > 2018. \end{cases}$$

Solution 2. Let $d_n = \gcd\big(f(n), f(1)\big)$. From $d_{n+1} \mid f(1)$ and $d_{n+1} \mid f(n+1) \mid f(n) + f(1)$, we can see that $d_{n+1} \mid f(n)$; then $d_{n+1} \mid \gcd\big(f(n), f(1)\big) = d_n$. So the sequence $d_1, d_2, \ldots$ is nonincreasing in the sense that every element is a divisor of the previous elements. Let $d = \min(d_1, d_2, \ldots) = \gcd(d_1.d_2, \ldots) = \gcd\big(f(1), f(2), \ldots\big)$; we have to prove $d \geqslant 2$.

For the sake of contradiction, suppose that the statement is wrong, so $d = 1$; that means there is some index n_0 such that $d_n = 1$ for every $n \geqslant n_0$, i.e., $f(n)$ is coprime with $f(1)$.

Claim 1. If $2^k \geqslant n_0$ then $f(2^k) \leqslant 2^k$.

Proof. By the condition, $f(2n) \mid 2f(n)$; a trivial induction yields $f(2^k) \mid 2^k f(1)$. If $2^k \geqslant n_0$ then $f(2^k)$ is coprime with $f(1)$, so $f(2^k)$ is a divisor of 2^k. □

Claim 2. There is a constant C such that $f(n) < n + C$ for every n.

Proof. Take the first power of 2 which is greater than or equal to n_0: let $K = 2^k \geqslant n_0$. By Claim 1, we have $f(K) \leqslant K$. Notice that $f(n+K) \mid f(n) + f(K)$ implies $f(n+K) \leqslant f(n) + f(K) \leqslant f(n) + K$. If $n = tK + r$ for some $t \geqslant 0$ and $1 \leqslant r \leqslant K$, then we conclude

$$f(n) \leqslant K + f(n-K) \leqslant 2K + f(n-2K) \leqslant \ldots \leqslant tK + f(r) < n + \max\big(f(1), f(2), \ldots, f(K)\big),$$

so the claim is true with $C = \max\big(f(1), \ldots, f(K)\big)$. □

Claim 3. If $a, b \in \mathbb{Z}_{>0}$ are coprime then $\gcd\big(f(a), f(b)\big) \mid f(1)$. In particular, if $a, b \geqslant n_0$ are coprime then $f(a)$ and $f(b)$ are coprime.

Proof. Let $d = \gcd\big(f(a), f(b)\big)$. We can replicate Euclid's algorithm. Formally, apply induction on $a + b$. If $a = 1$ or $b = 1$ then we already have $d \mid f(1)$.

Without loss of generality, suppose $1 < a < b$. Then $d \mid f(a)$ and $d \mid f(b) \mid f(a) + f(b-a)$, so $d \mid f(b-a)$. Therefore d divides $\gcd\big(f(a), f(b-a)\big)$ which is a divisor of $f(1)$ by the induction hypothesis. □

Let $p_1 < p_2 < \ldots$ be the sequence of all prime numbers; for every k, let q_k be the lowest power of p_k with $q_k \geqslant n_0$. (Notice that there are only finitely many positive integers with $q_k \neq p_k$.)

Take a positive integer N, and consider the numbers

$$f(1), f(q_1), f(q_2), \ldots, f(q_N).$$

Here we have $N+1$ numbers, each being greater than 1, and they are pairwise coprime by Claim 3. Therefore, they have at least $N+1$ different prime divisors in total, and their greatest prime divisor is at least p_{N+1}. Hence, $\max(f(1), f(q_1), \ldots, f(q_N)) \geqslant p_{N+1}$.

Choose N such that $\max(q_1, \ldots, q_N) = p_N$ (this is achieved if N is sufficiently large), and $p_{N+1} - p_N > C$ (that is possible, because there are arbitrarily long gaps between the primes). Then we establish a contradiction

$$p_{N+1} \leqslant \max(f(1), f(q_1), \ldots, f(q_N)) < \max(1 + C, q_1 + C, \ldots, q_N + C) = p_N + C < p_{N+1}$$

which proves the statement.

N7. Let $n \geqslant 2018$ be an integer, and let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be pairwise distinct positive integers not exceeding $5n$. Suppose that the sequence

$$\frac{a_1}{b_1}, \frac{a_2}{b_2}, \ldots, \frac{a_n}{b_n} \tag{1}$$

forms an arithmetic progression. Prove that the terms of the sequence are equal.

(Thailand)

Solution. Suppose that (1) is an arithmetic progression with nonzero difference. Let the difference be $\Delta = \frac{c}{d}$, where $d > 0$ and c, d are coprime.

We will show that too many denominators b_i should be divisible by d. To this end, for any $1 \leqslant i \leqslant n$ and any prime divisor p of d, say that *the index i is p-wrong*, if $v_p(b_i) < v_p(d)$. ($v_p(x)$ stands for the exponent of p in the prime factorisation of x.)

Claim 1. For any prime p, all p-wrong indices are congruent modulo p. In other words, the p-wrong indices (if they exist) are included in an arithmetic progression with difference p.

Proof. Let $\alpha = v_p(d)$. For the sake of contradiction, suppose that i and j are p-wrong indices (i.e., none of b_i and b_j is divisible by p^α) such that $i \not\equiv j \pmod p$. Then the least common denominator of $\frac{a_i}{b_i}$ and $\frac{a_j}{b_j}$ is not divisible by p^α. But this is impossible because in their difference, $(i-j)\Delta = \frac{(i-j)c}{d}$, the numerator is coprime to p, but p^α divides the denominator d. □

Claim 2. d has no prime divisors greater than 5.

Proof. Suppose that $p \geqslant 7$ is a prime divisor of d. Among the indices $1, 2, \ldots, n$, at most $\left\lceil \frac{n}{p} \right\rceil < \frac{n}{p} + 1$ are p-wrong, so p divides at least $\frac{p-1}{p}n - 1$ of $b_1, \ldots, b_n$. Since these denominators are distinct,

$$5n \geqslant \max\{b_i : \ p \mid b_i\} \geqslant \left(\frac{p-1}{p}n - 1\right) p = (p-1)(n-1) - 1 \geqslant 6(n-1) - 1 > 5n,$$

a contradiction. □

Claim 3. For every $0 \leqslant k \leqslant n - 30$, among the denominators $b_{k+1}, b_{k+2}, \ldots, b_{k+30}$, at least $\varphi(30) = 8$ are divisible by d.

Proof. By Claim 1, the 2-wrong, 3-wrong and 5-wrong indices can be covered by three arithmetic progressions with differences 2, 3 and 5. By a simple inclusion-exclusion, $(2-1)\cdot(3-1)\cdot(5-1) = 8$ indices are not covered; by Claim 2, we have $d \mid b_i$ for every uncovered index i. □

Claim 4. $|\Delta| < \frac{20}{n-2}$ and $d > \frac{n-2}{20}$.

Proof. From the sequence (1), remove all fractions with $b_n < \frac{n}{2}$, There remain at least $\frac{n}{2}$ fractions, and they cannot exceed $\frac{5n}{n/2} = 10$. So we have at least $\frac{n}{2}$ elements of the arithmetic progression (1) in the interval $(0, 10]$, hence the difference must be below $\frac{10}{n/2-1} = \frac{20}{n-2}$.

The second inequality follows from $\frac{1}{d} \leqslant \frac{|c|}{d} = |\Delta|$. □

Now we have everything to get the final contradiction. By Claim 3, we have $d \mid b_i$ for at least $\left\lfloor \frac{n}{30} \right\rfloor \cdot 8$ indices i. By Claim 4, we have $d \geqslant \frac{n-2}{20}$. Therefore,

$$5n \geqslant \max\{b_i : \ d \mid b_i\} \geqslant \left(\left\lfloor \frac{n}{30} \right\rfloor \cdot 8\right) \cdot d > \left(\frac{n}{30} - 1\right) \cdot 8 \cdot \frac{n-2}{20} > 5n.$$

Comment 1. It is possible that all terms in (1) are equal, for example with $a_i = 2i-1$ and $b_i = 4i-2$ we have $\frac{a_i}{b_i} = \frac{1}{2}$.

Comment 2. The bound $5n$ in the statement is far from sharp; the solution above can be modified to work for $9n$. For large n, the bound $5n$ can be replaced by $n^{\frac{3}{2}-\varepsilon}$.


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Shortlisted Problems
(with solutions)





Shortlisted Problems

(with solutions)

58th International Mathematical Olympiad

Rio de Janeiro, 12–23 July 2017

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Problems

Algebra

A1. Let $a_1, a_2, \ldots, a_n, k$, and M be positive integers such that

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} = k \quad \text{and} \quad a_1 a_2 \ldots a_n = M.$$

If $M > 1$, prove that the polynomial

$$P(x) = M(x+1)^k - (x+a_1)(x+a_2)\cdots(x+a_n)$$

has no positive roots.

(Trinidad and Tobago)

A2. Let q be a real number. Gugu has a napkin with ten distinct real numbers written on it, and he writes the following three lines of real numbers on the blackboard:

- In the first line, Gugu writes down every number of the form $a-b$, where a and b are two (not necessarily distinct) numbers on his napkin.
- In the second line, Gugu writes down every number of the form qab, where a and b are two (not necessarily distinct) numbers from the *first line*.
- In the third line, Gugu writes down every number of the form $a^2+b^2-c^2-d^2$, where a, b, c, d are four (not necessarily distinct) numbers from the *first line*.

Determine all values of q such that, regardless of the numbers on Gugu's napkin, every number in the second line is also a number in the third line.

(Austria)

A3. Let S be a finite set, and let $\mathcal{A}$ be the set of all functions from S to S. Let f be an element of $\mathcal{A}$, and let $T = f(S)$ be the image of S under f. Suppose that $f \circ g \circ f \neq g \circ f \circ g$ for every g in $\mathcal{A}$ with $g \neq f$. Show that $f(T) = T$.

(India)

A4. A sequence of real numbers $a_1, a_2, \ldots$ satisfies the relation

$$a_n = -\max_{i+j=n}(a_i + a_j) \qquad \text{for all } n > 2017.$$

Prove that this sequence is bounded, i.e., there is a constant M such that $|a_n| \leqslant M$ for all positive integers n.

(Russia)

A5. An integer $n \geqslant 3$ is given. We call an n-tuple of real numbers $(x_1, x_2, \ldots, x_n)$ *Shiny* if for each permutation $y_1, y_2, \ldots, y_n$ of these numbers we have

$$\sum_{i=1}^{n-1} y_i y_{i+1} = y_1y_2 + y_2y_3 + y_3y_4 + \cdots + y_{n-1}y_n \geqslant -1.$$

Find the largest constant $K = K(n)$ such that

$$\sum_{1 \leqslant i < j \leqslant n} x_i x_j \geqslant K$$

holds for every Shiny n-tuple $(x_1, x_2, \ldots, x_n)$.

(Serbia)

A6. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)f(y)) + f(x+y) = f(xy)$$

for all $x, y \in \mathbb{R}$.

(Albania)

A7. Let $a_0, a_1, a_2, \ldots$ be a sequence of integers and $b_0, b_1, b_2, \ldots$ be a sequence of *positive* integers such that $a_0 = 0, a_1 = 1$, and

$$a_{n+1} = \begin{cases} a_n b_n + a_{n-1}, & \text{if } b_{n-1} = 1 \\ a_n b_n - a_{n-1}, & \text{if } b_{n-1} > 1 \end{cases} \qquad \text{for } n = 1, 2, \ldots.$$

Prove that at least one of the two numbers a_{2017} and a_{2018} must be greater than or equal to 2017.

(Australia)

A8. Assume that a function $f\colon \mathbb{R} \to \mathbb{R}$ satisfies the following condition:

For every $x, y \in \mathbb{R}$ such that $\big(f(x)+y\big)\big(f(y)+x\big) > 0$, we have $f(x)+y = f(y)+x$.

Prove that $f(x) + y \leqslant f(y) + x$ whenever $x > y$.

(Netherlands)

Combinatorics

C1. A rectangle $\mathcal{R}$ with odd integer side lengths is divided into small rectangles with integer side lengths. Prove that there is at least one among the small rectangles whose distances from the four sides of $\mathcal{R}$ are either all odd or all even.

(Singapore)

C2. Let n be a positive integer. Define a *chameleon* to be any sequence of $3n$ letters, with exactly n occurrences of each of the letters a, b, and c. Define a *swap* to be the transposition of two adjacent letters in a chameleon. Prove that for any chameleon X, there exists a chameleon Y such that X cannot be changed to Y using fewer than $3n^2/2$ swaps.

(Australia)

C3. Sir Alex plays the following game on a row of 9 cells. Initially, all cells are empty. In each move, Sir Alex is allowed to perform exactly one of the following two operations:

(1) Choose any number of the form 2^j, where j is a non-negative integer, and put it into an empty cell.

(2) Choose two (not necessarily adjacent) cells with the same number in them; denote that number by 2^j. Replace the number in one of the cells with 2^{j+1} and erase the number in the other cell.

At the end of the game, one cell contains the number 2^n, where n is a given positive integer, while the other cells are empty. Determine the maximum number of moves that Sir Alex could have made, in terms of n.

(Thailand)

C4. Let $N \geqslant 2$ be an integer. $N(N+1)$ soccer players, no two of the same height, stand in a row in some order. Coach Ralph wants to remove $N(N-1)$ people from this row so that in the remaining row of $2N$ players, no one stands between the two tallest ones, no one stands between the third and the fourth tallest ones, ..., and finally no one stands between the two shortest ones. Show that this is always possible.

(Russia)

C5. A hunter and an invisible rabbit play a game in the Euclidean plane. The hunter's starting point H_0 coincides with the rabbit's starting point R_0. In the nth round of the game ($n \geqslant 1$), the following happens.

(1) First the invisible rabbit moves secretly and unobserved from its current point R_{n-1} to some new point R_n with $R_{n-1}R_n = 1$.

(2) The hunter has a tracking device (e.g. dog) that returns an approximate position R'_n of the rabbit, so that $R_nR'_n \leqslant 1$.

(3) The hunter then visibly moves from point H_{n-1} to a new point H_n with $H_{n-1}H_n = 1$.

Is there a strategy for the hunter that guarantees that after 10^9 such rounds the distance between the hunter and the rabbit is below 100?

(Austria)

C6. Let $n > 1$ be an integer. An $n \times n \times n$ cube is composed of n^3 unit cubes. Each unit cube is painted with one color. For each $n \times n \times 1$ box consisting of n^2 unit cubes (of any of the three possible orientations), we consider the set of the colors present in that box (each color is listed only once). This way, we get $3n$ sets of colors, split into three groups according to the orientation. It happens that for every set in any group, the same set appears in both of the other groups. Determine, in terms of n, the maximal possible number of colors that are present.

(Russia)

C7. For any finite sets X and Y of positive integers, denote by $f_X(k)$ the kth smallest positive integer not in X, and let

$$X * Y = X \cup \{f_X(y) \colon y \in Y\}.$$

Let A be a set of $a > 0$ positive integers, and let B be a set of $b > 0$ positive integers. Prove that if $A * B = B * A$, then

$$\underbrace{A * (A * \cdots * (A * (A * A)) \ldots)}_{A \text{ appears } b \text{ times}} = \underbrace{B * (B * \cdots * (B * (B * B)) \ldots)}_{B \text{ appears } a \text{ times}}.$$

(U.S.A.)

C8. Let n be a given positive integer. In the Cartesian plane, each lattice point with nonnegative coordinates initially contains a butterfly, and there are no other butterflies. The *neighborhood* of a lattice point c consists of all lattice points within the axis-aligned $(2n+1) \times (2n+1)$ square centered at c, apart from c itself. We call a butterfly *lonely*, *crowded*, or *comfortable*, depending on whether the number of butterflies in its neighborhood N is respectively less than, greater than, or equal to half of the number of lattice points in N.

Every minute, all lonely butterflies fly away simultaneously. This process goes on for as long as there are any lonely butterflies. Assuming that the process eventually stops, determine the number of comfortable butterflies at the final state.

(Bulgaria)

Geometry

G1. Let $ABCDE$ be a convex pentagon such that $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that the perpendicular line from E to BC and the line segments AC and BD are concurrent.

(Italy)

G2. Let R and S be distinct points on circle Ω, and let t denote the tangent line to Ω at R. Point R' is the reflection of R with respect to S. A point I is chosen on the smaller arc RS of Ω so that the circumcircle Γ of triangle ISR' intersects t at two different points. Denote by A the common point of Γ and t that is closest to R. Line AI meets Ω again at J. Show that JR' is tangent to Γ.

(Luxembourg)

G3. Let O be the circumcenter of an acute scalene triangle ABC. Line OA intersects the altitudes of ABC through B and C at P and Q, respectively. The altitudes meet at H. Prove that the circumcenter of triangle PQH lies on a median of triangle ABC.

(Ukraine)

G4. In triangle ABC, let ω be the excircle opposite A. Let D, E, and F be the points where ω is tangent to lines BC, CA, and AB, respectively. The circle AEF intersects line BC at P and Q. Let M be the midpoint of AD. Prove that the circle MPQ is tangent to ω.

(Denmark)

G5. Let $ABCC_1B_1A_1$ be a convex hexagon such that $AB = BC$, and suppose that the line segments AA_1, BB_1, and CC_1 have the same perpendicular bisector. Let the diagonals AC_1 and A_1C meet at D, and denote by ω the circle ABC. Let ω intersect the circle A_1BC_1 again at $E \neq B$. Prove that the lines BB_1 and DE intersect on ω.

(Ukraine)

G6. Let $n \geqslant 3$ be an integer. Two regular n-gons $\mathcal{A}$ and $\mathcal{B}$ are given in the plane. Prove that the vertices of $\mathcal{A}$ that lie inside $\mathcal{B}$ or on its boundary are consecutive.

(That is, prove that there exists a line separating those vertices of $\mathcal{A}$ that lie inside $\mathcal{B}$ or on its boundary from the other vertices of $\mathcal{A}$.)

(Czech Republic)

G7. A convex quadrilateral $ABCD$ has an inscribed circle with center I. Let I_a, I_b, I_c, and I_d be the incenters of the triangles DAB, ABC, BCD, and CDA, respectively. Suppose that the common external tangents of the circles AI_bI_d and CI_bI_d meet at X, and the common external tangents of the circles BI_aI_c and DI_aI_c meet at Y. Prove that $\angle XIY = 90^\circ$.

(Kazakhstan)

G8. There are 2017 mutually external circles drawn on a blackboard, such that no two are tangent and no three share a common tangent. A *tangent segment* is a line segment that is a common tangent to two circles, starting at one tangent point and ending at the other one. Luciano is drawing tangent segments on the blackboard, one at a time, so that no tangent segment intersects any other circles or previously drawn tangent segments. Luciano keeps drawing tangent segments until no more can be drawn. Find all possible numbers of tangent segments when he stops drawing.

(Australia)

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Number Theory

N1. The sequence $a_0, a_1, a_2, \ldots$ of positive integers satisfies

$$a_{n+1} = \begin{cases} \sqrt{a_n}, & \text{if } \sqrt{a_n} \text{ is an integer} \\ a_n + 3, & \text{otherwise} \end{cases} \qquad \text{for every } n \geqslant 0.$$

Determine all values of $a_0 > 1$ for which there is at least one number a such that $a_n = a$ for infinitely many values of n.

(South Africa)

N2. Let $p \geqslant 2$ be a prime number. Eduardo and Fernando play the following game making moves alternately: in each move, the current player chooses an index i in the set $\{0, 1, \ldots, p-1\}$ that was not chosen before by either of the two players and then chooses an element a_i of the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Eduardo has the first move. The game ends after all the indices $i \in \{0, 1, \ldots, p-1\}$ have been chosen. Then the following number is computed:

$$M = a_0 + 10 \cdot a_1 + \cdots + 10^{p-1} \cdot a_{p-1} = \sum_{j=0}^{p-1} a_j \cdot 10^j.$$

The goal of Eduardo is to make the number M divisible by p, and the goal of Fernando is to prevent this.

Prove that Eduardo has a winning strategy.

(Morocco)

N3. Determine all integers $n \geqslant 2$ with the following property: for any integers $a_1, a_2, \ldots, a_n$ whose sum is not divisible by n, there exists an index $1 \leqslant i \leqslant n$ such that none of the numbers

$$a_i, a_i + a_{i+1}, \ldots, a_i + a_{i+1} + \cdots + a_{i+n-1}$$

is divisible by n. (We let $a_i = a_{i-n}$ when $i > n$.)

(Thailand)

N4. Call a rational number *short* if it has finitely many digits in its decimal expansion. For a positive integer m, we say that a positive integer t is *m-tastic* if there exists a number $c \in \{1, 2, 3, \ldots, 2017\}$ such that $\dfrac{10^t - 1}{c \cdot m}$ is short, and such that $\dfrac{10^k - 1}{c \cdot m}$ is not short for any $1 \leqslant k < t$. Let $S(m)$ be the set of m-tastic numbers. Consider $S(m)$ for $m = 1, 2, \ldots$. What is the maximum number of elements in $S(m)$?

(Turkey)

N5. Find all pairs (p, q) of prime numbers with $p > q$ for which the number

$$\frac{(p+q)^{p+q}(p-q)^{p-q} - 1}{(p+q)^{p-q}(p-q)^{p+q} - 1}$$

is an integer.

(Japan)

N6. Find the smallest positive integer n, or show that no such n exists, with the following property: there are infinitely many distinct n-tuples of positive rational numbers $(a_1, a_2, \ldots, a_n)$ such that both

$$a_1 + a_2 + \cdots + a_n \qquad \text{and} \qquad \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}$$

are integers.

(Singapore)

N7. Say that an ordered pair (x, y) of integers is an *irreducible lattice point* if x and y are relatively prime. For any finite set S of irreducible lattice points, show that there is a homogenous polynomial in two variables, $f(x, y)$, with integer coefficients, of degree at least 1, such that $f(x, y) = 1$ for each (x, y) in the set S.

Note: A homogenous polynomial of degree n is any nonzero polynomial of the form

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots + a_{n-1} x y^{n-1} + a_n y^n.$$

(U.S.A.)

N8. Let p be an odd prime number and $\mathbb{Z}_{>0}$ be the set of positive integers. Suppose that a function $f\colon \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} \to \{0, 1\}$ satisfies the following properties:

- $f(1, 1) = 0$;
- $f(a, b) + f(b, a) = 1$ for any pair of relatively prime positive integers (a, b) not both equal to 1;
- $f(a + b, b) = f(a, b)$ for any pair of relatively prime positive integers (a, b).

Prove that

$$\sum_{n=1}^{p-1} f(n^2, p) \geqslant \sqrt{2p} - 2.$$

(Italy)

Solutions

Algebra

A1. Let $a_1, a_2, \ldots, a_n, k$, and M be positive integers such that

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} = k \quad \text{and} \quad a_1 a_2 \ldots a_n = M.$$

If $M > 1$, prove that the polynomial

$$P(x) = M(x+1)^k - (x + a_1)(x + a_2) \cdots (x + a_n)$$

has no positive roots.

(Trinidad and Tobago)

Solution 1. We first prove that, for $x > 0$,

$$a_i(x+1)^{1/a_i} \leqslant x + a_i, \tag{1}$$

with equality if and only if $a_i = 1$. It is clear that equality occurs if $a_i = 1$.

If $a_i > 1$, the AM–GM inequality applied to a single copy of $x + 1$ and $a_i - 1$ copies of 1 yields

$$\frac{(x+1) + \overbrace{1 + 1 + \cdots + 1}^{a_i - 1 \text{ ones}}}{a_i} \geqslant \sqrt[a_i]{(x+1) \cdot 1^{a_i - 1}} \implies a_i(x+1)^{1/a_i} \leqslant x + a_i.$$

Since $x + 1 > 1$, the inequality is strict for $a_i > 1$.

Multiplying the inequalities (1) for $i = 1, 2, \ldots, n$ yields

$$\prod_{i=1}^{n} a_i(x+1)^{1/a_i} \leqslant \prod_{i=1}^{n} (x + a_i) \iff M(x+1)^{\sum_{i=1}^{n} 1/a_i} - \prod_{i=1}^{n} (x + a_i) \leqslant 0 \iff P(x) \leqslant 0$$

with equality iff $a_i = 1$ for all $i \in \{1, 2, \ldots, n\}$. But this implies $M = 1$, which is not possible. Hence $P(x) < 0$ for all $x \in \mathbb{R}^+$, and P has no positive roots.

Comment 1. Inequality (1) can be obtained in several ways. For instance, we may also use the binomial theorem: since $a_i \geqslant 1$,

$$\left(1 + \frac{x}{a_i}\right)^{a_i} = \sum_{j=0}^{a_i} \binom{a_i}{j} \left(\frac{x}{a_i}\right)^j \geqslant \binom{a_i}{0} + \binom{a_i}{1} \cdot \frac{x}{a_i} = 1 + x.$$

Both proofs of (1) mimic proofs to Bernoulli's inequality for a positive integer exponent a_i; we can use this inequality directly:

$$\left(1 + \frac{x}{a_i}\right)^{a_i} \geqslant 1 + a_i \cdot \frac{x}{a_i} = 1 + x,$$

and so

$$x + a_i = a_i \left(1 + \frac{x}{a_i}\right) \geqslant a_i(1 + x)^{1/a_i},$$

or its (reversed) formulation, with exponent $1/a_i \leqslant 1$:

$$(1 + x)^{1/a_i} \leqslant 1 + \frac{1}{a_i} \cdot x = \frac{x + a_i}{a_i} \implies a_i(1 + x)^{1/a_i} \leqslant x + a_i.$$

Solution 2. We will prove that, in fact, all coefficients of the polynomial $P(x)$ are non-positive, and at least one of them is negative, which implies that $P(x) < 0$ for $x > 0$.

Indeed, since $a_j \geqslant 1$ for all j and $a_j > 1$ for some j (since $a_1 a_2 \dots a_n = M > 1$), we have $k = \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} < n$, so the coefficient of x^n in $P(x)$ is $-1 < 0$. Moreover, the coefficient of x^r in $P(x)$ is negative for $k < r \leqslant n = \deg(P)$.

For $0 \leqslant r \leqslant k$, the coefficient of x^r in $P(x)$ is

$$M \cdot \binom{k}{r} - \sum_{1 \leqslant i_1 < i_2 < \dots < i_{n-r} \leqslant n} a_{i_1} a_{i_2} \cdots a_{i_{n-r}} = a_1 a_2 \cdots a_n \cdot \binom{k}{r} - \sum_{1 \leqslant i_1 < i_2 < \dots < i_{n-r} \leqslant n} a_{i_1} a_{i_2} \cdots a_{i_{n-r}},$$

which is non-positive iff

$$\binom{k}{r} \leqslant \sum_{1 \leqslant j_1 < j_2 < \dots < j_r \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}}. \tag{2}$$

We will prove (2) by induction on r. For $r = 0$ it is an equality because the constant term of $P(x)$ is $P(0) = 0$, and if $r = 1$, (2) becomes $k = \sum_{i=1}^{n} \frac{1}{a_i}$. For $r > 1$, if (2) is true for a given $r < k$, we have

$$\binom{k}{r+1} = \frac{k-r}{r+1} \cdot \binom{k}{r} \leqslant \frac{k-r}{r+1} \cdot \sum_{1 \leqslant j_1 < j_2 < \dots < j_r \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}},$$

and it suffices to prove that

$$\frac{k-r}{r+1} \cdot \sum_{1 \leqslant j_1 < j_2 < \dots < j_r \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}} \leqslant \sum_{1 \leqslant j_1 < \dots < j_r < j_{r+1} \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r} a_{j_{r+1}}},$$

which is equivalent to

$$\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} - r\right) \sum_{1 \leqslant j_1 < j_2 < \dots < j_r \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}} \leqslant (r+1) \sum_{1 \leqslant j_1 < \dots < j_r < j_{r+1} \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r} a_{j_{r+1}}}.$$

Since there are $r+1$ ways to choose a fraction $\frac{1}{a_{j_i}}$ from $\frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r} a_{j_{r+1}}}$ to factor out, every term $\frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r} a_{j_{r+1}}}$ in the right hand side appears exactly $r+1$ times in the product

$$\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right) \sum_{1 \leqslant j_1 < j_2 < \dots < j_r \leqslant n} \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}}.$$

Hence all terms in the right hand side cancel out.

The remaining terms in the left hand side can be grouped in sums of the type

$$\frac{1}{a_{j_1}^2 a_{j_2} \cdots a_{j_r}} + \frac{1}{a_{j_1} a_{j_2}^2 \cdots a_{j_r}} + \dots + \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}^2} - \frac{r}{a_{j_1} a_{j_2} \cdots a_{j_r}}$$
$$= \frac{1}{a_{j_1} a_{j_2} \cdots a_{j_r}} \left(\frac{1}{a_{j_1}} + \frac{1}{a_{j_2}} + \dots + \frac{1}{a_{j_r}} - r\right),$$

which are all non-positive because $a_i \geqslant 1 \implies \frac{1}{a_i} \leqslant 1$, $i = 1, 2, \dots, n$.

Comment 2. The result is valid for any real numbers a_i, $i = 1, 2, \dots, n$ with $a_i \geqslant 1$ and product M greater than 1. A variation of Solution 1, namely using *weighted* AM–GM (or the Bernoulli inequality for real exponents), actually proves that $P(x) < 0$ for $x > -1$ and $x \neq 0$.

A2. Let q be a real number. Gugu has a napkin with ten distinct real numbers written on it, and he writes the following three lines of real numbers on the blackboard:

- In the first line, Gugu writes down every number of the form $a-b$, where a and b are two (not necessarily distinct) numbers on his napkin.
- In the second line, Gugu writes down every number of the form qab, where a and b are two (not necessarily distinct) numbers from the *first line*.
- In the third line, Gugu writes down every number of the form $a^2+b^2-c^2-d^2$, where a, b, c, d are four (not necessarily distinct) numbers from the *first line*.

Determine all values of q such that, regardless of the numbers on Gugu's napkin, every number in the second line is also a number in the third line.

(Austria)

Answer: $-2, 0, 2$.

Solution 1. Call a number q *good* if every number in the second line appears in the third line unconditionally. We first show that the numbers 0 and ± 2 are good. The third line necessarily contains 0, so 0 is good. For any two numbers a, b in the first line, write $a = x-y$ and $b = u-v$, where x, y, u, v are (not necessarily distinct) numbers on the napkin. We may now write

$$2ab = 2(x-y)(u-v) = (x-v)^2 + (y-u)^2 - (x-u)^2 - (y-v)^2,$$

which shows that 2 is good. By negating both sides of the above equation, we also see that -2 is good.

We now show that $-2, 0$, and 2 are the only good numbers. Assume for sake of contradiction that q is a good number, where $q \notin \{-2, 0, 2\}$. We now consider some particular choices of numbers on Gugu's napkin to arrive at a contradiction.

Assume that the napkin contains the integers $1, 2, \ldots, 10$. Then, the first line contains the integers $-9, -8, \ldots, 9$. The second line then contains q and $81q$, so the third line must also contain both of them. But the third line only contains integers, so q must be an integer. Furthermore, the third line contains no number greater than $162 = 9^2 + 9^2 - 0^2 - 0^2$ or less than -162, so we must have $-162 \leqslant 81q \leqslant 162$. This shows that the only possibilities for q are ± 1.

Now assume that $q = \pm 1$. Let the napkin contain $0, 1, 4, 8, 12, 16, 20, 24, 28, 32$. The first line contains ± 1 and ± 4, so the second line contains ± 4. However, for every number a in the first line, $a \not\equiv 2 \pmod 4$, so we may conclude that $a^2 \equiv 0, 1 \pmod 8$. Consequently, every number in the third line must be congruent to $-2, -1, 0, 1, 2 \pmod 8$; in particular, ± 4 cannot be in the third line, which is a contradiction.

Solution 2. Let q be a good number, as defined in the first solution, and define the polynomial $P(x_1, \ldots, x_{10})$ as

$$\prod_{i<j}(x_i - x_j) \prod_{a_i \in S} \big(q(x_1-x_2)(x_3-x_4) - (a_1-a_2)^2 - (a_3-a_4)^2 + (a_5-a_6)^2 + (a_7-a_8)^2\big),$$

where $S = \{x_1, \ldots, x_{10}\}$.

We claim that $P(x_1, \ldots, x_{10}) = 0$ for every choice of real numbers $(x_1, \ldots, x_{10})$. If any two of the x_i are equal, then $P(x_1, \ldots, x_{10}) = 0$ trivially. If no two are equal, assume that Gugu has those ten numbers $x_1, \ldots, x_{10}$ on his napkin. Then, the number $q(x_1-x_2)(x_3-x_4)$ is in the second line, so we must have some $a_1, \ldots, a_8$ so that

$$q(x_1-x_2)(x_3-x_4) - (a_1-a_2)^2 - (a_3-a_4)^2 + (a_5-a_6)^2 + (a_7-a_8)^2 = 0,$$

and hence $P(x_1, \ldots, x_{10}) = 0$.

Since every polynomial that evaluates to zero everywhere is the zero polynomial, and the product of two nonzero polynomials is necessarily nonzero, we may define F such that

$$F(x_1, \ldots, x_{10}) \equiv q(x_1 - x_2)(x_3 - x_4) - (a_1 - a_2)^2 - (a_3 - a_4)^2 + (a_5 - a_6)^2 + (a_7 - a_8)^2 \equiv 0 \quad (1)$$

for some particular choice $a_i \in S$.

Each of the sets $\{a_1, a_2\}$, $\{a_3, a_4\}$, $\{a_5, a_6\}$, and $\{a_7, a_8\}$ is equal to at most one of the four sets $\{x_1, x_3\}, \{x_2, x_3\}, \{x_1, x_4\}$, and $\{x_2, x_4\}$. Thus, without loss of generality, we may assume that at most one of the sets $\{a_1, a_2\}, \{a_3, a_4\}, \{a_5, a_6\}$, and $\{a_7, a_8\}$ is equal to $\{x_1, x_3\}$. Let u_1, u_3, u_5, u_7 be the indicator functions for this equality of sets: that is, $u_i = 1$ if and only if $\{a_i, a_{i+1}\} = \{x_1, x_3\}$. By assumption, at least three of the u_i are equal to 0.

We now compute the coefficient of $x_1 x_3$ in F. It is equal to $q + 2(u_1 + u_3 - u_5 - u_7) = 0$, and since at least three of the u_i are zero, we must have that $q \in \{-2, 0, 2\}$, as desired.

A3. Let S be a finite set, and let $\mathcal{A}$ be the set of all functions from S to S. Let f be an element of $\mathcal{A}$, and let $T = f(S)$ be the image of S under f. Suppose that $f \circ g \circ f \neq g \circ f \circ g$ for every g in $\mathcal{A}$ with $g \neq f$. Show that $f(T) = T$.

(India)

Solution. For $n \geqslant 1$, denote the n-th composition of f with itself by

$$f^n \stackrel{\text{def}}{=} \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}.$$

By hypothesis, if $g \in \mathcal{A}$ satisfies $f \circ g \circ f = g \circ f \circ g$, then $g = f$. A natural idea is to try to plug in $g = f^n$ for some n in the expression $f \circ g \circ f = g \circ f \circ g$ in order to get $f^n = f$, which solves the problem:

Claim. If there exists $n \geqslant 3$ such that $f^{n+2} = f^{2n+1}$, then the restriction $f\colon T \to T$ of f to T is a bijection.

Proof. Indeed, by hypothesis, $f^{n+2} = f^{2n+1} \iff f \circ f^n \circ f = f^n \circ f \circ f^n \implies f^n = f$. Since $n - 2 \geqslant 1$, the image of f^{n-2} is contained in $T = f(S)$, hence f^{n-2} restricts to a function $f^{n-2}\colon T \to T$. This is the inverse of $f\colon T \to T$. In fact, given $t \in T$, say $t = f(s)$ with $s \in S$, we have

$$t = f(s) = f^n(s) = f^{n-2}(f(t)) = f(f^{n-2}(t)), \qquad \text{i.e.,} \qquad f^{n-2} \circ f = f \circ f^{n-2} = \mathrm{id} \text{ on } T$$

(here id stands for the identity function). Hence, the restriction $f\colon T \to T$ of f to T is bijective with inverse given by $f^{n-2}\colon T \to T$. □

It remains to show that n as in the claim exists. For that, define

$$S_m \stackrel{\text{def}}{=} f^m(S) \qquad (S_m \text{ is image of } f^m)$$

Clearly the image of f^{m+1} is contained in the image of f^m, i.e., there is a descending chain of subsets of S

$$S \supseteq S_1 \supseteq S_2 \supseteq S_3 \supseteq S_4 \supseteq \cdots,$$

which must eventually stabilise since S is finite, i.e., there is a $k \geqslant 1$ such that

$$S_k = S_{k+1} = S_{k+2} = S_{k+3} = \cdots \stackrel{\text{def}}{=} S_\infty.$$

Hence f restricts to a *surjective* function $f\colon S_\infty \to S_\infty$, which is also *bijective* since $S_\infty \subseteq S$ is finite. To sum up, $f\colon S_\infty \to S_\infty$ is a permutation of the elements of the finite set S_∞, hence there exists an integer $r \geqslant 1$ such that $f^r = \mathrm{id}$ on S_∞ (for example, we may choose $r = |S_\infty|!$). In other words,

$$f^{m+r} = f^m \text{ on } S \text{ for all } m \geqslant k. \tag{$*$}$$

Clearly, $(*)$ also implies that $f^{m+tr} = f^m$ for all integers $t \geqslant 1$ and $m \geqslant k$. So, to find n as in the claim and finish the problem, it is enough to choose m and t in order to ensure that there exists $n \geqslant 3$ satisfying

$$\begin{cases} 2n + 1 = m + tr \\ n + 2 = m \end{cases} \iff \begin{cases} m = 3 + tr \\ n = m - 2. \end{cases}$$

This can be clearly done by choosing m large enough with $m \equiv 3 \pmod{r}$. For instance, we may take $n = 2kr + 1$, so that

$$f^{n+2} = f^{2kr+3} = f^{4kr+3} = f^{2n+1}$$

where the middle equality follows by $(*)$ since $2kr + 3 \geqslant k$.

A4. A sequence of real numbers $a_1, a_2, \ldots$ satisfies the relation

$$a_n = -\max_{i+j=n}(a_i + a_j) \qquad \text{for all } n > 2017.$$

Prove that this sequence is bounded, i.e., there is a constant M such that $|a_n| \leqslant M$ for all positive integers n.

(Russia)

Solution 1. Set $D = 2017$. Denote

$$M_n = \max_{k<n} a_k \qquad \text{and} \qquad m_n = -\min_{k<n} a_k = \max_{k<n}(-a_k).$$

Clearly, the sequences (m_n) and (M_n) are nondecreasing. We need to prove that both are bounded.

Consider an arbitrary $n > D$; our first aim is to bound a_n in terms of m_n and M_n.

(i) There exist indices p and q such that $a_n = -(a_p + a_q)$ and $p + q = n$. Since $a_p, a_q \leqslant M_n$, we have $a_n \geqslant -2M_n$.

(ii) On the other hand, choose an index $k < n$ such that $a_k = M_n$. Then, we have

$$a_n = -\max_{\ell<n}(a_{n-\ell} + a_\ell) \leqslant -(a_{n-k} + a_k) = -a_{n-k} - M_n \leqslant m_n - M_n.$$

Summarizing (i) and (ii), we get

$$-2M_n \leqslant a_n \leqslant m_n - M_n,$$

whence

$$m_n \leqslant m_{n+1} \leqslant \max\{m_n, 2M_n\} \quad \text{and} \quad M_n \leqslant M_{n+1} \leqslant \max\{M_n, m_n - M_n\}. \tag{1}$$

Now, say that an index $n > D$ is *lucky* if $m_n \leqslant 2M_n$. Two cases are possible.

Case 1. Assume that there exists a lucky index n. In this case, (1) yields $m_{n+1} \leqslant 2M_n$ and $M_n \leqslant M_{n+1} \leqslant M_n$. Therefore, $M_{n+1} = M_n$ and $m_{n+1} \leqslant 2M_n = 2M_{n+1}$. So, the index $n + 1$ is also lucky, and $M_{n+1} = M_n$. Applying the same arguments repeatedly, we obtain that all indices $k > n$ are lucky (i.e., $m_k \leqslant 2M_k$ for all these indices), and $M_k = M_n$ for all such indices. Thus, all of the m_k and M_k are bounded by $2M_n$.

Case 2. Assume now that there is no lucky index, i.e., $2M_n < m_n$ for all $n > D$. Then (1) shows that for all $n > D$ we have $m_n \leqslant m_{n+1} \leqslant m_n$, so $m_n = m_{D+1}$ for all $n > D$. Since $M_n < m_n/2$ for all such indices, all of the m_n and M_n are bounded by m_{D+1}.

Thus, in both cases the sequences (m_n) and (M_n) are bounded, as desired.

Solution 2. As in the previous solution, let $D = 2017$. If the sequence is bounded above, say, by Q, then we have that $a_n \geqslant \min\{a_1, \ldots, a_D, -2Q\}$ for all n, so the sequence is bounded. Assume for sake of contradiction that the sequence is not bounded above. Let $\ell = \min\{a_1, \ldots, a_D\}$, and $L = \max\{a_1, \ldots, a_D\}$. Call an index n *good* if the following criteria hold:

$$a_n > a_i \text{ for each } i < n, \quad a_n > -2\ell, \quad \text{and} \quad n > D \tag{2}$$

We first show that there must be some good index n. By assumption, we may take an index N such that $a_N > \max\{L, -2\ell\}$. Choose n minimally such that $a_n = \max\{a_1, a_2, \ldots, a_N\}$. Now, the first condition in (2) is satisfied because of the minimality of n, and the second and third conditions are satisfied because $a_n \geqslant a_N > L, -2\ell$, and $L \geqslant a_i$ for every i such that $1 \leqslant i \leqslant D$.

Let n be a good index. We derive a contradiction. We have that

$$a_n + a_u + a_v \leqslant 0, \tag{3}$$

whenever $u + v = n$.

We define the index u to maximize a_u over $1 \leqslant u \leqslant n-1$, and let $v = n - u$. Then, we note that $a_u \geqslant a_v$ by the maximality of a_u.

Assume first that $v \leqslant D$. Then, we have that

$$a_N + 2\ell \leqslant 0,$$

because $a_u \geqslant a_v \geqslant \ell$. But this contradicts our assumption that $a_n > -2\ell$ in the second criteria of (2).

Now assume that $v > D$. Then, there exist some indices w_1, w_2 summing up to v such that

$$a_v + a_{w_1} + a_{w_2} = 0.$$

But combining this with (3), we have

$$a_n + a_u \leqslant a_{w_1} + a_{w_2}.$$

Because $a_n > a_u$, we have that $\max\{a_{w_1}, a_{w_2}\} > a_u$. But since each of the w_i is less than v, this contradicts the maximality of a_u.

Comment 1. We present two harder versions of this problem below.

Version 1. Let $a_1, a_2, \ldots$ be a sequence of numbers that satisfies the relation

$$a_n = -\max_{i+j+k=n}(a_i + a_j + a_k) \qquad \text{for all } n > 2017.$$

Then, this sequence is bounded.

Proof. Set $D = 2017$. Denote

$$M_n = \max_{k<n} a_k \qquad \text{and} \qquad m_n = -\min_{k<n} a_k = \max_{k<n}(-a_k).$$

Clearly, the sequences (m_n) and (M_n) are nondecreasing. We need to prove that both are bounded.

Consider an arbitrary $n > 2D$; our first aim is to bound a_n in terms of m_i and M_i. Set $k = \lfloor n/2 \rfloor$.

(i) Choose indices p, q, and r such that $a_n = -(a_p + a_q + a_r)$ and $p + q + r = n$. Without loss of generality, $p \geqslant q \geqslant r$.

Assume that $p \geqslant k + 1 (> D)$; then $p > q + r$. Hence

$$-a_p = \max_{i_1+i_2+i_3=p}(a_{i_1} + a_{i_2} + a_{i_3}) \geqslant a_q + a_r + a_{p-q-r},$$

and therefore $a_n = -(a_p + a_q + a_r) \geqslant (a_q + a_r + a_{p-q-r}) - a_q - a_r = a_{p-q-r} \geqslant -m_n$.

Otherwise, we have $k \geqslant p \geqslant q \geqslant r$. Since $n < 3k$, we have $r < k$. Then $a_p, a_q \leqslant M_{k+1}$ and $a_r \leqslant M_k$, whence $a_n \geqslant -2M_{k+1} - M_k$.

Thus, in any case $a_n \geqslant -\max\{m_n, 2M_{k+1} + M_k\}$.

(ii) On the other hand, choose $p \leqslant k$ and $q \leqslant k-1$ such that $a_p = M_{k+1}$ and $a_q = M_k$. Then $p+q < n$, so $a_n \leqslant -(a_p + a_q + a_{n-p-q}) = -a_{n-p-q} - M_{k+1} - M_k \leqslant m_n - M_{k+1} - M_k$.

To summarize,

$$-\max\{m_n, 2M_{k+1} + M_k\} \leqslant a_n \leqslant m_n - M_{k+1} - M_k,$$

whence

$$m_n \leqslant m_{n+1} \leqslant \max\{m_n, 2M_{k+1} + M_k\} \quad \text{and} \quad M_n \leqslant M_{n+1} \leqslant \max\{M_n, m_n - M_{k+1} - M_k\}. \tag{4}$$

Now, say that an index $n > 2D$ is *lucky* if $m_n \leqslant 2M_{\lfloor n/2\rfloor+1} + M_{\lfloor n/2\rfloor}$. Two cases are possible.

Case 1. Assume that there exists a lucky index n; set $k = \lfloor n/2\rfloor$. In this case, (4) yields $m_{n+1} \leqslant 2M_{k+1} + M_k$ and $M_n \leqslant M_{n+1} \leqslant M_n$ (the last relation holds, since $m_n - M_{k+1} - M_k \leqslant (2M_{k+1} + M_k) - M_{k+1} - M_k = M_{k+1} \leqslant M_n$). Therefore, $M_{n+1} = M_n$ and $m_{n+1} \leqslant 2M_{k+1} + M_k$; the last relation shows that the index $n+1$ is also lucky.

Thus, all indices $N > n$ are lucky, and $M_N = M_n \geqslant m_N/3$, whence all the m_N and M_N are bounded by $3M_n$.

Case 2. Conversely, assume that there is no lucky index, i.e., $2M_{\lfloor n/2\rfloor+1} + M_{\lfloor n/2\rfloor} < m_n$ for all $n > 2D$. Then (4) shows that for all $n > 2D$ we have $m_n \leqslant m_{n+1} \leqslant m_n$, i.e., $m_N = m_{2D+1}$ for all $N > 2D$. Since $M_N < m_{2N+1}/3$ for all such indices, all the m_N and M_N are bounded by m_{2D+1}.

Thus, in both cases the sequences (m_n) and (M_n) are bounded, as desired. □

Version 2. Let $a_1, a_2, \ldots$ be a sequence of numbers that satisfies the relation

$$a_n = -\max_{i_1+\cdots+i_k=n}(a_{i_1} + \cdots + a_{i_k}) \qquad \text{for all } n > 2017.$$

Then, this sequence is bounded.

Proof. As in the solutions above, let $D = 2017$. If the sequence is bounded above, say, by Q, then we have that $a_n \geqslant \min\{a_1, \ldots, a_D, -kQ\}$ for all n, so the sequence is bounded. Assume for sake of contradiction that the sequence is not bounded above. Let $\ell = \min\{a_1, \ldots, a_D\}$, and $L = \max\{a_1, \ldots, a_D\}$. Call an index n *good* if the following criteria hold:

$$a_n > a_i \text{ for each } i < n, \quad a_n > -k\ell, \quad \text{and} \quad n > D \tag{5}$$

We first show that there must be some good index n. By assumption, we may take an index N such that $a_N > \max\{L, -k\ell\}$. Choose n minimally such that $a_n = \max\{a_1, a_2, \ldots, a_N\}$. Now, the first condition is satisfied because of the minimality of n, and the second and third conditions are satisfied because $a_n \geqslant a_N > L, -k\ell$, and $L \geqslant a_i$ for every i such that $1 \leqslant i \leqslant D$.

Let n be a good index. We derive a contradiction. We have that

$$a_n + a_{v_1} + \cdots + a_{v_k} \leqslant 0, \tag{6}$$

whenever $v_1 + \cdots + v_k = n$.

We define the sequence of indices $v_1, \ldots, v_{k-1}$ to greedily maximize a_{v_1}, then a_{v_2}, and so forth, selecting only from indices such that the equation $v_1 + \cdots + v_k = n$ can be satisfied by positive integers $v_1, \ldots, v_k$. More formally, we define them inductively so that the following criteria are satisfied by the v_i:

1. $1 \leqslant v_i \leqslant n - (k-i) - (v_1 + \cdots + v_{i-1})$.
2. a_{v_i} is maximal among all choices of v_i from the first criteria.

First of all, we note that for each i, the first criteria is always satisfiable by some v_i, because we are guaranteed that

$$v_{i-1} \leqslant n - (k - (i-1)) - (v_1 + \cdots + v_{i-2}),$$

which implies

$$1 \leqslant n - (k-i) - (v_1 + \cdots + v_{i-1}).$$

Secondly, the sum $v_1 + \cdots + v_{k-1}$ is at most $n-1$. Define $v_k = n - (v_1 + \cdots + v_{k-1})$. Then, (6) is satisfied by the v_i. We also note that $a_{v_i} \geqslant a_{v_j}$ for all $i < j$; otherwise, in the definition of v_i, we could have selected v_j instead.

Assume first that $v_k \leqslant D$. Then, from (6), we have that

$$a_n + k\ell \leqslant 0,$$

by using that $a_{v_1} \geqslant \cdots \geqslant a_{v_k} \geqslant \ell$. But this contradicts our assumption that $a_n > -k\ell$ in the second criteria of (5).

Now assume that $v_k > D$, and then we must have some indices $w_1, \ldots, w_k$ summing up to v_k such that

$$a_{v_k} + a_{w_1} + \cdots + a_{w_k} = 0.$$

But combining this with (6), we have

$$a_n + a_{v_1} + \cdots + a_{v_{k-1}} \leqslant a_{w_1} + \cdots + a_{w_k}.$$

Because $a_n > a_{v_1} \geqslant \cdots \geqslant a_{v_{k-1}}$, we have that $\max\{a_{w_1}, \ldots, a_{w_k}\} > a_{v_{k-1}}$. But since each of the w_i is less than v_k, in the definition of the v_{k-1} we could have chosen one of the w_i instead, which is a contradiction. □

Comment 2. It seems that each sequence satisfying the condition in Version 2 is eventually periodic, at least when its terms are integers.

However, up to this moment, the Problem Selection Committee is not aware of a proof for this fact (even in the case $k = 2$).

A5. An integer $n \geqslant 3$ is given. We call an n-tuple of real numbers $(x_1, x_2, \ldots, x_n)$ *Shiny* if for each permutation $y_1, y_2, \ldots, y_n$ of these numbers we have

$$\sum_{i=1}^{n-1} y_i y_{i+1} = y_1y_2 + y_2y_3 + y_3y_4 + \cdots + y_{n-1}y_n \geqslant -1.$$

Find the largest constant $K = K(n)$ such that

$$\sum_{1 \leqslant i < j \leqslant n} x_i x_j \geqslant K$$

holds for every Shiny n-tuple $(x_1, x_2, \ldots, x_n)$.

(Serbia)

Answer: $K = -(n-1)/2$.

Solution 1. First of all, we show that we may not take a larger constant K. Let t be a positive number, and take $x_2 = x_3 = \cdots = t$ and $x_1 = -1/(2t)$. Then, every product x_ix_j $(i \neq j)$ is equal to either t^2 or $-1/2$. Hence, for every permutation y_i of the x_i, we have

$$y_1y_2 + \cdots + y_{n-1}y_n \geqslant (n-3)t^2 - 1 \geqslant -1.$$

This justifies that the n-tuple $(x_1, \ldots, x_n)$ is Shiny. Now, we have

$$\sum_{i<j} x_ix_j = -\frac{n-1}{2} + \frac{(n-1)(n-2)}{2}t^2.$$

Thus, as t approaches 0 from above, $\sum_{i<j} x_ix_j$ gets arbitrarily close to $-(n-1)/2$. This shows that we may not take K any larger than $-(n-1)/2$. It remains to show that $\sum_{i<j} x_ix_j \geqslant -(n-1)/2$ for any Shiny choice of the x_i.

From now onward, assume that $(x_1, \ldots, x_n)$ is a Shiny n-tuple. Let the z_i $(1 \leqslant i \leqslant n)$ be some permutation of the x_i to be chosen later. The indices for z_i will always be taken modulo n. We will first split up the sum $\sum_{i<j} x_ix_j = \sum_{i<j} z_iz_j$ into $\lfloor (n-1)/2 \rfloor$ expressions, each of the form $y_1y_2 + \cdots + y_{n-1}y_n$ for some permutation y_i of the z_i, and some leftover terms. More specifically, write

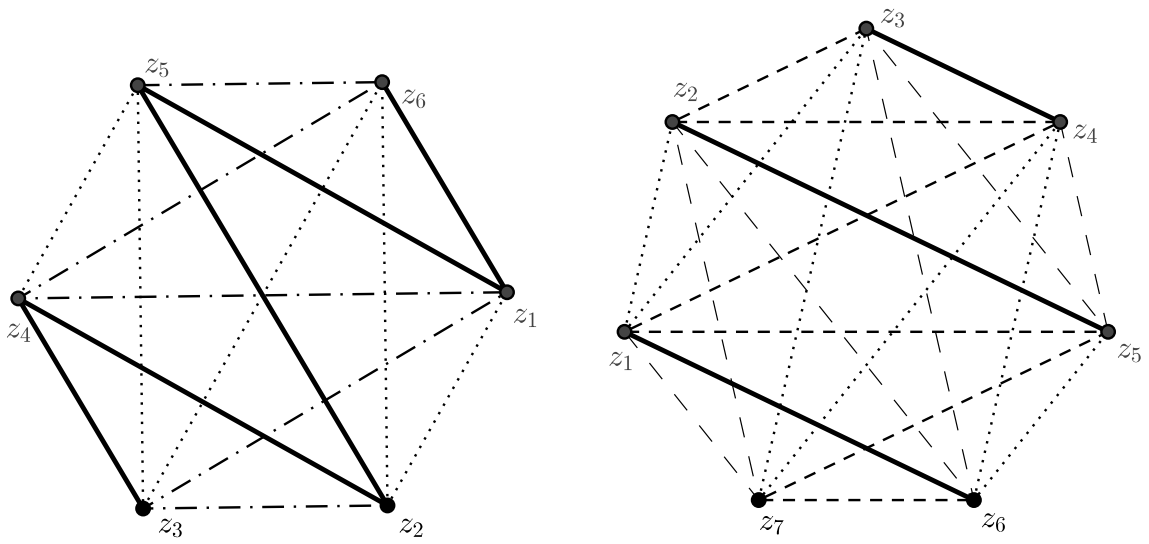
$$\sum_{i<j} z_iz_j = \sum_{q=0}^{n-1} \sum_{\substack{i+j \equiv q \pmod n \\ i \neq j \pmod n}} z_iz_j = \sum_{p=1}^{\lfloor \frac{n-1}{2} \rfloor} \sum_{\substack{i+j \equiv 2p-1, 2p \pmod n \\ i \neq j \pmod n}} z_iz_j + L, \tag{1}$$

where $L = z_1z_{-1} + z_2z_{-2} + \cdots + z_{(n-1)/2}z_{-(n-1)/2}$ if n is odd, and $L = z_1z_{-1} + z_1z_{-2} + z_2z_{-2} + \cdots + z_{(n-2)/2}z_{-n/2}$ if n is even. We note that for each $p = 1, 2, \ldots, \lfloor (n-1)/2 \rfloor$, there is some permutation y_i of the z_i such that

$$\sum_{\substack{i+j \equiv 2p-1, 2p \pmod n \\ i \neq j \pmod n}} z_iz_j = \sum_{k=1}^{n-1} y_ky_{k+1},$$

because we may choose $y_{2i-1} = z_{i+p-1}$ for $1 \leqslant i \leqslant (n+1)/2$ and $y_{2i} = z_{p-i}$ for $1 \leqslant i \leqslant n/2$.

We show (1) graphically for $n = 6, 7$ in the diagrams below. The edges of the graphs each represent a product z_iz_j, and the dashed and dotted series of lines represents the sum of the edges, which is of the form $y_1y_2 + \cdots + y_{n-1}y_n$ for some permutation y_i of the z_i precisely when the series of lines is a Hamiltonian path. The filled edges represent the summands of L.



Now, because the z_i are Shiny, we have that (1) yields the following bound:

$$\sum_{i<j} z_i z_j \geqslant -\left\lfloor \frac{n-1}{2} \right\rfloor + L.$$

It remains to show that, for each n, there exists some permutation z_i of the x_i such that $L \geqslant 0$ when n is odd, and $L \geqslant -1/2$ when n is even. We now split into cases based on the parity of n and provide constructions of the permutations z_i.

Since we have not made any assumptions yet about the x_i, we may now assume without loss of generality that

$$x_1 \leqslant x_2 \leqslant \cdots \leqslant x_k \leqslant 0 \leqslant x_{k+1} \leqslant \cdots \leqslant x_n. \tag{2}$$

Case 1: n is odd.

Without loss of generality, assume that k (from (2)) is even, because we may negate all the x_i if k is odd. We then have $x_1x_2, x_3x_4, \ldots, x_{n-2}x_{n-1} \geqslant 0$ because the factors are of the same sign. Let $L = x_1x_2 + x_3x_4 + \cdots + x_{n-2}x_{n-1} \geqslant 0$. We choose our z_i so that this definition of L agrees with the sum of the leftover terms in (1). Relabel the x_i as z_i such that

$$\{z_1, z_{n-1}\}, \{z_2, z_{n-2}\}, \ldots, \{z_{(n-1)/2}, z_{(n+1)/2}\}$$

are some permutation of

$$\{x_1, x_2\}, \{x_3, x_4\}, \ldots, \{x_{n-2}, x_{n-1}\},$$

and $z_n = x_n$. Then, we have $L = z_1z_{n-1} + \cdots + z_{(n-1)/2}z_{(n+1)/2}$, as desired.

Case 2: n is even.

Let $L = x_1x_2 + x_2x_3 + \cdots + x_{n-1}x_n$. Assume without loss of generality $k \neq 1$. Now, we have

$$2L = (x_1x_2 + \cdots + x_{n-1}x_n) + (x_1x_2 + \cdots + x_{n-1}x_n) \geqslant (x_2x_3 + \cdots + x_{n-1}x_n) + x_kx_{k+1}$$

$$\geqslant x_2x_3 + \cdots + x_{n-1}x_n + x_nx_1 \geqslant -1,$$

where the first inequality holds because the only negative term in L is x_kx_{k+1}, the second inequality holds because $x_1 \leqslant x_k \leqslant 0 \leqslant x_{k+1} \leqslant x_n$, and the third inequality holds because the x_i are assumed to be Shiny. We thus have that $L \geqslant -1/2$. We now choose a suitable z_i such that the definition of L matches the leftover terms in (1).

Relabel the x_i with z_i in the following manner: $x_{2i-1} = z_{-i}, x_{2i} = z_i$ (again taking indices modulo n). We have that

$$L = \sum_{\substack{i+j \equiv 0,-1 \pmod n \\ i \not\equiv j \pmod n}} z_i z_j,$$

as desired.

Solution 2. We present another proof that $\sum_{i<j} x_i x_j \geqslant -(n-1)/2$ for any Shiny n-tuple $(x_1, \ldots, x_n)$. Assume an ordering of the x_i as in (2), and let $\ell = n - k$. Assume without loss of generality that $k \geqslant \ell$. Also assume $k \neq n$, (as otherwise, all of the x_i are nonpositive, and so the inequality is trivial). Define the sets of indices $S = \{1, 2, \ldots, k\}$ and $T = \{k+1, \ldots, n\}$. Define the following sums:

$$K = \sum_{\substack{i<j \\ i,j \in S}} x_i x_j, \quad M = \sum_{\substack{i \in S \\ j \in T}} x_i x_j, \quad \text{and} \quad L = \sum_{\substack{i<j \\ i,j \in T}} x_i x_j$$

By definition, $K, L \geqslant 0$ and $M \leqslant 0$. We aim to show that $K + L + M \geqslant -(n-1)/2$.

We split into cases based on whether $k = \ell$ or $k > \ell$.

Case 1: $k > \ell$.

Consider all permutations $\phi\colon \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$ such that $\phi^{-1}(T) = \{2, 4, \ldots, 2\ell\}$. Note that there are $k!\ell!$ such permutations ϕ. Define

$$f(\phi) = \sum_{i=1}^{n-1} x_{\phi(i)} x_{\phi(i+1)}.$$

We know that $f(\phi) \geqslant -1$ for every permutation ϕ with the above property. Averaging $f(\phi)$ over all ϕ gives

$$-1 \leqslant \frac{1}{k!\ell!} \sum_{\phi} f(\phi) = \frac{2\ell}{k\ell} M + \frac{2(k-\ell-1)}{k(k-1)} K,$$

where the equality holds because there are $k\ell$ products in M, of which 2ℓ are selected for each ϕ, and there are $k(k-1)/2$ products in K, of which $k - \ell - 1$ are selected for each ϕ. We now have

$$K + L + M \geqslant K + L + \left(-\frac{k}{2} - \frac{k-\ell-1}{k-1} K \right) = -\frac{k}{2} + \frac{\ell}{k-1} K + L.$$

Since $k \leqslant n - 1$ and $K, L \geqslant 0$, we get the desired inequality.

Case 2: $k = \ell = n/2$.

We do a similar approach, considering all $\phi\colon \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$ such that $\phi^{-1}(T) = \{2, 4, \ldots, 2\ell\}$, and defining f the same way. Analogously to Case 1, we have

$$-1 \leqslant \frac{1}{k!\ell!} \sum_{\phi} f(\phi) = \frac{2\ell - 1}{k\ell} M,$$

because there are $k\ell$ products in M, of which $2\ell - 1$ are selected for each ϕ. Now, we have that

$$K + L + M \geqslant M \geqslant -\frac{n^2}{4(n-1)} \geqslant -\frac{n-1}{2},$$

where the last inequality holds because $n \geqslant 4$.

A6. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)f(y)) + f(x+y) = f(xy) \tag{$*$}$$

for all $x, y \in \mathbb{R}$.

(Albania)

Answer: There are 3 solutions:

$$x \mapsto 0 \qquad \text{or} \qquad x \mapsto x-1 \qquad \text{or} \qquad x \mapsto 1-x \qquad (x \in \mathbb{R}).$$

Solution. An easy check shows that all the 3 above mentioned functions indeed satisfy the original equation $(*)$.

In order to show that these are the only solutions, first observe that if $f(x)$ is a solution then $-f(x)$ is also a solution. Hence, without loss of generality we may (and will) assume that $f(0) \leqslant 0$ from now on. We have to show that either f is identically zero or $f(x) = x - 1$ $(\forall x \in \mathbb{R})$.

Observe that, for a fixed $x \neq 1$, we may choose $y \in \mathbb{R}$ so that $x + y = xy \iff y = \frac{x}{x-1}$, and therefore from the original equation $(*)$ we have

$$f\Big(f(x) \cdot f\Big(\frac{x}{x-1}\Big)\Big) = 0 \qquad (x \neq 1). \tag{1}$$

In particular, plugging in $x = 0$ in (1), we conclude that f has at least one zero, namely $(f(0))^2$:

$$f\big((f(0))^2\big) = 0. \tag{2}$$

We analyze two cases (recall that $f(0) \leqslant 0$):

Case 1: $f(0) = 0$.

Setting $y = 0$ in the original equation we get the identically zero solution:

$$f(f(x)f(0)) + f(x) = f(0) \implies f(x) = 0 \text{ for all } x \in \mathbb{R}.$$

From now on, we work on the main

Case 2: $f(0) < 0$.

We begin with the following

Claim 1.

$$f(1) = 0, \qquad f(a) = 0 \implies a = 1, \qquad \text{and} \qquad f(0) = -1. \tag{3}$$

Proof. We need to show that 1 is the *unique* zero of f. First, observe that f has at least one zero a by (2); if $a \neq 1$ then setting $x = a$ in (1) we get $f(0) = 0$, a contradiction. Hence from (2) we get $(f(0))^2 = 1$. Since we are assuming $f(0) < 0$, we conclude that $f(0) = -1$. □

Setting $y = 1$ in the original equation $(*)$ we get

$$f(f(x)f(1))+f(x+1) = f(x) \iff f(0)+f(x+1) = f(x) \iff f(x+1) = f(x)+1 \qquad (x \in \mathbb{R}).$$

An easy induction shows that

$$f(x+n) = f(x) + n \qquad (x \in \mathbb{R},\ n \in \mathbb{Z}). \tag{4}$$

Now we make the following

Claim 2. f is injective.

Proof. Suppose that $f(a) = f(b)$ with $a \neq b$. Then by (4), for all $N \in \mathbb{Z}$,

$$f(a + N + 1) = f(b + N) + 1.$$

Choose any integer $N < -b$; then there exist $x_0, y_0 \in \mathbb{R}$ with $x_0 + y_0 = a + N + 1$, $x_0 y_0 = b + N$. Since $a \neq b$, we have $x_0 \neq 1$ and $y_0 \neq 1$. Plugging in x_0 and y_0 in the original equation $(*)$ we get

$$\begin{aligned} f(f(x_0)f(y_0)) + f(a + N + 1) = f(b + N) &\iff f(f(x_0)f(y_0)) + 1 = 0 \\ &\iff f(f(x_0)f(y_0) + 1) = 0 \qquad \text{by (4)} \\ &\iff f(x_0)f(y_0) = 0 \qquad \text{by (3).} \end{aligned}$$

However, by Claim 1 we have $f(x_0) \neq 0$ and $f(y_0) \neq 0$ since $x_0 \neq 1$ and $y_0 \neq 1$, a contradiction. □

Now the end is near. For any $t \in \mathbb{R}$, plug in $(x, y) = (t, -t)$ in the original equation $(*)$ to get

$$\begin{aligned} f(f(t)f(-t)) + f(0) = f(-t^2) &\iff f(f(t)f(-t)) = f(-t^2) + 1 \qquad \text{by (3)} \\ &\iff f(f(t)f(-t)) = f(-t^2 + 1) \qquad \text{by (4)} \\ &\iff f(t)f(-t) = -t^2 + 1 \qquad \text{by injectivity of } f. \end{aligned}$$

Similarly, plugging in $(x, y) = (t, 1 - t)$ in $(*)$ we get

$$\begin{aligned} f(f(t)f(1 - t)) + f(1) = f(t(1 - t)) &\iff f(f(t)f(1 - t)) = f(t(1 - t)) \quad \text{by (3)} \\ &\iff f(t)f(1 - t) = t(1 - t) \qquad \text{by injectivity of } f. \end{aligned}$$

But since $f(1 - t) = 1 + f(-t)$ by (4), we get

$$\begin{aligned} f(t)f(1 - t) = t(1 - t) &\iff f(t)(1 + f(-t)) = t(1 - t) \iff f(t) + (-t^2 + 1) = t(1 - t) \\ &\iff f(t) = t - 1, \end{aligned}$$

as desired.

Comment. Other approaches are possible. For instance, after Claim 1, we may define

$$g(x) \stackrel{\text{def}}{=} f(x) + 1.$$

Replacing $x + 1$ and $y + 1$ in place of x and y in the original equation $(*)$, we get

$$f(f(x + 1)f(y + 1)) + f(x + y + 2) = f(xy + x + y + 1) \qquad (x, y \in \mathbb{R}),$$

and therefore, using (4) (so that in particular $g(x) = f(x + 1)$), we may rewrite $(*)$ as

$$g(g(x)g(y)) + g(x + y) = g(xy + x + y) \qquad (x, y \in \mathbb{R}). \qquad (**)$$

We are now to show that $g(x) = x$ for all $x \in \mathbb{R}$ under the assumption (Claim 1) that 0 is the unique zero of g.

Claim 3. Let $n \in \mathbb{Z}$ and $x \in \mathbb{R}$. Then

(a) $g(x + n) = x + n$, and the conditions $g(x) = n$ and $x = n$ are equivalent.

(b) $g(nx) = ng(x)$.

Proof. For part (a), just note that $g(x+n) = x+n$ is just a reformulation of (4). Then $g(x) = n \iff g(x-n) = 0 \iff x-n = 0$ since 0 is the unique zero of g. For part (b), we may assume that $x \neq 0$ since the result is obvious when $x = 0$. Plug in $y = n/x$ in $(**)$ and use part (a) to get

$$g\left(g(x)g\Big(\frac{n}{x}\Big)\right) + g\Big(x + \frac{n}{x}\Big) = g\Big(n + x + \frac{n}{x}\Big) \iff g\left(g(x)g\Big(\frac{n}{x}\Big)\right) = n \iff g(x)g\Big(\frac{n}{x}\Big) = n.$$

In other words, for $x \neq 0$ we have

$$g(x) = \frac{n}{g\big(n/x\big)}.$$

In particular, for $n = 1$, we get $g(1/x) = 1/g(x)$, and therefore replacing $x \leftarrow nx$ in the last equation we finally get

$$g(nx) = \frac{n}{g\big(1/x\big)} = ng(x),$$

as required.

Claim 4. The function g is *additive*, i.e., $g(a+b) = g(a) + g(b)$ for all $a, b \in \mathbb{R}$.

Proof. Set $x \leftarrow -x$ and $y \leftarrow -y$ in $(**)$; since g is an odd function (by Claim 3(b) with $n = -1$), we get

$$g(g(x)g(y)) - g(x+y) = -g(-xy + x + y).$$

Subtracting the last relation from $(**)$ we have

$$2g(x+y) = g(xy + x + y) + g(-xy + x + y)$$

and since by Claim 3(b) we have $2g(x+y) = g(2(x+y))$, we may rewrite the last equation as

$$g(\alpha + \beta) = g(\alpha) + g(\beta) \qquad \text{where} \qquad \begin{cases} \alpha = xy + x + y \\ \beta = -xy + x + y. \end{cases}$$

In other words, we have additivity for all $\alpha, \beta \in \mathbb{R}$ for which there are real numbers x and y satisfying

$$x + y = \frac{\alpha + \beta}{2} \qquad \text{and} \qquad xy = \frac{\alpha - \beta}{2},$$

i.e., for all $\alpha, \beta \in \mathbb{R}$ such that $(\frac{\alpha+\beta}{2})^2 - 4 \cdot \frac{\alpha-\beta}{2} \geqslant 0$. Therefore, given any $a, b \in \mathbb{R}$, we may choose $n \in \mathbb{Z}$ large enough so that we have additivity for $\alpha = na$ and $\beta = nb$, i.e.,

$$g(na) + g(nb) = g(na + nb) \iff ng(a) + ng(b) = ng(a+b)$$

by Claim 3(b). Cancelling n, we get the desired result. (Alternatively, setting either $(\alpha, \beta) = (a, b)$ or $(\alpha, \beta) = (-a, -b)$ will ensure that $(\frac{\alpha+\beta}{2})^2 - 4 \cdot \frac{\alpha-\beta}{2} \geqslant 0$). □

Now we may finish the solution. Set $y = 1$ in $(**)$, and use Claim 3 to get

$$g(g(x)g(1)) + g(x+1) = g(2x+1) \iff g(g(x)) + g(x) + 1 = 2g(x) + 1 \iff g(g(x)) = g(x).$$

By additivity, this is equivalent to $g(g(x) - x) = 0$. Since 0 is the unique zero of g by assumption, we finally get $g(x) - x = 0 \iff g(x) = x$ for all $x \in \mathbb{R}$.

A7. Let $a_0, a_1, a_2, \ldots$ be a sequence of integers and $b_0, b_1, b_2, \ldots$ be a sequence of *positive* integers such that $a_0 = 0, a_1 = 1$, and

$$a_{n+1} = \begin{cases} a_n b_n + a_{n-1}, & \text{if } b_{n-1} = 1 \\ a_n b_n - a_{n-1}, & \text{if } b_{n-1} > 1 \end{cases} \qquad \text{for } n = 1, 2, \ldots.$$

Prove that at least one of the two numbers a_{2017} and a_{2018} must be greater than or equal to 2017.

(Australia)

Solution 1. The value of b_0 is irrelevant since $a_0 = 0$, so we may assume that $b_0 = 1$.

Lemma. We have $a_n \geqslant 1$ for all $n \geqslant 1$.

Proof. Let us suppose otherwise in order to obtain a contradiction. Let

$$n \geqslant 1 \text{ be the smallest integer with } a_n \leqslant 0. \tag{1}$$

Note that $n \geqslant 2$. It follows that $a_{n-1} \geqslant 1$ and $a_{n-2} \geqslant 0$. Thus we cannot have $a_n = a_{n-1}b_{n-1} + a_{n-2}$, so we must have $a_n = a_{n-1}b_{n-1} - a_{n-2}$. Since $a_n \leqslant 0$, we have $a_{n-1} \leqslant a_{n-2}$. Thus we have $a_{n-2} \geqslant a_{n-1} \geqslant a_n$.

Let

$$r \text{ be the smallest index with } a_r \geqslant a_{r+1} \geqslant a_{r+2}. \tag{2}$$

Then $r \leqslant n-2$ by the above, but also $r \geqslant 2$: if $b_1 = 1$, then $a_2 = a_1 = 1$ and $a_3 = a_2b_2 + a_1 > a_2$; if $b_1 > 1$, then $a_2 = b_1 > 1 = a_1$.

By the minimal choice (2) of r, it follows that $a_{r-1} < a_r$. And since $2 \leqslant r \leqslant n-2$, by the minimal choice (1) of n we have $a_{r-1}, a_r, a_{r+1} > 0$. In order to have $a_{r+1} \geqslant a_{r+2}$, we must have $a_{r+2} = a_{r+1}b_{r+1} - a_r$ so that $b_r \geqslant 2$. Putting everything together, we conclude that

$$a_{r+1} = a_r b_r \pm a_{r-1} \geqslant 2a_r - a_{r-1} = a_r + (a_r - a_{r-1}) > a_r,$$

which contradicts (2). □

To complete the problem, we prove that $\max\{a_n, a_{n+1}\} \geqslant n$ by induction. The cases $n = 0, 1$ are given. Assume it is true for all non-negative integers strictly less than n, where $n \geqslant 2$. There are two cases:

Case 1: $b_{n-1} = 1$.

Then $a_{n+1} = a_n b_n + a_{n-1}$. By the inductive assumption one of a_{n-1}, a_n is at least $n-1$ and the other, by the lemma, is at least 1. Hence

$$a_{n+1} = a_n b_n + a_{n-1} \geqslant a_n + a_{n-1} \geqslant (n-1) + 1 = n.$$

Thus $\max\{a_n, a_{n+1}\} \geqslant n$, as desired.

Case 2: $b_{n-1} > 1$.

Since we defined $b_0 = 1$ there is an index r with $1 \leqslant r \leqslant n-1$ such that

$$b_{n-1}, b_{n-2}, \ldots, b_r \geqslant 2 \qquad \text{and} \qquad b_{r-1} = 1.$$

We have $a_{r+1} = a_r b_r + a_{r-1} \geqslant 2a_r + a_{r-1}$. Thus $a_{r+1} - a_r \geqslant a_r + a_{r-1}$.

Now we claim that $a_r + a_{r-1} \geqslant r$. Indeed, this holds by inspection for $r = 1$; for $r \geqslant 2$, one of a_r, a_{r-1} is at least $r-1$ by the inductive assumption, while the other, by the lemma, is at least 1. Hence $a_r + a_{r-1} \geqslant r$, as claimed, and therefore $a_{r+1} - a_r \geqslant r$ by the last inequality in the previous paragraph.

Since $r \geqslant 1$ and, by the lemma, $a_r \geqslant 1$, from $a_{r+1} - a_r \geqslant r$ we get the following two inequalities:

$$a_{r+1} \geqslant r + 1 \qquad \text{and} \qquad a_{r+1} > a_r.$$

Now observe that

$$a_m > a_{m-1} \implies a_{m+1} > a_m \text{ for } m = r+1, r+2, \ldots, n-1,$$

since $a_{m+1} = a_m b_m - a_{m-1} \geqslant 2a_m - a_{m-1} = a_m + (a_m - a_{m-1}) > a_m$. Thus

$$a_n > a_{n-1} > \cdots > a_{r+1} \geqslant r+1 \implies a_n \geqslant n.$$

So $\max\{a_n, a_{n+1}\} \geqslant n$, as desired.

Solution 2. We say that an index $n > 1$ is *bad* if $b_{n-1} = 1$ and $b_{n-2} > 1$; otherwise n is *good*. The value of b_0 is irrelevant to the definition of (a_n) since $a_0 = 0$; so we assume that $b_0 > 1$.

Lemma 1. (a) $a_n \geqslant 1$ for all $n > 0$.

(b) If $n > 1$ is good, then $a_n > a_{n-1}$.

Proof. Induction on n. In the base cases $n = 1, 2$ we have $a_1 = 1 \geqslant 1$, $a_2 = b_1 a_1 \geqslant 1$, and finally $a_2 > a_1$ if 2 is good, since in this case $b_1 > 1$.

Now we assume that the lemma statement is proved for $n = 1, 2, \ldots, k$ with $k \geqslant 2$, and prove it for $n = k+1$. Recall that a_k and a_{k-1} are positive by the induction hypothesis.

Case 1: k is bad.

We have $b_{k-1} = 1$, so $a_{k+1} = b_k a_k + a_{k-1} \geqslant a_k + a_{k-1} > a_k \geqslant 1$, as required.

Case 2: k is good.

We already have $a_k > a_{k-1} \geqslant 1$ by the induction hypothesis. We consider three easy subcases.

Subcase 2.1: $b_k > 1$.

Then $a_{k+1} \geqslant b_k a_k - a_{k-1} \geqslant a_k + (a_k - a_{k-1}) > a_k \geqslant 1$.

Subcase 2.2: $b_k = b_{k-1} = 1$.

Then $a_{k+1} = a_k + a_{k-1} > a_k \geqslant 1$.

Subcase 2.3: $b_k = 1$ but $b_{k-1} > 1$.

Then $k+1$ is bad, and we need to prove only (a), which is trivial: $a_{k+1} = a_k - a_{k-1} \geqslant 1$.

So, in all three subcases we have verified the required relations. □

Lemma 2. Assume that $n > 1$ is bad. Then there exists a $j \in \{1, 2, 3\}$ such that $a_{n+j} \geqslant a_{n-1} + j + 1$, and $a_{n+i} \geqslant a_{n-1} + i$ for all $1 \leqslant i < j$.

Proof. Recall that $b_{n-1} = 1$. Set

$$m = \inf\{i > 0 \colon b_{n+i-1} > 1\}$$

(possibly $m = +\infty$). We claim that $j = \min\{m, 3\}$ works. Again, we distinguish several cases, according to the value of m; in each of them we use Lemma 1 without reference.

Case 1: $m = 1$, so $b_n > 1$.

Then $a_{n+1} \geqslant 2a_n + a_{n-1} \geqslant a_{n-1} + 2$, as required.

Case 2: $m = 2$, so $b_n = 1$ and $b_{n+1} > 1$.

Then we successively get

$$a_{n+1} = a_n + a_{n-1} \geqslant a_{n-1} + 1,$$
$$a_{n+2} \geqslant 2a_{n+1} + a_n \geqslant 2(a_{n-1} + 1) + a_n = a_{n-1} + (a_{n-1} + a_n + 2) \geqslant a_{n-1} + 4,$$

which is even better than we need.

Case 3: $m > 2$, so $b_n = b_{n+1} = 1$.

Then we successively get

$$a_{n+1} = a_n + a_{n-1} \geqslant a_{n-1} + 1, \quad a_{n+2} = a_{n+1} + a_n \geqslant a_{n-1} + 1 + a_n \geqslant a_{n-1} + 2,$$
$$a_{n+3} \geqslant a_{n+2} + a_{n+1} \geqslant (a_{n-1} + 1) + (a_{n-1} + 2) \geqslant a_{n-1} + 4,$$

as required. □

Lemmas 1(b) and 2 provide enough information to prove that $\max\{a_n, a_{n+1}\} \geqslant n$ for all n and, moreover, that $a_n \geqslant n$ often enough. Indeed, assume that we have found some n with $a_{n-1} \geqslant n-1$. If n is good, then by Lemma 1(b) we have $a_n \geqslant n$ as well. If n is bad, then Lemma 2 yields $\max\{a_{n+i}, a_{n+i+1}\} \geqslant a_{n-1} + i + 1 \geqslant n + i$ for all $0 \leqslant i < j$ and $a_{n+j} \geqslant a_{n-1} + j + 1 \geqslant n + j$; so $n + j$ is the next index to start with.

A8. Assume that a function $f\colon \mathbb{R} \to \mathbb{R}$ satisfies the following condition:

For every $x, y \in \mathbb{R}$ such that $\big(f(x)+y\big)\big(f(y)+x\big) > 0$, we have $f(x)+y = f(y)+x$.

Prove that $f(x) + y \leqslant f(y) + x$ whenever $x > y$.

(Netherlands)

Solution 1. Define $g(x) = x - f(x)$. The condition on f then rewrites as follows:

For every $x, y \in \mathbb{R}$ such that $\big((x+y)-g(x)\big)\big((x+y)-g(y)\big) > 0$, we have $g(x) = g(y)$.

This condition may in turn be rewritten in the following form:

If $g(x) \neq g(y)$, then the number $x+y$ lies (non-strictly) between $g(x)$ and $g(y)$. $(*)$

Notice here that the function $g_1(x) = -g(-x)$ also satisfies $(*)$, since

$$g_1(x) \neq g_1(y) \implies g(-x) \neq g(-y) \implies -(x+y) \text{ lies between } g(-x) \text{ and } g(-y)$$
$$\implies x+y \text{ lies between } g_1(x) \text{ and } g_1(y).$$

On the other hand, the relation we need to prove reads now as

$$g(x) \leqslant g(y) \qquad \text{whenever } x < y. \tag{1}$$

Again, this condition is equivalent to the same one with g replaced by g_1.

If $g(x) = 2x$ for all $x \in \mathbb{R}$, then $(*)$ is obvious; so in what follows we consider the other case. We split the solution into a sequence of lemmas, strengthening one another. We always consider some value of x with $g(x) \neq 2x$ and denote $X = g(x)$.

Lemma 1. Assume that $X < 2x$. Then on the interval $(X-x; x]$ the function g attains at most two values — namely, X and, possibly, some $Y > X$. Similarly, if $X > 2x$, then g attains at most two values on $[x; X-x)$ — namely, X and, possibly, some $Y < X$.

Proof. We start with the first claim of the lemma. Notice that $X - x < x$, so the considered interval is nonempty.

Take any $a \in (X-x; x)$ with $g(a) \neq X$ (if it exists). If $g(a) < X$, then $(*)$ yields $g(a) \leqslant a + x \leqslant g(x) = X$, so $a \leqslant X - x$ which is impossible. Thus, $g(a) > X$ and hence by $(*)$ we get $X \leqslant a + x \leqslant g(a)$.

Now, for any $b \in (X-x; x)$ with $g(b) \neq X$ we similarly get $b + x \leqslant g(b)$. Therefore, the number $a + b$ (which is smaller than each of $a + x$ and $b + x$) cannot lie between $g(a)$ and $g(b)$, which by $(*)$ implies that $g(a) = g(b)$. Hence g may attain only two values on $(X-x; x]$, namely X and $g(a) > X$.

To prove the second claim, notice that $g_1(-x) = -X < 2 \cdot (-x)$, so g_1 attains at most two values on $(-X+x, -x]$, i.e., $-X$ and, possibly, some $-Y > -X$. Passing back to g, we get what we need. □

Lemma 2. If $X < 2x$, then g is constant on $(X-x; x)$. Similarly, if $X > 2x$, then g is constant on $(x; X-x)$.

Proof. Again, it suffices to prove the first claim only. Assume, for the sake of contradiction, that there exist $a, b \in (X-x; x)$ with $g(a) \neq g(b)$; by Lemma 1, we may assume that $g(a) = X$ and $Y = g(b) > X$.

Notice that $\min\{X-a, X-b\} > X - x$, so there exists a $u \in (X-x; x)$ such that $u < \min\{X-a, X-b\}$. By Lemma 1, we have either $g(u) = X$ or $g(u) = Y$. In the former case, by $(*)$ we have $X \leqslant u + b \leqslant Y$ which contradicts $u < X - b$. In the second case, by $(*)$ we have $X \leqslant u + a \leqslant Y$ which contradicts $u < X - a$. Thus the lemma is proved. □

Lemma 3. If $X < 2x$, then $g(a) = X$ for all $a \in (X-x; x)$. Similarly, if $X > 2x$, then $g(a) = X$ for all $a \in (x; X-x)$.

Proof. Again, we only prove the first claim.

By Lemmas 1 and 2, this claim may be violated only if g takes on a constant value $Y > X$ on $(X-x, x)$. Choose any $a, b \in (X-x; x)$ with $a < b$. By $(*)$, we have

$$Y \geqslant b + x \geqslant X. \tag{2}$$

In particular, we have $Y \geqslant b + x > 2a$. Applying Lemma 2 to a in place of x, we obtain that g is constant on $(a, Y-a)$. By (2) again, we have $x \leqslant Y - b < Y - a$; so $x, b \in (a; Y-a)$. But $X = g(x) \neq g(b) = Y$, which is a contradiction. □

Now we are able to finish the solution. Assume that $g(x) > g(y)$ for some $x < y$. Denote $X = g(x)$ and $Y = g(y)$; by $(*)$, we have $X \geqslant x + y \geqslant Y$, so $Y - y \leqslant x < y \leqslant X - x$, and hence $(Y-y; y) \cap (x; X-x) = (x, y) \neq \varnothing$. On the other hand, since $Y - y < y$ and $x < X - x$, Lemma 3 shows that g should attain a constant value X on $(x; X-x)$ and a constant value $Y \neq X$ on $(Y-y; y)$. Since these intervals overlap, we get the final contradiction.

Solution 2. As in the previous solution, we pass to the function g satisfying $(*)$ and notice that we need to prove the condition (1). We will also make use of the function g_1.

If g is constant, then (1) is clearly satisfied. So, in the sequel we assume that g takes on at least two different values. Now we collect some information about the function g.

Claim 1. For any $c \in \mathbb{R}$, all the solutions of $g(x) = c$ are bounded.

Proof. Fix any $y \in \mathbb{R}$ with $g(y) \neq c$. Assume first that $g(y) > c$. Now, for any x with $g(x) = c$, by $(*)$ we have $c \leqslant x + y \leqslant g(y)$, or $c - y \leqslant x \leqslant g(y) - y$. Since c and y are constant, we get what we need.

If $g(y) < c$, we may switch to the function g_1 for which we have $g_1(-y) > -c$. By the above arguments, we obtain that all the solutions of $g_1(-x) = -c$ are bounded, which is equivalent to what we need. □

As an immediate consequence, the function g takes on infinitely many values, which shows that the next claim is indeed widely applicable.

Claim 2. If $g(x) < g(y) < g(z)$, then $x < z$.

Proof. By $(*)$, we have $g(x) \leqslant x + y \leqslant g(y) \leqslant z + y \leqslant g(z)$, so $x + y \leqslant z + y$, as required. □

Claim 3. Assume that $g(x) > g(y)$ for some $x < y$. Then $g(a) \in \{g(x), g(y)\}$ for all $a \in [x; y]$.

Proof. If $g(y) < g(a) < g(x)$, then the triple (y, a, x) violates Claim 2. If $g(a) < g(y) < g(x)$, then the triple (a, y, x) violates Claim 2. If $g(y) < g(x) < g(a)$, then the triple (y, x, a) violates Claim 2. The only possible cases left are $g(a) \in \{g(x), g(y)\}$. □

In view of Claim 3, we say that an interval I (which may be open, closed, or semi-open) is a *Dirichlet interval** if the function g takes on just two values on I.

Assume now, for the sake of contradiction, that (1) is violated by some $x < y$. By Claim 3, $[x; y]$ is a Dirichlet interval. Set

$$r = \inf\{a\colon (a; y] \text{ is a Dirichlet interval}\} \quad \text{and} \quad s = \sup\{b\colon [x; b) \text{ is a Dirichlet interval}\}.$$

Clearly, $r \leqslant x < y \leqslant s$. By Claim 1, r and s are finite. Denote $X = g(x)$, $Y = g(y)$, and $\Delta = (y - x)/2$.

Suppose first that there exists a $t \in (r; r + \Delta)$ with $f(t) = Y$. By the definition of r, the interval $(r - \Delta; y]$ is not Dirichlet, so there exists an $r' \in (r - \Delta; r]$ such that $g(r') \notin \{X, Y\}$.

*The name *Dirichlet interval* is chosen for the reason that g theoretically might act similarly to the Dirichlet function on this interval.

The function g attains at least three distinct values on $[r';y]$, namely $g(r')$, $g(x)$, and $g(y)$. Claim 3 now yields $g(r') \leqslant g(y)$; the equality is impossible by the choice of r', so in fact $g(r') < Y$. Applying $(*)$ to the pairs (r', y) and (t, x) we obtain $r' + y \leqslant Y \leqslant t + x$, whence $r - \Delta + y < r' + y \leqslant t + x < r + \Delta + x$, or $y - x < 2\Delta$. This is a contradiction.

Thus, $g(t) = X$ for all $t \in (r; r + \Delta)$. Applying the same argument to g_1, we get $g(t) = Y$ for all $t \in (s - \Delta; s)$.

Finally, choose some $s_1, s_2 \in (s - \Delta; s)$ with $s_1 < s_2$ and denote $\delta = (s_2 - s_1)/2$. As before, we choose $r' \in (r - \delta; r)$ with $g(r') \notin \{X, Y\}$ and obtain $g(r') < Y$. Choose any $t \in (r; r + \delta)$; by the above arguments, we have $g(t) = X$ and $g(s_1) = g(s_2) = Y$. As before, we apply $(*)$ to the pairs (r', s_2) and (t, s_1) obtaining $r - \delta + s_2 < r' + s_2 \leqslant Y \leqslant t + s_1 < r + \delta + s_1$, or $s_2 - s_1 < 2\delta$. This is a final contradiction.

Comment 1. The original submission discussed the same functions f, but the question was different — namely, the following one:

Prove that the equation $f(x) = 2017x$ has at most one solution, and the equation $f(x) = -2017x$ has at least one solution.

The Problem Selection Committee decided that the question we are proposing is more natural, since it provides more natural information about the function g (which is indeed the main character in this story). On the other hand, the new problem statement is strong enough in order to imply the original one easily.

Namely, we will deduce from the new problem statement (along with the facts used in the solutions) that (i) for every $N > 0$ the equation $g(x) = -Nx$ has at most one solution, and (ii) for every $N > 1$ the equation $g(x) = Nx$ has at least one solution.

Claim (i) is now trivial. Indeed, g is proven to be non-decreasing, so $g(x) + Nx$ is strictly increasing and thus has at most one zero.

We proceed on claim (ii). If $g(0) = 0$, then the required root has been already found. Otherwise, we may assume that $g(0) > 0$ and denote $c = g(0)$. We intend to prove that $x = c/N$ is the required root. Indeed, by monotonicity we have $g(c/N) \geqslant g(0) = c$; if we had $g(c/N) > c$, then $(*)$ would yield $c \leqslant 0 + c/N \leqslant g(c/N)$ which is false. Thus, $g(x) = c = Nx$.

Comment 2. There are plenty of functions g satisfying $(*)$ (and hence of functions f satisfying the problem conditions). One simple example is $g_0(x) = 2x$. Next, for any increasing sequence $A = (\dots, a_{-1}, a_0, a_1, \dots)$ which is unbounded in both directions (i.e., for every N this sequence contains terms greater than N, as well as terms smaller than $-N$), the function g_A defined by

$$g_A(x) = a_i + a_{i+1} \qquad \text{whenever } x \in [a_i; a_{i+1})$$

satisfies $(*)$. Indeed, pick any $x < y$ with $g(x) \neq g(y)$; this means that $x \in [a_i; a_{i+1})$ and $y \in [a_j; a_{j+1})$ for some $i < j$. Then we have $g(x) = a_i + a_{i+1} \leqslant x + y < a_j + a_{j+1} = g(y)$, as required.

There also exist examples of the mixed behavior; e.g., for an arbitrary sequence A as above and an arbitrary subset $I \subseteq \mathbb{Z}$ the function

$$g_{A,I}(x) = \begin{cases} g_0(x), & x \in [a_i; a_{i+1}) \quad \text{with } i \in I; \\ g_A(x), & x \in [a_i; a_{i+1}) \quad \text{with } i \notin I \end{cases}$$

also satisfies $(*)$.

Finally, it is even possible to provide a complete description of all functions g satisfying $(*)$ (and hence of all functions f satisfying the problem conditions); however, it seems to be far out of scope for the IMO. This description looks as follows.

Let A be any closed subset of $\mathbb{R}$ which is unbounded in both directions. Define the functions i_A, s_A, and g_A as follows:

$$i_A(x) = \inf\{a \in A \colon a \geqslant x\}, \quad s_A(x) = \sup\{a \in A \colon a \leqslant x\}, \quad g_A(x) = i_A(x) + s_A(x).$$

It is easy to see that for different sets A and B the functions g_A and g_B are also different (since, e.g., for any $a \in A \setminus B$ the function g_B is constant in a small neighborhood of a, but the function g_A is not). One may check, similarly to the arguments above, that each such function satisfies $(*)$.

Finally, one more modification is possible. Namely, for any $x \in A$ one may redefine $g_A(x)$ (which is $2x$) to be any of the numbers

$$g_{A+}(x) = i_{A+}(x) + x \quad \text{or} \quad g_{A-}(x) = x + s_{A-}(x),$$
$$\text{where} \qquad i_{A+}(x) = \inf\{a \in A\colon a > x\} \quad \text{and} \quad s_{A-}(x) = \sup\{a \in A\colon a < x\}.$$

This really changes the value if x has some right (respectively, left) semi-neighborhood disjoint from A, so there are at most countably many possible changes; all of them can be performed independently.

With some effort, one may show that the construction above provides all functions g satisfying $(*)$.

Combinatorics

C1. A rectangle $\mathcal{R}$ with odd integer side lengths is divided into small rectangles with integer side lengths. Prove that there is at least one among the small rectangles whose distances from the four sides of $\mathcal{R}$ are either all odd or all even.

(Singapore)

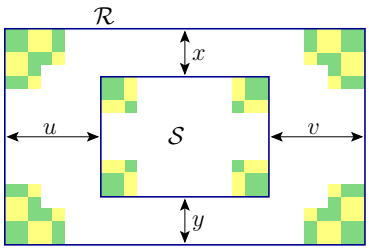
Solution. Let the width and height of $\mathcal{R}$ be odd numbers a and b. Divide $\mathcal{R}$ into ab unit squares and color them green and yellow in a checkered pattern. Since the side lengths of a and b are odd, the corner squares of $\mathcal{R}$ will all have the same color, say green.

Call a rectangle (either $\mathcal{R}$ or a small rectangle) *green* if its corners are all green; call it *yellow* if the corners are all yellow, and call it *mixed* if it has both green and yellow corners. In particular, $\mathcal{R}$ is a green rectangle.

We will use the following trivial observations.

- Every mixed rectangle contains the same number of green and yellow squares;
- Every green rectangle contains one more green square than yellow square;
- Every yellow rectangle contains one more yellow square than green square.

The rectangle $\mathcal{R}$ is green, so it contains more green unit squares than yellow unit squares. Therefore, among the small rectangles, at least one is green. Let $\mathcal{S}$ be such a small green rectangle, and let its distances from the sides of $\mathcal{R}$ be x, y, u and v, as shown in the picture. The top-left corner of $\mathcal{R}$ and the top-left corner of $\mathcal{S}$ have the same color, which happen if and only if x and u have the same parity. Similarly, the other three green corners of $\mathcal{S}$ indicate that x and v have the same parity, y and u have the same parity, i.e. x, y, u and v are all odd or all even.



C2. Let n be a positive integer. Define a *chameleon* to be any sequence of $3n$ letters, with exactly n occurrences of each of the letters a, b, and c. Define a *swap* to be the transposition of two adjacent letters in a chameleon. Prove that for any chameleon X, there exists a chameleon Y such that X cannot be changed to Y using fewer than $3n^2/2$ swaps.

(Australia)

Solution 1. To start, notice that the swap of two identical letters does not change a chameleon, so we may assume there are no such swaps.

For any two chameleons X and Y, define their *distance* $d(X,Y)$ to be the minimal number of swaps needed to transform X into Y (or vice versa). Clearly, $d(X,Y)+d(Y,Z)\geqslant d(X,Z)$ for any three chameleons X, Y, and Z.

Lemma. Consider two chameleons

$$P=\underbrace{aa\ldots a}_{n}\underbrace{bb\ldots b}_{n}\underbrace{cc\ldots c}_{n}\quad\text{and}\quad Q=\underbrace{cc\ldots c}_{n}\underbrace{bb\ldots b}_{n}\underbrace{aa\ldots a}_{n}.$$

Then $d(P,Q)\geqslant 3n^2$.

Proof. For any chameleon X and any pair of distinct letters $u,v\in\{a,b,c\}$, we define $f_{u,v}(X)$ to be the number of pairs of positions in X such that the left one is occupied by u, and the right one is occupied by v. Define $f(X)=f_{a,b}(X)+f_{a,c}(X)+f_{b,c}(X)$. Notice that $f_{a,b}(P)=f_{a,c}(P)=f_{b,c}(P)=n^2$ and $f_{a,b}(Q)=f_{a,c}(Q)=f_{b,c}(Q)=0$, so $f(P)=3n^2$ and $f(Q)=0$.

Now consider some swap changing a chameleon X to X'; say, the letters a and b are swapped. Then $f_{a,b}(X)$ and $f_{a,b}(X')$ differ by exactly 1, while $f_{a,c}(X)=f_{a,c}(X')$ and $f_{b,c}(X)=f_{b,c}(X')$. This yields $|f(X)-f(X')|=1$, i.e., on any swap the value of f changes by 1. Hence $d(X,Y)\geqslant|f(X)-f(Y)|$ for any two chameleons X and Y. In particular, $d(P,Q)\geqslant|f(P)-f(Q)|=3n^2$, as desired. □

Back to the problem, take any chameleon X and notice that $d(X,P)+d(X,Q)\geqslant d(P,Q)\geqslant 3n^2$ by the lemma. Consequently, $\max\{d(X,P),d(X,Q)\}\geqslant\frac{3n^2}{2}$, which establishes the problem statement.

Comment 1. The problem may be reformulated in a graph language. Construct a graph G with the chameleons as vertices, two vertices being connected with an edge if and only if these chameleons differ by a single swap. Then $d(X,Y)$ is the usual distance between the vertices X and Y in this graph. Recall that the *radius* of a connected graph G is defined as

$$r(G)=\min_{v\in V}\max_{u\in V}d(u,v).$$

So we need to prove that the radius of the constructed graph is at least $3n^2/2$.

It is well-known that the radius of any connected graph is at least the half of its diameter (which is simply $\max_{u,v\in V}d(u,v)$). Exactly this fact has been used above in order to finish the solution.

Solution 2. We use the notion of distance from Solution 1, but provide a different lower bound for it.

In any chameleon X, we enumerate the positions in it from left to right by $1,2,\ldots,3n$. Define $s_c(X)$ as the sum of positions occupied by c. The value of s_c changes by at most 1 on each swap, but this fact alone does not suffice to solve the problem; so we need an improvement.

For every chameleon X, denote by $X_{\overline{c}}$ the sequence obtained from X by removing all n letters c. Enumerate the positions in $X_{\overline{c}}$ from left to right by $1,2,\ldots,2n$, and define $s_{\overline{c},b}(X)$ as the sum of positions in $X_{\overline{c}}$ occupied by b. (In other words, here we consider the positions of the b's relatively to the a's only.) Finally, denote

$$d'(X,Y):=|s_c(X)-s_c(Y)|+|s_{\overline{c},b}(X)-s_{\overline{c},b}(Y)|.$$

Now consider any swap changing a chameleon X to X'. If no letter c is involved into this swap, then $s_c(X) = s_c(X')$; on the other hand, exactly one letter b changes its position in $X_{\overline{c}}$, so $|s_{\overline{c},b}(X) - s_{\overline{c},b}(X')| = 1$. If a letter c is involved into a swap, then $X_{\overline{c}} = X'_{\overline{c}}$, so $s_{\overline{c},b}(X) = s_{\overline{c},b}(X')$ and $|s_c(X) - s_c(X')| = 1$. Thus, in all cases we have $d'(X, X') = 1$.

As in the previous solution, this means that $d(X, Y) \geqslant d'(X, Y)$ for any two chameleons X and Y. Now, for any chameleon X we will indicate a chameleon Y with $d'(X, Y) \geqslant 3n^2/2$, thus finishing the solution.

The function s_c attains all integer values from $1 + \cdots + n = \frac{n(n+1)}{2}$ to $(2n+1) + \cdots + 3n = 2n^2 + \frac{n(n+1)}{2}$. If $s_c(X) \leqslant n^2 + \frac{n(n+1)}{2}$, then we put the letter c into the last n positions in Y; otherwise we put the letter c into the first n positions in Y. In either case we already have $|s_c(X) - s_c(Y)| \geqslant n^2$.

Similarly, $s_{\overline{c},b}$ ranges from $\frac{n(n+1)}{2}$ to $n^2 + \frac{n(n+1)}{2}$. So, if $s_{\overline{c},b}(X) \leqslant \frac{n^2}{2} + \frac{n(n+1)}{2}$, then we put the letter b into the last n positions in Y which are still free; otherwise, we put the letter b into the first n such positions. The remaining positions are occupied by a. In any case, we have $|s_{\overline{c},b}(X) - s_{\overline{c},b}(Y)| \geqslant \frac{n^2}{2}$, thus $d'(X, Y) \geqslant n^2 + \frac{n^2}{2} = \frac{3n^2}{2}$, as desired.

Comment 2. The two solutions above used two lower bounds $|f(X) - f(Y)|$ and $d'(X, Y)$ for the number $d(X, Y)$. One may see that these bounds are closely related to each other, as

$$f_{a,c}(X) + f_{b,c}(X) = s_c(X) - \frac{n(n+1)}{2} \quad \text{and} \quad f_{a,b}(X) = s_{\overline{c},b}(X) - \frac{n(n+1)}{2}.$$

One can see that, e.g., the bound $d'(X, Y)$ could as well be used in the proof of the lemma in Solution 1.

Let us describe here an even sharper bound which also can be used in different versions of the solutions above.

In each chameleon X, enumerate the occurrences of a from the left to the right as $a_1, a_2, \ldots, a_n$. Since we got rid of swaps of identical letters, the relative order of these letters remains the same during the swaps. Perform the same operation with the other letters, obtaining new letters $b_1, \ldots, b_n$ and $c_1, \ldots, c_n$. Denote by A the set of the $3n$ obtained letters.

Since all $3n$ letters became different, for any chameleon X and any $s \in A$ we may define the position $N_s(X)$ of s in X (thus $1 \leqslant N_s(X) \leqslant 3n$). Now, for any two chameleons X and Y we say that a pair of letters $(s, t) \in A \times A$ is an (X, Y)-*inversion* if $N_s(X) < N_t(X)$ but $N_s(Y) > N_t(Y)$, and define $d^*(X, Y)$ to be the number of (X, Y)-inversions. Then for any two chameleons Y and Y' differing by a single swap, we have $|d^*(X, Y) - d^*(X, Y')| = 1$. Since $d^*(X, X) = 0$, this yields $d(X, Y) \geqslant d^*(X, Y)$ for any pair of chameleons X and Y. The bound d^* may also be used in both Solution 1 and Solution 2.

Comment 3. In fact, one may prove that the distance d^* defined in the previous comment *coincides* with d. Indeed, if $X \neq Y$, then there exist an (X, Y)-inversion (s, t). One can show that such s and t may be chosen to occupy consecutive positions in Y. Clearly, s and t correspond to different letters among $\{a, b, c\}$. So, swapping them in Y we get another chameleon Y' with $d^*(X, Y') = d^*(X, Y) - 1$. Proceeding in this manner, we may change Y to X in $d^*(X, Y)$ steps.

Using this fact, one can show that the estimate in the problem statement is sharp for all $n \geqslant 2$. (For $n = 1$ it is not sharp, since any permutation of three letters can be changed to an opposite one in no less than three swaps.) We outline the proof below.

For any $k \geqslant 0$, define

$$X_{2k} = \underbrace{abc\,abc\ldots abc}_{3k \text{ letters}}\underbrace{cba\,cba\ldots cba}_{3k \text{ letters}} \quad \text{and} \quad X_{2k+3} = \underbrace{abc\,abc\ldots abc}_{3k \text{ letters}}abc\,bca\,cab\underbrace{cba\,cba\ldots cba}_{3k \text{ letters}}.$$

We claim that for every $n \geqslant 2$ and every chameleon Y, we have $d^*(X_n, Y) \leqslant \lceil 3n^2/2 \rceil$. This will mean that for every $n \geqslant 2$ the number $3n^2/2$ in the problem statement cannot be changed by any number larger than $\lceil 3n^2/2 \rceil$.

For any distinct letters $u, v \in \{a, b, c\}$ and any two chameleons X and Y, we define $d^*_{u,v}(X, Y)$ as the number of (X, Y)-inversions (s, t) such that s and t are instances of u and v (in any of the two possible orders). Then $d^*(X, Y) = d^*_{a,b}(X, Y) + d^*_{b,c}(X, Y) + d^*_{c,a}(X, Y)$.

We start with the case when $n = 2k$ is even; denote $X = X_{2k}$. We show that $d^*_{a,b}(X, Y) \leqslant 2k^2$ for any chameleon Y; this yields the required estimate. Proceed by the induction on k with the trivial base case $k = 0$. To perform the induction step, notice that $d^*_{a,b}(X, Y)$ is indeed the minimal number of swaps needed to change $Y_{\overline{c}}$ into $X_{\overline{c}}$. One may show that moving a_1 and a_{2k} in Y onto the first and the last positions in Y, respectively, takes at most $2k$ swaps, and that subsequent moving b_1 and b_{2k} onto the second and the second last positions takes at most $2k - 2$ swaps. After performing that, one may delete these letters from both $X_{\overline{c}}$ and $Y_{\overline{c}}$ and apply the induction hypothesis; so $X_{\overline{c}}$ can be obtained from $Y_{\overline{c}}$ using at most $2(k-1)^2 + 2k + (2k-2) = 2k^2$ swaps, as required.

If $n = 2k + 3$ is odd, the proof is similar but more technically involved. Namely, we claim that $d^*_{a,b}(X_{2k+3}, Y) \leqslant 2k^2 + 6k + 5$ for any chameleon Y, and that the equality is achieved only if $Y_{\overline{c}} = bb\ldots b\,aa\ldots a$. The proof proceeds by a similar induction, with some care taken of the base case, as well as of extracting the equality case. Similar estimates hold for $d^*_{b,c}$ and $d^*_{c,a}$. Summing three such estimates, we obtain

$$d^*(X_{2k+3}, Y) \leqslant 3(2k^2 + 6k + 5) = \left\lceil \frac{3n^2}{2} \right\rceil + 1,$$

which is by 1 more than we need. But the equality could be achieved only if $Y_{\overline{c}} = bb\ldots b\,aa\ldots a$ and, similarly, $Y_{\overline{b}} = aa\ldots a\,cc\ldots c$ and $Y_{\overline{a}} = cc\ldots c\,bb\ldots b$. Since these three equalities cannot hold simultaneously, the proof is finished.

C3. Sir Alex plays the following game on a row of 9 cells. Initially, all cells are empty. In each move, Sir Alex is allowed to perform exactly one of the following two operations:

(1) Choose any number of the form 2^j, where j is a non-negative integer, and put it into an empty cell.

(2) Choose two (not necessarily adjacent) cells with the same number in them; denote that number by 2^j. Replace the number in one of the cells with 2^{j+1} and erase the number in the other cell.

At the end of the game, one cell contains the number 2^n, where n is a given positive integer, while the other cells are empty. Determine the maximum number of moves that Sir Alex could have made, in terms of n.

(Thailand)

Answer: $2\sum_{j=0}^{8}\binom{n}{j}-1$.

Solution 1. We will solve a more general problem, replacing the row of 9 cells with a row of k cells, where k is a positive integer. Denote by $m(n,k)$ the maximum possible number of moves Sir Alex can make starting with a row of k empty cells, and ending with one cell containing the number 2^n and all the other $k-1$ cells empty. Call an operation of type (1) an *insertion*, and an operation of type (2) a *merge*.

Only one move is possible when $k=1$, so we have $m(n,1)=1$. From now on we consider $k\geqslant 2$, and we may assume Sir Alex's last move was a merge. Then, just before the last move, there were exactly two cells with the number 2^{n-1}, and the other $k-2$ cells were empty.

Paint one of those numbers 2^{n-1} blue, and the other one red. Now trace back Sir Alex's moves, always painting the numbers blue or red following this rule: *if a and b merge into c, paint a and b with the same color as c.* Notice that in this backward process new numbers are produced only by reversing merges, since reversing an insertion simply means deleting one of the numbers. Therefore, all numbers appearing in the whole process will receive one of the two colors.

Sir Alex's first move is an insertion. Without loss of generality, assume this first number inserted is blue. Then, from this point on, until the last move, there is always at least one cell with a blue number.

Besides the last move, there is no move involving a blue and a red number, since all merges involves numbers with the same color, and insertions involve only one number. Call an insertion of a blue number or merge of two blue numbers a *blue move*, and define a *red move* analogously.

The whole sequence of blue moves could be repeated on another row of k cells to produce one cell with the number 2^{n-1} and all the others empty, so there are at most $m(n-1,k)$ blue moves.

Now we look at the red moves. Since every time we perform a red move there is at least one cell occupied with a blue number, the whole sequence of red moves could be repeated on a row of $k-1$ cells to produce one cell with the number 2^{n-1} and all the others empty, so there are at most $m(n-1,k-1)$ red moves. This proves that

$$m(n,k)\leqslant m(n-1,k)+m(n-1,k-1)+1.$$

On the other hand, we can start with an empty row of k cells and perform $m(n-1,k)$ moves to produce one cell with the number 2^{n-1} and all the others empty, and after that perform $m(n-1,k-1)$ moves on those $k-1$ empty cells to produce the number 2^{n-1} in one of them, leaving $k-2$ empty. With one more merge we get one cell with 2^n and the others empty, proving that

$$m(n,k)\geqslant m(n-1,k)+m(n-1,k-1)+1.$$

It follows that

$$m(n,k) = m(n-1,k) + m(n-1,k-1) + 1, \tag{1}$$

for $n \geqslant 1$ and $k \geqslant 2$.

If $k = 1$ or $n = 0$, we must insert 2^n on our first move and immediately get the final configuration, so $m(0,k) = 1$ and $m(n,1) = 1$, for $n \geqslant 0$ and $k \geqslant 1$. These initial values, together with the recurrence relation (1), determine $m(n,k)$ uniquely.

Finally, we show that

$$m(n,k) = 2\sum_{j=0}^{k-1} \binom{n}{j} - 1, \tag{2}$$

for all integers $n \geqslant 0$ and $k \geqslant 1$.

We use induction on n. Since $m(0,k) = 1$ for $k \geqslant 1$, (2) is true for the base case. We make the induction hypothesis that (2) is true for some fixed positive integer n and all $k \geqslant 1$. We have $m(n+1,1) = 1 = 2\binom{n+1}{0} - 1$, and for $k \geqslant 2$ the recurrence relation (1) and the induction hypothesis give us

$$\begin{aligned} m(n+1,k) &= m(n,k) + m(n,k-1) + 1 = 2\sum_{j=0}^{k-1} \binom{n}{j} - 1 + 2\sum_{j=0}^{k-2} \binom{n}{j} - 1 + 1 \\ &= 2\sum_{j=0}^{k-1} \binom{n}{j} + 2\sum_{j=0}^{k-1} \binom{n}{j-1} - 1 = 2\sum_{j=0}^{k-1} \left(\binom{n}{j} + \binom{n}{j-1} \right) - 1 = 2\sum_{j=0}^{k-1} \binom{n+1}{j} - 1, \end{aligned}$$

which completes the proof.

Comment 1. After deducing the recurrence relation (1), it may be convenient to homogenize the recurrence relation by defining $h(n,k) = m(n,k) + 1$. We get the new relation

$$h(n,k) = h(n-1,k) + h(n-1,k), \tag{3}$$

for $n \geqslant 1$ and $k \geqslant 2$, with initial values $h(0,k) = h(n,1) = 2$, for $n \geqslant 0$ and $k \geqslant 1$.

This may help one to guess the answer, and also with other approaches like the one we develop next.

Comment 2. We can use a generating function to find the answer without guessing. We work with the homogenized recurrence relation (3). Define $h(n,0) = 0$ so that (3) is valid for $k = 1$ as well. Now we set up the generating function $f(x,y) = \sum_{n,k\geqslant 0} h(n,k)x^n y^k$. Multiplying the recurrence relation (3) by $x^n y^k$ and summing over $n, k \geqslant 1$, we get

$$\sum_{n,k\geqslant 1} h(n,k)x^n y^k = x \sum_{n,k\geqslant 1} h(n-1,k)x^{n-1}y^k + xy \sum_{n,k\geqslant 1} h(n-1,k-1)x^{n-1}y^{k-1}.$$

Completing the missing terms leads to the following equation on $f(x,y)$:

$$f(x,y) - \sum_{n\geqslant 0} h(n,0)x^n - \sum_{k\geqslant 1} h(0,k)y^k = xf(x,y) - x\sum_{n\geqslant 0} h(n,0)x^n + xyf(x,y).$$

Substituting the initial values, we obtain

$$f(x,y) = \frac{2y}{1-y} \cdot \frac{1}{1-x(1+y)}.$$

Developing as a power series, we get

$$f(x,y) = 2\sum_{j\geqslant 1} y^j \cdot \sum_{n\geqslant 0} (1+y)^n x^n.$$

The coefficient of x^n in this power series is

$$2\sum_{j\geqslant 1} y^j \cdot (1+y)^n = 2\sum_{j\geqslant 1} y^j \cdot \sum_{i\geqslant 0} \binom{n}{i} y^i,$$

and extracting the coefficient of y^k in this last expression we finally obtain the value for $h(n,k)$,

$$h(n,k) = 2\sum_{j=0}^{k-1} \binom{n}{j}.$$

This proves that

$$m(n,k) = 2\sum_{j=0}^{k-1} \binom{n}{j} - 1.$$

The generating function approach also works if applied to the non-homogeneous recurrence relation (1), but the computations are less straightforward.

Solution 2. Define merges and insertions as in Solution 1. After each move made by Sir Alex we compute the number N of empty cells, and the sum S of all the numbers written in the cells. Insertions always increase S by some power of 2, and increase N exactly by 1. Merges do not change S and decrease N exactly by 1. Since the initial value of N is 0 and its final value is 1, the total number of insertions exceeds that of merges by exactly one. So, to maximize the number of moves, we need to maximize the number of insertions.

We will need the following lemma.

Lemma. If the binary representation of a positive integer A has d nonzero digits, then A cannot be represented as a sum of fewer than d powers of 2. Moreover, any representation of A as a sum of d powers of 2 must coincide with its binary representation.

Proof. Let s be the minimum number of summands in all possible representations of A as sum of powers of 2. Suppose there is such a representation with s summands, where two of the summands are equal to each other. Then, replacing those two summands with the result of their sum, we obtain a representation with fewer than s summands, which is a contradiction. We deduce that in any representation with s summands, the summands are all distinct, so any such representation must coincide with the unique binary representation of A, and $s = d$. □

Now we split the solution into a sequence of claims.

Claim 1. After every move, the number S is the sum of at most $k-1$ distinct powers of 2.

Proof. If S is the sum of k (or more) distinct powers of 2, the Lemma implies that the k cells are filled with these numbers. This is a contradiction since no more merges or insertions can be made. □

Let $A(n,k-1)$ denote the set of all positive integers not exceeding 2^n with at most $k-1$ nonzero digits in its base 2 representation. Since every insertion increases the value of S, by Claim 1, the total number of insertions is at most $|A(n,k-1)|$. We proceed to prove that it is possible to achieve this number of insertions.

Claim 2. Let $A(n,k-1) = \{a_1, a_2, \ldots, a_m\}$, with $a_1 < a_2 < \cdots < a_m$. If after some of Sir Alex's moves the value of S is a_j, with $j \in \{1,2,\ldots,m-1\}$, then there is a sequence of moves after which the value of S is exactly a_{j+1}.

Proof. Suppose $S = a_j$. Performing all possible merges, we eventually get different powers of 2 in all nonempty cells. After that, by Claim 1 there will be at least one empty cell, in which we want to insert $a_{j+1} - a_j$. It remains to show that $a_{j+1} - a_j$ is a power of 2.

For this purpose, we notice that if a_j has less than $k-1$ nonzero digits in base 2 then $a_{j+1} = a_j + 1$. Otherwise, we have $a_j = 2^{b_{k-1}} + \cdots + 2^{b_2} + 2^{b_1}$ with $b_1 < b_2 < \cdots < b_{k-1}$. Then, adding any number less than 2^{b_1} to a_j will result in a number with more than $k-1$ nonzero

binary digits. On the other hand, $a_j + 2^{b_1}$ is a sum of k powers of 2, not all distinct, so by the Lemma it will be a sum of less then k distinct powers of 2. This means that $a_{j+1} - a_j = 2^{b_1}$, completing the proof. □

Claims 1 and 2 prove that the maximum number of insertions is $|A(n, k-1)|$. We now compute this number.

Claim 3. $|A(n, k-1)| = \sum_{j=0}^{k-1} \binom{n}{j}$.

Proof. The number 2^n is the only element of $A(n, k-1)$ with $n+1$ binary digits. Any other element has at most n binary digits, at least one and at most $k-1$ of them are nonzero (so they are ones). For each $j \in \{1, 2, \ldots, k-1\}$, there are $\binom{n}{j}$ such elements with exactly j binary digits equal to one. We conclude that $|A(n, k-1)| = 1 + \sum_{j=1}^{k-1} \binom{n}{j} = \sum_{j=0}^{k-1} \binom{n}{j}$. □

Recalling that the number of insertions exceeds that of merges by exactly 1, we deduce that the maximum number of moves is $2\sum_{j=0}^{k-1} \binom{n}{j} - 1$.

C4. Let $N \geqslant 2$ be an integer. $N(N+1)$ soccer players, no two of the same height, stand in a row in some order. Coach Ralph wants to remove $N(N-1)$ people from this row so that in the remaining row of $2N$ players, no one stands between the two tallest ones, no one stands between the third and the fourth tallest ones, ..., and finally no one stands between the two shortest ones. Show that this is always possible.

(Russia)

Solution 1. Split the row into N blocks with $N+1$ consecutive people each. We will show how to remove $N-1$ people from each block in order to satisfy the coach's wish.

First, construct a $(N+1) \times N$ matrix where $x_{i,j}$ is the height of the ith tallest person of the jth block—in other words, each column lists the heights within a single block, sorted in decreasing order from top to bottom.

We will reorder this matrix by repeatedly swapping whole columns. First, by column permutation, make sure that $x_{2,1} = \max\{x_{2,i} \colon i = 1, 2, \ldots, N\}$ (the first column contains the largest height of the *second* row). With the first column fixed, permute the other ones so that $x_{3,2} = \max\{x_{3,i} \colon i = 2, \ldots, N\}$ (the second column contains the tallest person of the *third* row, first column excluded). In short, at step k ($k = 1, 2, \ldots, N-1$), we permute the columns from k to N so that $x_{k+1,k} = \max\{x_{i,k} \colon i = k, k+1, \ldots, N\}$, and end up with an array like this:

$$\begin{array}{ccccccccc}
\boldsymbol{x_{1,1}} & & x_{1,2} & & x_{1,3} & \cdots & x_{1,N-1} & & x_{1,N} \\
\vee & & \vee & & \vee & & \vee & & \vee \\
\boldsymbol{x_{2,1}} & > & \boldsymbol{x_{2,2}} & & x_{2,3} & \cdots & x_{2,N-1} & & x_{2,N} \\
\vee & & \vee & & \vee & & \vee & & \vee \\
x_{3,1} & & \boldsymbol{x_{3,2}} & > & \boldsymbol{x_{3,3}} & \cdots & x_{3,N-1} & & x_{3,N} \\
\vee & & \vee & & \vee & & \vee & & \vee \\
\vdots & & \vdots & & \vdots & \ddots & \vdots & & \vdots \\
\vee & & \vee & & \vee & & \vee & & \vee \\
x_{N,1} & & x_{N,2} & & x_{N,3} & \cdots & \boldsymbol{x_{N,N-1}} & > & \boldsymbol{x_{N,N}} \\
\vee & & \vee & & \vee & & \vee & & \vee \\
x_{N+1,1} & & x_{N+1,2} & & x_{N+1,3} & \cdots & x_{N+1,N-1} & & \boldsymbol{x_{N+1,N}}
\end{array}$$

Now we make the bold choice: from the original row of people, remove everyone but those with heights

$$x_{1,1} > x_{2,1} > x_{2,2} > x_{3,2} > \cdots > x_{N,N-1} > x_{N,N} > x_{N+1,N} \qquad (*)$$

Of course this *height* order $(*)$ is not necessarily their *spatial* order in the new row. We now need to convince ourselves that each pair $(x_{k,k}; x_{k+1,k})$ remains spatially together in this new row. But $x_{k,k}$ and $x_{k+1,k}$ belong to the same column/block of consecutive $N+1$ people; the only people that could possibly stand between them were also in this block, and they are all gone.

Solution 2. Split the people into N groups *by height*: group G_1 has the $N+1$ tallest ones, group G_2 has the next $N+1$ tallest, and so on, up to group G_N with the $N+1$ shortest people.

Now scan the original row from left to right, stopping as soon as you have scanned two people (consecutively or not) from the same group, say, G_i. Since we have N groups, this must happen before or at the $(N+1)^{\text{th}}$ person of the row. Choose this pair of people, removing all the other people from the same group G_i and also all people that have been scanned so far. The only people that could separate this pair's heights were in group G_i (and they are gone); the only people that could separate this pair's positions were already scanned (and they are gone too).

We are now left with $N-1$ groups (all except G_i). Since each of them lost at most one person, each one has at least N unscanned people left in the row. Repeat the scanning process from left to right, choosing the next two people from the same group, removing this group and

everyone scanned up to that point. Once again we end up with two people who are next to each other in the remaining row and whose heights cannot be separated by anyone else who remains (since the rest of their group is gone). After picking these 2 pairs, we still have $N-2$ groups with at least $N-1$ people each.

If we repeat the scanning process a total of N times, it is easy to check that we will end up with 2 people from each group, for a total of $2N$ people remaining. The height order is guaranteed by the grouping, and the scanning construction from left to right guarantees that each pair from a group stand next to each other in the final row. We are done.

Solution 3. This is essentially the same as solution 1, but presented inductively. The essence of the argument is the following lemma.

Lemma. Assume that we have N disjoint groups of at least $N+1$ people in each, all people have distinct heights. Then one can choose two people from each group so that among the chosen people, the two tallest ones are in one group, the third and the fourth tallest ones are in one group, ..., and the two shortest ones are in one group.

Proof. Induction on $N \geqslant 1$; for $N=1$, the statement is trivial.

Consider now N groups $G_1, \ldots, G_N$ with at least $N+1$ people in each for $N \geqslant 2$. Enumerate the people by $1, 2, \ldots, N(N+1)$ according to their height, say, from tallest to shortest. Find the least s such that two people among $1, 2, \ldots, s$ are in one group (without loss of generality, say this group is G_N). By the minimality of s, the two mentioned people in G_N are s and some $i < s$.

Now we choose people i and s in G_N, forget about this group, and remove the people $1, 2, \ldots, s$ from $G_1, \ldots, G_{N-1}$. Due to minimality of s again, each of the obtained groups $G'_1, \ldots, G'_{N-1}$ contains at least N people. By the induction hypothesis, one can choose a pair of people from each of $G'_1, \ldots, G'_{N-1}$ so as to satisfy the required conditions. Since all these people have numbers greater than s, addition of the pair (s, i) from G_N does not violate these requirements. □

To solve the problem, it suffices now to split the row into N contiguous groups with $N+1$ people in each and apply the Lemma to those groups.

Comment 1. One can identify each person with a pair of indices (p, h) $(p, h \in \{1, 2, \ldots, N(N+1)\})$ so that the pth person in the row (say, from left to right) is the hth tallest person in the group. Say that (a, b) *separates* (x_1, y_1) and (x_2, y_2) whenever a is strictly between x_1 and y_1, *or* b is strictly between x_2 and y_2. So the coach wants to pick $2N$ people $(p_i, h_i)(i = 1, 2, \ldots, 2N)$ such that no chosen person separates (p_1, h_1) from (p_2, h_2), no chosen person separates (p_3, h_3) and (p_4, h_4), and so on. This formulation reveals a duality between positions and heights. In that sense, solutions 1 and 2 are dual of each other.

Comment 2. The number $N(N+1)$ is sharp for $N=2$ and $N=3$, due to arrangements $1, 5, 3, 4, 2$ and $1, 10, 6, 4, 3, 9, 5, 8, 7, 2, 11$.

C5. A hunter and an invisible rabbit play a game in the Euclidean plane. The hunter's starting point H_0 coincides with the rabbit's starting point R_0. In the nth round of the game ($n \geqslant 1$), the following happens.

(1) First the invisible rabbit moves secretly and unobserved from its current point R_{n-1} to some new point R_n with $R_{n-1}R_n = 1$.

(2) The hunter has a tracking device (e.g. dog) that returns an approximate position R'_n of the rabbit, so that $R_nR'_n \leqslant 1$.

(3) The hunter then visibly moves from point H_{n-1} to a new point H_n with $H_{n-1}H_n = 1$.

Is there a strategy for the hunter that guarantees that after 10^9 such rounds the distance between the hunter and the rabbit is below 100?

(Austria)

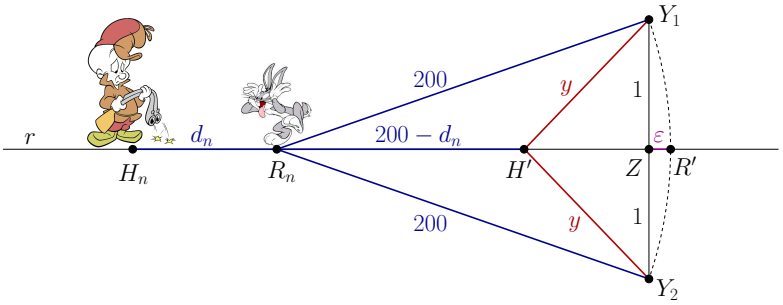
Answer: There is no such strategy for the hunter. The rabbit "wins".

Solution. If the answer were "yes", the hunter would have a strategy that would "work", no matter how the rabbit moved or where the radar pings R'_n appeared. We will show the opposite: with bad luck from the radar pings, there is no strategy for the hunter that guarantees that the distance stays below 100 in 10^9 rounds.

So, let d_n be the distance between the hunter and the rabbit after n rounds. Of course, if $d_n \geqslant 100$ for any $n < 10^9$, the rabbit has won — it just needs to move straight away from the hunter, and the distance will be kept at or above 100 thereon.

We will now show that, while $d_n < 100$, whatever given strategy the hunter follows, the rabbit has a way of increasing d_n^2 by at least $\frac{1}{2}$ every 200 rounds (as long as the radar pings are lucky enough for the rabbit). This way, d_n^2 will reach 10^4 in less than $2 \cdot 10^4 \cdot 200 = 4 \cdot 10^6 < 10^9$ rounds, and the rabbit wins.

Suppose the hunter is at H_n and the rabbit is at R_n. Suppose even that the rabbit *reveals* its position at this moment to the hunter (this allows us to ignore all information from previous radar pings). Let r be the line H_nR_n, and Y_1 and Y_2 be points which are 1 unit away from r and 200 units away from R_n, as in the figure below.



The rabbit's plan is simply to choose one of the points Y_1 or Y_2 and hop 200 rounds straight towards it. Since all hops stay within 1 distance unit from r, it is possible that all radar pings stay on r. In particular, in this case, the hunter has no way of knowing whether the rabbit chose Y_1 or Y_2.

Looking at such pings, what is the hunter going to do? If the hunter's strategy tells him to go 200 rounds straight to the right, he ends up at point H' in the figure. Note that the hunter does not have a better alternative! Indeed, after these 200 rounds he will always end up at a point to the left of H'. If his strategy took him to a point above r, he would end up even further from Y_2; and if his strategy took him below r, he would end up even further from Y_1. In other words, no matter what strategy the hunter follows, he can never be sure his distance to the rabbit will be less than $y \stackrel{\text{def}}{=} H'Y_1 = H'Y_2$ after these 200 rounds.

To estimate y^2, we take Z as the midpoint of segment Y_1Y_2, we take R' as a point 200 units to the right of R_n and we define $\varepsilon = ZR'$ (note that $H'R' = d_n$). Then

$$y^2 = 1 + (H'Z)^2 = 1 + (d_n - \varepsilon)^2$$

where

$$\varepsilon = 200 - R_n Z = 200 - \sqrt{200^2 - 1} = \frac{1}{200 + \sqrt{200^2 - 1}} > \frac{1}{400}.$$

In particular, $\varepsilon^2 + 1 = 400\varepsilon$, so

$$y^2 = d_n^2 - 2\varepsilon d_n + \varepsilon^2 + 1 = d_n^2 + \varepsilon(400 - 2d_n).$$

Since $\varepsilon > \frac{1}{400}$ and we assumed $d_n < 100$, this shows that $y^2 > d_n^2 + \frac{1}{2}$. So, as we claimed, with this list of radar pings, no matter what the hunter does, the rabbit might achieve $d_{n+200}^2 > d_n^2 + \frac{1}{2}$. The wabbit wins.

Comment 1. Many different versions of the solution above can be found by replacing 200 with some other number N for the number of hops the rabbit takes between reveals. If this is done, we have:

$$\varepsilon = N - \sqrt{N^2 - 1} > \frac{1}{N + \sqrt{N^2 - 1}} > \frac{1}{2N}$$

and

$$\varepsilon^2 + 1 = 2N\varepsilon,$$

so, as long as $N > d_n$, we would find

$$y^2 = d_n^2 + \varepsilon(2N - 2d_n) > d_n^2 + \frac{N - d_n}{N}.$$

For example, taking $N = 101$ is already enough—the squared distance increases by at least $\frac{1}{101}$ every 101 rounds, and $101^2 \cdot 10^4 = 1.0201 \cdot 10^8 < 10^9$ rounds are enough for the rabbit. If the statement is made sharper, some such versions might not work any longer.

Comment 2. The original statement asked whether the distance could be kept under 10^{10} in 10^{100} rounds.

C6. Let $n > 1$ be an integer. An $n \times n \times n$ cube is composed of n^3 unit cubes. Each unit cube is painted with one color. For each $n \times n \times 1$ box consisting of n^2 unit cubes (of any of the three possible orientations), we consider the set of the colors present in that box (each color is listed only once). This way, we get $3n$ sets of colors, split into three groups according to the orientation. It happens that for every set in any group, the same set appears in both of the other groups. Determine, in terms of n, the maximal possible number of colors that are present.

(Russia)

Answer: The maximal number is $\frac{n(n+1)(2n+1)}{6}$.

Solution 1. Call a $n \times n \times 1$ box an *x-box*, a *y-box*, or a *z-box*, according to the direction of its short side. Let C be the number of colors in a valid configuration. We start with the upper bound for C.

Let $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_3$ be the sets of colors which appear in the big cube exactly once, exactly twice, and at least thrice, respectively. Let M_i be the set of unit cubes whose colors are in $\mathcal{C}_i$, and denote $n_i = |M_i|$.

Consider any x-box X, and let Y and Z be a y- and a z-box containing the same set of colors as X does.

Claim. $4|X \cap M_1| + |X \cap M_2| \leqslant 3n + 1$.

Proof. We distinguish two cases.

Case 1: $X \cap M_1 \neq \varnothing$.

A cube from $X \cap M_1$ should appear in all three boxes X, Y, and Z, so it should lie in $X \cap Y \cap Z$. Thus $X \cap M_1 = X \cap Y \cap Z$ and $|X \cap M_1| = 1$.

Consider now the cubes in $X \cap M_2$. There are at most $2(n-1)$ of them lying in $X \cap Y$ or $X \cap Z$ (because the cube from $X \cap Y \cap Z$ is in M_1). Let a be some *other* cube from $X \cap M_2$. Recall that there is just one other cube a' sharing a color with a. But both Y and Z should contain such cube, so $a' \in Y \cap Z$ (but $a' \notin X \cap Y \cap Z$). The map $a \mapsto a'$ is clearly injective, so the number of cubes a we are interested in does not exceed $|(Y \cap Z) \setminus X| = n - 1$. Thus $|X \cap M_2| \leqslant 2(n-1) + (n-1) = 3(n-1)$, and hence $4|X \cap M_1| + |X \cap M_2| \leqslant 4 + 3(n-1) = 3n+1$.

Case 2: $X \cap M_1 = \varnothing$.

In this case, the same argument applies with several changes. Indeed, $X \cap M_2$ contains at most $2n - 1$ cubes from $X \cap Y$ or $X \cap Z$. Any other cube a in $X \cap M_2$ corresponds to some $a' \in Y \cap Z$ (possibly with $a' \in X$), so there are at most n of them. All this results in $|X \cap M_2| \leqslant (2n-1) + n = 3n - 1$, which is even better than we need (by the assumptions of our case). □

Summing up the inequalities from the Claim over all x-boxes X, we obtain

$$4n_1 + n_2 \leqslant n(3n + 1).$$

Obviously, we also have $n_1 + n_2 + n_3 = n^3$.

Now we are prepared to estimate C. Due to the definition of the M_i, we have $n_i \geqslant i|\mathcal{C}_i|$, so

$$C \leqslant n_1 + \frac{n_2}{2} + \frac{n_3}{3} = \frac{n_1 + n_2 + n_3}{3} + \frac{4n_1 + n_2}{6} \leqslant \frac{n^3}{3} + \frac{3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6}.$$

It remains to present an example of an appropriate coloring in the above-mentioned number of colors. For each color, we present the set of all cubes of this color. These sets are:

1. n singletons of the form $S_i = \{(i, i, i)\}$ (with $1 \leqslant i \leqslant n$);
2. $3\binom{n}{2}$ doubletons of the forms $D^1_{i,j} = \{(i, j, j), (j, i, i)\}$, $D^2_{i,j} = \{(j, i, j), (i, j, i)\}$, and $D^3_{i,j} = \{(j, j, i), (i, i, j)\}$ (with $1 \leqslant i < j \leqslant n$);

3. $2\binom{n}{3}$ triplets of the form $T_{i,j,k} = \{(i,j,k),(j,k,i),(k,i,j)\}$ (with $1 \leqslant i < j < k \leqslant n$ or $1 \leqslant i < k < j \leqslant n$).

One may easily see that the ith boxes of each orientation contain the same set of colors, and that

$$n + \frac{3n(n-1)}{2} + \frac{n(n-1)(n-2)}{3} = \frac{n(n+1)(2n+1)}{6}$$

colors are used, as required.

Solution 2. We will approach a new version of the original problem. In this new version, each cube may have a color, or be *invisible* (not both). Now we make sets of colors for each $n \times n \times 1$ box as before (where "invisible" is not considered a color) and group them by orientation, also as before. Finally, we require that, for every *non-empty* set in any group, the same set must appear in the other 2 groups. What is the maximum number of colors present with these new requirements?

Let us call *strange* a big $n \times n \times n$ cube whose painting scheme satisfies the new requirements, and let D be the number of colors in a strange cube. Note that any cube that satisfies the original requirements is also strange, so $\max(D)$ is an upper bound for the original answer.

Claim. $D \leqslant \frac{n(n+1)(2n+1)}{6}$.

Proof. The proof is by induction on n. If $n = 1$, we must paint the cube with at most 1 color.

Now, pick a $n \times n \times n$ strange cube A, where $n \geqslant 2$. If A is completely invisible, $D = 0$ and we are done. Otherwise, pick a non-empty set of colors $\mathcal{S}$ which corresponds to, say, the boxes X, Y and Z of different orientations.

Now find all cubes in A whose colors are in $\mathcal{S}$ and make them invisible. Since X, Y and Z are now completely invisible, we can throw them away and focus on the remaining $(n-1) \times (n-1) \times (n-1)$ cube B. The sets of colors in all the groups for B are the same as the sets for A, removing exactly the colors in $\mathcal{S}$, and no others! Therefore, every nonempty set that appears in one group for B still shows up in all possible orientations (it is possible that an empty set of colors in B only matched X, Y or Z before these were thrown away, but remember we do not require empty sets to match anyway). In summary, B is also strange.

By the induction hypothesis, we may assume that B has at most $\frac{(n-1)n(2n-1)}{6}$ colors. Since there were at most n^2 different colors in $\mathcal{S}$, we have that A has at most $\frac{(n-1)n(2n-1)}{6} + n^2 = \frac{n(n+1)(2n+1)}{6}$ colors. □

Finally, the construction in the previous solution shows a painting scheme (with no invisible cubes) that reaches this maximum, so we are done.

C7. For any finite sets X and Y of positive integers, denote by $f_X(k)$ the kth smallest positive integer not in X, and let

$$X * Y = X \cup \{f_X(y) \colon y \in Y\}.$$

Let A be a set of $a > 0$ positive integers, and let B be a set of $b > 0$ positive integers. Prove that if $A * B = B * A$, then

$$\underbrace{A * (A * \cdots * (A * (A * A)) \ldots)}_{A \text{ appears } b \text{ times}} = \underbrace{B * (B * \cdots * (B * (B * B)) \ldots)}_{B \text{ appears } a \text{ times}}.$$

(U.S.A.)

Solution 1. For any function $g \colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ and any subset $X \subset \mathbb{Z}_{>0}$, we define $g(X) = \{g(x) \colon x \in X\}$. We have that the image of f_X is $f_X(\mathbb{Z}_{>0}) = \mathbb{Z}_{>0} \setminus X$. We now show a general lemma about the operation $*$, with the goal of showing that $*$ is associative.

Lemma 1. Let X and Y be finite sets of positive integers. The functions f_{X*Y} and $f_X \circ f_Y$ are equal.

Proof. We have

$$f_{X*Y}(\mathbb{Z}_{>0}) = \mathbb{Z}_{>0} \setminus (X*Y) = (\mathbb{Z}_{>0} \setminus X) \setminus f_X(Y) = f_X(\mathbb{Z}_{>0}) \setminus f_X(Y) = f_X(\mathbb{Z}_{>0} \setminus Y) = f_X(f_Y(\mathbb{Z}_{>0})).$$

Thus, the functions f_{X*Y} and $f_X \circ f_Y$ are strictly increasing functions with the same range. Because a strictly function is uniquely defined by its range, we have $f_{X*Y} = f_X \circ f_Y$. □

Lemma 1 implies that $*$ is associative, in the sense that $(A * B) * C = A * (B * C)$ for any finite sets A, B, and C of positive integers. We prove the associativity by noting

$$\mathbb{Z}_{>0} \setminus ((A * B) * C) = f_{(A*B)*C}(\mathbb{Z}_{>0}) = f_{A*B}(f_C(\mathbb{Z}_{>0})) = f_A(f_B(f_C(\mathbb{Z}_{>0})))$$

$$= f_A(f_{B*C}(\mathbb{Z}_{>0}) = f_{A*(B*C)}(\mathbb{Z}_{>0}) = \mathbb{Z}_{>0} \setminus (A * (B * C)).$$

In light of the associativity of $*$, we may drop the parentheses when we write expressions like $A * (B * C)$. We also introduce the notation

$$X^{*k} = \underbrace{X * (X * \cdots * (X * (X * X)) \ldots)}_{X \text{ appears } k \text{ times}}.$$

Our goal is then to show that $A * B = B * A$ implies $A^{*b} = B^{*a}$. We will do so via the following general lemma.

Lemma 2. Suppose that X and Y are finite sets of positive integers satisfying $X * Y = Y * X$ and $|X| = |Y|$. Then, we must have $X = Y$.

Proof. Assume that X and Y are not equal. Let s be the largest number in exactly one of X and Y. Without loss of generality, say that $s \in X \setminus Y$. The number $f_X(s)$ counts the sth number not in X, which implies that

$$f_X(s) = s + \left|X \cap \{1, 2, \ldots, f_X(s)\}\right|. \tag{1}$$

Since $f_X(s) \geqslant s$, we have that

$$\left\{f_X(s) + 1, f_X(s) + 2, \ldots\right\} \cap X = \left\{f_X(s) + 1, f_X(s) + 2, \ldots\right\} \cap Y,$$

which, together with the assumption that $|X| = |Y|$, gives

$$\left|X \cap \{1, 2, \ldots, f_X(s)\}\right| = \left|Y \cap \{1, 2, \ldots, f_X(s)\}\right|. \tag{2}$$

Now consider the equation

$$t - \big|Y \cap \{1, 2, \ldots, t\}\big| = s.$$

This equation is satisfied only when $t \in \big[f_Y(s), f_Y(s+1)\big)$, because the left hand side counts the number of elements up to t that are not in Y. We have that the value $t = f_X(s)$ satisfies the above equation because of (1) and (2). Furthermore, since $f_X(s) \notin X$ and $f_X(s) \geqslant s$, we have that $f_X(s) \notin Y$ due to the maximality of s. Thus, by the above discussion, we must have $f_X(s) = f_Y(s)$.

Finally, we arrive at a contradiction. The value $f_X(s)$ is neither in X nor in $f_X(Y)$, because s is not in Y by assumption. Thus, $f_X(s) \notin X * Y$. However, since $s \in X$, we have $f_Y(s) \in Y * X$, a contradiction. □

We are now ready to finish the proof. Note first of all that $|A^{*b}| = ab = |B^{*a}|$. Moreover, since $A * B = B * A$, and $*$ is associative, it follows that $A^{*b} * B^{*a} = B^{*a} * A^{*b}$. Thus, by Lemma 2, we have $A^{*b} = B^{*a}$, as desired.

Comment 1. Taking $A = X^{*k}$ and $B = X^{*l}$ generates many non-trivial examples where $A*B = B*A$. There are also other examples not of this form. For example, if $A = \{1, 2, 4\}$ and $B = \{1, 3\}$, then $A * B = \{1, 2, 3, 4, 6\} = B * A$.

Solution 2. We will use Lemma 1 from Solution 1. Additionally, let X^{*k} be defined as in Solution 1. If X and Y are finite sets, then

$$f_X = f_Y \iff f_X(\mathbb{Z}_{>0}) = f_Y(\mathbb{Z}_{>0}) \iff (\mathbb{Z}_{>0} \setminus X) = (\mathbb{Z}_{>0} \setminus Y) \iff X = Y, \tag{3}$$

where the first equivalence is because f_X and f_Y are strictly increasing functions, and the second equivalence is because $f_X(\mathbb{Z}_{>0}) = \mathbb{Z}_{>0} \setminus X$ and $f_Y(\mathbb{Z}_{>0}) = \mathbb{Z}_{>0} \setminus Y$.

Denote $g = f_A$ and $h = f_B$. The given relation $A * B = B * A$ is equivalent to $f_{A*B} = f_{B*A}$ because of (3), and by Lemma 1 of the first solution, this is equivalent to $g \circ h = h \circ g$. Similarly, the required relation $A^{*b} = B^{*a}$ is equivalent to $g^b = h^a$. We will show that

$$g^b(n) = h^a(n) \tag{4}$$

for all $n \in \mathbb{Z}_{>0}$, which suffices to solve the problem.

To start, we claim that (4) holds for all sufficiently large n. Indeed, let p and q be the maximal elements of A and B, respectively; we may assume that $p \geqslant q$. Then, for every $n \geqslant p$ we have $g(n) = n + a$ and $h(n) = n + b$, whence $g^b(n) = n + ab = h^a(n)$, as was claimed.

In view of this claim, if (4) is not identically true, then there exists a maximal s with $g^b(s) \neq h^a(s)$. Without loss of generality, we may assume that $g(s) \neq s$, for if we had $g(s) = h(s) = s$, then s would satisfy (4). As g is increasing, we then have $g(s) > s$, so (4) holds for $n = g(s)$. But then we have

$$g(g^b(s)) = g^{b+1}(s) = g^b(n) = h^a(n) = h^a(g(s)) = g(h^a(s)),$$

where the last equality holds in view of $g \circ h = h \circ g$. By the injectivity of g, the above equality yields $g^b(s) = h^a(s)$, which contradicts the choice of s. Thus, we have proved that (4) is identically true on $\mathbb{Z}_{>0}$, as desired.

Comment 2. We present another proof of Lemma 2 of the first solution.

Let $x = |X| = |Y|$. Say that u is the smallest number in X and v is the smallest number in Y; assume without loss of generality that $u \leqslant v$.

Let T be any finite set of positive integers, and define $t = |T|$. Enumerate the elements of X as $x_1 < x_2 < \cdots < x_n$. Define $S_m = f_{(T*X^{*(m-1)})}(X)$, and enumerate its elements $s_{m,1} < s_{m,2} < \cdots < s_{m,n}$. Note that the S_m are pairwise disjoint; indeed, if we have $m < m'$, then

$$S_m \subset T * X^{*m} \subset T * X^{*(m'-1)} \quad \text{and} \quad S_{m'} = (T * X^{*m'}) \setminus (T * X^{*(m'-1)})$$

We claim the following statement, which essentially says that the S_m are eventually linear translates of each other:

Claim. For every i, there exists some m_i and c_i such that for all $m > m_i$, we have that $s_{m,i} = t+mn-c_i$. Furthermore, the c_i do not depend on the choice of T.

First, we show that this claim implies Lemma 2. We may choose $T = X$ and $T = Y$. Then, there is some m' such that for all $m \geqslant m'$, we have

$$f_{X^{*m}}(X) = f_{(Y*X^{*(m-1)})}(X). \tag{5}$$

Because u is the minimum element of X, v is the minimum element of Y, and $u \leqslant v$, we have that

$$\left(\bigcup_{m=m'}^{\infty} f_{X^{*m}}(X)\right) \cup X^{*m'} = \left(\bigcup_{m=m'}^{\infty} f_{(Y*X^{*(m-1)})}(X)\right) \cup \left(Y * X^{*(m'-1)}\right) = \{u, u+1, \dots\},$$

and in both the first and second expressions, the unions are of pairwise distinct sets. By (5), we obtain $X^{*m'} = Y * X^{*(m'-1)}$. Now, because X and Y commute, we get $X^{*m'} = X^{*(m'-1)} * Y$, and so $X = Y$.

We now prove the claim.

Proof of the claim. We induct downwards on i, first proving the statement for $i = n$, and so on.

Assume that m is chosen so that all elements of S_m are greater than all elements of T (which is possible because T is finite). For $i = n$, we have that $s_{m,n} > s_{k,n}$ for every $k < m$. Thus, all $(m-1)n$ numbers of the form $s_{k,u}$ for $k < m$ and $1 \leqslant u \leqslant n$ are less than $s_{m,n}$. We then have that $s_{m,n}$ is the $((m-1)n+x_n)^{th}$ number not in T, which is equal to $t+(m-1)n+x_n$. So we may choose $c_n = x_n - n$, which does not depend on T, which proves the base case for the induction.

For $i < n$, we have again that all elements $s_{m,j}$ for $j < i$ and $s_{p,i}$ for $p < m$ are less than $s_{m,i}$, so $s_{m,i}$ is the $((m-1)i + x_i)^{th}$ element not in T or of the form $s_{p,j}$ for $j > i$ and $p < m$. But by the inductive hypothesis, each of the sequences $s_{p,j}$ is eventually periodic with period n, and thus the sequence $s_{m,i}$ such must be as well. Since each of the sequences $s_{p,j} - t$ with $j > i$ eventually do not depend on T, the sequence $s_{m,i} - t$ eventually does not depend on T either, so the inductive step is complete. This proves the claim and thus Lemma 2. □

C8. Let n be a given positive integer. In the Cartesian plane, each lattice point with nonnegative coordinates initially contains a butterfly, and there are no other butterflies. The *neighborhood* of a lattice point c consists of all lattice points within the axis-aligned $(2n+1)\times(2n+1)$ square centered at c, apart from c itself. We call a butterfly *lonely*, *crowded*, or *comfortable*, depending on whether the number of butterflies in its neighborhood N is respectively less than, greater than, or equal to half of the number of lattice points in N.

Every minute, all lonely butterflies fly away simultaneously. This process goes on for as long as there are any lonely butterflies. Assuming that the process eventually stops, determine the number of comfortable butterflies at the final state.

(Bulgaria)

Answer: n^2+1.

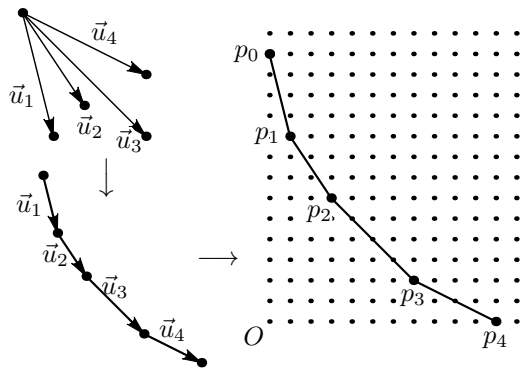
Solution. We always identify a butterfly with the lattice point it is situated at. For two points p and q, we write $p\geqslant q$ if each coordinate of p is at least the corresponding coordinate of q. Let O be the origin, and let $\mathcal{Q}$ be the set of initially occupied points, i.e., of all lattice points with nonnegative coordinates. Let $\mathcal{R}_{\mathrm{H}}=\{(x,0)\colon x\geqslant 0\}$ and $\mathcal{R}_{\mathrm{V}}=\{(0,y)\colon y\geqslant 0\}$ be the sets of the lattice points lying on the horizontal and vertical boundary rays of $\mathcal{Q}$. Denote by $N(a)$ the neighborhood of a lattice point a.

1. Initial observations. We call a set of lattice points *up-right closed* if its points stay in the set after being shifted by any lattice vector (i,j) with $i,j\geqslant 0$. Whenever the butterflies form a up-right closed set $\mathcal{S}$, we have $|N(p)\cap\mathcal{S}|\geqslant|N(q)\cap\mathcal{S}|$ for any two points $p,q\in\mathcal{S}$ with $p\geqslant q$. So, since $\mathcal{Q}$ is up-right closed, the set of butterflies at any moment also preserves this property. We assume all forthcoming sets of lattice points to be up-right closed.

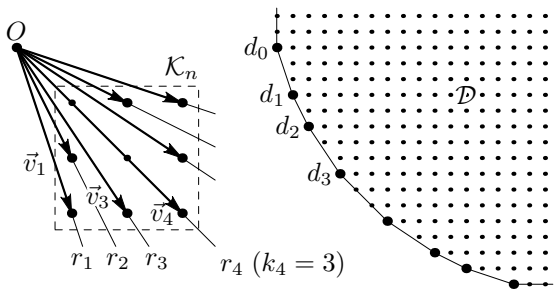
When speaking of some set $\mathcal{S}$ of lattice points, we call its points *lonely*, *comfortable*, or *crowded* with respect to this set (i.e., as if the butterflies were exactly at all points of $\mathcal{S}$). We call a set $\mathcal{S}\subset\mathcal{Q}$ *stable* if it contains no lonely points. In what follows, we are interested only in those stable sets whose complements in $\mathcal{Q}$ are finite, because one can easily see that only a finite number of butterflies can fly away on each minute.

If the initial set $\mathcal{Q}$ of butterflies contains some stable set $\mathcal{S}$, then, clearly no butterfly of this set will fly away. On the other hand, the set $\mathcal{F}$ of all butterflies in the end of the process is stable. This means that $\mathcal{F}$ is the largest (with respect to inclusion) stable set within $\mathcal{Q}$, and we are about to describe this set.

2. A description of a final set. The following notion will be useful. Let $\mathcal{U}=\{\vec{u}_1,\vec{u}_2,\ldots,\vec{u}_d\}$ be a set of d pairwise non-parallel lattice vectors, each having a positive x- and a negative y-coordinate. Assume that they are numbered in increasing order according to slope. We now define a *$\mathcal{U}$-curve* to be the broken line $p_0p_1\ldots p_d$ such that $p_0\in\mathcal{R}_{\mathrm{V}}$, $p_d\in\mathcal{R}_{\mathrm{H}}$, and $\overrightarrow{p_{i-1}p_i}=\vec{u}_i$ for all $i=1,2,\ldots,m$ (see the Figure below to the left).



Construction of $\mathcal{U}$-curve



Construction of $\mathcal{D}$

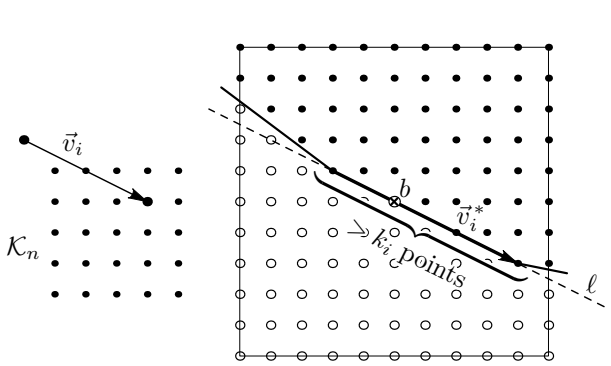
Now, let $\mathcal{K}_n = \{(i,j)\colon 1 \leqslant i \leqslant n,\ -n \leqslant j \leqslant -1\}$. Consider all the rays emerging at O and passing through a point from $\mathcal{K}_n$; number them as $r_1, \dots, r_m$ in increasing order according to slope. Let A_i be the farthest from O lattice point in $r_i \cap \mathcal{K}_n$, set $k_i = |r_i \cap \mathcal{K}_n|$, let $\vec{v}_i = \overrightarrow{OA_i}$, and finally denote $\mathcal{V} = \{\vec{v}_i\colon 1 \leqslant i \leqslant m\}$; see the Figure above to the right. We will concentrate on the $\mathcal{V}$-curve $d_0 d_1 \dots d_m$; let $\mathcal{D}$ be the set of all lattice points p such that $p \geqslant p'$ for some (not necessarily lattice) point p' on the $\mathcal{V}$-curve. In fact, we will show that $\mathcal{D} = \mathcal{F}$.

Clearly, the $\mathcal{V}$-curve is symmetric in the line $y = x$. Denote by D the convex hull of $\mathcal{D}$.

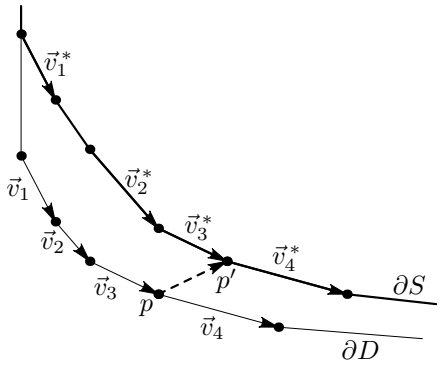
3. We prove that the set $\mathcal{D}$ contains all stable sets. Let $\mathcal{S} \subset \mathcal{Q}$ be a stable set (recall that it is assumed to be up-right closed and to have a finite complement in $\mathcal{Q}$). Denote by S its convex hull; clearly, the vertices of S are lattice points. The boundary of S consists of two rays (horizontal and vertical ones) along with some $\mathcal{V}_*$-curve for some set of lattice vectors $\mathcal{V}_*$.

Claim 1. For every $\vec{v}_i \in \mathcal{V}$, there is a $\vec{v}_i^{\,*} \in \mathcal{V}_*$ co-directed with $\vec{v}$ with $|\vec{v}_i^{\,*}| \geqslant |\vec{v}|$.

Proof. Let ℓ be the supporting line of S parallel to $\vec{v}_i$ (i.e., ℓ contains some point of S, and the set S lies on one side of ℓ). Take any point $b \in \ell \cap \mathcal{S}$ and consider $N(b)$. The line ℓ splits the set $N(b) \setminus \ell$ into two congruent parts, one having an empty intersection with $\mathcal{S}$. Hence, in order for b not to be lonely, at least half of the set $\ell \cap N(b)$ (which contains $2k_i$ points) should lie in S. Thus, the boundary of S contains a segment $\ell \cap S$ with at least $k_i + 1$ lattice points (including b) on it; this segment corresponds to the required vector $\vec{v}_i^{\,*} \in \mathcal{V}_*$. □



Proof of Claim 1



Proof of Claim 2

Claim 2. Each stable set $\mathcal{S} \subseteq \mathcal{Q}$ lies in $\mathcal{D}$.

Proof. To show this, it suffices to prove that the $\mathcal{V}_*$-curve lies in D, i.e., that all its vertices do so. Let p' be an arbitrary vertex of the $\mathcal{V}_*$-curve; p' partitions this curve into two parts, $\mathcal{X}$ (being down-right of p) and $\mathcal{Y}$ (being up-left of p). The set $\mathcal{V}$ is split now into two parts: $\mathcal{V}_\mathcal{X}$ consisting of those $\vec{v}_i \in \mathcal{V}$ for which $\vec{v}_i^{\,*}$ corresponds to a segment in $\mathcal{X}$, and a similar part $\mathcal{V}_\mathcal{Y}$. Notice that the $\mathcal{V}$-curve consists of several segments corresponding to $\mathcal{V}_\mathcal{X}$, followed by those corresponding to $\mathcal{V}_\mathcal{Y}$. Hence there is a vertex p of the $\mathcal{V}$-curve separating $\mathcal{V}_\mathcal{X}$ from $\mathcal{V}_\mathcal{Y}$. Claim 1 now yields that $p' \geqslant p$, so $p' \in \mathcal{D}$, as required. □

Claim 2 implies that the final set $\mathcal{F}$ is contained in $\mathcal{D}$.

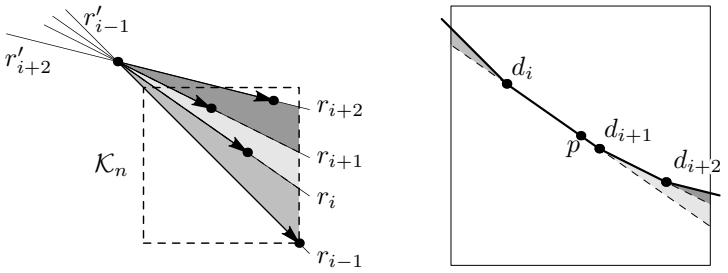
4. $\mathcal{D}$ is stable, and its comfortable points are known. Recall the definitions of r_i; let r_i' be the ray complementary to r_i. By our definitions, the set $N(O)$ contains no points between the rays r_i and r_{i+1}, as well as between r_i' and r_{i+1}'.

Claim 3. In the set $\mathcal{D}$, all lattice points of the $\mathcal{V}$-curve are comfortable.

Proof. Let p be any lattice point of the $\mathcal{V}$-curve, belonging to some segment $d_i d_{i+1}$. Draw the line ℓ containing this segment. Then $\ell \cap \mathcal{D}$ contains exactly $k_i + 1$ lattice points, all of which lie in $N(p)$ except for p. Thus, exactly half of the points in $N(p) \cap \ell$ lie in $\mathcal{D}$. It remains to show that all points of $N(p)$ above ℓ lie in $\mathcal{D}$ (recall that all the points below ℓ lack this property).

Notice that each vector in $\mathcal{V}$ has one coordinate greater than $n/2$; thus the neighborhood of p contains parts of at most two segments of the $\mathcal{V}$-curve succeeding d_id_{i+1}, as well as at most two of those preceding it.

The angles formed by these consecutive segments are obtained from those formed by r_j and r'_{j-1} (with $i-1 \leqslant j \leqslant i+2$) by shifts; see the Figure below. All the points in $N(p)$ above ℓ which could lie outside $\mathcal{D}$ lie in shifted angles between r_j, r_{j+1} or r'_j, r'_{j-1}. But those angles, restricted to $N(p)$, have no lattice points due to the above remark. The claim is proved. □

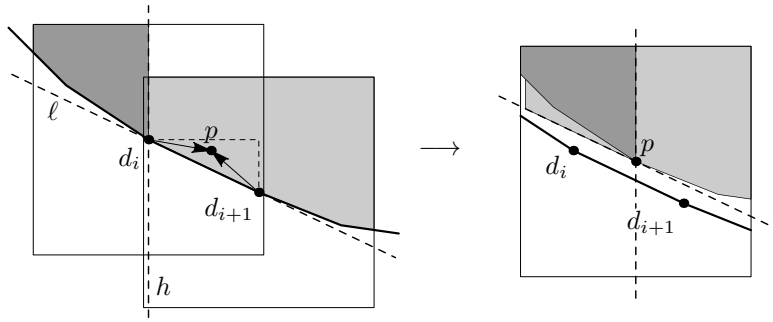


Proof of Claim 3

Claim 4. All the points of $\mathcal{D}$ which are not on the boundary of D are crowded.

Proof. Let $p \in \mathcal{D}$ be such a point. If it is to the up-right of some point p' on the curve, then the claim is easy: the shift of $N(p') \cap \mathcal{D}$ by $\overrightarrow{p'p}$ is still in $\mathcal{D}$, and $N(p)$ contains at least one more point of $\mathcal{D}$ — either below or to the left of p. So, we may assume that p lies in a right triangle constructed on some hypothenuse d_id_{i+1}. Notice here that $d_i, d_{i+1} \in N(p)$.

Draw a line $\ell \parallel d_id_{i+1}$ through p, and draw a vertical line h through d_i; see Figure below. Let $\mathcal{D}_{\mathrm{L}}$ and $\mathcal{D}_{\mathrm{R}}$ be the parts of $\mathcal{D}$ lying to the left and to the right of h, respectively (points of $\mathcal{D} \cap h$ lie in both parts).



Proof of Claim 4

Notice that the vectors $\overrightarrow{d_ip}$, $\overrightarrow{d_{i+1}d_{i+2}}$, $\overrightarrow{d_id_{i+1}}$, $\overrightarrow{d_{i-1}d_i}$, and $\overrightarrow{pd_{i+1}}$ are arranged in non-increasing order by slope. This means that $\mathcal{D}_{\mathrm{L}}$ shifted by $\overrightarrow{d_ip}$ still lies in $\mathcal{D}$, as well as $\mathcal{D}_{\mathrm{R}}$ shifted by $\overrightarrow{d_{i+1}p}$. As we have seen in the proof of Claim 3, these two shifts cover all points of $N(p)$ above ℓ, along with those on ℓ to the left of p. Since $N(p)$ contains also d_i and d_{i+1}, the point p is crowded. □

Thus, we have proved that $\mathcal{D} = \mathcal{F}$, and have shown that the lattice points on the $\mathcal{V}$-curve are exactly the comfortable points of $\mathcal{D}$. It remains to find their number.

Recall the definition of $\mathcal{K}_n$ (see Figure on the first page of the solution). Each segment d_id_{i+1} contains k_i lattice points different from d_i. Taken over all i, these points exhaust all the lattice points in the $\mathcal{V}$-curve, except for d_1, and thus the number of lattice points on the $\mathcal{V}$-curve is $1 + \sum_{i=1}^{m} k_i$. On the other hand, $\sum_{i=1}^{m} k_i$ is just the number of points in $\mathcal{K}_n$, so it equals n^2. Hence the answer to the problem is $n^2 + 1$.

Comment 1. The assumption that the process eventually stops is unnecessary for the problem, as one can see that, in fact, the process stops for every $n \geqslant 1$. Indeed, the proof of Claims 3 and 4 do not rely essentially on this assumption, and they together yield that the set $\mathcal{D}$ is stable. So, only butterflies that are not in $\mathcal{D}$ may fly away, and this takes only a finite time.

This assumption has been inserted into the problem statement in order to avoid several technical details regarding finiteness issues. It may also simplify several other arguments.

Comment 2. The description of the final set $\mathcal{F}(= \mathcal{D})$ seems to be crucial for the solution; the Problem Selection Committee is not aware of any solution that completely avoids such a description.

On the other hand, after the set $\mathcal{D}$ has been defined, the further steps may be performed in several ways. For example, in order to prove that all butterflies outside $\mathcal{D}$ will fly away, one may argue as follows. (Here we will also make use of the assumption that the process eventually stops.)

First of all, notice that the process can be modified in the following manner: *Each minute, exactly one of the lonely butterflies flies away, until there are no more lonely butterflies.* The modified process necessarily stops at the same state as the initial one. Indeed, one may observe, as in solution above, that the (unique) largest stable set is still the final set for the modified process.

Thus, in order to prove our claim, it suffices to indicate an order in which the butterflies should fly away in the new process; if we are able to exhaust the whole set $\mathcal{Q} \setminus \mathcal{D}$, we are done.

Let $\mathcal{C}_0 = d_0 d_1 \ldots d_m$ be the $\mathcal{V}$-curve. Take its copy $\mathcal{C}$ and shift it downwards so that d_0 comes to some point below the origin O. Now we start moving $\mathcal{C}$ upwards continuously, until it comes back to its initial position $\mathcal{C}_0$. At each moment when $\mathcal{C}$ meets some lattice points, we convince all the butterflies at those points to fly away in a certain order. We will now show that we always have enough arguments for butterflies to do so, which will finish our argument for the claim..

Let $\mathcal{C}' = d'_0 d'_1 \ldots d'_m$ be a position of $\mathcal{C}$ when it meets some butterflies. We assume that all butterflies under this current position of $\mathcal{C}$ were already convinced enough and flied away. Consider the lowest butterfly b on $\mathcal{C}'$. Let $d'_i d'_{i+1}$ be the segment it lies on; we choose i so that $b \neq d'_{i+1}$ (this is possible because $\mathcal{C}$ as not yet reached $\mathcal{C}_0$).

Draw a line ℓ containing the segment $d'_i d'_{i+1}$. Then all the butterflies in $N(b)$ are situated on or above ℓ; moreover, those on ℓ all lie on the segment $d_i d_{i+1}$. But this segment now contains at most k_i butterflies (including b), since otherwise some butterfly had to occupy d'_{i+1} which is impossible by the choice of b. Thus, b is lonely and hence may be convinced to fly away.

After b has flied away, we switch to the lowest of the remaining butterflies on $\mathcal{C}'$, and so on.

Claims 3 and 4 also allow some different proofs which are not presented here.

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Geometry

G1. Let $ABCDE$ be a convex pentagon such that $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that the perpendicular line from E to BC and the line segments AC and BD are concurrent.

(Italy)

Solution 1. Throughout the solution, we refer to $\angle A$, $\angle B$, $\angle C$, $\angle D$, and $\angle E$ as internal angles of the pentagon $ABCDE$. Let the perpendicular bisectors of AC and BD, which pass respectively through B and C, meet at point I. Then $BD \perp CI$ and, similarly, $AC \perp BI$. Hence AC and BD meet at the orthocenter H of the triangle BIC, and $IH \perp BC$. It remains to prove that E lies on the line IH or, equivalently, $EI \perp BC$.

Lines IB and IC bisect $\angle B$ and $\angle C$, respectively. Since $IA = IC$, $IB = ID$, and $AB = BC = CD$, the triangles IAB, ICB and ICD are congruent. Hence $\angle IAB = \angle ICB = \angle C/2 = \angle A/2$, so the line IA bisects $\angle A$. Similarly, the line ID bisects $\angle D$. Finally, the line IE bisects $\angle E$ because I lies on all the other four internal bisectors of the angles of the pentagon.

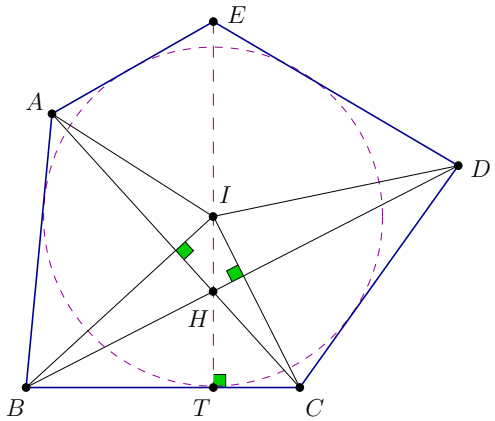
The sum of the internal angles in a pentagon is 540°, so

$$\angle E = 540^\circ - 2\angle A + 2\angle B.$$

In quadrilateral $ABIE$,

$$\begin{aligned}\angle BIE &= 360^\circ - \angle EAB - \angle ABI - \angle AEI = 360^\circ - \angle A - \frac{1}{2}\angle B - \frac{1}{2}\angle E \\ &= 360^\circ - \angle A - \frac{1}{2}\angle B - (270^\circ - \angle A - \angle B) \\ &= 90^\circ + \frac{1}{2}\angle B = 90^\circ + \angle IBC,\end{aligned}$$

which means that $EI \perp BC$, completing the proof.



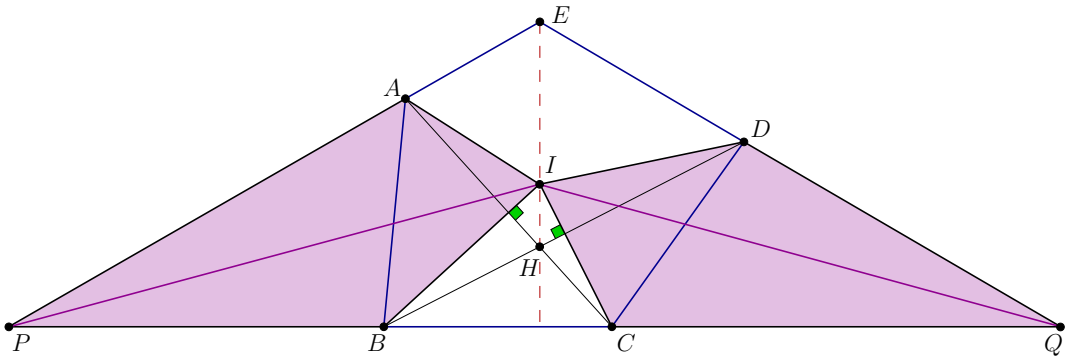
Solution 2. We present another proof of the fact that E lies on line IH. Since all five internal bisectors of $ABCDE$ meet at I, this pentagon has an inscribed circle with center I. Let this circle touch side BC at T.

Applying Brianchon's theorem to the (degenerate) hexagon $ABTCDE$ we conclude that AC, BD and ET are concurrent, so point E also lies on line IHT, completing the proof.

Solution 3. We present yet another proof that $EI \perp BC$. In pentagon $ABCDE$, $\angle E < 180^\circ \iff \angle A + \angle B + \angle C + \angle D > 360^\circ$. Then $\angle A + \angle B = \angle C + \angle D > 180^\circ$, so rays EA and CB meet at a point P, and rays BC and ED meet at a point Q. Now,

$$\angle PBA = 180^\circ - \angle B = 180^\circ - \angle D = \angle QDC$$

and, similarly, $\angle PAB = \angle QCD$. Since $AB = CD$, the triangles PAB and QCD are congruent with the same orientation. Moreover, PQE is isosceles with $EP = EQ$.



In Solution 1 we have proved that triangles IAB and ICD are also congruent with the same orientation. Then we conclude that quadrilaterals $PBIA$ and $QDIC$ are congruent, which implies $IP = IQ$. Then EI is the perpendicular bisector of PQ and, therefore, $EI \perp PQ \iff EI \perp BC$.

Comment. Even though all three solutions used the point I, there are solutions that do not need it. We present an outline of such a solution: if J is the incenter of $\triangle QCD$ (with P and Q as defined in Solution 3), then a simple angle chasing shows that triangles CJD and BHC are congruent. Then if S is the projection of J onto side CD and T is the orthogonal projection of H onto side BC, one can verify that

$$QT = QC + CT = QC + DS = QC + \frac{CD + DQ - QC}{2} = \frac{PB + BC + QC}{2} = \frac{PQ}{2},$$

so T is the midpoint of PQ, and E, H and T all lie on the perpendicular bisector of PQ.

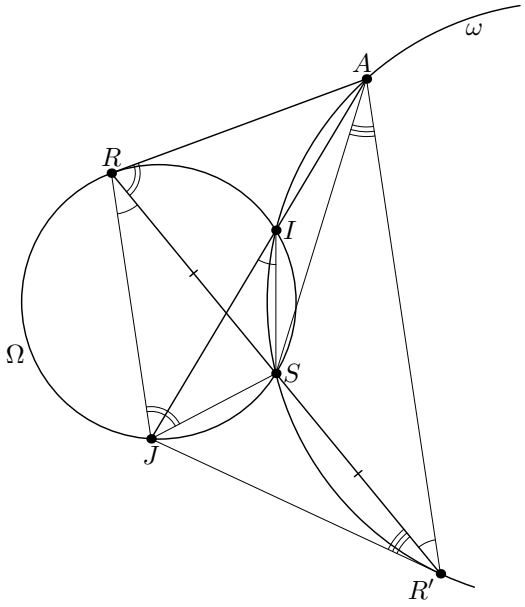
G2. Let R and S be distinct points on circle Ω, and let t denote the tangent line to Ω at R. Point R' is the reflection of R with respect to S. A point I is chosen on the smaller arc RS of Ω so that the circumcircle Γ of triangle ISR' intersects t at two different points. Denote by A the common point of Γ and t that is closest to R. Line AI meets Ω again at J. Show that JR' is tangent to Γ.

(Luxembourg)

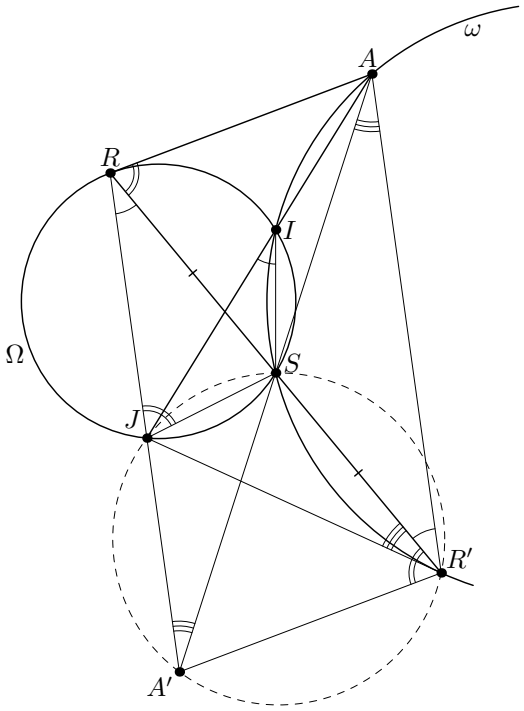
Solution 1. In the circles Ω and Γ we have $\angle JRS = \angle JIS = \angle AR'S$. On the other hand, since RA is tangent to Ω, we get $\angle SJR = \angle SRA$. So the triangles ARR' and SJR are similar, and

$$\frac{R'R}{RJ} = \frac{AR'}{SR} = \frac{AR'}{SR'}.$$

The last relation, together with $\angle AR'S = \angle JRR'$, yields $\triangle ASR' \sim \triangle R'JR$, hence $\angle SAR' = \angle RR'J$. It follows that JR' is tangent to Γ at R'.



Solution 1



Solution 2

Solution 2. As in Solution 1, we notice that $\angle JRS = \angle JIS = \angle AR'S$, so we have $RJ \parallel AR'$.

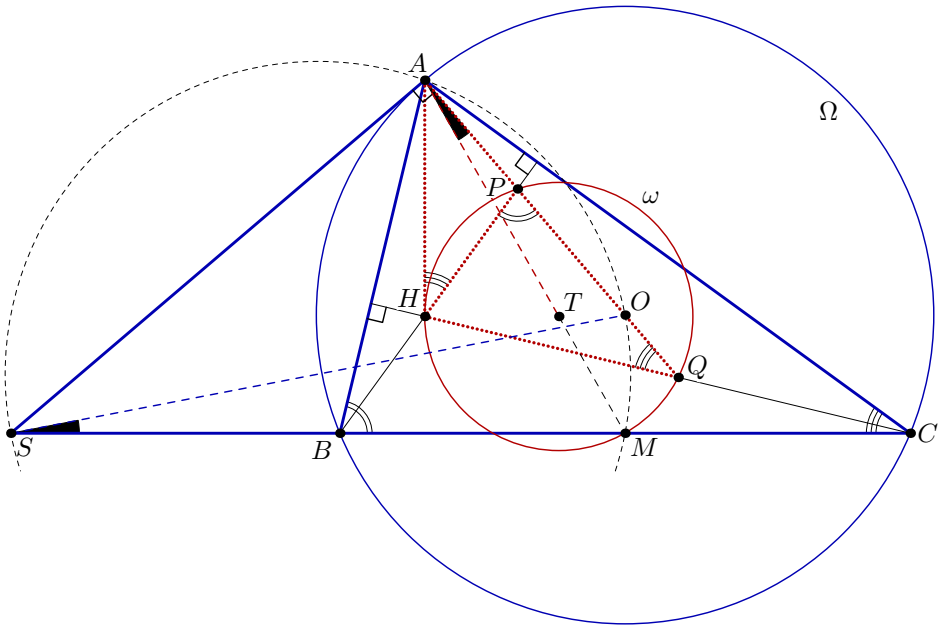
Let A' be the reflection of A about S; then $ARA'R'$ is a parallelogram with center S, and hence the point J lies on the line RA'.

From $\angle SR'A' = \angle SRA = \angle SJR$ we get that the points S, J, A', R' are concyclic. This proves that $\angle SR'J = \angle SA'J = \angle SA'R = \angle SAR'$, so JR' is tangent to Γ at R'.

G3. Let O be the circumcenter of an acute scalene triangle ABC. Line OA intersects the altitudes of ABC through B and C at P and Q, respectively. The altitudes meet at H. Prove that the circumcenter of triangle PQH lies on a median of triangle ABC.

(Ukraine)

Solution. Suppose, without loss of generality, that $AB < AC$. We have $\angle PQH = 90^\circ - \angle QAB = 90^\circ - \angle OAB = \frac{1}{2}\angle AOB = \angle ACB$, and similarly $\angle QPH = \angle ABC$. Thus triangles ABC and HPQ are similar. Let Ω and ω be the circumcircles of ABC and HPQ, respectively. Since $\angle AHP = 90^\circ - \angle HAC = \angle ACB = \angle HQP$, line AH is tangent to ω.



Let T be the center of ω and let lines AT and BC meet at M. We will take advantage of the similarity between ABC and HPQ and the fact that AH is tangent to ω at H, with A on line PQ. Consider the corresponding tangent AS to Ω, with $S \in BC$. Then S and A correspond to each other in $\triangle ABC \sim \triangle HPQ$, and therefore $\angle OSM = \angle OAT = \angle OAM$. Hence quadrilateral $SAOM$ is cyclic, and since the tangent line AS is perpendicular to AO, $\angle OMS = 180^\circ - \angle OAS = 90^\circ$. This means that M is the orthogonal projection of O onto BC, which is its midpoint. So T lies on median AM of triangle ABC.

G4. In triangle ABC, let ω be the excircle opposite A. Let D, E, and F be the points where ω is tangent to lines BC, CA, and AB, respectively. The circle AEF intersects line BC at P and Q. Let M be the midpoint of AD. Prove that the circle MPQ is tangent to ω.

(Denmark)

Solution 1. Denote by Ω the circle $AEFPQ$, and denote by γ the circle PQM. Let the line AD meet ω again at $T \neq D$. We will show that γ is tangent to ω at T.

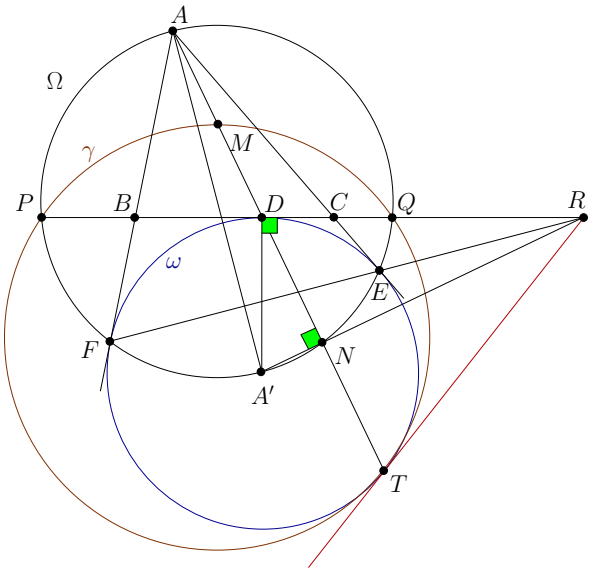
We first prove that points P, Q, M, T are concyclic. Let A' be the center of ω. Since $A'E \perp AE$ and $A'F \perp AF$, AA' is a diameter in Ω. Let N be the midpoint of DT; from $A'D = A'T$ we can see that $\angle A'NA = 90^\circ$ and therefore N also lies on the circle Ω. Now, from the power of D with respect to the circles γ and Ω we get

$$DP \cdot DQ = DA \cdot DN = 2DM \cdot \frac{DT}{2} = DM \cdot DT,$$

so P, Q, M, T are concyclic.

If $EF \parallel BC$, then ABC is isosceles and the problem is now immediate by symmetry. Otherwise, let the tangent line to ω at T meet line BC at point R. The tangent line segments RD and RT have the same length, so $A'R$ is the perpendicular bisector of DT; since $ND = NT$, N lies on this perpendicular bisector.

In right triangle $A'RD$, $RD^2 = RN \cdot RA' = RP \cdot RQ$, in which the last equality was obtained from the power of R with respect to Ω. Hence $RT^2 = RP \cdot RQ$, which implies that RT is also tangent to γ. Because RT is a common tangent to ω and γ, these two circles are tangent at T.



Solution 2. After proving that P, Q, M, T are concyclic, we finish the problem in a different fashion. We only consider the case in which EF and BC are not parallel. Let lines PQ and EF meet at point R. Since PQ and EF are radical axes of Ω, γ and ω, γ, respectively, R is the radical center of these three circles.

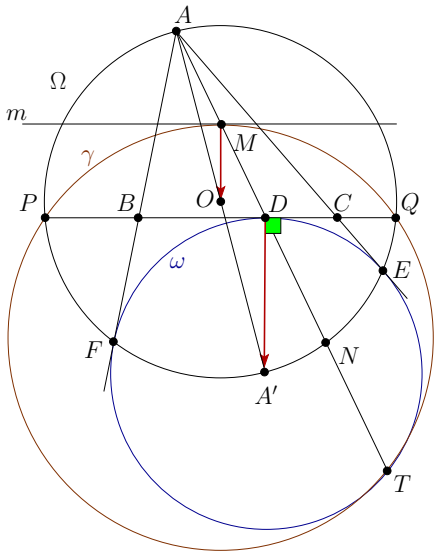
With respect to the circle ω, the line DR is the polar of D, and the line EF is the polar of A. So the pole of line ADT is $DR \cap EF = R$, and therefore RT is tangent to ω.

Finally, since T belongs to γ and ω and R is the radical center of γ, ω and Ω, line RT is the radical axis of γ and ω, and since it is tangent to ω, it is also tangent to γ. Because RT is a common tangent to ω and γ, these two circles are tangent at T.

Comment. In Solution 2 we defined the point R from Solution 1 in a different way.

Solution 3. We give an alternative proof that the circles are tangent at the common point T. Again, we start from the fact that P, Q, M, T are concyclic. Let point O be the midpoint of diameter AA'. Then MO is the midline of triangle ADA', so $MO \parallel A'D$. Since $A'D \perp PQ$, MO is perpendicular to PQ as well.

Looking at circle Ω, which has center O, $MO \perp PQ$ implies that MO is the perpendicular bisector of the chord PQ. Thus M is the midpoint of arc $\widehat{PQ}$ from γ, and the tangent line m to γ at M is parallel to PQ.



Consider the homothety with center T and ratio $\frac{TD}{TM}$. It takes D to M, and the line PQ to the line m. Since the circle that is tangent to a line at a given point and that goes through another given point is unique, this homothety also takes ω (tangent to PQ and going through T) to γ (tangent to m and going through T). We conclude that ω and γ are tangent at T.

G5. Let $ABCC_1B_1A_1$ be a convex hexagon such that $AB = BC$, and suppose that the line segments AA_1, BB_1, and CC_1 have the same perpendicular bisector. Let the diagonals AC_1 and A_1C meet at D, and denote by ω the circle ABC. Let ω intersect the circle A_1BC_1 again at $E \neq B$. Prove that the lines BB_1 and DE intersect on ω.

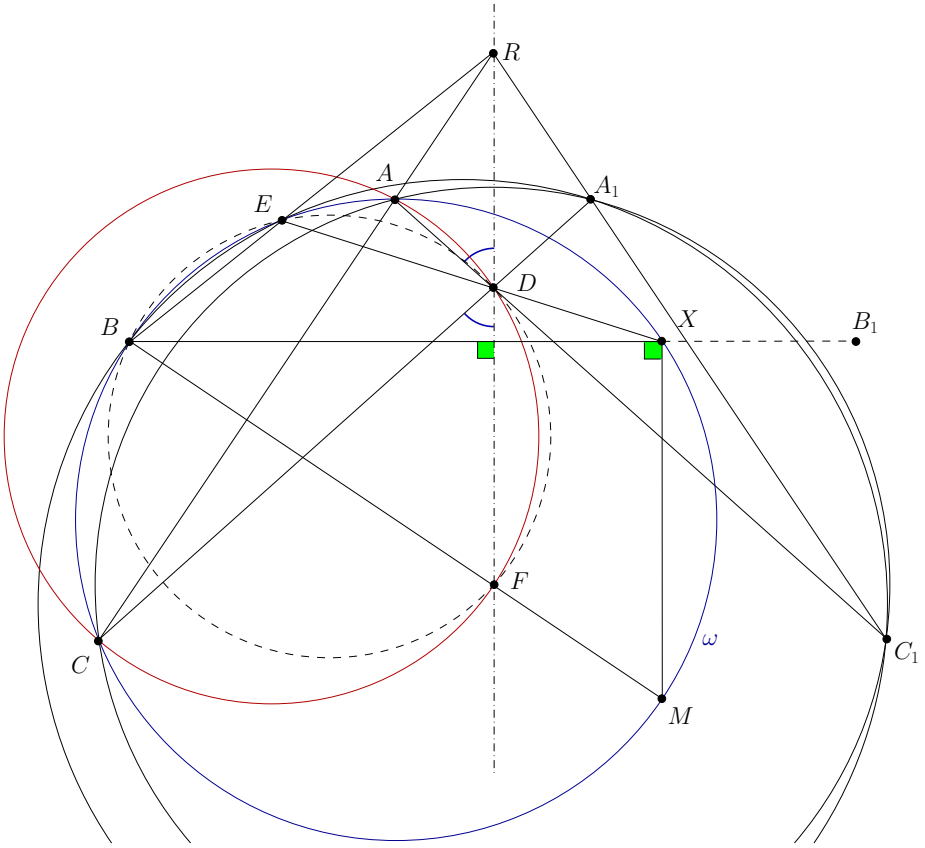
(Ukraine)

Solution 1. If $AA_1 = CC_1$, then the hexagon is symmetric about the line BB_1; in particular the circles ABC and A_1BC_1 are tangent to each other. So AA_1 and CC_1 must be different. Since the points A and A_1 can be interchanged with C and C_1, respectively, we may assume $AA_1 < CC_1$.

Let R be the radical center of the circles $AEBC$ and A_1EBC_1, and the circumcircle of the symmetric trapezoid ACC_1A_1; that is the common point of the pairwise radical axes AC, A_1C_1, and BE. By the symmetry of AC and A_1C_1, the point R lies on the common perpendicular bisector of AA_1 and CC_1, which is the external bisector of $\angle ADC$.

Let F be the second intersection of the line DR and the circle ACD. From the power of R with respect to the circles ω and $ACFD$ we have $RB \cdot RE = RA \cdot RC = RD \cdot DF$, so the points B, E, D and F are concyclic.

The line RDF is the external bisector of $\angle ADC$, so the point F bisects the arc $\overset{\frown}{CDA}$. By $AB = BC$, on circle ω, the point B is the midpoint of arc $\overset{\frown}{AEC}$; let M be the point diametrically opposite to B, that is the midpoint of the opposite arc $\overset{\frown}{CA}$ of ω. Notice that the points B, F and M lie on the perpendicular bisector of AC, so they are collinear.



Finally, let X be the second intersection point of ω and the line DE. Since BM is a diameter in ω, we have $\angle BXM = 90^\circ$. Moreover,

$$\angle EXM = 180^\circ - \angle MBE = 180^\circ - \angle FBE = \angle EDF,$$

so MX and FD are parallel. Since BX is perpendicular to MX and BB_1 is perpendicular to FD, this shows that X lies on line BB_1.

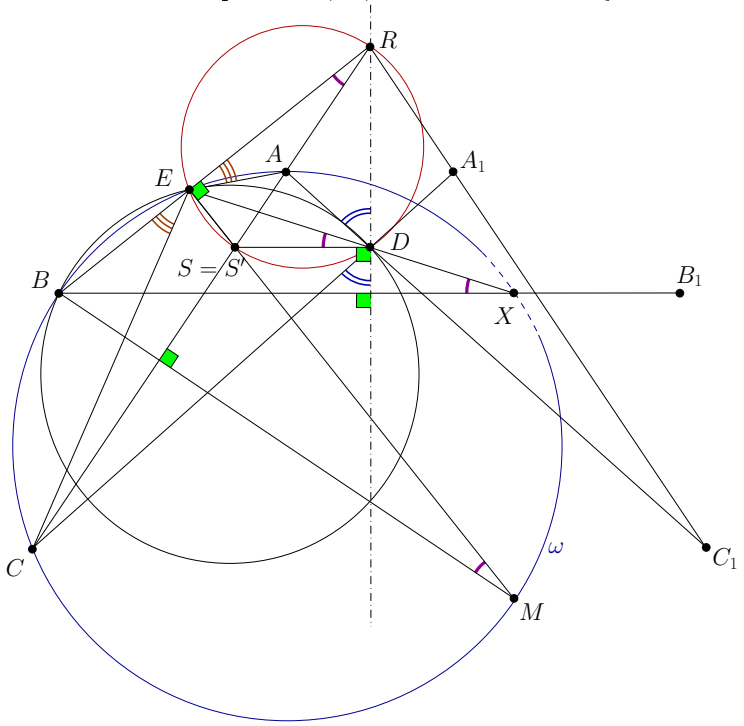
Solution 2. Define point M as the point opposite to B on circle ω, and point R as the intersection of lines AC, A_1C_1 and BE, and show that R lies on the external bisector of $\angle ADC$, like in the first solution.

Since B is the midpoint of the arc $\overset{\frown}{AEC}$, the line BER is the external bisector of $\angle CEA$. Now we show that the internal angle bisectors of $\angle ADC$ and $\angle CEA$ meet on the segment AC. Let the angle bisector of $\angle ADC$ meet AC at S, and let the angle bisector of $\angle CEA$, which is line EM, meet AC at S'. By applying the angle bisector theorem to both internal and external bisectors of $\angle ADC$ and $\angle CEA$,

$$AS : CS = AD : CD = AR : CR = AE : CE = AS' : CS',$$

so indeed $S = S'$.

By $\angle RDS = \angle SER = 90^\circ$ the points R, S, D and E are concyclic.



Now let the lines BB_1 and DE meet at point X. Notice that $\angle EXB = \angle EDS$ because both BB_1 and DS are perpendicular to the line DR, we have that $\angle EDS = \angle ERS$ in circle $SRDE$, and $\angle ERS = \angle EMB$ because $SR \perp BM$ and $ER \perp ME$. Therefore, $\angle EXB = \angle EMB$, so indeed, the point X lies on ω.

G6. Let $n \geqslant 3$ be an integer. Two regular n-gons $\mathcal{A}$ and $\mathcal{B}$ are given in the plane. Prove that the vertices of $\mathcal{A}$ that lie inside $\mathcal{B}$ or on its boundary are consecutive.

(That is, prove that there exists a line separating those vertices of $\mathcal{A}$ that lie inside $\mathcal{B}$ or on its boundary from the other vertices of $\mathcal{A}$.)

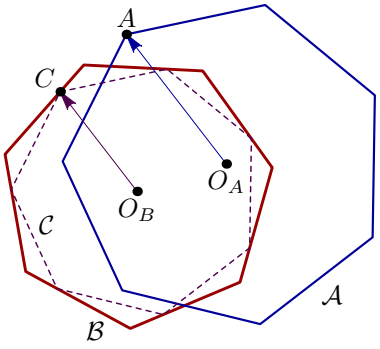
(Czech Republic)

Solution 1. In both solutions, by a *polygon* we always mean its interior together with its boundary.

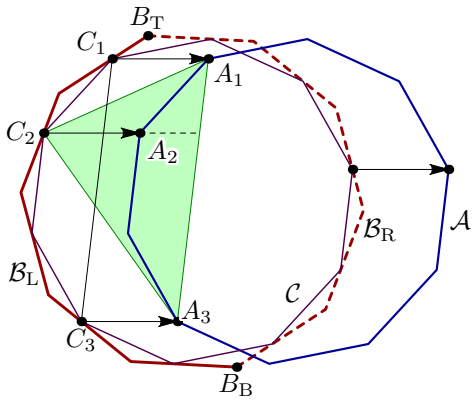
We start with finding a regular n-gon $\mathcal{C}$ which (i) is inscribed into $\mathcal{B}$ (that is, all vertices of $\mathcal{C}$ lie on the perimeter of $\mathcal{B}$); and (ii) is either a translation of $\mathcal{A}$, or a homothetic image of $\mathcal{A}$ with a positive factor.

Such a polygon may be constructed as follows. Let O_A and O_B be the centers of $\mathcal{A}$ and $\mathcal{B}$, respectively, and let A be an arbitrary vertex of $\mathcal{A}$. Let $\overrightarrow{O_BC}$ be the vector co-directional to $\overrightarrow{O_AA}$, with C lying on the perimeter of $\mathcal{B}$. The rotations of C around O_B by multiples of $2\pi/n$ form the required polygon. Indeed, it is regular, inscribed into $\mathcal{B}$ (due to the rotational symmetry of $\mathcal{B}$), and finally the translation/homothety mapping $\overrightarrow{O_AA}$ to $\overrightarrow{O_BC}$ maps $\mathcal{A}$ to $\mathcal{C}$.

Now we separate two cases.



Construction of $\mathcal{C}$



Case 1: Translation

Case 1: $\mathcal{C}$ is a translation of $\mathcal{A}$ by a vector $\vec{v}$.

Denote by t the translation transform by vector $\vec{v}$. We need to prove that the vertices of $\mathcal{C}$ which stay in $\mathcal{B}$ under t are consecutive. To visualize the argument, we refer the plane to Cartesian coordinates so that the x-axis is co-directional with $\vec{v}$. This way, the notions of right/left and top/bottom are also introduced, according to the x- and y-coordinates, respectively.

Let B_T and B_B be the top and the bottom vertices of $\mathcal{B}$ (if several vertices are extremal, we take the rightmost of them). They split the perimeter of $\mathcal{B}$ into the *right part* $\mathcal{B}_R$ and the *left part* $\mathcal{B}_L$ (the vertices B_T and B_B are assumed to lie in both parts); each part forms a connected subset of the perimeter of $\mathcal{B}$. So the vertices of $\mathcal{C}$ are also split into two parts $\mathcal{C}_L \subset \mathcal{B}_L$ and $\mathcal{C}_R \subset \mathcal{B}_R$, each of which consists of consecutive vertices.

Now, all the points in $\mathcal{B}_R$ (and hence in $\mathcal{C}_R$) move out from $\mathcal{B}$ under t, since they are the rightmost points of $\mathcal{B}$ on the corresponding horizontal lines. It remains to prove that the vertices of $\mathcal{C}_L$ which stay in $\mathcal{B}$ under t are consecutive.

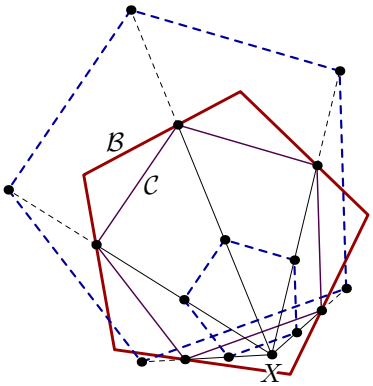
For this purpose, let C_1, C_2, and C_3 be three vertices in $\mathcal{C}_L$ such that C_2 is between C_1 and C_3, and $t(C_1)$ and $t(C_3)$ lie in $\mathcal{B}$; we need to prove that $t(C_2) \in \mathcal{B}$ as well. Let $A_i = t(C_i)$. The line through C_2 parallel to $\vec{v}$ crosses the segment C_1C_3 to the right of C_2; this means that this line crosses A_1A_3 to the right of A_2, so A_2 lies inside the triangle $A_1C_2A_3$ which is contained in $\mathcal{B}$. This yields the desired result.

Case 2: $\mathcal{C}$ is a homothetic image of $\mathcal{A}$ centered at X with factor $k > 0$.

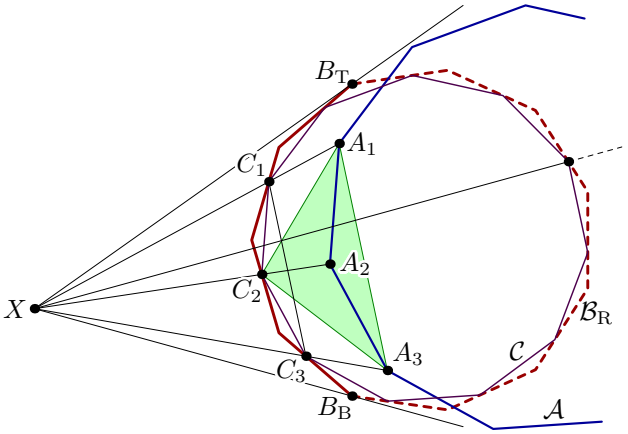
Denote by h the homothety mapping $\mathcal{C}$ to $\mathcal{A}$. We need now to prove that the vertices of $\mathcal{C}$ which stay in $\mathcal{B}$ after applying h are consecutive. If $X \in \mathcal{B}$, the claim is easy. Indeed, if $k < 1$, then the vertices of $\mathcal{A}$ lie on the segments of the form XC (C being a vertex of $\mathcal{C}$) which lie in $\mathcal{B}$. If $k > 1$, then the vertices of $\mathcal{A}$ lie on the extensions of such segments XC beyond C, and almost all these extensions lie outside $\mathcal{B}$. The exceptions may occur only in case when X lies on the boundary of $\mathcal{B}$, and they may cause one or two vertices of $\mathcal{A}$ stay on the boundary of $\mathcal{B}$. But even in this case those vertices are still consecutive.

So, from now on we assume that $X \notin \mathcal{B}$.

Now, there are two vertices B_{T} and $\mathcal{B}_{\mathrm{B}}$ of $\mathcal{B}$ such that $\mathcal{B}$ is contained in the angle $\angle B_{\mathrm{T}} X B_{\mathrm{B}}$; if there are several options, say, for B_{T}, then we choose the farthest one from X if $k > 1$, and the nearest one if $k < 1$. For the visualization purposes, we refer the plane to Cartesian coordinates so that the y-axis is co-directional with $\overrightarrow{B_{\mathrm{B}}B_{\mathrm{T}}}$, and X lies to the left of the line $B_{\mathrm{T}}B_{\mathrm{B}}$. Again, the perimeter of $\mathcal{B}$ is split by B_{T} and B_{B} into the *right part* $\mathcal{B}_{\mathrm{R}}$ and the *left part* $\mathcal{B}_{\mathrm{L}}$, and the set of vertices of $\mathcal{C}$ is split into two subsets $\mathcal{C}_{\mathrm{R}} \subset \mathcal{B}_{\mathrm{R}}$ and $\mathcal{C}_{\mathrm{L}} \subset \mathcal{B}_{\mathrm{L}}$.



Case 2, X inside $\mathcal{B}$

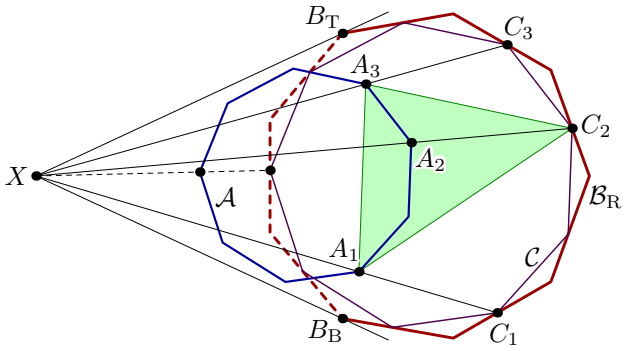


Subcase 2.1: $k > 1$

Subcase 2.1: $k > 1$.

In this subcase, all points from $\mathcal{B}_{\mathrm{R}}$ (and hence from $\mathcal{C}_{\mathrm{R}}$) move out from $\mathcal{B}$ under h, because they are the farthest points of $\mathcal{B}$ on the corresponding rays emanated from X. It remains to prove that the vertices of $\mathcal{C}_{\mathrm{L}}$ which stay in $\mathcal{B}$ under h are consecutive.

Again, let C_1, C_2, C_3 be three vertices in $\mathcal{C}_{\mathrm{L}}$ such that C_2 is between C_1 and C_3, and $h(C_1)$ and $h(C_3)$ lie in $\mathcal{B}$. Let $A_i = h(C_i)$. Then the ray XC_2 crosses the segment C_1C_3 beyond C_2, so this ray crosses A_1A_3 beyond A_2; this implies that A_2 lies in the triangle $A_1C_2A_3$, which is contained in $\mathcal{B}$.



Subcase 2.2: $k < 1$

Subcase 2.2: $k < 1$.

This case is completely similar to the previous one. All points from $\mathcal{B}_{\mathrm{L}}$ (and hence from $\mathcal{C}_{\mathrm{L}}$ move out from $\mathcal{B}$ under h, because they are the nearest points of $\mathcal{B}$ on the corresponding

rays emanated from X. Assume that C_1, C_2, and C_3 are three vertices in $\mathcal{C}_{\mathrm{R}}$ such that C_2 lies between C_1 and C_3, and $h(C_1)$ and $h(C_3)$ lie in $\mathcal{B}$; let $A_i = h(C_i)$. Then A_2 lies on the segment XC_2, and the segments XA_2 and A_1A_3 cross each other. Thus A_2 lies in the triangle $A_1C_2A_3$, which is contained in $\mathcal{B}$.

Comment 1. In fact, Case 1 can be reduced to Case 2 via the following argument.

Assume that $\mathcal{A}$ and $\mathcal{C}$ are congruent. Apply to $\mathcal{A}$ a homothety centered at O_B with a factor slightly smaller than 1 to obtain a polygon $\mathcal{A}'$. With appropriately chosen factor, the vertices of $\mathcal{A}$ which were outside/inside $\mathcal{B}$ stay outside/inside it, so it suffices to prove our claim for $\mathcal{A}'$ instead of $\mathcal{A}$. And now, the polygon $\mathcal{A}'$ is a homothetic image of $\mathcal{C}$, so the arguments from Case 2 apply.

Comment 2. After the polygon $\mathcal{C}$ has been found, the rest of the solution uses only the convexity of the polygons, instead of regularity. Thus, it proves a more general statement:

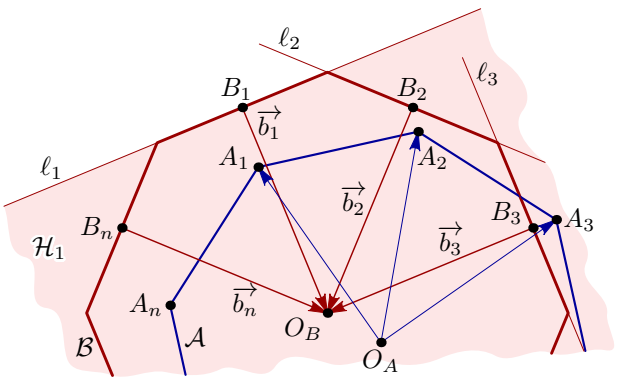
Assume that $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are three convex polygons in the plane such that $\mathcal{C}$ is inscribed into $\mathcal{B}$, and $\mathcal{A}$ can be obtained from it via either translation or positive homothety. Then the vertices of $\mathcal{A}$ that lie inside $\mathcal{B}$ or on its boundary are consecutive.

Solution 2. Let O_A and O_B be the centers of $\mathcal{A}$ and $\mathcal{B}$, respectively. Denote $[n] = \{1, 2, \ldots, n\}$.

We start with introducing appropriate enumerations and notations. Enumerate the *sidelines* of $\mathcal{B}$ clockwise as $\ell_1, \ell_2, \ldots, \ell_n$. Denote by $\mathcal{H}_i$ the half-plane of ℓ_i that contains $\mathcal{B}$ ($\mathcal{H}_i$ is assumed to contain ℓ_i); by B_i the *midpoint* of the side belonging to ℓ_i; and finally denote $\vec{b_i} = \overrightarrow{B_iO_B}$. (As usual, the numbering is cyclic modulo n, so $\ell_{n+i} = \ell_i$ etc.)

Now, choose a vertex A_1 of $\mathcal{A}$ such that the vector $\overrightarrow{O_AA_1}$ points "mostly outside $\mathcal{H}_1$"; strictly speaking, this means that the scalar product $\langle \overrightarrow{O_AA_1}, \vec{b_1} \rangle$ is minimal. Starting from A_1, enumerate the vertices of $\mathcal{A}$ clockwise as $A_1, A_2, \ldots, A_n$; by the rotational symmetry, the choice of A_1 yields that the vector $\overrightarrow{O_AA_i}$ points "mostly outside $\mathcal{H}_i$", i.e.,

$$\langle \overrightarrow{O_AA_i}, \vec{b_i} \rangle = \min_{j \in [n]} \langle \overrightarrow{O_AA_j}, \vec{b_i} \rangle. \tag{1}$$



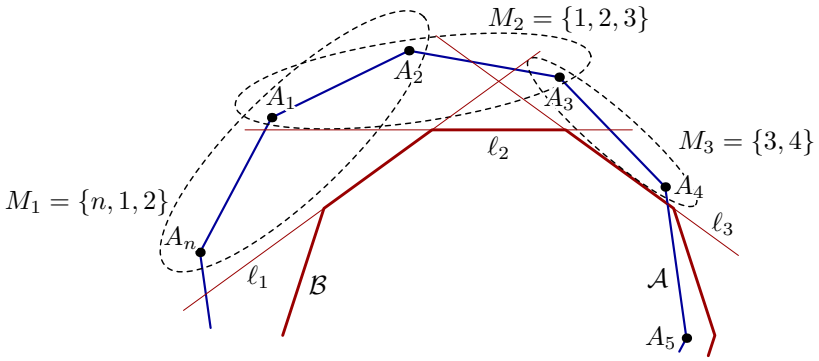
Enumerations and notations

We intend to reformulate the problem in more combinatorial terms, for which purpose we introduce the following notion. Say that a subset $I \subseteq [n]$ is *connected* if the elements of this set are consecutive in the cyclic order (in other words, if we join each i with $i+1 \mod n$ by an edge, this subset is connected in the usual graph sense). Clearly, the union of two connected subsets sharing at least one element is connected too. Next, for any half-plane $\mathcal{H}$ the indices of vertices of, say, $\mathcal{A}$ that lie in $\mathcal{H}$ form a connected set.

To access the problem, we denote

$$M = \{j \in [n] \colon A_j \notin \mathcal{B}\}, \qquad M_i = \{j \in [n] \colon A_j \notin \mathcal{H}_i\} \quad \text{for } i \in [n].$$

We need to prove that $[n] \setminus M$ is connected, which is equivalent to M being connected. On the other hand, since $\mathcal{B} = \bigcap_{i \in [n]} \mathcal{H}_i$, we have $M = \bigcup_{i \in [n]} M_i$, where the sets M_i are easier to investigate. We will utilize the following properties of these sets; the first one holds by the definition of M_i, along with the above remark.



The sets M_i

Property 1: Each set M_i is connected. □

Property 2: If M_i is nonempty, then $i \in M_i$.

Proof. Indeed, we have

$$j \in M_i \iff A_j \notin \mathcal{H}_i \iff \langle \overrightarrow{B_iA_j}, \vec{b_i} \rangle < 0 \iff \langle \overrightarrow{O_AA_j}, \vec{b_i} \rangle < \langle \overrightarrow{O_AB_i}, \vec{b_i} \rangle. \tag{2}$$

The right-hand part of the last inequality does not depend on j. Therefore, if some j lies in M_i, then by (1) so does i. □

In view of Property 2, it is useful to define the set

$$M' = \{i \in [n] \colon i \in M_i\} = \{i \in [n] \colon M_i \neq \varnothing\}.$$

Property 3: The set M' is connected.

Proof. To prove this property, we proceed on with the investigation started in (2) to write

$$i \in M' \iff A_i \in M_i \iff \langle \overrightarrow{B_iA_i}, \vec{b_i} \rangle < 0 \iff \langle \overrightarrow{O_BO_A}, \vec{b_i} \rangle < \langle \overrightarrow{O_BB_i}, \vec{b_i} \rangle + \langle \overrightarrow{A_iO_A}, \vec{b_i} \rangle.$$

The right-hand part of the obtained inequality does not depend on i, due to the rotational symmetry; denote its constant value by μ. Thus, $i \in M'$ if and only if $\langle \overrightarrow{O_BO_A}, \vec{b_i} \rangle < \mu$. This condition is in turn equivalent to the fact that B_i lies in a certain (open) half-plane whose boundary line is orthogonal to O_BO_A; thus, it defines a connected set. □

Now we can finish the solution. Since $M' \subseteq M$, we have

$$M = \bigcup_{i \in [n]} M_i = M' \cup \bigcup_{i \in [n]} M_i,$$

so M can be obtained from M' by adding all the sets M_i one by one. All these sets are connected, and each nonempty M_i contains an element of M' (namely, i). Thus their union is also connected.

Comment 3. Here we present a way in which one can come up with a solution like the one above.

Assume, for sake of simplicity, that O_A lies inside $\mathcal{B}$. Let us first put onto the plane a very small regular n-gon $\mathcal{A}'$ centered at O_A and aligned with $\mathcal{A}$; all its vertices lie inside $\mathcal{B}$. Now we start blowing it up, looking at the order in which the vertices leave $\mathcal{B}$. To go out of $\mathcal{B}$, a vertex should cross a certain side of $\mathcal{B}$ (which is hard to describe), or, equivalently, to cross at least one *sideline* of $\mathcal{B}$ — and this event is easier to describe. Indeed, the first vertex of $\mathcal{A}'$ to cross ℓ_i is the vertex A_i' (corresponding to A_i in $\mathcal{A}$); more generally, the vertices A_j' cross ℓ_i in such an order that the scalar product $\langle \overrightarrow{O_AA_j}, \vec{b_i} \rangle$ does not increase. For different indices i, these orders are just cyclic shifts of each other; and this provides some intuition for the notions and claims from Solution 2.

G7. A convex quadrilateral $ABCD$ has an inscribed circle with center I. Let I_a, I_b, I_c, and I_d be the incenters of the triangles DAB, ABC, BCD, and CDA, respectively. Suppose that the common external tangents of the circles AI_bI_d and CI_bI_d meet at X, and the common external tangents of the circles BI_aI_c and DI_aI_c meet at Y. Prove that $\angle XIY = 90^\circ$.

(Kazakhstan)

Solution. Denote by ω_a, ω_b, ω_c and ω_d the circles AI_bI_d, BI_aI_c, CI_bI_d, and DI_aI_c, let their centers be O_a, O_b, O_c and O_d, and let their radii be r_a, r_b, r_c and r_d, respectively.

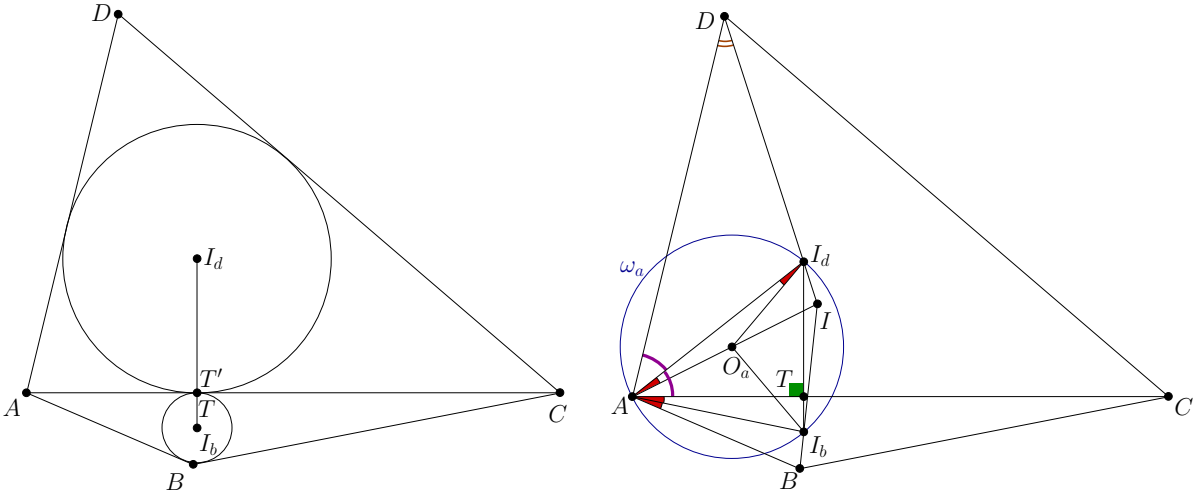
Claim 1. $I_bI_d \perp AC$ and $I_aI_c \perp BD$.

Proof. Let the incircles of triangles ABC and ACD be tangent to the line AC at T and T', respectively. (See the figure to the left.) We have $AT = \frac{AB+AC-BC}{2}$ in triangle ABC, $AT' = \frac{AD+AC-CD}{2}$ in triangle ACD, and $AB - BC = AD - CD$ in quadrilateral $ABCD$, so

$$AT = \frac{AC + AB - BC}{2} = \frac{AC + AD - CD}{2} = AT'.$$

This shows $T = T'$. As an immediate consequence, $I_bI_d \perp AC$.

The second statement can be shown analogously. □



Claim 2. The points O_a, O_b, O_c and O_d lie on the lines AI, BI, CI and DI, respectively.

Proof. By symmetry it suffices to prove the claim for O_a. (See the figure to the right above.)

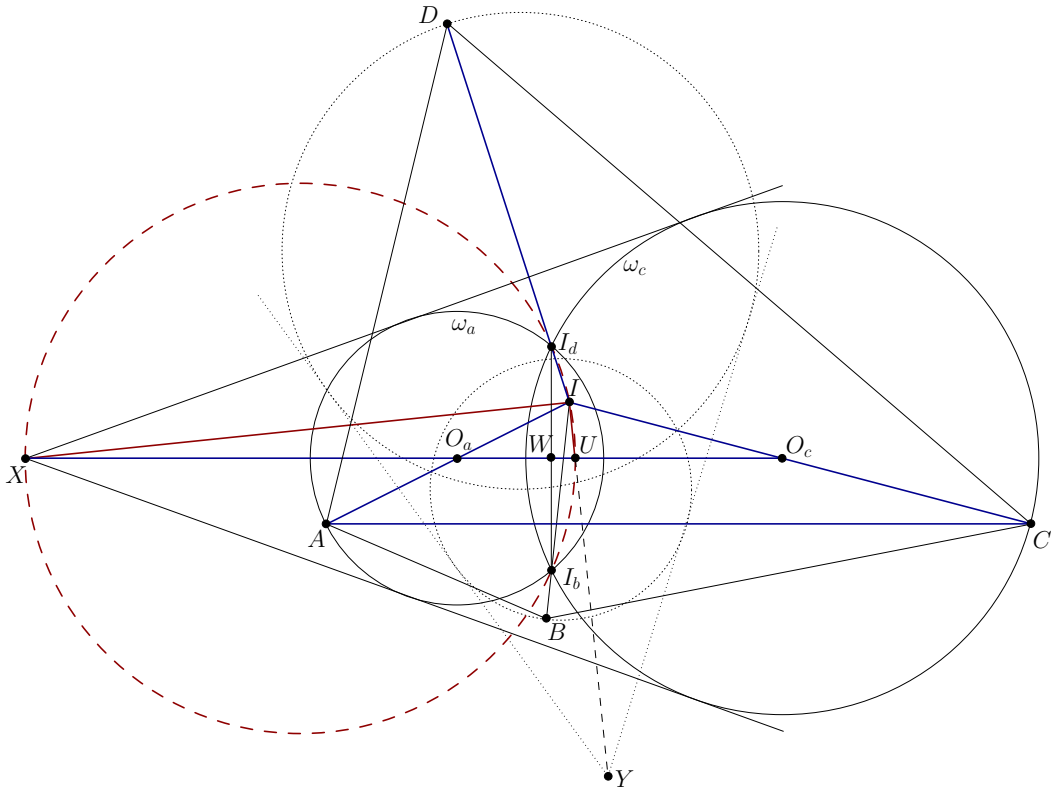
Notice first that the incircles of triangles ABC and ACD can be obtained from the incircle of the quadrilateral $ABCD$ with homothety centers B and D, respectively, and homothety factors less than 1, therefore the points I_b and I_d lie on the line segments BI and DI, respectively.

As is well-known, in every triangle the altitude and the diameter of the circumcircle starting from the same vertex are symmetric about the angle bisector. By Claim 1, in triangle AI_dI_b, the segment AT is the altitude starting from A. Since the foot T lies inside the segment I_bI_d, the circumcenter O_a of triangle AI_dI_b lies in the angle domain I_bAI_d in such a way that $\angle I_bAT = \angle O_aAI_d$. The points I_b and I_d are the incenters of triangles ABC and ACD, so the lines AI_b and AI_d bisect the angles $\angle BAC$ and $\angle CAD$, respectively. Then

$$\angle O_aAD = \angle O_aAI_d + \angle I_dAD = \angle I_bAT + \angle I_dAD = \tfrac{1}{2}\angle BAC + \tfrac{1}{2}\angle CAD = \tfrac{1}{2}\angle BAD,$$

so O_a lies on the angle bisector of $\angle BAD$, that is, on line AI. □

The point X is the external similitude center of ω_a and ω_c; let U be their internal similitude center. The points O_a and O_c lie on the perpendicular bisector of the common chord I_bI_d of ω_a and ω_c, and the two similitude centers X and U lie on the same line; by Claim 2, that line is parallel to AC.



From the similarity of the circles ω_a and ω_c, from $O_aI_b = O_aI_d = O_aA = r_a$ and $O_cI_b = O_cI_d = O_cC = r_c$, and from $AC \parallel O_aO_c$ we can see that

$$\frac{O_aX}{O_cX} = \frac{O_aU}{O_cU} = \frac{r_a}{r_c} = \frac{O_aI_b}{O_cI_b} = \frac{O_aI_d}{O_cI_d} = \frac{O_aA}{O_cC} = \frac{O_aI}{O_cI}.$$

So the points X, U, I_b, I_d, I lie on the Apollonius circle of the points O_a, O_c with ratio $r_a : r_c$. In this Apollonius circle XU is a diameter, and the lines IU and IX are respectively the internal and external bisectors of $\angle O_aIO_c = \angle AIC$, according to the angle bisector theorem. Moreover, in the Apollonius circle the diameter UX is the perpendicular bisector of I_bI_d, so the lines IX and IU are the internal and external bisectors of $\angle I_bII_d = \angle BID$, respectively.

Repeating the same argument for the points B, D instead of A, C, we get that the line IY is the internal bisector of $\angle AIC$ and the external bisector of $\angle BID$. Therefore, the lines IX and IY respectively are the internal and external bisectors of $\angle BID$, so they are perpendicular.

Comment. In fact the points O_a, O_b, O_c and O_d lie on the *line segments* AI, BI, CI and DI, respectively. For the point O_a this can be shown for example by $\angle I_dO_aA + \angle AO_aI_b = (180^\circ - 2\angle O_aAI_d) + (180^\circ - 2\angle I_bAO_a) = 360^\circ - \angle BAD = \angle ADI + \angle DIA + \angle AIB + \angle IBA > \angle I_dIA + \angle AII_b$.

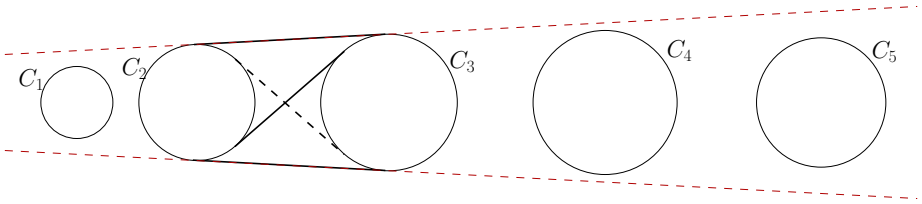
The solution also shows that the line IY passes through the point U, and analogously, IX passes through the internal similitude center of ω_b and ω_d.

G8. There are 2017 mutually external circles drawn on a blackboard, such that no two are tangent and no three share a common tangent. A *tangent segment* is a line segment that is a common tangent to two circles, starting at one tangent point and ending at the other one. Luciano is drawing tangent segments on the blackboard, one at a time, so that no tangent segment intersects any other circles or previously drawn tangent segments. Luciano keeps drawing tangent segments until no more can be drawn. Find all possible numbers of tangent segments when he stops drawing.

(Australia)

Answer: If there were n circles, there would always be exactly $3(n-1)$ segments; so the only possible answer is $3 \cdot 2017 - 3 = 6048$.

Solution 1. First, consider a particular arrangement of circles $C_1, C_2, \ldots, C_n$ where all the centers are aligned and each C_i is eclipsed from the other circles by its neighbors – for example, taking C_i with center $(i^2, 0)$ and radius $i/2$ works. Then the only tangent segments that can be drawn are between adjacent circles C_i and C_{i+1}, and exactly three segments can be drawn for each pair. So Luciano will draw exactly $3(n-1)$ segments in this case.



For the general case, start from a *final* configuration (that is, an arrangement of circles and segments in which no further segments can be drawn). The idea of the solution is to continuously resize and move the circles around the plane, one by one (in particular, making sure we never have 4 circles with a common tangent line), and show that the number of segments drawn remains constant as the picture changes. This way, we can reduce any circle/segment configuration to the particular one mentioned above, and the final number of segments must remain at $3n-3$.

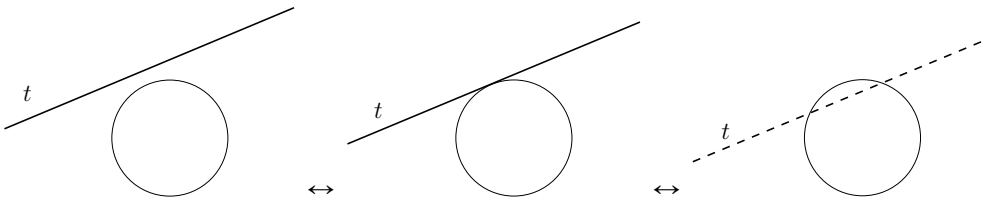
Some preliminary considerations: look at all possible tangent segments joining any two circles. A segment that is tangent to a circle A can do so in two possible orientations — it may come out of A in clockwise or counterclockwise orientation. Two segments touching the same circle with the same orientation will never intersect each other. Each pair (A, B) of circles has 4 choices of tangent segments, which can be identified by their orientations — for example, $(A+, B-)$ would be the segment which comes out of A in clockwise orientation and comes out of B in counterclockwise orientation. In total, we have $2n(n-1)$ possible segments, disregarding intersections.

Now we pick a circle C and start to continuously move and resize it, maintaining all existing tangent segments according to their identifications, including those involving C. We can keep our choice of tangent segments until the configuration reaches a *transition*. We lose nothing if we assume that C is kept at least ε units away from any other circle, where ε is a positive, fixed constant; therefore at a transition either: (1) a currently drawn tangent segment t suddenly becomes obstructed; or (2) a currently absent tangent segment t suddenly becomes unobstructed and available.

Claim. A transition can only occur when three circles C_1, C_2, C_3 are tangent to a common line ℓ containing t, in a way such that the three tangent segments lying on ℓ (joining the three circles pairwise) are not obstructed by any other circles or tangent segments (other than C_1, C_2, C_3).

Proof. Since (2) is effectively the reverse of (1), it suffices to prove the claim for (1). Suppose t has suddenly become obstructed, and let us consider two cases.

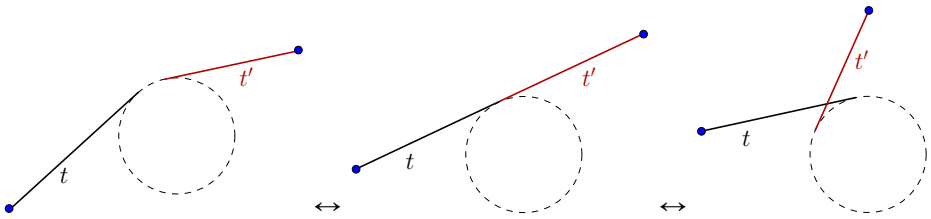
Case 1: t becomes obstructed by a circle



Then the new circle becomes the third circle tangent to ℓ, and no other circles or tangent segments are obstructing t.

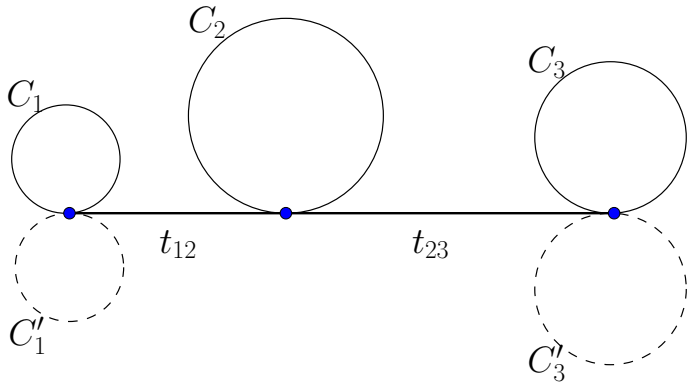
Case 2: t becomes obstructed by another tangent segment t'

When two segments t and t' first intersect each other, they must do so at a vertex of one of them. But if a vertex of t' first crossed an interior point of t, the circle associated to this vertex was already blocking t (absurd), or is about to (we already took care of this in case 1). So we only have to analyze the possibility of t and t' suddenly having a common vertex. However, if that happens, this vertex must belong to a single circle (remember we are keeping different circles at least ε units apart from each other throughout the moving/resizing process), and therefore they must have different orientations with respect to that circle.



Thus, at the transition moment, both t and t' are tangent to the same circle at a common point, that is, they must be on the same line ℓ and hence we again have three circles simultaneously tangent to ℓ. Also no other circles or tangent segments are obstructing t or t' (otherwise, they would have disappeared before this transition). □

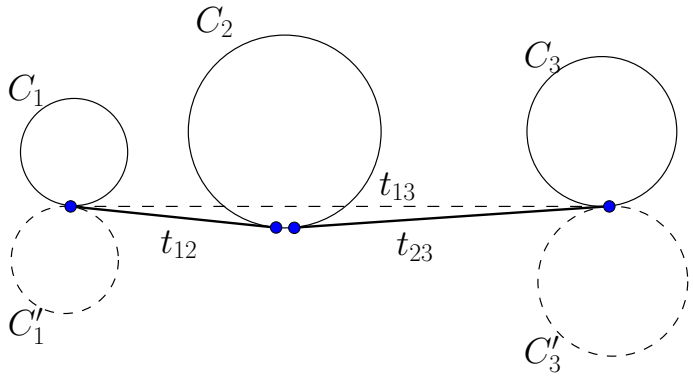
Next, we focus on the maximality of a configuration immediately before and after a transition, where three circles share a common tangent line ℓ. Let the three circles be C_1, C_2, C_3, ordered by their tangent points. The only possibly affected segments are the ones lying on ℓ, namely t_{12}, t_{23} and t_{13}. Since C_2 is in the middle, t_{12} and t_{23} must have different orientations with respect to C_2. For C_1, t_{12} and t_{13} must have the same orientation, while for C_3, t_{13} and t_{23} must have the same orientation. The figure below summarizes the situation, showing alternative positions for C_1 (namely, C_1 and C_1') and for C_3 (C_3 and C_3').



Now perturb the diagram slightly so the three circles no longer have a common tangent, while preserving the definition of t_{12}, t_{23} and t_{13} according to their identifications. First note that no other circles or tangent segments can obstruct any of these segments. Also recall that tangent segments joining the same circle at the same orientation will never obstruct each other.

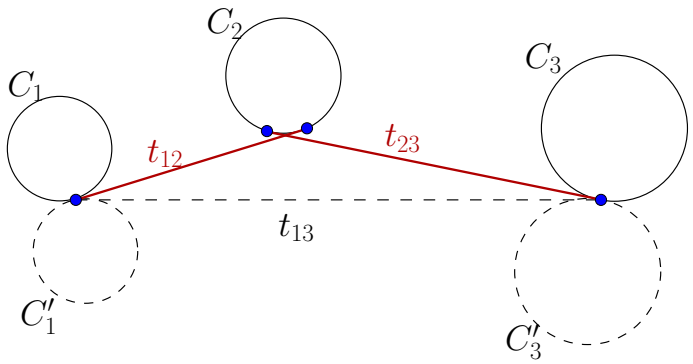
The availability of the tangent segments can now be checked using simple diagrams.

Case 1: t_{13} passes through C_2



In this case, t_{13} is not available, but both t_{12} and t_{23} are.

Case 2: t_{13} does not pass through C_2

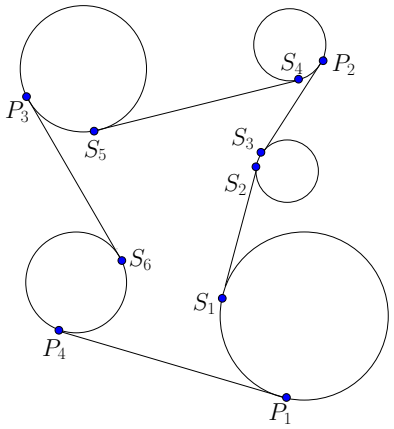


Now t_{13} is available, but t_{12} and t_{23} obstruct each other, so only one can be drawn.

In any case, exactly 2 out of these 3 segments can be drawn. Thus the maximal number of segments remains constant as we move or resize the circles, and we are done.

Solution 2. First note that all tangent segments lying on the boundary of the convex hull of the circles are always drawn since they do not intersect anything else. Now in the final picture, aside from the n circles, the blackboard is divided into regions. We can consider the picture as a plane (multi-)graph G in which the circles are the vertices and the tangent segments are the edges. The idea of this solution is to find a relation between the number of edges and the number of regions in G; then, once we prove that G is connected, we can use Euler's formula to finish the problem.

The boundary of each region consists of 1 or more (for now) simple closed curves, each made of arcs and tangent segments. The segment and the arc might meet smoothly (as in S_i, $i = 1, 2, \ldots, 6$ in the figure below) or not (as in P_1, P_2, P_3, P_4; call such points *sharp corners* of the boundary). In other words, if a person walks along the border, her direction would suddenly turn an angle of π at a sharp corner.



Claim 1. The outer boundary B_1 of any internal region has at least 3 sharp corners.

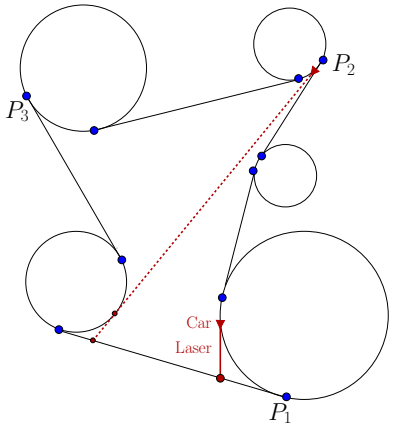
Proof. Let a person walk one lap along B_1 in the counterclockwise orientation. As she does so, she will turn *clockwise* as she moves along the circle arcs, and not turn at all when moving along the lines. On the other hand, her total rotation after one lap is 2π in the *counterclockwise* direction! Where could she be turning counterclockwise? She can only do so at sharp corners, and, even then, she turns only an angle of π there. But two sharp corners are not enough, since at least one arc must be present—so she must have gone through at least 3 sharp corners. □

Claim 2. Each internal region is simply connected, that is, has only one boundary curve.

Proof. Suppose, by contradiction, that some region has an outer boundary B_1 and inner boundaries $B_2, B_3, \ldots, B_m$ ($m \geqslant 2$). Let P_1 be one of the sharp corners of B_1.

Now consider a car starting at P_1 and traveling counterclockwise along B_1. It starts in reverse, i.e., it is initially facing the corner P_1. Due to the tangent conditions, the car may travel in a way so that its orientation only changes when it is moving along an arc. In particular, this means the car will sometimes travel forward. For example, if the car approaches a sharp corner when driving in reverse, it would continue travel forward after the corner, instead of making an immediate half-turn. This way, the orientation of the car only changes in a clockwise direction since the car always travels clockwise around each arc.

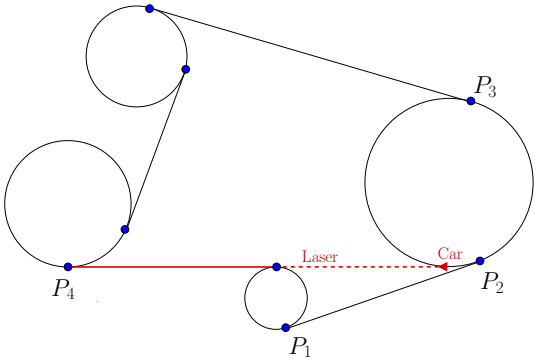
Now imagine there is a laser pointer at the front of the car, pointing directly ahead. Initially, the laser endpoint hits P_1, but, as soon as the car hits an arc, the endpoint moves clockwise around B_1. In fact, the laser endpoint must move continuously along B_1! Indeed, if the endpoint ever jumped (within B_1, or from B_1 to one of the inner boundaries), at the moment of the jump the interrupted laser would be a drawable tangent segment that Luciano missed (see figure below for an example).



Now, let P_2 and P_3 be the next two sharp corners the car goes through, after P_1 (the previous lemma assures their existence). At P_2 the car starts moving forward, and at P_3 it will start to move in reverse again. So, at P_3, the laser endpoint is at P_3 itself. So while the car moved counterclockwise between P_1 and P_3, the laser endpoint moved clockwise between P_1 and P_3. That means the laser beam itself scanned the whole region within B_1, and it should have crossed some of the inner boundaries. □

Claim 3. Each region has exactly 3 sharp corners.

Proof. Consider again the car of the previous claim, with its laser still firmly attached to its front, traveling the same way as before and going through the same *consecutive* sharp corners P_1, P_2 and P_3. As we have seen, as the car goes counterclockwise from P_1 to P_3, the laser endpoint goes clockwise from P_1 to P_3, so together they cover the whole boundary. If there were a fourth sharp corner P_4, at some moment the laser endpoint would pass through it. But, since P_4 is a sharp corner, this means the car must be on the extension of a tangent segment going through P_4. Since the car is not on that segment itself (the car never goes through P_4), we would have 3 circles with a common tangent line, which is not allowed.



□

We are now ready to finish the solution. Let r be the number of internal regions, and s be the number of tangent segments. Since each tangent segment contributes exactly 2 sharp corners to the diagram, and each region has exactly 3 sharp corners, we must have $2s = 3r$. Since the graph corresponding to the diagram is connected, we can use Euler's formula $n - s + r = 1$ and find $s = 3n - 3$ and $r = 2n - 2$.

Number Theory

N1. The sequence $a_0, a_1, a_2, \ldots$ of positive integers satisfies

$$a_{n+1} = \begin{cases} \sqrt{a_n}, & \text{if } \sqrt{a_n} \text{ is an integer} \\ a_n + 3, & \text{otherwise} \end{cases} \qquad \text{for every } n \geqslant 0.$$

Determine all values of $a_0 > 1$ for which there is at least one number a such that $a_n = a$ for infinitely many values of n.

(South Africa)

Answer: All positive multiples of 3.

Solution. Since the value of a_{n+1} only depends on the value of a_n, if $a_n = a_m$ for two different indices n and m, then the sequence is eventually periodic. So we look for the values of a_0 for which the sequence is eventually periodic.

Claim 1. If $a_n \equiv -1 \pmod 3$, then, for all $m > n$, a_m is not a perfect square. It follows that the sequence is eventually strictly increasing, so it is not eventually periodic.

Proof. A square cannot be congruent to -1 modulo 3, so $a_n \equiv -1 \pmod 3$ implies that a_n is not a square, therefore $a_{n+1} = a_n + 3 > a_n$. As a consequence, $a_{n+1} \equiv a_n \equiv -1 \pmod 3$, so a_{n+1} is not a square either. By repeating the argument, we prove that, from a_n on, all terms of the sequence are not perfect squares and are greater than their predecessors, which completes the proof. □

Claim 2. If $a_n \not\equiv -1 \pmod 3$ and $a_n > 9$ then there is an index $m > n$ such that $a_m < a_n$.

Proof. Let t^2 be the largest perfect square which is less than a_n. Since $a_n > 9$, t is at least 3. The first square in the sequence $a_n, a_n + 3, a_n + 6, \ldots$ will be $(t+1)^2$, $(t+2)^2$ or $(t+3)^2$, therefore there is an index $m > n$ such that $a_m \leqslant t + 3 < t^2 < a_n$, as claimed. □

Claim 3. If $a_n \equiv 0 \pmod 3$, then there is an index $m > n$ such that $a_m = 3$.

Proof. First we notice that, by the definition of the sequence, a multiple of 3 is always followed by another multiple of 3. If $a_n \in \{3, 6, 9\}$ the sequence will eventually follow the periodic pattern $3, 6, 9, 3, 6, 9, \ldots$. If $a_n > 9$, let j be an index such that a_j is equal to the minimum value of the set $\{a_{n+1}, a_{n+2}, \ldots\}$. We must have $a_j \leqslant 9$, otherwise we could apply Claim 2 to a_j and get a contradiction on the minimality hypothesis. It follows that $a_j \in \{3, 6, 9\}$, and the proof is complete. □

Claim 4. If $a_n \equiv 1 \pmod 3$, then there is an index $m > n$ such that $a_m \equiv -1 \pmod 3$.

Proof. In the sequence, 4 is always followed by $2 \equiv -1 \pmod 3$, so the claim is true for $a_n = 4$. If $a_n = 7$, the next terms will be $10, 13, 16, 4, 2, \ldots$ and the claim is also true. For $a_n \geqslant 10$, we again take an index $j > n$ such that a_j is equal to the minimum value of the set $\{a_{n+1}, a_{n+2}, \ldots\}$, which by the definition of the sequence consists of non-multiples of 3. Suppose $a_j \equiv 1 \pmod 3$. Then we must have $a_j \leqslant 9$ by Claim 2 and the minimality of a_j. It follows that $a_j \in \{4, 7\}$, so $a_m = 2 < a_j$ for some $m > j$, contradicting the minimality of a_j. Therefore, we must have $a_j \equiv -1 \pmod 3$. □

It follows from the previous claims that if a_0 is a multiple of 3 the sequence will eventually reach the periodic pattern $3, 6, 9, 3, 6, 9, \ldots$; if $a_0 \equiv -1 \pmod 3$ the sequence will be strictly increasing; and if $a_0 \equiv 1 \pmod 3$ the sequence will be eventually strictly increasing.

So the sequence will be eventually periodic if, and only if, a_0 is a multiple of 3.

N2. Let $p \geqslant 2$ be a prime number. Eduardo and Fernando play the following game making moves alternately: in each move, the current player chooses an index i in the set $\{0, 1, \ldots, p-1\}$ that was not chosen before by either of the two players and then chooses an element a_i of the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Eduardo has the first move. The game ends after all the indices $i \in \{0, 1, \ldots, p-1\}$ have been chosen. Then the following number is computed:

$$M = a_0 + 10 \cdot a_1 + \cdots + 10^{p-1} \cdot a_{p-1} = \sum_{j=0}^{p-1} a_j \cdot 10^j.$$

The goal of Eduardo is to make the number M divisible by p, and the goal of Fernando is to prevent this.

Prove that Eduardo has a winning strategy.

(Morocco)

Solution. We say that a player makes the move (i, a_i) if he chooses the index i and then the element a_i of the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ in this move.

If $p = 2$ or $p = 5$ then Eduardo chooses $i = 0$ and $a_0 = 0$ in the first move, and wins, since, independently of the next moves, M will be a multiple of 10.

Now assume that the prime number p does not belong to $\{2, 5\}$. Eduardo chooses $i = p-1$ and $a_{p-1} = 0$ in the first move. By Fermat's Little Theorem, $(10^{(p-1)/2})^2 = 10^{p-1} \equiv 1 \pmod{p}$, so $p \mid (10^{(p-1)/2})^2 - 1 = (10^{(p-1)/2} + 1)(10^{(p-1)/2} - 1)$. Since p is prime, either $p \mid 10^{(p-1)/2} + 1$ or $p \mid 10^{(p-1)/2} - 1$. Thus we have two cases:

Case a: $10^{(p-1)/2} \equiv -1 \pmod{p}$

In this case, for each move (i, a_i) of Fernando, Eduardo immediately makes the move $(j, a_j) = (i + \frac{p-1}{2}, a_i)$, if $0 \leqslant i \leqslant \frac{p-3}{2}$, or $(j, a_j) = (i - \frac{p-1}{2}, a_i)$, if $\frac{p-1}{2} \leqslant i \leqslant p-2$. We will have $10^j \equiv -10^i \pmod{p}$, and so $a_j \cdot 10^j = a_i \cdot 10^j \equiv -a_i \cdot 10^i \pmod{p}$. Notice that this move by Eduardo is always possible. Indeed, immediately before a move by Fernando, for any set of the type $\{r, r + (p-1)/2\}$ with $0 \leqslant r \leqslant (p-3)/2$, either no element of this set was chosen as an index by the players in the previous moves or else both elements of this set were chosen as indices by the players in the previous moves. Therefore, after each of his moves, Eduardo always makes the sum of the numbers $a_k \cdot 10^k$ corresponding to the already chosen pairs (k, a_k) divisible by p, and thus wins the game.

Case b: $10^{(p-1)/2} \equiv 1 \pmod{p}$

In this case, for each move (i, a_i) of Fernando, Eduardo immediately makes the move $(j, a_j) = (i + \frac{p-1}{2}, 9 - a_i)$, if $0 \leqslant i \leqslant \frac{p-3}{2}$, or $(j, a_j) = (i - \frac{p-1}{2}, 9 - a_i)$, if $\frac{p-1}{2} \leqslant i \leqslant p-2$. The same argument as above shows that Eduardo can always make such move. We will have $10^j \equiv 10^i \pmod{p}$, and so $a_j \cdot 10^j + a_i \cdot 10^i \equiv (a_i + a_j) \cdot 10^i = 9 \cdot 10^i \pmod{p}$. Therefore, at the end of the game, the sum of all terms $a_k \cdot 10^k$ will be congruent to

$$\sum_{i=0}^{\frac{p-3}{2}} 9 \cdot 10^i = 10^{(p-1)/2} - 1 \equiv 0 \pmod{p},$$

and Eduardo wins the game.

N3. Determine all integers $n \geqslant 2$ with the following property: for any integers $a_1, a_2, \ldots, a_n$ whose sum is not divisible by n, there exists an index $1 \leqslant i \leqslant n$ such that none of the numbers

$$a_i, a_i + a_{i+1}, \ldots, a_i + a_{i+1} + \cdots + a_{i+n-1}$$

is divisible by n. (We let $a_i = a_{i-n}$ when $i > n$.)

(Thailand)

Answer: These integers are exactly the prime numbers.

Solution. Let us first show that, if $n = ab$, with $a, b \geqslant 2$ integers, then the property in the statement of the problem does not hold. Indeed, in this case, let $a_k = a$ for $1 \leqslant k \leqslant n-1$ and $a_n = 0$. The sum $a_1 + a_2 + \cdots + a_n = a \cdot (n-1)$ is not divisible by n. Let i with $1 \leqslant i \leqslant n$ be an arbitrary index. Taking $j = b$ if $1 \leqslant i \leqslant n-b$, and $j = b+1$ if $n-b < i \leqslant n$, we have

$$a_i + a_{i+1} + \cdots + a_{i+j-1} = a \cdot b = n \equiv 0 \pmod{n}.$$

It follows that the given example is indeed a counterexample to the property of the statement.

Now let n be a prime number. Suppose by contradiction that the property in the statement of the problem does not hold. Then there are integers $a_1, a_2, \ldots, a_n$ whose sum is not divisible by n such that for each i, $1 \leqslant i \leqslant n$, there is j, $1 \leqslant j \leqslant n$, for which the number $a_i + a_{i+1} + \cdots + a_{i+j-1}$ is divisible by n. Notice that, in any such case, we should have $1 \leqslant j \leqslant n-1$, since $a_1 + a_2 + \cdots + a_n$ is not divisible by n. So we may construct recursively a finite sequence of integers $0 = i_0 < i_1 < i_2 < \cdots < i_n$ with $i_{s+1} - i_s \leqslant n-1$ for $0 \leqslant s \leqslant n-1$ such that, for $0 \leqslant s \leqslant n-1$,

$$a_{i_s+1} + a_{i_s+2} + \cdots + a_{i_{s+1}} \equiv 0 \pmod{n}$$

(where we take indices modulo n). Indeed, for $0 \leqslant s < n$, we apply the previous observation to $i = i_s + 1$ in order to define $i_{s+1} = i_s + j$.

In the sequence of $n+1$ indices $i_0, i_1, i_2, \ldots, i_n$, by the pigeonhole principle, we have two distinct elements which are congruent modulo n. So there are indices r, s with $0 \leqslant r < s \leqslant n$ such that $i_s \equiv i_r \pmod{n}$ and

$$a_{i_r+1} + a_{i_r+2} + \cdots + a_{i_s} = \sum_{j=r}^{s-1} (a_{i_j+1} + a_{i_j+2} + \cdots + a_{i_{j+1}}) \equiv 0 \pmod{n}.$$

Since $i_s \equiv i_r \pmod{n}$, we have $i_s - i_r = k \cdot n$ for some positive integer k, and, since $i_{j+1} - i_j \leqslant n-1$ for $0 \leqslant j \leqslant n-1$, we have $i_s - i_r \leqslant (n-1) \cdot n$, so $k \leqslant n-1$. But in this case

$$a_{i_r+1} + a_{i_r+2} + \cdots + a_{i_s} = k \cdot (a_1 + a_2 + \cdots + a_n)$$

cannot be a multiple of n, since n is prime and neither k nor $a_1 + a_2 + \cdots + a_n$ is a multiple of n. A contradiction.

N4. Call a rational number *short* if it has finitely many digits in its decimal expansion. For a positive integer m, we say that a positive integer t is m-*tastic* if there exists a number $c \in \{1, 2, 3, \ldots, 2017\}$ such that $\dfrac{10^t - 1}{c \cdot m}$ is short, and such that $\dfrac{10^k - 1}{c \cdot m}$ is not short for any $1 \leqslant k < t$. Let $S(m)$ be the set of m-tastic numbers. Consider $S(m)$ for $m = 1, 2, \ldots$. What is the maximum number of elements in $S(m)$?

(Turkey)

Answer: 807.

Solution. First notice that $x \in \mathbb{Q}$ is short if and only if there are exponents $a, b \geqslant 0$ such that $2^a \cdot 5^b \cdot x \in \mathbb{Z}$. In fact, if x is short, then $x = \frac{n}{10^k}$ for some k and we can take $a = b = k$; on the other hand, if $2^a \cdot 5^b \cdot x = q \in \mathbb{Z}$ then $x = \frac{2^b \cdot 5^a q}{10^{a+b}}$, so x is short.

If $m = 2^a \cdot 5^b \cdot s$, with $\gcd(s, 10) = 1$, then $\frac{10^t - 1}{m}$ is short if and only if s divides $10^t - 1$. So we may (and will) suppose without loss of generality that $\gcd(m, 10) = 1$. Define

$$C = \{1 \leqslant c \leqslant 2017 \colon \gcd(c, 10) = 1\}.$$

The m-tastic numbers are then precisely the smallest exponents $t > 0$ such that $10^t \equiv 1 \pmod{cm}$ for some integer $c \in C$, that is, the set of *orders of* 10 *modulo* cm. In other words,

$$S(m) = \{\operatorname{ord}_{cm}(10) \colon c \in C\}.$$

Since there are $4 \cdot 201 + 3 = 807$ numbers c with $1 \leqslant c \leqslant 2017$ and $\gcd(c, 10) = 1$, namely those such that $c \equiv 1, 3, 7, 9 \pmod{10}$,

$$|S(m)| \leqslant |C| = 807.$$

Now we find m such that $|S(m)| = 807$. Let

$$P = \{1 < p \leqslant 2017 \colon p \text{ is prime}, p \neq 2, 5\}$$

and choose a positive integer α such that every $p \in P$ divides $10^\alpha - 1$ (e.g. $\alpha = \varphi(T)$, T being the product of all primes in P), and let $m = 10^\alpha - 1$.

Claim. For every $c \in C$, we have

$$\operatorname{ord}_{cm}(10) = c\alpha.$$

As an immediate consequence, this implies $|S(m)| = |C| = 807$, finishing the problem.

Proof. Obviously $\operatorname{ord}_m(10) = \alpha$. Let $t = \operatorname{ord}_{cm}(10)$. Then

$$cm \mid 10^t - 1 \quad \Longrightarrow \quad m \mid 10^t - 1 \quad \Longrightarrow \quad \alpha \mid t.$$

Hence $t = k\alpha$ for some $k \in \mathbb{Z}_{>0}$. We will show that $k = c$.

Denote by $\nu_p(n)$ the number of prime factors p in n, that is, the maximum exponent β for which $p^\beta \mid n$. For every $\ell \geqslant 1$ and $p \in P$, the Lifting the Exponent Lemma provides

$$\nu_p(10^{\ell\alpha} - 1) = \nu_p((10^\alpha)^\ell - 1) = \nu_p(10^\alpha - 1) + \nu_p(\ell) = \nu_p(m) + \nu_p(\ell),$$

so

$$\begin{aligned} cm \mid 10^{k\alpha} - 1 &\iff \forall p \in P;\ \nu_p(cm) \leqslant \nu_p(10^{k\alpha} - 1) \\ &\iff \forall p \in P;\ \nu_p(m) + \nu_p(c) \leqslant \nu_p(m) + \nu_p(k) \\ &\iff \forall p \in P;\ \nu_p(c) \leqslant \nu_p(k) \\ &\iff c \mid k. \end{aligned}$$

The first such k is $k = c$, so $\operatorname{ord}_{cm}(10) = c\alpha$. □

Comment. The Lifting the Exponent Lemma states that, for any odd prime p, any integers a, b coprime with p such that $p \mid a - b$, and any positive integer exponent n,

$$\nu_p(a^n - b^n) = \nu_p(a - b) + \nu_p(n),$$

and, for $p = 2$,

$$\nu_2(a^n - b^n) = \nu_2(a^2 - b^2) + \nu_p(n) - 1.$$

Both claims can be proved by induction on n.

N5. Find all pairs (p, q) of prime numbers with $p > q$ for which the number

$$\frac{(p+q)^{p+q}(p-q)^{p-q}-1}{(p+q)^{p-q}(p-q)^{p+q}-1}$$

is an integer.

(Japan)

Answer: The only such pair is $(3, 2)$.

Solution. Let $M = (p+q)^{p-q}(p-q)^{p+q} - 1$, which is relatively prime with both $p+q$ and $p-q$. Denote by $(p-q)^{-1}$ the multiplicative inverse of $(p-q)$ modulo M.

By eliminating the term -1 in the numerator,

$$(p+q)^{p+q}(p-q)^{p-q} - 1 \equiv (p+q)^{p-q}(p-q)^{p+q} - 1 \pmod{M}$$
$$(p+q)^{2q} \equiv (p-q)^{2q} \pmod{M} \tag{1}$$
$$\left((p+q)\cdot(p-q)^{-1}\right)^{2q} \equiv 1 \pmod{M}. \tag{2}$$

Case 1: $q \geqslant 5$.

Consider an arbitrary prime divisor r of M. Notice that M is odd, so $r \geqslant 3$. By (2), the multiplicative order of $\left((p+q)\cdot(p-q)^{-1}\right)$ modulo r is a divisor of the exponent $2q$ in (2), so it can be 1, 2, q or $2q$.

By Fermat's theorem, the order divides $r-1$. So, if the order is q or $2q$ then $r \equiv 1 \pmod{q}$. If the order is 1 or 2 then $r \mid (p+q)^2 - (p-q)^2 = 4pq$, so $r = p$ or $r = q$. The case $r = p$ is not possible, because, by applying Fermat's theorem,

$$M = (p+q)^{p-q}(p-q)^{p+q} - 1 \equiv q^{p-q}(-q)^{p+q} - 1 = \left(q^2\right)^p - 1 \equiv q^2 - 1 = (q+1)(q-1) \pmod{p}$$

and the last factors $q-1$ and $q+1$ are less than p and thus $p \nmid M$. Hence, all prime divisors of M are either q or of the form $kq+1$; it follows that all positive divisors of M are congruent to 0 or 1 modulo q.

Now notice that

$$M = \left((p+q)^{\frac{p-q}{2}}(p-q)^{\frac{p+q}{2}} - 1\right)\left((p+q)^{\frac{p-q}{2}}(p-q)^{\frac{p+q}{2}} + 1\right)$$

is the product of two consecutive positive odd numbers; both should be congruent to 0 or 1 modulo q. But this is impossible by the assumption $q \geqslant 5$. So, there is no solution in Case 1.

Case 2: $q = 2$.

By (1), we have $M \mid (p+q)^{2q} - (p-q)^{2q} = (p+2)^4 - (p-2)^4$, so

$$(p+2)^{p-2}(p-2)^{p+2} - 1 = M \leqslant (p+2)^4 - (p-2)^4 \leqslant (p+2)^4 - 1,$$
$$(p+2)^{p-6}(p-2)^{p+2} \leqslant 1.$$

If $p \geqslant 7$ then the left-hand side is obviously greater than 1. For $p = 5$ we have $(p+2)^{p-6}(p-2)^{p+2} = 7^{-1}\cdot 3^7$ which is also too large.

There remains only one candidate, $p = 3$, which provides a solution:

$$\frac{(p+q)^{p+q}(p-q)^{p-q}-1}{(p+q)^{p-q}(p-q)^{p+q}-1} = \frac{5^5\cdot 1^1 - 1}{5^1\cdot 1^5 - 1} = \frac{3124}{4} = 781.$$

So in Case 2 the only solution is $(p, q) = (3, 2)$.

Case 3: $q = 3$.

Similarly to Case 2, we have

$$M \mid (p+q)^{2q} - (p-q)^{2q} = 64 \cdot \left(\left(\frac{p+3}{2} \right)^6 - \left(\frac{p-3}{2} \right)^6 \right).$$

Since M is odd, we conclude that

$$M \mid \left(\frac{p+3}{2} \right)^6 - \left(\frac{p-3}{2} \right)^6$$

and

$$(p+3)^{p-3}(p-3)^{p+3} - 1 = M \leqslant \left(\frac{p+3}{2} \right)^6 - \left(\frac{p-3}{2} \right)^6 \leqslant \left(\frac{p+3}{2} \right)^6 - 1,$$
$$64(p+3)^{p-9}(p-3)^{p+3} \leqslant 1.$$

If $p \geqslant 11$ then the left-hand side is obviously greater than 1. If $p = 7$ then the left-hand side is $64 \cdot 10^{-2} \cdot 4^{10} > 1$. If $p = 5$ then the left-hand side is $64 \cdot 8^{-4} \cdot 2^8 = 2^2 > 1$. Therefore, there is no solution in Case 3.

N6. Find the smallest positive integer n, or show that no such n exists, with the following property: there are infinitely many distinct n-tuples of positive rational numbers $(a_1, a_2, \ldots, a_n)$ such that both

$$a_1 + a_2 + \cdots + a_n \qquad \text{and} \qquad \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}$$

are integers.

(Singapore)

Answer: $n = 3$.

Solution 1. For $n = 1$, $a_1 \in \mathbb{Z}_{>0}$ and $\frac{1}{a_1} \in \mathbb{Z}_{>0}$ if and only if $a_1 = 1$. Next we show that

(i) *There are finitely many $(x, y) \in \mathbb{Q}^2_{>0}$ satisfying $x + y \in \mathbb{Z}$ and $\frac{1}{x} + \frac{1}{y} \in \mathbb{Z}$*

Write $x = \frac{a}{b}$ and $y = \frac{c}{d}$ with $a, b, c, d \in \mathbb{Z}_{>0}$ and $\gcd(a, b) = \gcd(c, d) = 1$. Then $x + y \in \mathbb{Z}$ and $\frac{1}{x} + \frac{1}{y} \in \mathbb{Z}$ is equivalent to the two divisibility conditions

$$bd \mid ad + bc \quad (1) \qquad \text{and} \qquad ac \mid ad + bc \quad (2)$$

Condition (1) implies that $d \mid ad + bc \iff d \mid bc \iff d \mid b$ since $\gcd(c, d) = 1$. Still from (1) we get $b \mid ad + bc \iff b \mid ad \iff b \mid d$ since $\gcd(a, b) = 1$. From $b \mid d$ and $d \mid b$ we have $b = d$.

An analogous reasoning with condition (2) shows that $a = c$. Hence $x = \frac{a}{b} = \frac{c}{d} = y$, i.e., the problem amounts to finding all $x \in \mathbb{Q}_{>0}$ such that $2x \in \mathbb{Z}_{>0}$ and $\frac{2}{x} \in \mathbb{Z}_{>0}$. Letting $n = 2x \in \mathbb{Z}_{>0}$, we have that $\frac{2}{x} \in \mathbb{Z}_{>0} \iff \frac{4}{n} \in \mathbb{Z}_{>0} \iff n = 1$, 2 or 4, and there are finitely many solutions, namely $(x, y) = (\frac{1}{2}, \frac{1}{2})$, $(1, 1)$ or $(2, 2)$.

(ii) *There are infinitely many triples $(x, y, z) \in \mathbb{Q}^2_{>0}$ such that $x + y + z \in \mathbb{Z}$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \in \mathbb{Z}$.* We will look for triples such that $x + y + z = 1$, so we may write them in the form

$$(x, y, z) = \left(\frac{a}{a+b+c}, \frac{b}{a+b+c}, \frac{c}{a+b+c}\right) \qquad \text{with } a, b, c \in \mathbb{Z}_{>0}$$

We want these to satisfy

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{a+b+c}{a} + \frac{a+b+c}{b} + \frac{a+b+c}{c} \in \mathbb{Z} \iff \frac{b+c}{a} + \frac{a+c}{b} + \frac{a+b}{c} \in \mathbb{Z}$$

Fixing $a = 1$, it suffices to find infinitely many pairs $(b, c) \in \mathbb{Z}^2_{>0}$ such that

$$\frac{1}{b} + \frac{1}{c} + \frac{c}{b} + \frac{b}{c} = 3 \iff b^2 + c^2 - 3bc + b + c = 0 \qquad (*)$$

To show that equation $(*)$ has infinitely many solutions, we use *Vieta jumping* (also known as *root flipping*): starting with $b = 2$, $c = 3$, the following algorithm generates infinitely many solutions. Let $c \geqslant b$, and view $(*)$ as a quadratic equation in b for c fixed:

$$b^2 - (3c - 1) \cdot b + (c^2 + c) = 0 \qquad (**)$$

Then there exists another root $b_0 \in \mathbb{Z}$ of $(**)$ which satisfies $b + b_0 = 3c - 1$ and $b \cdot b_0 = c^2 + c$. Since $c \geqslant b$ by assumption,

$$b_0 = \frac{c^2 + c}{b} \geqslant \frac{c^2 + c}{c} > c$$

Hence from the solution (b, c) we obtain another one (c, b_0) with $b_0 > c$, and we can then "jump" again, this time with c as the "variable" in the quadratic $(*)$. This algorithm will generate an infinite sequence of distinct solutions, whose first terms are

$$(2, 3),\ (3, 6),\ (6, 14),\ (14, 35),\ (35, 90),\ (90, 234),\ (234, 611),\ (611, 1598),\ (1598, 4182), \ldots$$

Comment. Although not needed for solving this problem, we may also explicitly solve the recursion given by the Vieta jumping. Define the sequence (x_n) as follows:

$$x_0 = 2, \quad x_1 = 3 \qquad \text{and} \qquad x_{n+2} = 3x_{n+1} - x_n - 1 \text{ for } n \geqslant 0$$

Then the triple

$$(x, y, z) = \left(\frac{1}{1 + x_n + x_{n+1}}, \frac{x_n}{1 + x_n + x_{n+1}}, \frac{x_{n+1}}{1 + x_n + x_{n+1}} \right)$$

satisfies the problem conditions for all $n \in \mathbb{N}$. It is easy to show that $x_n = F_{2n+1} + 1$, where F_n denotes the n-th term of the Fibonacci sequence ($F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for $n \geqslant 0$).

Solution 2. Call the n-tuples $(a_1, a_2, \ldots, a_n) \in \mathbb{Q}_{>0}^n$ satisfying the conditions of the problem statement *good*, and those for which

$$f(a_1, \ldots, a_n) \stackrel{\text{def}}{=} (a_1 + a_2 + \cdots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \right)$$

is an integer *pretty*. Then good n-tuples are pretty, and if $(b_1, \ldots, b_n)$ is pretty then

$$\left(\frac{b_1}{b_1 + b_2 + \cdots + b_n}, \frac{b_2}{b_1 + b_2 + \cdots + b_n}, \ldots, \frac{b_n}{b_1 + b_2 + \cdots + b_n} \right)$$

is good since the sum of its components is 1, and the sum of the reciprocals of its components equals $f(b_1, \ldots, b_n)$. We declare pretty n-tuples proportional to each other *equivalent* since they are precisely those which give rise to the same good n-tuple. Clearly, each such equivalence class contains exactly one n-tuple of positive integers having no common prime divisors. Call such n-tuple a *primitive* pretty tuple. Our task is to find infinitely many primitive pretty n-tuples.

For $n = 1$, there is clearly a single primitive 1-tuple. For $n = 2$, we have $f(a, b) = \frac{(a+b)^2}{ab}$, which can be integral (for coprime $a, b \in \mathbb{Z}_{>0}$) only if $a = b = 1$ (see for instance (i) in the first solution).

Now we construct infinitely many primitive pretty triples for $n = 3$. Fix $b, c, k \in \mathbb{Z}_{>0}$; we will try to find sufficient conditions for the existence of an $a \in \mathbb{Q}_{>0}$ such that $f(a, b, c) = k$. Write $\sigma = b + c$, $\tau = bc$. From $f(a, b, c) = k$, we have that a should satisfy the quadratic equation

$$a^2 \cdot \sigma + a \cdot (\sigma^2 - (k-1)\tau) + \sigma\tau = 0 \tag{1}$$

whose discriminant is

$$\Delta = (\sigma^2 - (k-1)\tau)^2 - 4\sigma^2\tau = ((k+1)\tau - \sigma^2)^2 - 4k\tau^2.$$

We need it to be a square of an integer, say, $\Delta = M^2$ for some $M \in \mathbb{Z}$, i.e., we want

$$((k+1)\tau - \sigma^2)^2 - M^2 = 2k \cdot 2\tau^2$$

so that it suffices to set

$$(k+1)\tau - \sigma^2 = \tau^2 + k, \qquad M = \tau^2 - k.$$

The first relation reads $\sigma^2 = (\tau - 1)(k - \tau)$, so if b and c satisfy

$$\tau - 1 \mid \sigma^2 \qquad \text{i.e.} \qquad bc - 1 \mid (b + c)^2 \tag{2}$$

then $k = \frac{\sigma^2}{\tau - 1} + \tau$ will be integral, and we find rational solutions to (1), namely

$$a = \frac{\sigma}{\tau - 1} = \frac{b + c}{bc - 1} \qquad \text{or} \qquad a = \frac{\tau^2 - \tau}{\sigma} = \frac{bc \cdot (bc - 1)}{b + c}$$

We can now find infinitely many pairs (b, c) satisfying (2) by Vieta jumping. For example, if we impose

$$(b + c)^2 = 5 \cdot (bc - 1)$$

then all pairs $(b, c) = (v_i, v_{i+1})$ satisfy the above condition, where

$$v_1 = 2,\ v_2 = 3, \qquad v_{i+2} = 3v_{i+1} - v_i \quad \text{for } i \geqslant 0$$

For $(b, c) = (v_i, v_{i+1})$, one of the solutions to (1) will be $a = (b + c)/(bc - 1) = 5/(b + c) = 5/(v_i + v_{i+1})$. Then the pretty triple (a, b, c) will be equivalent to the integral pretty triple

$$(5, v_i(v_i + v_{i+1}), v_{i+1}(v_i + v_{i+1}))$$

After possibly dividing by 5, we obtain infinitely many primitive pretty triples, as required.

Comment. There are many other infinite series of $(b, c) = (v_i, v_{i+1})$ with $bc - 1 \mid (b + c)^2$. Some of them are:

$$\begin{array}{llll} v_1 = 1, & v_2 = 3, & v_{i+1} = 6v_i - v_{i-1}, & (v_i + v_{i+1})^2 = 8 \cdot (v_i v_{i+1} - 1); \\ v_1 = 1, & v_2 = 2, & v_{i+1} = 7v_i - v_{i-1}, & (v_i + v_{i+1})^2 = 9 \cdot (v_i v_{i+1} - 1); \\ v_1 = 1, & v_2 = 5, & v_{i+1} = 7v_i - v_{i-1}, & (v_i + v_{i+1})^2 = 9 \cdot (v_i v_{i+1} - 1) \end{array}$$

(the last two are in fact one sequence prolonged in two possible directions).

N7. Say that an ordered pair (x, y) of integers is an *irreducible lattice point* if x and y are relatively prime. For any finite set S of irreducible lattice points, show that there is a homogenous polynomial in two variables, $f(x, y)$, with integer coefficients, of degree at least 1, such that $f(x, y) = 1$ for each (x, y) in the set S.

Note: A homogenous polynomial of degree n is any nonzero polynomial of the form

$$f(x, y) = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \cdots + a_{n-1}xy^{n-1} + a_ny^n.$$

(U.S.A.)

Solution 1. First of all, we note that finding a homogenous polynomial $f(x, y)$ such that $f(x, y) = \pm 1$ is enough, because we then have $f^2(x, y) = 1$. Label the irreducible lattice points (x_1, y_1) through (x_n, y_n). If any two of these lattice points (x_i, y_i) and (x_j, y_j) lie on the same line through the origin, then $(x_j, y_j) = (-x_i, -y_i)$ because both of the points are irreducible. We then have $f(x_j, y_j) = \pm f(x_i, y_i)$ whenever f is homogenous, so we can assume that no two of the lattice points are collinear with the origin by ignoring the extra lattice points.

Consider the homogenous polynomials $\ell_i(x, y) = y_ix - x_iy$ and define

$$g_i(x, y) = \prod_{j\neq i} \ell_j(x, y).$$

Then $\ell_i(x_j, y_j) = 0$ if and only if $j = i$, because there is only one lattice point on each line through the origin. Thus, $g_i(x_j, y_j) = 0$ for all $j \neq i$. Define $a_i = g_i(x_i, y_i)$, and note that $a_i \neq 0$.

Note that $g_i(x, y)$ is a degree $n - 1$ polynomial with the following two properties:

1. $g_i(x_j, y_j) = 0$ if $j \neq i$.
2. $g_i(x_i, y_i) = a_i$.

For any $N \geqslant n - 1$, there also exists a polynomial of degree N with the same two properties. Specifically, let $I_i(x, y)$ be a degree 1 homogenous polynomial such that $I_i(x_i, y_i) = 1$, which exists since (x_i, y_i) is irreducible. Then $I_i(x, y)^{N-(n-1)}g_i(x, y)$ satisfies both of the above properties and has degree N.

We may now reduce the problem to the following claim:

Claim: For each positive integer a, there is a homogenous polynomial $f_a(x, y)$, with integer coefficients, of degree at least 1, such that $f_a(x, y) \equiv 1 \pmod a$ for all relatively prime (x, y).

To see that this claim solves the problem, take a to be the least common multiple of the numbers a_i $(1 \leqslant i \leqslant n)$. Take f_a given by the claim, choose some power $f_a(x, y)^k$ that has degree at least $n - 1$, and subtract appropriate multiples of the g_i constructed above to obtain the desired polynomial.

We prove the claim by factoring a. First, if a is a power of a prime $(a = p^k)$, then we may choose either:

- $f_a(x, y) = (x^{p-1} + y^{p-1})^{\phi(a)}$ if p is odd;
- $f_a(x, y) = (x^2 + xy + y^2)^{\phi(a)}$ if $p = 2$.

Now suppose a is any positive integer, and let $a = q_1q_2 \cdots q_k$, where the q_i are prime powers, pairwise relatively prime. Let f_{q_i} be the polynomials just constructed, and let F_{q_i} be powers of these that all have the same degree. Note that

$$\frac{a}{q_i}F_{q_i}(x, y) \equiv \frac{a}{q_i} \pmod a$$

for any relatively prime x, y. By Bézout's lemma, there is an integer linear combination of the $\frac{a}{q_i}$ that equals 1. Thus, there is a linear combination of the F_{q_i} such that $F_{q_i}(x, y) \equiv 1 \pmod a$ for any relatively prime (x, y); and this polynomial is homogenous because all the F_{q_i} have the same degree.

Solution 2. As in the previous solution, label the irreducible lattice points $(x_1, y_1), \ldots, (x_n, y_n)$ and assume without loss of generality that no two of the points are collinear with the origin. We induct on n to construct a homogenous polynomial $f(x, y)$ such that $f(x_i, y_i) = 1$ for all $1 \leqslant i \leqslant n$.

If $n = 1$: Since x_1 and y_1 are relatively prime, there exist some integers c, d such that $cx_1 + dy_1 = 1$. Then $f(x, y) = cx + dy$ is suitable.

If $n \geqslant 2$: By the induction hypothesis we already have a homogeneous polynomial $g(x, y)$ with $g(x_1, y_1) = \ldots = g(x_{n-1}, y_{n-1}) = 1$. Let $j = \deg g$,

$$g_n(x, y) = \prod_{k=1}^{n-1} (y_k x - x_k y),$$

and $a_n = g_n(x_n, y_n)$. By assumption, $a_n \neq 0$. Take some integers c, d such that $cx_n + dy_n = 1$. We will construct $f(x, y)$ in the form

$$f(x, y) = g(x, y)^K - C \cdot g_n(x, y) \cdot (cx + dy)^L,$$

where K and L are some positive integers and C is some integer. We assume that $L = Kj - n + 1$ so that f is homogenous.

Due to $g(x_1, y_1) = \ldots = g(x_{n-1}, y_{n-1}) = 1$ and $g_n(x_1, y_1) = \ldots = g_n(x_{n-1}, y_{n-1}) = 0$, the property $f(x_1, y_1) = \ldots = f(x_{n-1}, y_{n-1}) = 1$ is automatically satisfied with any choice of K, L, and C.

Furthermore,

$$f(x_n, y_n) = g(x_n, y_n)^K - C \cdot g_n(x_n, y_n) \cdot (cx_n + dy_n)^L = g(x_n, y_n)^K - Ca_n.$$

If we have an exponent K such that $g(x_n, y_n)^K \equiv 1 \pmod{a_n}$, then we may choose C such that $f(x_n, y_n) = 1$. We now choose such a K.

Consider an arbitrary prime divisor p of a_n. By

$$p \mid a_n = g_n(x_n, y_n) = \prod_{k=1}^{n-1} (y_k x_n - x_k y_n),$$

there is some $1 \leqslant k < n$ such that $x_k y_n \equiv x_n y_k \pmod{p}$. We first show that $x_k x_n$ or $y_k y_n$ is relatively prime with p. This is trivial in the case $x_k y_n \equiv x_n y_k \not\equiv 0 \pmod{p}$. In the other case, we have $x_k y_n \equiv x_n y_k \equiv 0 \pmod{p}$, If, say $p \mid x_k$, then $p \nmid y_k$ because (x_k, y_k) is irreducible, so $p \mid x_n$; then $p \nmid y_n$ because (x_k, y_k) is irreducible. In summary, $p \mid x_k$ implies $p \nmid y_k y_n$. Similarly, $p \mid y_n$ implies $p \nmid x_k x_n$.

By the homogeneity of g we have the congruences

$$x_k^d \cdot g(x_n, y_n) = g(x_k x_n, x_k y_n) \equiv g(x_k x_n, y_k x_n) = x_n^d \cdot g(x_k, y_k) = x_n^d \pmod{p} \tag{1.1}$$

and

$$y_k^d \cdot g(x_n, y_n) = g(y_k x_n, y_k y_n) \equiv g(x_k y_n, y_k y_n) = y_n^d \cdot g(x_k, y_k) = y_n^d \pmod{p}. \tag{1.2}$$

If $p \nmid x_k x_n$, then take the $(p-1)^{st}$ power of (1.1); otherwise take the $(p-1)^{st}$ power of (1.2); by Fermat's theorem, in both cases we get

$$g(x_n, y_n)^{p-1} \equiv 1 \pmod{p}.$$

If $p^\alpha \mid m$, then we have

$$g(x_n, y_n)^{p^{\alpha-1}(p-1)} \equiv 1 \pmod{p^\alpha},$$

which implies that the exponent $K = n \cdot \varphi(a_n)$, which is a multiple of all $p^{\alpha-1}(p-1)$, is a suitable choice. (The factor n is added only so that $K \geqslant n$ and so $L > 0$.)

Comment. It is possible to show that there is no constant C for which, given any two irreducible lattice points, there is some homogenous polynomial f of degree at most C with integer coefficients that takes the value 1 on the two points. Indeed, if one of the points is $(1,0)$ and the other is (a,b), the polynomial $f(x,y) = a_0x^n + a_1x^{n-1}y + \cdots + a_ny^n$ should satisfy $a_0 = 1$, and so $a^n \equiv 1 \pmod{b}$. If $a = 3$ and $b = 2^k$ with $k \geqslant 3$, then $n \geqslant 2^{k-2}$. If we choose $2^{k-2} > C$, this gives a contradiction.

N8. Let p be an odd prime number and $\mathbb{Z}_{>0}$ be the set of positive integers. Suppose that a function $f\colon \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} \to \{0,1\}$ satisfies the following properties:

- $f(1,1) = 0$;
- $f(a,b) + f(b,a) = 1$ for any pair of relatively prime positive integers (a,b) not both equal to 1;
- $f(a+b,b) = f(a,b)$ for any pair of relatively prime positive integers (a,b).

Prove that

$$\sum_{n=1}^{p-1} f(n^2,p) \geqslant \sqrt{2p} - 2.$$

(Italy)

Solution 1. Denote by $\mathbb{A}$ the set of all pairs of coprime positive integers. Notice that for every $(a,b) \in \mathbb{A}$ there exists a pair $(u,v) \in \mathbb{Z}^2$ with $ua + vb = 1$. Moreover, if (u_0,v_0) is one such pair, then all such pairs are of the form $(u,v) = (u_0 + kb, v_0 - ka)$, where $k \in \mathbb{Z}$. So there exists a unique such pair (u,v) with $-b/2 < u \leqslant b/2$; we denote this pair by $(u,v) = g(a,b)$.

Lemma. Let $(a,b) \in \mathbb{A}$ and $(u,v) = g(a,b)$. Then $f(a,b) = 1 \iff u > 0$.

Proof. We induct on $a+b$. The base case is $a+b=2$. In this case, we have that $a = b = 1$, $g(a,b) = g(1,1) = (0,1)$ and $f(1,1) = 0$, so the claim holds.

Assume now that $a+b > 2$, and so $a \neq b$, since a and b are coprime. Two cases are possible.

Case 1: $a > b$.

Notice that $g(a-b,b) = (u, v+u)$, since $u(a-b) + (v+u)b = 1$ and $u \in (-b/2, b/2]$. Thus $f(a,b) = 1 \iff f(a-b,b) = 1 \iff u > 0$ by the induction hypothesis.

Case 2: $a < b$. (Then, clearly, $b \geqslant 2$.)

Now we estimate v. Since $vb = 1 - ua$, we have

$$1 + \frac{ab}{2} > vb \geqslant 1 - \frac{ab}{2}, \quad \text{so} \quad \frac{1+a}{2} \geqslant \frac{1}{b} + \frac{a}{2} > v \geqslant \frac{1}{b} - \frac{a}{2} > -\frac{a}{2}.$$

Thus $1 + a > 2v > -a$, so $a \geqslant 2v > -a$, hence $a/2 \geqslant v > -a/2$, and thus $g(b,a) = (v,u)$.

Observe that $f(a,b) = 1 \iff f(b,a) = 0 \iff f(b-a,a) = 0$. We know from Case 1 that $g(b-a,a) = (v, u+v)$. We have $f(b-a,a) = 0 \iff v \leqslant 0$ by the inductive hypothesis. Then, since $b > a \geqslant 1$ and $ua + vb = 1$, we have $v \leqslant 0 \iff u > 0$, and we are done. □

The Lemma proves that, for all $(a,b) \in \mathbb{A}$, $f(a,b) = 1$ if and only if the inverse of a modulo b, taken in $\{1,2,\ldots,b-1\}$, is at most $b/2$. Then, for any odd prime p and integer n such that $n \not\equiv 0 \pmod p$, $f(n^2,p) = 1$ iff the inverse of $n^2 \bmod p$ is less than $p/2$. Since $\{n^2 \bmod p\colon 1 \leqslant n \leqslant p-1\} = \{n^{-2} \bmod p\colon 1 \leqslant n \leqslant p-1\}$, including multiplicities (two for each quadratic residue in each set), we conclude that the desired sum is twice the number of quadratic residues that are less than $p/2$, i.e.,

$$\sum_{n=1}^{p-1} f(n^2,p) = 2\left|\left\{k\colon 1 \leqslant k \leqslant \frac{p-1}{2} \text{ and } k^2 \bmod p < \frac{p}{2}\right\}\right|. \tag{1}$$

Since the number of perfect squares in the interval $[1, p/2)$ is $\lfloor\sqrt{p/2}\rfloor > \sqrt{p/2} - 1$, we conclude that

$$\sum_{n=1}^{p-1} f(n^2,p) > 2\left(\sqrt{\frac{p}{2}} - 1\right) = \sqrt{2p} - 2.$$

Solution 2. We provide a different proof for the Lemma. For this purpose, we use continued fractions to find $g(a,b)=(u,v)$ explicitly.

The function f is completely determined on $\mathbb{A}$ by the following

Claim. Represent a/b as a continued fraction; that is, let a_0 be an integer and $a_1,\dots,a_k$ be positive integers such that $a_k\geqslant 2$ and

$$\frac{a}{b}=a_0+\cfrac{1}{a_1+\cfrac{1}{a_2+\cfrac{1}{\dots+\cfrac{1}{a_k}}}}=[a_0;a_1,a_2,\dots,a_k].$$

Then $f(a,b)=0 \iff k$ is even.

Proof. We induct on b. If $b=1$, then $a/b=[a]$ and $k=0$. Then, for $a\geqslant 1$, an easy induction shows that $f(a,1)=f(1,1)=0$.

Now consider the case $b>1$. Perform the Euclidean division $a=qb+r$, with $0\leqslant r<b$. We have $r\neq 0$ because $\gcd(a,b)=1$. Hence

$$f(a,b)=f(r,b)=1-f(b,r),\quad \frac{a}{b}=[q;a_1,\dots,a_k],\quad \text{and}\quad \frac{b}{r}=[a_1;a_2,\dots,a_k].$$

Then the number of terms in the continued fraction representations of a/b and b/r differ by one. Since $r<b$, the inductive hypothesis yields

$$f(b,r)=0 \iff k-1 \text{ is even},$$

and thus

$$f(a,b)=0 \iff f(b,r)=1 \iff k-1 \text{ is odd} \iff k \text{ is even}. \qquad \square$$

Now we use the following well-known properties of continued fractions to prove the Lemma:

Let p_i and q_i be coprime positive integers with $[a_0;a_1,a_2,\dots,a_i]=p_i/q_i$, with the notation borrowed from the Claim. In particular, $a/b=[a_0;a_1,a_2,\dots,a_k]=p_k/q_k$. Assume that $k>0$ and define $q_{-1}=0$ if necessary. Then

- $q_k=a_kq_{k-1}+q_{k-2}$, and
- $aq_{k-1}-bp_{k-1}=p_kq_{k-1}-q_kp_{k-1}=(-1)^{k-1}$.

Assume that $k>0$. Then $a_k\geqslant 2$, and

$$b=q_k=a_kq_{k-1}+q_{k-2}\geqslant a_kq_{k-1}\geqslant 2q_{k-1} \implies q_{k-1}\leqslant\frac{b}{2},$$

with strict inequality for $k>1$, and

$$(-1)^{k-1}q_{k-1}a+(-1)^kp_{k-1}b=1.$$

Now we finish the proof of the Lemma. It is immediate for $k=0$. If $k=1$, then $(-1)^{k-1}=1$, so

$$-b/2<0\leqslant(-1)^{k-1}q_{k-1}\leqslant b/2.$$

If $k>1$, we have $q_{k-1}<b/2$, so

$$-b/2<(-1)^{k-1}q_{k-1}<b/2.$$

Thus, for any $k>0$, we find that $g(a,b)=((-1)^{k-1}q_{k-1},(-1)^kp_{k-1})$, and so

$$f(a,b)=1 \iff k \text{ is odd} \iff u=(-1)^{k-1}q_{k-1}>0.$$

Comment 1. The Lemma can also be established by observing that f is uniquely defined on $\mathbb{A}$, defining $f_1(a,b) = 1$ if $u > 0$ in $g(a,b) = (u,v)$ and $f_1(a,b) = 0$ otherwise, and verifying that f_1 satisfies all the conditions from the statement.

It seems that the main difficulty of the problem is in conjecturing the Lemma.

Comment 2. The case $p \equiv 1 \pmod 4$ is, in fact, easier than the original problem. We have, in general, for $1 \leqslant a \leqslant p-1$,

$$f(a,p) = 1 - f(p,a) = 1 - f(p-a,a) = f(a,p-a) = f(a+(p-a),p-a) = f(p,p-a) = 1 - f(p-a,p).$$

If $p \equiv 1 \pmod 4$, then a is a quadratic residue modulo p if and only if $p-a$ is a quadratic residue modulo p. Therefore, denoting by r_k (with $1 \leqslant r_k \leqslant p-1$) the remainder of the division of k^2 by p, we get

$$\sum_{n=1}^{p-1} f(n^2,p) = \sum_{n=1}^{p-1} f(r_n,p) = \frac{1}{2}\sum_{n=1}^{p-1}(f(r_n,p) + f(p-r_n,p)) = \frac{p-1}{2}.$$

Comment 3. The estimate for the sum $\sum_{n=1}^{p} f(n^2,p)$ can be improved by refining the final argument in Solution 1. In fact, one can prove that

$$\sum_{n=1}^{p-1} f(n^2,p) \geqslant \frac{p-1}{16}.$$

By counting the number of perfect squares in the intervals $[kp,(k+1/2)p)$, we find that

$$\sum_{n=1}^{p-1} f(n^2,p) = \sum_{k=0}^{p-1}\left(\left\lfloor\sqrt{\left(k+\frac{1}{2}\right)p}\right\rfloor - \left\lfloor\sqrt{kp}\right\rfloor\right). \tag{2}$$

Each summand of (2) is non-negative. We now estimate the number of positive summands. Suppose that a summand is zero, i.e.,

$$\left\lfloor\sqrt{\left(k+\frac{1}{2}\right)p}\right\rfloor = \left\lfloor\sqrt{kp}\right\rfloor =: q.$$

Then both of the numbers kp and $kp+p/2$ lie within the interval $[q^2,(q+1)^2)$. Hence

$$\frac{p}{2} < (q+1)^2 - q^2,$$

which implies

$$q \geqslant \frac{p-1}{4}.$$

Since $q \leqslant \sqrt{kp}$, if the kth summand of (2) is zero, then

$$k \geqslant \frac{q^2}{p} \geqslant \frac{(p-1)^2}{16p} > \frac{p-2}{16} \implies k \geqslant \frac{p-1}{16}.$$

So at least the first $\lceil\frac{p-1}{16}\rceil$ summands (from $k=0$ to $k=\lceil\frac{p-1}{16}\rceil - 1$) are positive, and the result follows.

Comment 4. The bound can be further improved by using different methods. In fact, we prove that

$$\sum_{n=1}^{p-1} f(n^2,p) \geqslant \frac{p-3}{4}.$$

To that end, we use the Legendre symbol

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a \\ 1 & \text{if } a \text{ is a nonzero quadratic residue mod } p \\ -1 & \text{otherwise.} \end{cases}$$

We start with the following Claim, which tells us that there are not too many consecutive quadratic residues or consecutive quadratic non-residues.

Claim. $\sum_{n=1}^{p-1}\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1$.

Proof. We have $\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = \left(\frac{n(n+1)}{p}\right)$. For $1 \leqslant n \leqslant p-1$, we get that $n(n+1) \equiv n^2(1+n^{-1}) \pmod p$, hence $\left(\frac{n(n+1)}{p}\right) = \left(\frac{1+n^{-1}}{p}\right)$. Since $\{1+n^{-1} \bmod p\colon 1 \leqslant n \leqslant p-1\} = \{0, 2, 3, \ldots, p-1 \bmod p\}$, we find

$$\sum_{n=1}^{p-1}\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = \sum_{n=1}^{p-1}\left(\frac{1+n^{-1}}{p}\right) = \sum_{n=1}^{p-1}\left(\frac{n}{p}\right) - 1 = -1,$$

because $\sum_{n=1}^{p}\left(\frac{n}{p}\right) = 0$. □

Observe that (1) becomes

$$\sum_{n=1}^{p-1} f(n^2, p) = 2\,|S|\,, \quad S = \left\{r\colon 1 \leqslant r \leqslant \frac{p-1}{2} \text{ and } \left(\frac{r}{p}\right) = 1\right\}.$$

We connect S with the sum from the claim by pairing quadratic residues and quadratic non-residues. To that end, define

$$S' = \left\{r\colon 1 \leqslant r \leqslant \frac{p-1}{2} \text{ and } \left(\frac{r}{p}\right) = -1\right\}$$
$$T = \left\{r\colon \frac{p+1}{2} \leqslant r \leqslant p-1 \text{ and } \left(\frac{r}{p}\right) = 1\right\}$$
$$T' = \left\{r\colon \frac{p+1}{2} \leqslant r \leqslant p-1 \text{ and } \left(\frac{r}{p}\right) = -1\right\}$$

Since there are exactly $(p-1)/2$ nonzero quadratic residues modulo p, $|S| + |T| = (p-1)/2$. Also we obviously have $|T| + |T'| = (p-1)/2$. Then $|S| = |T'|$.

For the sake of brevity, define $t = |S| = |T'|$. If $\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1$, then exactly of one the numbers $\left(\frac{n}{p}\right)$ and $\left(\frac{n+1}{p}\right)$ is equal to 1, so

$$\left|\left\{n\colon 1 \leqslant n \leqslant \frac{p-3}{2} \text{ and } \left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1\right\}\right| \leqslant |S| + |S-1| = 2t.$$

On the other hand, if $\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1$, then exactly one of $\left(\frac{n}{p}\right)$ and $\left(\frac{n+1}{p}\right)$ is equal to -1, and

$$\left|\left\{n\colon \frac{p+1}{2} \leqslant n \leqslant p-2 \text{ and } \left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1\right\}\right| \leqslant |T'| + |T'-1| = 2t.$$

Thus, taking into account that the middle term $\left(\frac{(p-1)/2}{p}\right)\left(\frac{(p+1)/2}{p}\right)$ may happen to be -1,

$$\left|\left\{n\colon 1 \leqslant n \leqslant p-2 \text{ and } \left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = -1\right\}\right| \leqslant 4t+1.$$

This implies that

$$\left|\left\{n\colon 1 \leqslant n \leqslant p-2 \text{ and } \left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) = 1\right\}\right| \geqslant (p-2) - (4t+1) = p - 4t - 3,$$

and so

$$-1 = \sum_{n=1}^{p-1}\left(\frac{n}{p}\right)\left(\frac{n+1}{p}\right) \geqslant p - 4t - 3 - (4t+1) = p - 8t - 4,$$

which implies $8t \geqslant p-3$, and thus

$$\sum_{n=1}^{p-1} f(n^2, p) = 2t \geqslant \frac{p-3}{4}.$$

Comment 5. It is possible to prove that

$$\sum_{n=1}^{p-1} f(n^2, p) \geqslant \frac{p-1}{2}.$$

The case $p \equiv 1 \pmod 4$ was already mentioned, and it is the equality case. If $p \equiv 3 \pmod 4$, then, by a theorem of Dirichlet, we have

$$\left|\left\{r \colon 1 \leqslant r \leqslant \frac{p-1}{2} \text{ and } \left(\frac{r}{p}\right) = 1\right\}\right| > \frac{p-1}{4},$$

which implies the result.

See https://en.wikipedia.org/wiki/Quadratic_residue#Dirichlet.27s_formulas for the full statement of the theorem. It seems that no elementary proof of it is known; a proof using complex analysis is available, for instance, in Chapter 7 of the book *Quadratic Residues and Non-Residues: Selected Topics*, by Steve Wright, available in https://arxiv.org/abs/1408.0235.





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Problems

Algebra

A1. Let a, b and c be positive real numbers such that $\min\{ab, bc, ca\} \geqslant 1$. Prove that

$$\sqrt[3]{(a^2+1)(b^2+1)(c^2+1)} \leqslant \left(\frac{a+b+c}{3}\right)^2 + 1.$$

A2. Find the smallest real constant C such that for any positive real numbers a_1, a_2, a_3, a_4 and a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k and l such that

$$\left|\frac{a_i}{a_j} - \frac{a_k}{a_l}\right| \leqslant C.$$

A3. Find all integers $n \geqslant 3$ with the following property: for all real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ satisfying $|a_k| + |b_k| = 1$ for $1 \leqslant k \leqslant n$, there exist $x_1, x_2, \ldots, x_n$, each of which is either -1 or 1, such that

$$\left|\sum_{k=1}^{n} x_k a_k\right| + \left|\sum_{k=1}^{n} x_k b_k\right| \leqslant 1.$$

A4. Denote by $\mathbb{R}^+$ the set of all positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)\left(f(f(x^2)) + f(f(y^2))\right)$$

for all positive real numbers x and y.

A5.

(a) Prove that for every positive integer n, there exists a fraction $\frac{a}{b}$ where a and b are integers satisfying $0 < b \leqslant \sqrt{n}+1$ and $\sqrt{n} \leqslant \frac{a}{b} \leqslant \sqrt{n+1}$.

(b) Prove that there are infinitely many positive integers n such that there is no fraction $\frac{a}{b}$ where a and b are integers satisfying $0 < b \leqslant \sqrt{n}$ and $\sqrt{n} \leqslant \frac{a}{b} \leqslant \sqrt{n+1}$.

A6. The equation

$$(x-1)(x-2)\cdots(x-2016)=(x-1)(x-2)\cdots(x-2016)$$

is written on the board. One tries to erase some linear factors from both sides so that each side still has at least one factor, and the resulting equation has no real roots. Find the least number of linear factors one needs to erase to achieve this.

A7. Denote by $\mathbb{R}$ the set of all real numbers. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that $f(0)\neq 0$ and

$$f(x+y)^2=2f(x)f(y)+\max\{f(x^2)+f(y^2),f(x^2+y^2)\}$$

for all real numbers x and y.

A8. Determine the largest real number a such that for all $n\geqslant 1$ and for all real numbers $x_0,x_1,\ldots,x_n$ satisfying $0=x_0<x_1<x_2<\cdots<x_n$, we have

$$\frac{1}{x_1-x_0}+\frac{1}{x_2-x_1}+\cdots+\frac{1}{x_n-x_{n-1}}\geqslant a\left(\frac{2}{x_1}+\frac{3}{x_2}+\cdots+\frac{n+1}{x_n}\right).$$

Combinatorics

C1. The leader of an IMO team chooses positive integers n and k with $n > k$, and announces them to the deputy leader and a contestant. The leader then secretly tells the deputy leader an n-digit binary string, and the deputy leader writes down all n-digit binary strings which differ from the leader's in exactly k positions. (For example, if $n = 3$ and $k = 1$, and if the leader chooses 101, the deputy leader would write down 001, 111 and 100.) The contestant is allowed to look at the strings written by the deputy leader and guess the leader's string. What is the minimum number of guesses (in terms of n and k) needed to guarantee the correct answer?

C2. Find all positive integers n for which all positive divisors of n can be put into the cells of a rectangular table under the following constraints:

- each cell contains a distinct divisor;
- the sums of all rows are equal; and
- the sums of all columns are equal.

C3. Let n be a positive integer relatively prime to 6. We paint the vertices of a regular n-gon with three colours so that there is an odd number of vertices of each colour. Show that there exists an isosceles triangle whose three vertices are of different colours.

C4. Find all positive integers n for which we can fill in the entries of an $n \times n$ table with the following properties:

- each entry can be one of I, M and O;
- in each row and each column, the letters I, M and O occur the same number of times; and
- in any diagonal whose number of entries is a multiple of three, the letters I, M and O occur the same number of times.

C5. Let $n \geqslant 3$ be a positive integer. Find the maximum number of diagonals of a regular n-gon one can select, so that any two of them do not intersect in the interior or they are perpendicular to each other.

C6. There are $n \geqslant 3$ islands in a city. Initially, the ferry company offers some routes between some pairs of islands so that it is impossible to divide the islands into two groups such that no two islands in different groups are connected by a ferry route.

After each year, the ferry company will close a ferry route between some two islands X and Y. At the same time, in order to maintain its service, the company will open new routes according to the following rule: for any island which is connected by a ferry route to exactly one of X and Y, a new route between this island and the other of X and Y is added.

Suppose at any moment, if we partition all islands into two nonempty groups in any way, then it is known that the ferry company will close a certain route connecting two islands from the two groups after some years. Prove that after some years there will be an island which is connected to all other islands by ferry routes.

C7. Let $n \geqslant 2$ be an integer. In the plane, there are n segments given in such a way that any two segments have an intersection point in the interior, and no three segments intersect at a single point. Jeff places a snail at one of the endpoints of each of the segments and claps his hands $n-1$ times. Each time when he claps his hands, all the snails move along their own segments and stay at the next intersection points until the next clap. Since there are $n-1$ intersection points on each segment, all snails will reach the furthest intersection points from their starting points after $n-1$ claps.

(a) Prove that if n is odd then Jeff can always place the snails so that no two of them ever occupy the same intersection point.

(b) Prove that if n is even then there must be a moment when some two snails occupy the same intersection point no matter how Jeff places the snails.

C8. Let n be a positive integer. Determine the smallest positive integer k with the following property: it is possible to mark k cells on a $2n \times 2n$ board so that there exists a unique partition of the board into 1×2 and 2×1 dominoes, none of which contains two marked cells.

Geometry

G1. In a convex pentagon $ABCDE$, let F be a point on AC such that $\angle FBC = 90°$. Suppose triangles ABF, ACD and ADE are similar isosceles triangles with

$$\angle FAB = \angle FBA = \angle DAC = \angle DCA = \angle EAD = \angle EDA.$$

Let M be the midpoint of CF. Point X is chosen such that $AMXE$ is a parallelogram. Show that BD, EM and FX are concurrent.

G2. Let ABC be a triangle with circumcircle Γ and incentre I. Let M be the midpoint of side BC. Denote by D the foot of perpendicular from I to side BC. The line through I perpendicular to AI meets sides AB and AC at F and E respectively. Suppose the circumcircle of triangle AEF intersects Γ at a point X other than A. Prove that lines XD and AM meet on Γ.

G3. Let $B = (-1, 0)$ and $C = (1, 0)$ be fixed points on the coordinate plane. A nonempty, bounded subset S of the plane is said to be *nice* if

(i) there is a point T in S such that for every point Q in S, the segment TQ lies entirely in S; and

(ii) for any triangle $P_1P_2P_3$, there exists a unique point A in S and a permutation σ of the indices $\{1, 2, 3\}$ for which triangles ABC and $P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$ are similar.

Prove that there exist two distinct nice subsets S and S' of the set $\{(x, y) : x \geqslant 0, y \geqslant 0\}$ such that if $A \in S$ and $A' \in S'$ are the unique choices of points in (ii), then the product $BA \cdot BA'$ is a constant independent of the triangle $P_1P_2P_3$.

G4. Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D, and the line through D perpendicular to AC meets AI at E. Prove that the reflection of I in AC lies on the circumcircle of triangle BDE.

G5. Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC. A circle ω with centre S passes through A and D, and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC, and let M be the midpoint of BC. Prove that the circumcentre of triangle XSY is equidistant from P and M.

G6. Let $ABCD$ be a convex quadrilateral with $\angle ABC = \angle ADC < 90°$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at E and F respectively, and meet each other at point P. Let M be the midpoint of AC and let ω be the circumcircle of triangle BPD. Segments BM and DM intersect ω again at X and Y respectively. Denote by Q the intersection point of lines XE and YF. Prove that $PQ \perp AC$.

G7. Let I be the incentre of a non-equilateral triangle ABC, I_A be the A-excentre, I'_A be the reflection of I_A in BC, and l_A be the reflection of line AI'_A in AI. Define points I_B, I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B.

(a) Prove that P lies on line OI where O is the circumcentre of triangle ABC.

(b) Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y. Show that $\angle XIY = 120°$.

G8. Let A_1, B_1 and C_1 be points on sides BC, CA and AB of an acute triangle ABC respectively, such that AA_1, BB_1 and CC_1 are the internal angle bisectors of triangle ABC. Let I be the incentre of triangle ABC, and H be the orthocentre of triangle $A_1B_1C_1$. Show that

$$AH + BH + CH \geqslant AI + BI + CI.$$

Number Theory

N1. For any positive integer k, denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer $n \geqslant 2016$, the integer $P(n)$ is positive and

$$S(P(n)) = P(S(n)).$$

N2. Let $\tau(n)$ be the number of positive divisors of n. Let $\tau_1(n)$ be the number of positive divisors of n which have remainders 1 when divided by 3. Find all possible integral values of the fraction $\frac{\tau(10n)}{\tau_1(10n)}$.

N3. Define $P(n) = n^2 + n + 1$. For any positive integers a and b, the set

$$\{P(a), P(a+1), P(a+2), \ldots, P(a+b)\}$$

is said to be *fragrant* if none of its elements is relatively prime to the product of the other elements. Determine the smallest size of a fragrant set.

N4. Let n, m, k and l be positive integers with $n \neq 1$ such that $n^k + mn^l + 1$ divides $n^{k+l} - 1$. Prove that

- $m = 1$ and $l = 2k$; or
- $l|k$ and $m = \frac{n^{k-l}-1}{n^l-1}$.

N5. Let a be a positive integer which is not a square number. Denote by A the set of all positive integers k such that

$$k = \frac{x^2 - a}{x^2 - y^2} \tag{1}$$

for some integers x and y with $x > \sqrt{a}$. Denote by B the set of all positive integers k such that (1) is satisfied for some integers x and y with $0 \leqslant x < \sqrt{a}$. Prove that $A = B$.

N6. Denote by $\mathbb{N}$ the set of all positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all positive integers m and n, the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

N7. Let n be an odd positive integer. In the Cartesian plane, a cyclic polygon P with area S is chosen. All its vertices have integral coordinates, and all squares of its side lengths are divisible by n. Prove that $2S$ is an integer divisible by n.

N8. Find all polynomials $P(x)$ of odd degree d and with integer coefficients satisfying the following property: for each positive integer n, there exist n positive integers $x_1, x_2, \ldots, x_n$ such that $\frac{1}{2} < \frac{P(x_i)}{P(x_j)} < 2$ and $\frac{P(x_i)}{P(x_j)}$ is the d-th power of a rational number for every pair of indices i and j with $1 \leqslant i, j \leqslant n$.

Solutions

Algebra

A1. Let a, b and c be positive real numbers such that $\min\{ab, bc, ca\} \geqslant 1$. Prove that

$$\sqrt[3]{(a^2+1)(b^2+1)(c^2+1)} \leqslant \left(\frac{a+b+c}{3}\right)^2 + 1. \tag{1}$$

Solution 1. We first show the following.

• **Claim.** For any positive real numbers x, y with $xy \geqslant 1$, we have

$$(x^2+1)(y^2+1) \leqslant \left(\left(\frac{x+y}{2}\right)^2 + 1\right)^2. \tag{2}$$

Proof. Note that $xy \geqslant 1$ implies $(\frac{x+y}{2})^2 - 1 \geqslant xy - 1 \geqslant 0$. We find that

$$(x^2+1)(y^2+1) = (xy-1)^2 + (x+y)^2 \leqslant \left(\left(\frac{x+y}{2}\right)^2 - 1\right)^2 + (x+y)^2 = \left(\left(\frac{x+y}{2}\right)^2 + 1\right)^2.$$

□

Without loss of generality, assume $a \geqslant b \geqslant c$. This implies $a \geqslant 1$. Let $d = \frac{a+b+c}{3}$. Note that

$$ad = \frac{a(a+b+c)}{3} \geqslant \frac{1+1+1}{3} = 1.$$

Then we can apply (2) to the pair (a, d) and the pair (b, c). We get

$$(a^2+1)(d^2+1)(b^2+1)(c^2+1) \leqslant \left(\left(\frac{a+d}{2}\right)^2 + 1\right)^2 \left(\left(\frac{b+c}{2}\right)^2 + 1\right)^2. \tag{3}$$

Next, from

$$\frac{a+d}{2} \cdot \frac{b+c}{2} \geqslant \sqrt{ad} \cdot \sqrt{bc} \geqslant 1,$$

we can apply (2) again to the pair $(\frac{a+d}{2}, \frac{b+c}{2})$. Together with (3), we have

$$(a^2+1)(d^2+1)(b^2+1)(c^2+1) \leqslant \left(\left(\frac{a+b+c+d}{4}\right)^2 + 1\right)^4 = (d^2+1)^4.$$

Therefore, $(a^2+1)(b^2+1)(c^2+1) \leqslant (d^2+1)^3$, and (1) follows by taking cube root of both sides.

Comment. After justifying the Claim, one may also obtain (1) by mixing variables. Indeed, the function involved is clearly continuous, and hence it suffices to check that the condition $xy \geqslant 1$ is preserved under each mixing step. This is true since whenever $ab, bc, ca \geqslant 1$, we have

$$\frac{a+b}{2} \cdot \frac{a+b}{2} \geqslant ab \geqslant 1 \quad \text{and} \quad \frac{a+b}{2} \cdot c \geqslant \frac{1+1}{2} = 1.$$

Solution 2. Let $f(x) = \ln(1+x^2)$. Then the inequality (1) to be shown is equivalent to

$$\frac{f(a)+f(b)+f(c)}{3} \leqslant f\left(\frac{a+b+c}{3}\right),$$

while (2) becomes

$$\frac{f(x)+f(y)}{2} \leqslant f\left(\frac{x+y}{2}\right)$$

for $xy \geqslant 1$.

Without loss of generality, assume $a \geqslant b \geqslant c$. From the Claim in Solution 1, we have

$$\frac{f(a)+f(b)+f(c)}{3} \leqslant \frac{f(a)+2f(\frac{b+c}{2})}{3}.$$

Note that $a \geqslant 1$ and $\frac{b+c}{2} \geqslant \sqrt{bc} \geqslant 1$. Since

$$f''(x) = \frac{2(1-x^2)}{(1+x^2)^2},$$

we know that f is concave on $[1, \infty)$. Then we can apply Jensen's Theorem to get

$$\frac{f(a)+2f(\frac{b+c}{2})}{3} \leqslant f\left(\frac{a+2\cdot\frac{b+c}{2}}{3}\right) = f\left(\frac{a+b+c}{3}\right).$$

This completes the proof.

A2. Find the smallest real constant C such that for any positive real numbers a_1, a_2, a_3, a_4 and a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k and l such that

$$\left|\frac{a_i}{a_j} - \frac{a_k}{a_l}\right| \leqslant C. \tag{1}$$

Answer. The smallest C is $\frac{1}{2}$.

Solution. We first show that $C \leqslant \frac{1}{2}$. For any positive real numbers $a_1 \leqslant a_2 \leqslant a_3 \leqslant a_4 \leqslant a_5$, consider the five fractions

$$\frac{a_1}{a_2}, \frac{a_3}{a_4}, \frac{a_1}{a_5}, \frac{a_2}{a_3}, \frac{a_4}{a_5}. \tag{2}$$

Each of them lies in the interval $(0, 1]$. Therefore, by the Pigeonhole Principle, at least three of them must lie in $(0, \frac{1}{2}]$ or lie in $(\frac{1}{2}, 1]$ simultaneously. In particular, there must be two consecutive terms in (2) which belong to an interval of length $\frac{1}{2}$ (here, we regard $\frac{a_1}{a_2}$ and $\frac{a_4}{a_5}$ as consecutive). In other words, the difference of these two fractions is less than $\frac{1}{2}$. As the indices involved in these two fractions are distinct, we can choose them to be i, j, k, l and conclude that $C \leqslant \frac{1}{2}$.

Next, we show that $C = \frac{1}{2}$ is best possible. Consider the numbers $1, 2, 2, 2, n$ where n is a large real number. The fractions formed by two of these numbers in ascending order are $\frac{1}{n}, \frac{2}{n}, \frac{1}{2}, \frac{2}{2}, \frac{2}{1}, \frac{n}{2}, \frac{n}{1}$. Since the indices i, j, k, l are distinct, $\frac{1}{n}$ and $\frac{2}{n}$ cannot be chosen simultaneously. Therefore the minimum value of the left-hand side of (1) is $\frac{1}{2} - \frac{2}{n}$. When n tends to infinity, this value approaches $\frac{1}{2}$, and so C cannot be less than $\frac{1}{2}$.

These conclude that $C = \frac{1}{2}$ is the smallest possible choice.

Comment. The conclusion still holds if $a_1, a_2, \ldots, a_5$ are pairwise distinct, since in the construction, we may replace the 2's by real numbers sufficiently close to 2.

There are two possible simplifications for this problem:

(i) the answer $C = \frac{1}{2}$ is given to the contestants; or

(ii) simply ask the contestants to prove the inequality (1) for $C = \frac{1}{2}$.

A3. Find all integers $n \geqslant 3$ with the following property: for all real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ satisfying $|a_k| + |b_k| = 1$ for $1 \leqslant k \leqslant n$, there exist $x_1, x_2, \ldots, x_n$, each of which is either -1 or 1, such that

$$\left|\sum_{k=1}^{n} x_k a_k\right| + \left|\sum_{k=1}^{n} x_k b_k\right| \leqslant 1. \tag{1}$$

Answer. n can be any odd integer greater than or equal to 3.

Solution 1. For any even integer $n \geqslant 4$, we consider the case

$$a_1 = a_2 = \cdots = a_{n-1} = b_n = 0 \quad \text{and} \quad b_1 = b_2 = \cdots = b_{n-1} = a_n = 1.$$

The condition $|a_k| + |b_k| = 1$ is satisfied for each $1 \leqslant k \leqslant n$. No matter how we choose each x_k, both sums $\sum_{k=1}^{n} x_k a_k$ and $\sum_{k=1}^{n} x_k b_k$ are odd integers. This implies $|\sum_{k=1}^{n} x_k a_k| \geqslant 1$ and $|\sum_{k=1}^{n} x_k b_k| \geqslant 1$, which shows (1) cannot hold.

For any odd integer $n \geqslant 3$, we may assume without loss of generality $b_k \geqslant 0$ for $1 \leqslant k \leqslant n$ (this can be done by flipping the pair (a_k, b_k) to $(-a_k, -b_k)$ and x_k to $-x_k$ if necessary) and $a_1 \geqslant a_2 \geqslant \cdots \geqslant a_m \geqslant 0 > a_{m+1} \geqslant \cdots \geqslant a_n$. We claim that the choice $x_k = (-1)^{k+1}$ for $1 \leqslant k \leqslant n$ will work. Define

$$s = \sum_{k=1}^{m} x_k a_k \quad \text{and} \quad t = -\sum_{k=m+1}^{n} x_k a_k.$$

Note that

$$s = (a_1 - a_2) + (a_3 - a_4) + \cdots \geqslant 0$$

by the assumption $a_1 \geqslant a_2 \geqslant \cdots \geqslant a_m$ (when m is odd, there is a single term a_m at the end, which is also positive). Next, we have

$$s = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots \leqslant a_1 \leqslant 1.$$

Similarly,

$$t = (-a_n + a_{n-1}) + (-a_{n-2} + a_{n-3}) + \cdots \geqslant 0$$

and

$$t = -a_n + (a_{n-1} - a_{n-2}) + (a_{n-3} - a_{n-4}) + \cdots \leqslant -a_n \leqslant 1.$$

From the condition, we have $a_k + b_k = 1$ for $1 \leqslant k \leqslant m$ and $-a_k + b_k = 1$ for $m+1 \leqslant k \leqslant n$. It follows that $\sum_{k=1}^{n} x_k a_k = s - t$ and $\sum_{k=1}^{n} x_k b_k = 1 - s - t$. Hence it remains to prove

$$|s - t| + |1 - s - t| \leqslant 1$$

under the constraint $0 \leqslant s, t \leqslant 1$. By symmetry, we may assume $s \geqslant t$. If $1 - s - t \geqslant 0$, then we have

$$|s - t| + |1 - s - t| = s - t + 1 - s - t = 1 - 2t \leqslant 1.$$

If $1 - s - t \leqslant 0$, then we have

$$|s - t| + |1 - s - t| = s - t - 1 + s + t = 2s - 1 \leqslant 1.$$

Hence, the inequality is true in both cases.

These show n can be any odd integer greater than or equal to 3.

Solution 2. The even case can be handled in the same way as Solution 1. For the odd case, we prove by induction on n.

Firstly, for $n = 3$, we may assume without loss of generality $a_1 \geqslant a_2 \geqslant a_3 \geqslant 0$ and $b_1 = a_1 - 1$ (if $b_1 = 1 - a_1$, we may replace each b_k by $-b_k$).

- **Case 1.** $b_2 = a_2 - 1$ and $b_3 = a_3 - 1$, in which case we take $(x_1, x_2, x_3) = (1, -1, 1)$.

  Let $c = a_1 - a_2 + a_3$ so that $0 \leqslant c \leqslant 1$. Then $|b_1 - b_2 + b_3| = |a_1 - a_2 + a_3 - 1| = 1 - c$ and hence $|c| + |b_1 - b_2 + b_3| = 1$.

- **Case 2.** $b_2 = 1 - a_2$ and $b_3 = 1 - a_3$, in which case we take $(x_1, x_2, x_3) = (1, -1, 1)$.

  Let $c = a_1 - a_2 + a_3$ so that $0 \leqslant c \leqslant 1$. Since $a_3 \leqslant a_2$ and $a_1 \leqslant 1$, we have

$$c - 1 \leqslant b_1 - b_2 + b_3 = a_1 + a_2 - a_3 - 1 \leqslant 1 - c.$$

  This gives $|b_1 - b_2 + b_3| \leqslant 1 - c$ and hence $|c| + |b_1 - b_2 + b_3| \leqslant 1$.

- **Case 3.** $b_2 = a_2 - 1$ and $b_3 = 1 - a_3$, in which case we take $(x_1, x_2, x_3) = (-1, 1, 1)$.

  Let $c = -a_1 + a_2 + a_3$. If $c \geqslant 0$, then $a_3 \leqslant 1$ and $a_2 \leqslant a_1$ imply

$$c - 1 \leqslant -b_1 + b_2 + b_3 = -a_1 + a_2 - a_3 + 1 \leqslant 1 - c.$$

  If $c < 0$, then $a_1 \leqslant a_2 + 1$ and $a_3 \geqslant 0$ imply

$$-c - 1 \leqslant -b_1 + b_2 + b_3 = -a_1 + a_2 - a_3 + 1 \leqslant 1 + c.$$

  In both cases, we get $| - b_1 + b_2 + b_3| \leqslant 1 - |c|$ and hence $|c| + | - b_1 + b_2 + b_3| \leqslant 1$.

- **Case 4.** $b_2 = 1 - a_2$ and $b_3 = a_3 - 1$, in which case we take $(x_1, x_2, x_3) = (-1, 1, 1)$.

  Let $c = -a_1 + a_2 + a_3$. If $c \geqslant 0$, then $a_2 \leqslant 1$ and $a_3 \leqslant a_1$ imply

$$c - 1 \leqslant -b_1 + b_2 + b_3 = -a_1 - a_2 + a_3 + 1 \leqslant 1 - c.$$

  If $c < 0$, then $a_1 \leqslant a_3 + 1$ and $a_2 \geqslant 0$ imply

$$-c - 1 \leqslant -b_1 + b_2 + b_3 = -a_1 - a_2 + a_3 + 1 \leqslant 1 + c.$$

  In both cases, we get $| - b_1 + b_2 + b_3| \leqslant 1 - |c|$ and hence $|c| + | - b_1 + b_2 + b_3| \leqslant 1$.

We have found x_1, x_2, x_3 satisfying (1) in each case for $n = 3$.

Now, let $n \geqslant 5$ be odd and suppose the result holds for any smaller odd cases. Again we may assume $a_k \geqslant 0$ for each $1 \leqslant k \leqslant n$. By the Pigeonhole Principle, there are at least three indices k for which $b_k = a_k - 1$ or $b_k = 1 - a_k$. Without loss of generality, suppose $b_k = a_k - 1$ for $k = 1, 2, 3$. Again by the Pigeonhole Principle, as a_1, a_2, a_3 lies between 0 and 1, the difference of two of them is at most $\frac{1}{2}$. By changing indices if necessary, we may assume $0 \leqslant d = a_1 - a_2 \leqslant \frac{1}{2}$.

By the inductive hypothesis, we can choose $x_3, x_4, \ldots, x_n$ such that $a' = \sum_{k=3}^{n} x_k a_k$ and $b' = \sum_{k=3}^{n} x_k b_k$ satisfy $|a'| + |b'| \leqslant 1$. We may further assume $a' \geqslant 0$.

• **Case 1.** $b' \geqslant 0$, in which case we take $(x_1, x_2) = (-1, 1)$.
We have $|-a_1+a_2+a'|+|-(a_1-1)+(a_2-1)+b'| = |-d+a'|+|-d+b'| \leqslant \max\{a'+b'-2d, a'-b', b'-a', 2d-a'-b'\} \leqslant 1$ since $0 \leqslant a', b', a'+b' \leqslant 1$ and $0 \leqslant d \leqslant \frac{1}{2}$.

• **Case 2.** $0 > b' \geqslant -a'$, in which case we take $(x_1, x_2) = (-1, 1)$.
We have $|-a_1+a_2+a'|+|-(a_1-1)+(a_2-1)+b'| = |-d+a'|+|-d+b'|$. If $-d+a' \geqslant 0$, this equals $a'-b' = |a'|+|b'| \leqslant 1$. If $-d+a' < 0$, this equals $2d-a'-b' \leqslant 2d \leqslant 1$.

• **Case 3.** $b' < -a'$, in which case we take $(x_1, x_2) = (1, -1)$.
We have $|a_1-a_2+a'|+|(a_1-1)-(a_2-1)+b'| = |d+a'|+|d+b'|$. If $d+b' \geqslant 0$, this equals $2d+a'+b' < 2d \leqslant 1$. If $d+b' < 0$, this equals $a'-b' = |a'|+|b'| \leqslant 1$.

Therefore, we have found $x_1, x_2, \ldots, x_n$ satisfying (1) in each case. By induction, the property holds for all odd integers $n \geqslant 3$.

A4. Denote by $\mathbb{R}^+$ the set of all positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)\left(f(f(x^2)) + f(f(y^2))\right) \tag{1}$$

for all positive real numbers x and y.

Answer. $f(x) = \frac{1}{x}$ for any $x \in \mathbb{R}^+$.

Solution 1. Taking $x = y = 1$ in (1), we get $f(1)f(f(1)) + f(f(1)) = 2f(1)f(f(1))$ and hence $f(1) = 1$. Swapping x and y in (1) and comparing with (1) again, we find

$$xf(x^2)f(f(y)) + f(yf(x)) = yf(y^2)f(f(x)) + f(xf(y)). \tag{2}$$

Taking $y = 1$ in (2), we have $xf(x^2) + f(f(x)) = f(f(x)) + f(x)$, that is,

$$f(x^2) = \frac{f(x)}{x}. \tag{3}$$

Take $y = 1$ in (1) and apply (3) to $xf(x^2)$. We get $f(x) + f(f(x)) = f(x)(f(f(x^2)) + 1)$, which implies

$$f(f(x^2)) = \frac{f(f(x))}{f(x)}. \tag{4}$$

For any $x \in \mathbb{R}^+$, we find that

$$f(f(x)^2) \overset{(3)}{=} \frac{f(f(x))}{f(x)} \overset{(4)}{=} f(f(x^2)) \overset{(3)}{=} f\left(\frac{f(x)}{x}\right). \tag{5}$$

It remains to show the following key step.

• **Claim.** The function f is injective.

Proof. Using (3) and (4), we rewrite (1) as

$$f(x)f(f(y)) + f(yf(x)) = f(xy)\left(\frac{f(f(x))}{f(x)} + \frac{f(f(y))}{f(y)}\right). \tag{6}$$

Take $x = y$ in (6) and apply (3). This gives $f(x)f(f(x)) + f(xf(x)) = 2\frac{f(f(x))}{x}$, which means

$$f(xf(x)) = f(f(x))\left(\frac{2}{x} - f(x)\right). \tag{7}$$

Using (3), equation (2) can be rewritten as

$$f(x)f(f(y)) + f(yf(x)) = f(y)f(f(x)) + f(xf(y)). \tag{8}$$

Suppose $f(x) = f(y)$ for some $x, y \in \mathbb{R}^+$. Then (8) implies

$$f(yf(y)) = f(yf(x)) = f(xf(y)) = f(xf(x)).$$

Using (7), this gives

$$f(f(y))\left(\frac{2}{y} - f(y)\right) = f(f(x))\left(\frac{2}{x} - f(x)\right).$$

Noting $f(x) = f(y)$, we find $x = y$. This establishes the injectivity. □

By the Claim and (5), we get the only possible solution $f(x) = \frac{1}{x}$. It suffices to check that this is a solution. Indeed, the left-hand side of (1) becomes

$$x \cdot \frac{1}{x^2} \cdot y + \frac{x}{y} = \frac{y}{x} + \frac{x}{y},$$

while the right-hand side becomes

$$\frac{1}{xy}(x^2 + y^2) = \frac{x}{y} + \frac{y}{x}.$$

The two sides agree with each other.

Solution 2. Taking $x = y = 1$ in (1), we get $f(1)f(f(1)) + f(f(1)) = 2f(1)f(f(1))$ and hence $f(1) = 1$. Putting $x = 1$ in (1), we have $f(f(y)) + f(y) = f(y)(1 + f(f(y^2)))$ so that

$$f(f(y)) = f(y)f(f(y^2)). \tag{9}$$

Putting $y = 1$ in (1), we get $xf(x^2) + f(f(x)) = f(x)(f(f(x^2)) + 1)$. Using (9), this gives

$$xf(x^2) = f(x). \tag{10}$$

Replace y by $\frac{1}{x}$ in (1). Then we have

$$xf(x^2)f\left(f\left(\frac{1}{x}\right)\right) + f\left(\frac{f(x)}{x}\right) = f(f(x^2)) + f\left(f\left(\frac{1}{x^2}\right)\right).$$

The relation (10) shows $f(\frac{f(x)}{x}) = f(f(x^2))$. Also, using (9) with $y = \frac{1}{x}$ and using (10) again, the last equation reduces to

$$f(x)f\left(\frac{1}{x}\right) = 1. \tag{11}$$

Replace x by $\frac{1}{x}$ and y by $\frac{1}{y}$ in (1) and apply (11). We get

$$\frac{1}{xf(x^2)f(f(y))} + \frac{1}{f(yf(x))} = \frac{1}{f(xy)}\left(\frac{1}{f(f(x^2))} + \frac{1}{f(f(y^2))}\right).$$

Clearing denominators, we can use (1) to simplify the numerators and obtain

$$f(xy)^2 f(f(x^2))f(f(y^2)) = xf(x^2)f(f(y))f(yf(x)).$$

Using (9) and (10), this is the same as

$$f(xy)^2 f(f(x)) = f(x)^2 f(y)f(yf(x)). \tag{12}$$

Substitute $y = f(x)$ in (12) and apply (10) (with x replaced by $f(x)$). We have

$$f(xf(x))^2 = f(x)f(f(x)). \tag{13}$$

Taking $y = x$ in (12), squaring both sides, and using (10) and (13), we find that

$$f(f(x)) = x^4 f(x)^3. \tag{14}$$

Finally, we combine (9), (10) and (14) to get

$$y^4 f(y)^3 \overset{(14)}{=} f(f(y)) \overset{(9)}{=} f(y)f(f(y^2)) \overset{(14)}{=} f(y)y^8 f(y^2)^3 \overset{(10)}{=} y^5 f(y)^4,$$

which implies $f(y) = \frac{1}{y}$. This is a solution by the checking in Solution 1.

A5.

(a) Prove that for every positive integer n, there exists a fraction $\frac{a}{b}$ where a and b are integers satisfying $0 < b \leqslant \sqrt{n} + 1$ and $\sqrt{n} \leqslant \frac{a}{b} \leqslant \sqrt{n+1}$.

(b) Prove that there are infinitely many positive integers n such that there is no fraction $\frac{a}{b}$ where a and b are integers satisfying $0 < b \leqslant \sqrt{n}$ and $\sqrt{n} \leqslant \frac{a}{b} \leqslant \sqrt{n+1}$.

Solution.

(a) Let r be the unique positive integer for which $r^2 \leqslant n < (r+1)^2$. Write $n = r^2 + s$. Then we have $0 \leqslant s \leqslant 2r$. We discuss in two cases according to the parity of s.

• **Case 1.** s is even.

Consider the number $(r + \frac{s}{2r})^2 = r^2 + s + (\frac{s}{2r})^2$. We find that

$$n = r^2 + s \leqslant r^2 + s + \left(\frac{s}{2r}\right)^2 \leqslant r^2 + s + 1 = n + 1.$$

It follows that

$$\sqrt{n} \leqslant r + \frac{s}{2r} \leqslant \sqrt{n+1}.$$

Since s is even, we can choose the fraction $r + \frac{s}{2r} = \frac{r^2 + (s/2)}{r}$ since $r \leqslant \sqrt{n}$.

• **Case 2.** s is odd.

Consider the number $(r + 1 - \frac{2r+1-s}{2(r+1)})^2 = (r+1)^2 - (2r+1-s) + (\frac{2r+1-s}{2(r+1)})^2$. We find that

$$\begin{aligned} n = r^2 + s = (r+1)^2 - (2r+1-s) &\leqslant (r+1)^2 - (2r+1-s) + \left(\frac{2r+1-s}{2(r+1)}\right)^2 \\ &\leqslant (r+1)^2 - (2r+1-s) + 1 = n + 1. \end{aligned}$$

It follows that

$$\sqrt{n} \leqslant r + 1 - \frac{2r+1-s}{2(r+1)} \leqslant \sqrt{n+1}.$$

Since s is odd, we can choose the fraction $(r+1) - \frac{2r+1-s}{2(r+1)} = \frac{(r+1)^2 - r + ((s-1)/2)}{r+1}$ since $r + 1 \leqslant \sqrt{n} + 1$.

(b) We show that for every positive integer r, there is no fraction $\frac{a}{b}$ with $b \leqslant \sqrt{r^2+1}$ such that $\sqrt{r^2+1} \leqslant \frac{a}{b} \leqslant \sqrt{r^2+2}$. Suppose on the contrary that such a fraction exists. Since $b \leqslant \sqrt{r^2+1} < r + 1$ and b is an integer, we have $b \leqslant r$. Hence,

$$(br)^2 < b^2(r^2+1) \leqslant a^2 \leqslant b^2(r^2+2) \leqslant b^2r^2 + 2br < (br+1)^2.$$

This shows the square number a^2 is strictly bounded between the two consecutive squares $(br)^2$ and $(br+1)^2$, which is impossible. Hence, we have found infinitely many $n = r^2 + 1$ for which there is no fraction of the desired form.

A6. The equation

$$(x-1)(x-2)\cdots(x-2016)=(x-1)(x-2)\cdots(x-2016)$$

is written on the board. One tries to erase some linear factors from both sides so that each side still has at least one factor, and the resulting equation has no real roots. Find the least number of linear factors one needs to erase to achieve this.

Answer. 2016.

Solution. Since there are 2016 common linear factors on both sides, we need to erase at least 2016 factors. We claim that the equation has no real roots if we erase all factors $(x-k)$ on the left-hand side with $k\equiv 2,3 \pmod 4$, and all factors $(x-m)$ on the right-hand side with $m\equiv 0,1 \pmod 4$. Therefore, it suffices to show that no real number x satisfies

$$\prod_{j=0}^{503}(x-4j-1)(x-4j-4)=\prod_{j=0}^{503}(x-4j-2)(x-4j-3). \tag{1}$$

- **Case 1.** $x=1,2,\ldots,2016$.

  In this case, one side of (1) is zero while the other side is not. This shows x cannot satisfy (1).

- **Case 2.** $4k+1<x<4k+2$ or $4k+3<x<4k+4$ for some $k=0,1,\ldots,503$.

  For $j=0,1,\ldots,503$ with $j\neq k$, the product $(x-4j-1)(x-4j-4)$ is positive. For $j=k$, the product $(x-4k-1)(x-4k-4)$ is negative. This shows the left-hand side of (1) is negative. On the other hand, each product $(x-4j-2)(x-4j-3)$ on the right-hand side of (1) is positive. This yields a contradiction.

- **Case 3.** $x<1$ or $x>2016$ or $4k<x<4k+1$ for some $k=1,2,\ldots,503$.

  The equation (1) can be rewritten as

$$1=\prod_{j=0}^{503}\frac{(x-4j-1)(x-4j-4)}{(x-4j-2)(x-4j-3)}=\prod_{j=0}^{503}\left(1-\frac{2}{(x-4j-2)(x-4j-3)}\right).$$

Note that $(x-4j-2)(x-4j-3)>2$ for $0\leqslant j\leqslant 503$ in this case. So each term in the product lies strictly between 0 and 1, and the whole product must be less than 1, which is impossible.

- **Case 4.** $4k+2<x<4k+3$ for some $k=0,1,\ldots,503$.

  This time we rewrite (1) as

$$\begin{aligned}1&=\frac{x-1}{x-2}\cdot\frac{x-2016}{x-2015}\prod_{j=1}^{503}\frac{(x-4j)(x-4j-1)}{(x-4j+1)(x-4j-2)}\\&=\frac{x-1}{x-2}\cdot\frac{x-2016}{x-2015}\prod_{j=1}^{503}\left(1+\frac{2}{(x-4j+1)(x-4j-2)}\right).\end{aligned}$$

Clearly, $\frac{x-1}{x-2}$ and $\frac{x-2016}{x-2015}$ are both greater than 1. For the range of x in this case, each term in the product is also greater than 1. Then the right-hand side must be greater than 1 and hence a contradiction arises.

From the four cases, we conclude that (1) has no real roots. Hence, the minimum number of linear factors to be erased is 2016.

Comment. We discuss the general case when 2016 is replaced by a positive integer n. The above solution works equally well when n is divisible by 4.

If $n \equiv 2 \pmod 4$, one may leave $l(x) = (x-1)(x-2)\cdots(x-\frac{n}{2})$ on the left-hand side and $r(x) = (x-\frac{n}{2}-1)(x-\frac{n}{2}-2)\cdots(x-n)$ on the right-hand side. One checks that for $x < \frac{n+1}{2}$, we have $|l(x)| < |r(x)|$, while for $x > \frac{n+1}{2}$, we have $|l(x)| > |r(x)|$.

If $n \equiv 3 \pmod 4$, one may leave $l(x) = (x-1)(x-2)\cdots(x-\frac{n+1}{2})$ on the left-hand side and $r(x) = (x-\frac{n+3}{2})(x-\frac{x+5}{2})\cdots(x-n)$ on the right-hand side. For $x < 1$ or $\frac{n+1}{2} < x < \frac{n+3}{2}$, we have $l(x) > 0 > r(x)$. For $1 < x < \frac{n+1}{2}$, we have $|l(x)| < |r(x)|$. For $x > \frac{n+3}{2}$, we have $|l(x)| > |r(x)|$.

If $n \equiv 1 \pmod 4$, as the proposer mentioned, the situation is a bit more out of control. Since the construction for $n-1 \equiv 0 \pmod 4$ works, the answer can be either n or $n-1$. For $n = 5$, we can leave the products $(x-1)(x-2)(x-3)(x-4)$ and $(x-5)$. For $n = 9$, the only example that works is $l(x) = (x-1)(x-2)(x-9)$ and $r(x) = (x-3)(x-4)\cdots(x-8)$, while there seems to be no such partition for $n = 13$.

A7. Denote by $\mathbb{R}$ the set of all real numbers. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(0) \neq 0$ and

$$f(x+y)^2 = 2f(x)f(y) + \max\{f(x^2) + f(y^2), f(x^2+y^2)\} \tag{1}$$

for all real numbers x and y.

Answer.

- $f(x) = -1$ for any $x \in \mathbb{R}$; or
- $f(x) = x - 1$ for any $x \in \mathbb{R}$.

Solution 1. Taking $x = y = 0$ in (1), we get $f(0)^2 = 2f(0)^2 + \max\{2f(0), f(0)\}$. If $f(0) > 0$, then $f(0)^2 + 2f(0) = 0$ gives no positive solution. If $f(0) < 0$, then $f(0)^2 + f(0) = 0$ gives $f(0) = -1$. Putting $y = 0$ in (1), we have $f(x)^2 = -2f(x) + f(x^2)$, which is the same as $(f(x)+1)^2 = f(x^2) + 1$. Let $g(x) = f(x) + 1$. Then for any $x \in \mathbb{R}$, we have

$$g(x^2) = g(x)^2 \geqslant 0. \tag{2}$$

From (1), we find that $f(x+y)^2 \geqslant 2f(x)f(y) + f(x^2) + f(y^2)$. In terms of g, this becomes $(g(x+y)-1)^2 \geqslant 2(g(x)-1)(g(y)-1) + g(x^2) + g(y^2) - 2$. Using (2), this means

$$(g(x+y)-1)^2 \geqslant (g(x)+g(y)-1)^2 - 1. \tag{3}$$

Putting $x = 1$ in (2), we get $g(1) = 0$ or 1. The two cases are handled separately.

• **Case 1.** $g(1) = 0$, which is the same as $f(1) = -1$.

We put $x = -1$ and $y = 0$ in (1). This gives $f(-1)^2 = -2f(-1) - 1$, which forces $f(-1) = -1$. Next, we take $x = -1$ and $y = 1$ in (1) to get $1 = 2 + \max\{-2, f(2)\}$. This clearly implies $1 = 2 + f(2)$ and hence $f(2) = -1$, that is, $g(2) = 0$. From (2), we can prove inductively that $g(2^{2^n}) = g(2)^{2^n} = 0$ for any $n \in \mathbb{N}$. Substitute $y = 2^{2^n} - x$ in (3). We obtain

$$(g(x) + g(2^{2^n} - x) - 1)^2 \leqslant (g(2^{2^n}) - 1)^2 + 1 = 2.$$

For any fixed $x \geqslant 0$, we consider n to be sufficiently large so that $2^{2^n} - x > 0$. From (2), this implies $g(2^{2^n} - x) \geqslant 0$ so that $g(x) \leqslant 1 + \sqrt{2}$. Using (2) again, we get

$$g(x)^{2^n} = g(x^{2^n}) \leqslant 1 + \sqrt{2}$$

for any $n \in \mathbb{N}$. Therefore, $|g(x)| \leqslant 1$ for any $x \geqslant 0$.

If there exists $a \in \mathbb{R}$ for which $g(a) \neq 0$, then for sufficiently large n we must have $g((a^2)^{\frac{1}{2^n}}) = g(a^2)^{\frac{1}{2^n}} > \frac{1}{2}$. By taking $x = -y = -(a^2)^{\frac{1}{2^n}}$ in (1), we obtain

$$\begin{aligned} 1 &= 2f(x)f(-x) + \max\{2f(x^2), f(2x^2)\} \\ &= 2(g(x)-1)(g(-x)-1) + \max\{2(g(x^2)-1), g(2x^2)-1\} \\ &\leqslant 2\left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) + 0 = \frac{1}{2} \end{aligned}$$

since $|g(-x)| = |g(x)| \in (\frac{1}{2}, 1]$ by (2) and the choice of x, and since $g(z) \leqslant 1$ for $z \geqslant 0$. This yields a contradiction and hence $g(x) = 0$ must hold for any x. This means $f(x) = -1$ for any $x \in \mathbb{R}$, which clearly satisfies (1).

• **Case 2.** $g(1) = 1$, which is the same as $f(1) = 0$.

We put $x = -1$ and $y = 1$ in (1) to get $1 = \max\{0, f(2)\}$. This clearly implies $f(2) = 1$ and hence $g(2) = 2$. Setting $x = 2n$ and $y = 2$ in (3), we have

$$(g(2n+2)-1)^2 \geqslant (g(2n)+1)^2 - 1.$$

By induction on n, it is easy to prove that $g(2n) \geqslant n+1$ for all $n \in \mathbb{N}$. For any real number $a > 1$, we choose a large $n \in \mathbb{N}$ and take k to be the positive integer such that $2k \leqslant a^{2^n} < 2k+2$. From (2) and (3), we have

$$(g(a)^{2^n}-1)^2 + 1 = (g(a^{2^n})-1)^2 + 1 \geqslant (g(2k) + g(a^{2^n}-2k) - 1)^2 \geqslant k^2 > \frac{1}{4}(a^{2^n}-2)^2$$

since $g(a^{2^n} - 2k) \geqslant 0$. For large n, this clearly implies $g(a)^{2^n} > 1$. Thus,

$$(g(a)^{2^n})^2 > (g(a)^{2^n}-1)^2 + 1 > \frac{1}{4}(a^{2^n}-2)^2.$$

This yields

$$g(a)^{2^n} > \frac{1}{2}(a^{2^n}-2). \tag{4}$$

Note that

$$\frac{a^{2^n}}{a^{2^n}-2} = 1 + \frac{2}{a^{2^n}-2} \leqslant \left(1 + \frac{2}{2^n(a^{2^n}-2)}\right)^{2^n}$$

by binomial expansion. This can be rewritten as

$$(a^{2^n}-2)^{\frac{1}{2^n}} \geqslant \frac{a}{1 + \frac{2}{2^n(a^{2^n}-2)}}.$$

Together with (4), we conclude $g(a) \geqslant a$ by taking n sufficiently large.

Consider $x = na$ and $y = a > 1$ in (3). This gives $(g((n+1)a)-1)^2 \geqslant (g(na)+g(a)-1)^2 - 1$. By induction on n, it is easy to show $g(na) \geqslant (n-1)(g(a)-1) + a$ for any $n \in \mathbb{N}$. We choose a large $n \in \mathbb{N}$ and take k to be the positive integer such that $ka \leqslant 2^{2^n} < (k+1)a$. Using (2) and (3), we have

$$2^{2^{n+1}} > (2^{2^n}-1)^2 + 1 = (g(2^{2^n})-1)^2 + 1 \geqslant (g(2^{2^n}-ka) + g(ka) - 1)^2 \geqslant ((k-1)(g(a)-1) + a - 1)^2,$$

from which it follows that

$$2^{2^n} \geqslant (k-1)(g(a)-1) + a - 1 > \frac{2^{2^n}}{a}(g(a)-1) - 2(g(a)-1) + a - 1$$

holds for sufficiently large n. Hence, we must have $\frac{g(a)-1}{a} \leqslant 1$, which implies $g(a) \leqslant a+1$ for any $a > 1$. Then for large $n \in \mathbb{N}$, from (3) and (2) we have

$$4a^{2^{n+1}} = (2a^{2^n})^2 \geqslant (g(2a^{2^n})-1)^2 \geqslant (2g(a^{2^n})-1)^2 - 1 = (2g(a)^{2^n}-1)^2 - 1.$$

This implies

$$2a^{2^n} > \frac{1}{2}(1+\sqrt{4a^{2^{n+1}}+1}) \geqslant g(a)^{2^n}.$$

When n tends to infinity, this forces $g(a) \leqslant a$. Together with $g(a) \geqslant a$, we get $g(a) = a$ for all real numbers $a > 1$, that is, $f(a) = a - 1$ for all $a > 1$.

Finally, for any $x \in \mathbb{R}$, we choose y sufficiently large in (1) so that $y, x+y > 1$. This gives $(x+y-1)^2 = 2f(x)(y-1) + \max\{f(x^2)+y^2-1, x^2+y^2-1\}$, which can be rewritten as

$$2(x-1-f(x))y = -x^2+2x-2-2f(x)+\max\{f(x^2), x^2\}.$$

As the right-hand side is fixed, this can only hold for all large y when $f(x) = x-1$. We now check that this function satisfies (1). Indeed, we have

$$\begin{aligned} f(x+y)^2 &= (x+y-1)^2 = 2(x-1)(y-1) + (x^2+y^2-1) \\ &= 2f(x)f(y) + \max\{f(x^2)+f(y^2), f(x^2+y^2)\}. \end{aligned}$$

Solution 2. Taking $x = y = 0$ in (1), we get $f(0)^2 = 2f(0)^2 + \max\{2f(0), f(0)\}$. If $f(0) > 0$, then $f(0)^2 + 2f(0) = 0$ gives no positive solution. If $f(0) < 0$, then $f(0)^2 + f(0) = 0$ gives $f(0) = -1$. Putting $y = 0$ in (1), we have

$$f(x)^2 = -2f(x) + f(x^2). \tag{5}$$

Replace x by $-x$ in (5) and compare with (5) again. We get $f(x)^2 + 2f(x) = f(-x)^2 + 2f(-x)$, which implies

$$f(x) = f(-x) \quad \text{or} \quad f(x) + f(-x) = -2. \tag{6}$$

Taking $x = y$ and $x = -y$ respectively in (1) and comparing the two equations obtained, we have

$$f(2x)^2 - 2f(x)^2 = 1 - 2f(x)f(-x). \tag{7}$$

Combining (6) and (7) to eliminate $f(-x)$, we find that $f(2x)$ can be ± 1 (when $f(x) = f(-x)$) or $\pm(2f(x)+1)$ (when $f(x) + f(-x) = -2$).

We prove the following.

• **Claim.** $f(x) + f(-x) = -2$ for any $x \in \mathbb{R}$.

Proof. Suppose there exists $a \in \mathbb{R}$ such that $f(a) + f(-a) \neq -2$. Then $f(a) = f(-a) \neq -1$ and we may assume $a > 0$. We first show that $f(a) \neq 1$. Suppose $f(a) = 1$. Consider $y = a$ in (7). We get $f(2a)^2 = 1$. Taking $x = y = a$ in (1), we have $1 = 2 + \max\{2f(a^2), f(2a^2)\}$. From (5), $f(a^2) = 3$ so that $1 \geqslant 2 + 6$. This is impossible, and thus $f(a) \neq 1$.

As $f(a) \neq \pm 1$, we have $f(a) = \pm(2f(\frac{a}{2}) + 1)$. Similarly, $f(-a) = \pm(2f(-\frac{a}{2}) + 1)$. These two expressions are equal since $f(a) = f(-a)$. If $f(\frac{a}{2}) = f(-\frac{a}{2})$, then the above argument works when we replace a by $\frac{a}{2}$. In particular, we have $f(a)^2 = f(2 \cdot \frac{a}{2})^2 = 1$, which is a contradiction. Therefore, (6) forces $f(\frac{a}{2}) + f(-\frac{a}{2}) = -2$. Then we get

$$\pm\left(2f\left(\frac{a}{2}\right)+1\right) = \pm\left(-2f\left(\frac{a}{2}\right)-3\right).$$

For any choices of the two signs, we either get a contradiction or $f(\frac{a}{2}) = -1$, in which case $f(\frac{a}{2}) = f(-\frac{a}{2})$ and hence $f(a) = \pm 1$ again. Therefore, there is no such real number a and the Claim follows. □

Replace x and y by $-x$ and $-y$ in (1) respectively and compare with (1). We get

$$f(x+y)^2 - 2f(x)f(y) = f(-x-y)^2 - 2f(-x)f(-y).$$

Using the Claim, this simplifies to $f(x+y) = f(x)+f(y)+1$. In addition, (5) can be rewritten as $(f(x)+1)^2 = f(x^2)+1$. Therefore, the function g defined by $g(x) = f(x)+1$ satisfies $g(x+y) = g(x)+g(y)$ and $g(x)^2 = g(x^2)$. The latter relation shows $g(y)$ is nonnegative for $y \geqslant 0$. For such a function satisfying the Cauchy Equation $g(x+y) = g(x)+g(y)$, it must be monotonic increasing and hence $g(x) = cx$ for some constant c.

From $(cx)^2 = g(x)^2 = g(x^2) = cx^2$, we get $c = 0$ or 1, which corresponds to the two functions $f(x) = -1$ and $f(x) = x - 1$ respectively, both of which are solutions to (1) as checked in Solution 1.

Solution 3. As in Solution 2, we find that $f(0) = -1$,

$$(f(x)+1)^2 = f(x^2)+1 \tag{8}$$

and

$$f(x) = f(-x) \quad \text{or} \quad f(x)+f(-x) = -2 \tag{9}$$

for any $x \in \mathbb{R}$. We shall show that one of the statements in (9) holds for all $x \in \mathbb{R}$. Suppose $f(a) = f(-a)$ but $f(a)+f(-a) \neq -2$, while $f(b) \neq f(-b)$ but $f(b)+f(-b) = -2$. Clearly, $a, b \neq 0$ and $f(a), f(b) \neq -1$.

Taking $y = a$ and $y = -a$ in (1) respectively and comparing the two equations obtained, we have $f(x+a)^2 = f(x-a)^2$, that is, $f(x+a) = \pm f(x-a)$. This implies $f(x+2a) = \pm f(x)$ for all $x \in \mathbb{R}$. Putting $x = b$ and $x = -2a-b$ respectively, we find $f(2a+b) = \pm f(b)$ and $f(-2a-b) = \pm f(-b) = \pm(-2-f(b))$. Since $f(b) \neq -1$, the term $\pm(-2-f(b))$ is distinct from $\pm f(b)$ in any case. So $f(2a+b) \neq f(-2a-b)$. From (9), we must have $f(2a+b)+f(-2a-b) = -2$. Note that we also have $f(b)+f(-b) = -2$ where $|f(b)|, |f(-b)|$ are equal to $|f(2a+b)|, |f(-2a-b)|$ respectively. The only possible case is $f(2a+b) = f(b)$ and $f(-2a-b) = f(-b)$.

Applying the argument to $-a$ instead of a and using induction, we have $f(2ka+b) = f(b)$ and $f(2ka-b) = f(-b)$ for any integer k. Note that $f(b)+f(-b) = -2$ and $f(b) \neq -1$ imply one of $f(b), f(-b)$ is less than -1. Without loss of generality, assume $f(b) < -1$. We consider $x = \sqrt{2ka+b}$ in (8) for sufficiently large k so that

$$(f(x)+1)^2 = f(2ka+b)+1 = f(b)+1 < 0$$

yields a contradiction. Therefore, one of the statements in (9) must hold for all $x \in \mathbb{R}$.

• **Case 1.** $f(x) = f(-x)$ for any $x \in \mathbb{R}$.

For any $a \in \mathbb{R}$, setting $x = y = \frac{a}{2}$ and $x = -y = \frac{a}{2}$ in (1) respectively and comparing these, we obtain $f(a)^2 = f(0)^2 = 1$, which means $f(a) = \pm 1$ for all $a \in \mathbb{R}$. If $f(a) = 1$ for some a, we may assume $a > 0$ since $f(a) = f(-a)$. Taking $x = y = \sqrt{a}$ in (1), we get

$$f(2\sqrt{a})^2 = 2f(\sqrt{a})^2 + \max\{2, f(2a)\} = 2f(\sqrt{a})^2 + 2.$$

Note that the left-hand side is ± 1 while the right-hand side is an even integer. This is a contradiction. Therefore, $f(x) = -1$ for all $x \in \mathbb{R}$, which is clearly a solution.

• **Case 2.** $f(x) + f(-x) = -2$ for any $x \in \mathbb{R}$.

This case can be handled in the same way as in Solution 2, which yields another solution $f(x) = x - 1$.

A8. Determine the largest real number a such that for all $n \geqslant 1$ and for all real numbers $x_0, x_1, \ldots, x_n$ satisfying $0 = x_0 < x_1 < x_2 < \cdots < x_n$, we have

$$\frac{1}{x_1 - x_0} + \frac{1}{x_2 - x_1} + \cdots + \frac{1}{x_n - x_{n-1}} \geqslant a\left(\frac{2}{x_1} + \frac{3}{x_2} + \cdots + \frac{n+1}{x_n}\right). \tag{1}$$

Answer. The largest a is $\frac{4}{9}$.

Solution 1. We first show that $a = \frac{4}{9}$ is admissible. For each $2 \leqslant k \leqslant n$, by the Cauchy-Schwarz Inequality, we have

$$(x_{k-1} + (x_k - x_{k-1}))\left(\frac{(k-1)^2}{x_{k-1}} + \frac{3^2}{x_k - x_{k-1}}\right) \geqslant (k - 1 + 3)^2,$$

which can be rewritten as

$$\frac{9}{x_k - x_{k-1}} \geqslant \frac{(k+2)^2}{x_k} - \frac{(k-1)^2}{x_{k-1}}. \tag{2}$$

Summing (2) over $k = 2, 3, \ldots, n$ and adding $\frac{9}{x_1}$ to both sides, we have

$$9\sum_{k=1}^{n} \frac{1}{x_k - x_{k-1}} \geqslant 4\sum_{k=1}^{n} \frac{k+1}{x_k} + \frac{n^2}{x_n} > 4\sum_{k=1}^{n} \frac{k+1}{x_k}.$$

This shows (1) holds for $a = \frac{4}{9}$.

Next, we show that $a = \frac{4}{9}$ is the optimal choice. Consider the sequence defined by $x_0 = 0$ and $x_k = x_{k-1} + k(k+1)$ for $k \geqslant 1$, that is, $x_k = \frac{1}{3}k(k+1)(k+2)$. Then the left-hand side of (1) equals

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1 - \frac{1}{n+1},$$

while the right-hand side equals

$$a\sum_{k=1}^{n} \frac{k+1}{x_k} = 3a\sum_{k=1}^{n} \frac{1}{k(k+2)} = \frac{3}{2}a\sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+2}\right) = \frac{3}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right)a.$$

When n tends to infinity, the left-hand side tends to 1 while the right-hand side tends to $\frac{9}{4}a$. Therefore a has to be at most $\frac{4}{9}$.

Hence the largest value of a is $\frac{4}{9}$.

Solution 2. We shall give an alternative method to establish (1) with $a = \frac{4}{9}$. We define $y_k = x_k - x_{k-1} > 0$ for $1 \leqslant k \leqslant n$. By the Cauchy-Schwarz Inequality, for $1 \leqslant k \leqslant n$, we have

$$(y_1 + y_2 + \cdots + y_k)\left(\sum_{j=1}^{k} \frac{1}{y_j}\binom{j+1}{2}^2\right) \geqslant \left(\binom{2}{2} + \binom{3}{2} + \cdots + \binom{k+1}{2}\right)^2 = \binom{k+2}{3}^2.$$

This can be rewritten as

$$\frac{k+1}{y_1+y_2+\cdots+y_k} \leqslant \frac{36}{k^2(k+1)(k+2)^2}\left(\sum_{j=1}^{k}\frac{1}{y_j}\binom{j+1}{2}^2\right). \tag{3}$$

Summing (3) over $k = 1, 2, \ldots, n$, we get

$$\frac{2}{y_1}+\frac{3}{y_1+y_2}+\cdots+\frac{n+1}{y_1+y_2+\cdots+y_n} \leqslant \frac{c_1}{y_1}+\frac{c_2}{y_2}+\cdots+\frac{c_n}{y_n} \tag{4}$$

where for $1 \leqslant m \leqslant n$,

$$\begin{aligned} c_m &= 36\binom{m+1}{2}^2 \sum_{k=m}^{n}\frac{1}{k^2(k+1)(k+2)^2} \\ &= \frac{9m^2(m+1)^2}{4}\sum_{k=m}^{n}\left(\frac{1}{k^2(k+1)^2}-\frac{1}{(k+1)^2(k+2)^2}\right) \\ &= \frac{9m^2(m+1)^2}{4}\left(\frac{1}{m^2(m+1)^2}-\frac{1}{(n+1)^2(n+2)^2}\right) < \frac{9}{4}. \end{aligned}$$

From (4), the inequality (1) holds for $a = \frac{4}{9}$. This is also the upper bound as can be verified in the same way as Solution 1.

Combinatorics

C1. The leader of an IMO team chooses positive integers n and k with $n > k$, and announces them to the deputy leader and a contestant. The leader then secretly tells the deputy leader an n-digit binary string, and the deputy leader writes down all n-digit binary strings which differ from the leader's in exactly k positions. (For example, if $n = 3$ and $k = 1$, and if the leader chooses 101, the deputy leader would write down 001, 111 and 100.) The contestant is allowed to look at the strings written by the deputy leader and guess the leader's string. What is the minimum number of guesses (in terms of n and k) needed to guarantee the correct answer?

Answer. The minimum number of guesses is 2 if $n = 2k$ and 1 if $n \neq 2k$.

Solution 1. Let X be the binary string chosen by the leader and let X' be the binary string of length n every digit of which is different from that of X. The strings written by the deputy leader are the same as those in the case when the leader's string is X' and k is changed to $n - k$. In view of this, we may assume $k \geqslant \frac{n}{2}$. Also, for the particular case $k = \frac{n}{2}$, this argument shows that the strings X and X' cannot be distinguished, and hence in that case the contestant has to guess at least twice.

It remains to show that the number of guesses claimed suffices. Consider any string Y which differs from X in m digits where $0 < m < 2k$. Without loss of generality, assume the first m digits of X and Y are distinct. Let Z be the binary string obtained from X by changing its first k digits. Then Z is written by the deputy leader. Note that Z differs from Y by $|m - k|$ digits where $|m - k| < k$ since $0 < m < 2k$. From this observation, the contestant must know that Y is not the desired string.

As we have assumed $k \geqslant \frac{n}{2}$, when $n < 2k$, every string $Y \neq X$ differs from X in fewer than $2k$ digits. When $n = 2k$, every string except X and X' differs from X in fewer than $2k$ digits. Hence, the answer is as claimed.

Solution 2. Firstly, assume $n \neq 2k$. Without loss of generality suppose the first digit of the leader's string is 1. Then among the $\binom{n}{k}$ strings written by the deputy leader, $\binom{n-1}{k}$ will begin with 1 and $\binom{n-1}{k-1}$ will begin with 0. Since $n \neq 2k$, we have $k + (k - 1) \neq n - 1$ and so $\binom{n-1}{k} \neq \binom{n-1}{k-1}$. Thus, by counting the number of strings written by the deputy leader that start with 0 and 1, the contestant can tell the first digit of the leader's string. The same can be done on the other digits, so 1 guess suffices when $n \neq 2k$.

Secondly, for the case $n = 2$ and $k = 1$, the answer is clearly 2. For the remaining cases where $n = 2k > 2$, the deputy leader would write down the same strings if the leader's string X is replaced by X' obtained by changing each digit of X. This shows at least 2 guesses are needed. We shall show that 2 guesses suffice in this case. Suppose the first two digits of the leader's string are the same. Then among the strings written by the deputy leader, the prefices 01 and 10 will occur $\binom{2k-2}{k-1}$ times each, while the prefices 00 and 11 will occur $\binom{2k-2}{k}$ times each. The two numbers are interchanged if the first two digits of the leader's string are different. Since $\binom{2k-2}{k-1} \neq \binom{2k-2}{k}$, the contestant can tell whether the first two digits of the leader's string are the same or not. He can work out the relation of the first digit and the

other digits in the same way and reduce the leader's string to only 2 possibilities. The proof is complete.

C2. Find all positive integers n for which all positive divisors of n can be put into the cells of a rectangular table under the following constraints:

- each cell contains a distinct divisor;
- the sums of all rows are equal; and
- the sums of all columns are equal.

Answer. 1.

Solution 1. Suppose all positive divisors of n can be arranged into a rectangular table of size $k \times l$ where the number of rows k does not exceed the number of columns l. Let the sum of numbers in each column be s. Since n belongs to one of the columns, we have $s \geqslant n$, where equality holds only when $n = 1$.

For $j = 1, 2, \ldots, l$, let d_j be the largest number in the j-th column. Without loss of generality, assume $d_1 > d_2 > \cdots > d_l$. Since these are divisors of n, we have

$$d_l \leqslant \frac{n}{l}. \tag{1}$$

As d_l is the maximum entry of the l-th column, we must have

$$d_l \geqslant \frac{s}{k} \geqslant \frac{n}{k}. \tag{2}$$

The relations (1) and (2) combine to give $\frac{n}{l} \geqslant \frac{n}{k}$, that is, $k \geqslant l$. Together with $k \leqslant l$, we conclude that $k = l$. Then all inequalities in (1) and (2) are equalities. In particular, $s = n$ and so $n = 1$, in which case the conditions are clearly satisfied.

Solution 2. Clearly $n = 1$ works. Then we assume $n > 1$ and let its prime factorization be $n = p_1^{r_1} p_2^{r_2} \cdots p_t^{r_t}$. Suppose the table has k rows and l columns with $1 < k \leqslant l$. Note that kl is the number of positive divisors of n and the sum of all entries is the sum of positive divisors of n, which we denote by $\sigma(n)$. Consider the column containing n. Since the column sum is $\frac{\sigma(n)}{l}$, we must have $\frac{\sigma(n)}{l} > n$. Therefore, we have

$$\begin{aligned}(r_1 + 1)(r_2 + 1) \cdots (r_t + 1) = kl \leqslant l^2 &< \left(\frac{\sigma(n)}{n}\right)^2 \\ &= \left(1 + \frac{1}{p_1} + \cdots + \frac{1}{p_1^{r_1}}\right)^2 \cdots \left(1 + \frac{1}{p_t} + \cdots + \frac{1}{p_t^{r_t}}\right)^2.\end{aligned}$$

This can be rewritten as

$$f(p_1, r_1) f(p_2, r_2) \cdots f(p_t, r_t) < 1 \tag{3}$$

where

$$f(p, r) = \frac{r + 1}{\left(1 + \frac{1}{p} + \cdots + \frac{1}{p^r}\right)^2} = \frac{(r + 1)\left(1 - \frac{1}{p}\right)^2}{\left(1 - \frac{1}{p^{r+1}}\right)^2}.$$

Direct computation yields

$$f(2,1)=\frac{8}{9},\quad f(2,2)=\frac{48}{49},\quad f(3,1)=\frac{9}{8}.$$

Also, we find that

$$\begin{aligned}
f(2,r) &\geqslant \left(1-\frac{1}{2^{r+1}}\right)^{-2} > 1 \quad \text{for } r \geqslant 3,\\
f(3,r) &\geqslant \frac{4}{3}\left(1-\frac{1}{3^{r+1}}\right)^{-2} > \frac{4}{3} > \frac{9}{8} \quad \text{for } r \geqslant 2, \text{ and}\\
f(p,r) &\geqslant \frac{32}{25}\left(1-\frac{1}{p^{r+1}}\right)^{-2} > \frac{32}{25} > \frac{9}{8} \quad \text{for } p \geqslant 5.
\end{aligned}$$

From these values and bounds, it is clear that (3) holds only when $n=2$ or 4. In both cases, it is easy to see that the conditions are not satisfied. Hence, the only possible n is 1.

C3. Let n be a positive integer relatively prime to 6. We paint the vertices of a regular n-gon with three colours so that there is an odd number of vertices of each colour. Show that there exists an isosceles triangle whose three vertices are of different colours.

Solution. For $k = 1, 2, 3$, let a_k be the number of isosceles triangles whose vertices contain exactly k colours. Suppose on the contrary that $a_3 = 0$. Let b, c, d be the number of vertices of the three different colours respectively. We now count the number of pairs $(\triangle, E)$ where $\triangle$ is an isosceles triangle and E is a side of $\triangle$ whose endpoints are of different colours.

On the one hand, since we have assumed $a_3 = 0$, each triangle in the pair must contain exactly two colours, and hence each triangle contributes twice. Thus the number of pairs is $2a_2$.

On the other hand, if we pick any two vertices A, B of distinct colours, then there are three isosceles triangles having these as vertices, two when AB is not the base and one when AB is the base since n is odd. Note that the three triangles are all distinct as $(n, 3) = 1$. In this way, we count the number of pairs to be $3(bc + cd + db)$. However, note that $2a_2$ is even while $3(bc + cd + db)$ is odd, as each of b, c, d is. This yields a contradiction and hence $a_3 \geqslant 1$.

Comment. A slightly stronger version of this problem is to replace the condition $(n, 6) = 1$ by n being odd (where equilateral triangles are regarded as isosceles triangles). In that case, the only difference in the proof is that by fixing any two vertices A, B, one can find exactly one or three isosceles triangles having these as vertices. But since only parity is concerned in the solution, the proof goes the same way.

The condition that there is an odd number of vertices of each colour is necessary, as can be seen from the following example. Consider $n = 25$ and we label the vertices $A_0, A_1, \ldots, A_{24}$. Suppose colour 1 is used for A_0, colour 2 is used for $A_5, A_{10}, A_{15}, A_{20}$, while colour 3 is used for the remaining vertices. Then any isosceles triangle having colours 1 and 2 must contain A_0 and one of $A_5, A_{10}, A_{15}, A_{20}$. Clearly, the third vertex must have index which is a multiple of 5 so it is not of colour 3.

C4. Find all positive integers n for which we can fill in the entries of an $n \times n$ table with the following properties:

- each entry can be one of I, M and O;
- in each row and each column, the letters I, M and O occur the same number of times; and
- in any diagonal whose number of entries is a multiple of three, the letters I, M and O occur the same number of times.

Answer. n can be any multiple of 9.

Solution. We first show that such a table exists when n is a multiple of 9. Consider the following 9×9 table.

$$\begin{pmatrix} I & I & I & M & M & M & O & O & O \\ M & M & M & O & O & O & I & I & I \\ O & O & O & I & I & I & M & M & M \\ I & I & I & M & M & M & O & O & O \\ M & M & M & O & O & O & I & I & I \\ O & O & O & I & I & I & M & M & M \\ I & I & I & M & M & M & O & O & O \\ M & M & M & O & O & O & I & I & I \\ O & O & O & I & I & I & M & M & M \end{pmatrix} \tag{1}$$

It is a direct checking that the table (1) satisfies the requirements. For $n = 9k$ where k is a positive integer, we form an $n \times n$ table using $k \times k$ copies of (1). For each row and each column of the table of size n, since there are three I's, three M's and three O's for any nine consecutive entries, the numbers of I, M and O are equal. In addition, every diagonal of the large table whose number of entries is divisible by 3 intersects each copy of (1) at a diagonal with number of entries divisible by 3 (possibly zero). Therefore, every such diagonal also contains the same number of I, M and O.

Next, consider any $n \times n$ table for which the requirements can be met. As the number of entries of each row should be a multiple of 3, we let $n = 3k$ where k is a positive integer. We divide the whole table into $k \times k$ copies of 3×3 blocks. We call the entry at the centre of such a 3×3 square a *vital entry*. We also call any row, column or diagonal that contains at least one vital entry a *vital line*. We compute the number of pairs (l, c) where l is a vital line and c is an entry belonging to l that contains the letter M. We let this number be N.

On the one hand, since each vital line contains the same number of I, M and O, it is obvious that each vital row and each vital column contain k occurrences of M. For vital diagonals in either direction, we count there are exactly

$$1 + 2 + \cdots + (k-1) + k + (k-1) + \cdots + 2 + 1 = k^2$$

occurrences of M. Therefore, we have $N = 4k^2$.

On the other hand, there are $3k^2$ occurrences of M in the whole table. Note that each entry belongs to exactly 1 or 4 vital lines. Therefore, N must be congruent to $3k^2$ mod 3.

From the double counting, we get $4k^2 \equiv 3k^2 \pmod 3$, which forces k to be a multiple of 3. Therefore, n has to be a multiple of 9 and the proof is complete.

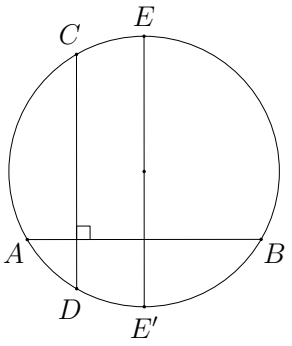
C5. Let $n \geqslant 3$ be a positive integer. Find the maximum number of diagonals of a regular n-gon one can select, so that any two of them do not intersect in the interior or they are perpendicular to each other.

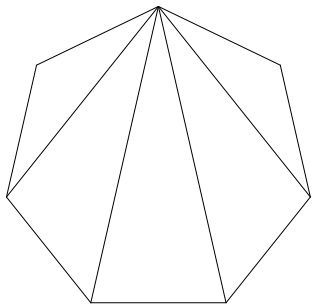
Answer. $n-2$ if n is even and $n-3$ if n is odd.

Solution 1. We consider two cases according to the parity of n.

• **Case 1.** n is odd.

We first claim that no pair of diagonals is perpendicular. Suppose A, B, C, D are vertices where AB and CD are perpendicular, and let E be the vertex lying on the perpendicular bisector of AB. Let E' be the opposite point of E on the circumcircle of the regular polygon. Since $EC = E'D$ and C, D, E are vertices of the regular polygon, E' should also belong to the polygon. This contradicts the fact that a regular polygon with an odd number of vertices does not contain opposite points on the circumcircle.



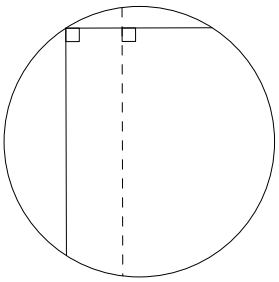


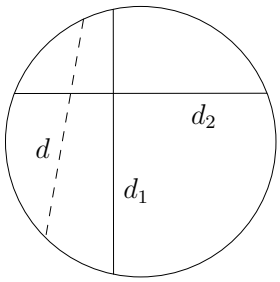
Therefore in the odd case we can only select diagonals which do not intersect. In the maximal case these diagonals should divide the regular n-gon into $n-2$ triangles, so we can select at most $n-3$ diagonals. This can be done, for example, by selecting all diagonals emanated from a particular vertex.

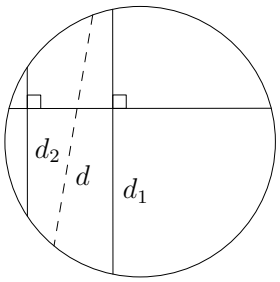
• **Case 2.** n is even.

If there is no intersection, then the proof in the odd case works. Suppose there are two perpendicular diagonals selected. We consider the set S of all selected diagonals parallel to one of them which intersect with some selected diagonals. Suppose S contains k diagonals and the number of distinct endpoints of the k diagonals is l.

Firstly, consider the longest diagonal in one of the two directions in S. No other diagonal in S can start from either endpoint of that diagonal, since otherwise it has to meet another longer diagonal in S. The same holds true for the other direction. Ignoring these two longest diagonals and their four endpoints, the remaining $k-2$ diagonals share $l-4$ endpoints where each endpoint can belong to at most two diagonals. This gives $2(l-4) \geqslant 2(k-2)$, so that $k \leqslant l-2$.





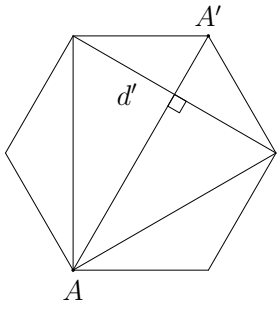


Consider a group of consecutive vertices of the regular n-gon so that each of the two outermost vertices is an endpoint of a diagonal in S, while the interior points are not. There are l such groups. We label these groups $P_1, P_2, \ldots, P_l$ in this order. We claim that each selected diagonal outside S must connect vertices of the same group P_i. Consider any diagonal d joining vertices from distinct groups P_i and P_j. Let d_1 and d_2 be two diagonals in S each having one of the outermost points of P_i as endpoint. Then d must meet either d_1, d_2 or a diagonal in S which is perpendicular to both d_1 and d_2. In any case d should belong to S by definition, which is a contradiction.

Within the same group P_i, there are no perpendicular diagonals since the vertices belong to the same side of a diameter of the circumcircle. Hence there can be at most $|P_i|-2$ selected diagonals within P_i, including the one joining the two outermost points of P_i when $|P_i| > 2$. Therefore, the maximum number of diagonals selected is

$$\sum_{i=1}^{l} (|P_i| - 2) + k = \sum_{i=1}^{l} |P_i| - 2l + k = (n + l) - 2l + k = n - l + k \leqslant n - 2.$$

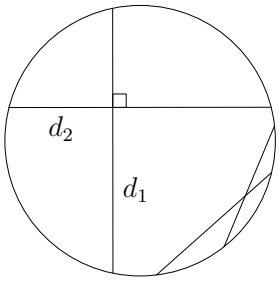
This upper bound can be attained as follows. We take any vertex A and let A' be the vertex for which AA' is a diameter of the circumcircle. If we select all diagonals emanated from A together with the diagonal d' joining the two neighbouring vertices of A', then the only pair of diagonals that meet each other is AA' and d', which are perpendicular to each other. In total we can take $n - 2$ diagonals.

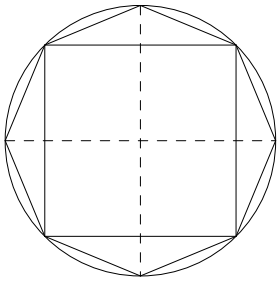


Solution 2. The constructions and the odd case are the same as Solution 1. Instead of dealing separately with the case where n is even, we shall prove by induction more generally that we can select at most $n - 2$ diagonals for any cyclic n-gon with circumcircle Γ.

The base case $n = 3$ is trivial since there is no diagonal at all. Suppose the upper bound holds for any cyclic polygon with fewer than n sides. For a cyclic n-gon, if there is a selected diagonal which does not intersect any other selected diagonal, then this diagonal divides the n-gon into an m-gon and an l-gon (with $m+l = n+2$) so that each selected diagonal belongs to one of them. Without loss of generality, we may assume the m-gon lies on the same side of a diameter of Γ. Then no two selected diagonals of the m-gon can intersect, and hence we can select at most $m-3$ diagonals. Also, we can apply the inductive hypothesis to the l-gon. This shows the maximum number of selected diagonals is $(m-3)+(l-2)+1 = n-2$.

It remains to consider the case when all selected diagonals meet at least one other selected diagonal. Consider a pair of selected perpendicular diagonals d_1, d_2. They divide the circumference of Γ into four arcs, each of which lies on the same side of a diameter of Γ. If there are two selected diagonals intersecting each other and neither is parallel to d_1 or d_2, then their endpoints must belong to the same arc determined by d_1, d_2, and hence they cannot be perpendicular. This violates the condition, and hence all selected diagonals must have the same direction as one of d_1, d_2.





Take the longest selected diagonal in one of the two directions. We argue as in Solution 1 that its endpoints do not belong to any other selected diagonal. The same holds true for the longest diagonal in the other direction. Apart from these four endpoints, each of the remaining $n-4$ vertices can belong to at most two selected diagonals. Thus we can select at most $\frac{1}{2}(2(n-4)+4) = n-2$ diagonals. Then the proof follows by induction.

C6. There are $n \geqslant 3$ islands in a city. Initially, the ferry company offers some routes between some pairs of islands so that it is impossible to divide the islands into two groups such that no two islands in different groups are connected by a ferry route.

After each year, the ferry company will close a ferry route between some two islands X and Y. At the same time, in order to maintain its service, the company will open new routes according to the following rule: for any island which is connected by a ferry route to exactly one of X and Y, a new route between this island and the other of X and Y is added.

Suppose at any moment, if we partition all islands into two nonempty groups in any way, then it is known that the ferry company will close a certain route connecting two islands from the two groups after some years. Prove that after some years there will be an island which is connected to all other islands by ferry routes.

Solution. Initially, we pick any pair of islands A and B which are connected by a ferry route and put A in set $\mathcal{A}$ and B in set $\mathcal{B}$. From the condition, without loss of generality there must be another island which is connected to A. We put such an island C in set $\mathcal{B}$. We say that two sets of islands form a *network* if each island in one set is connected to each island in the other set.

Next, we shall included all islands to $\mathcal{A} \cup \mathcal{B}$ one by one. Suppose we have two sets $\mathcal{A}$ and $\mathcal{B}$ which form a network where $3 \leqslant |\mathcal{A} \cup \mathcal{B}| < n$. This relation no longer holds only when a ferry route between islands $A \in \mathcal{A}$ and $B \in \mathcal{B}$ is closed. In that case, we define $\mathcal{A}' = \{A, B\}$, and $\mathcal{B}' = (\mathcal{A} \cup \mathcal{B}) - \{A, B\}$. Note that $\mathcal{B}'$ is nonempty. Consider any island $C \in \mathcal{A} - \{A\}$. From the relation of $\mathcal{A}$ and $\mathcal{B}$, we know that C is connected to B. If C was not connected to A before the route between A and B closes, then there will be a route added between C and A afterwards. Hence, C must now be connected to both A and B. The same holds true for any island in $\mathcal{B} - \{B\}$. Therefore, $\mathcal{A}'$ and $\mathcal{B}'$ form a network, and $\mathcal{A}' \cup \mathcal{B}' = \mathcal{A} \cup \mathcal{B}$. Hence these islands can always be partitioned into sets $\mathcal{A}$ and $\mathcal{B}$ which form a network.

As $|\mathcal{A} \cup \mathcal{B}| < n$, there are some islands which are not included in $\mathcal{A} \cup \mathcal{B}$. From the condition, after some years there must be a ferry route between an island A in $\mathcal{A} \cup \mathcal{B}$ and an island D outside $\mathcal{A} \cup \mathcal{B}$ which closes. Without loss of generality assume $A \in \mathcal{A}$. Then each island in $\mathcal{B}$ must then be connected to D, no matter it was or not before. Hence, we can put D in set $\mathcal{A}$ so that the new sets $\mathcal{A}$ and $\mathcal{B}$ still form a network and the size of $\mathcal{A} \cup \mathcal{B}$ is increased by 1. The same process can be done to increase the size of $\mathcal{A} \cup \mathcal{B}$. Eventually, all islands are included in this way so we may now assume $|\mathcal{A} \cup \mathcal{B}| = n$.

Suppose a ferry route between $A \in \mathcal{A}$ and $B \in \mathcal{B}$ is closed after some years. We put A and B in set $\mathcal{A}'$ and all remaining islands in set $\mathcal{B}'$. Then $\mathcal{A}'$ and $\mathcal{B}'$ form a network. This relation no longer holds only when a route between A, without loss of generality, and $C \in \mathcal{B}'$ is closed. Since this must eventually occur, at that time island B will be connected to all other islands and the result follows.

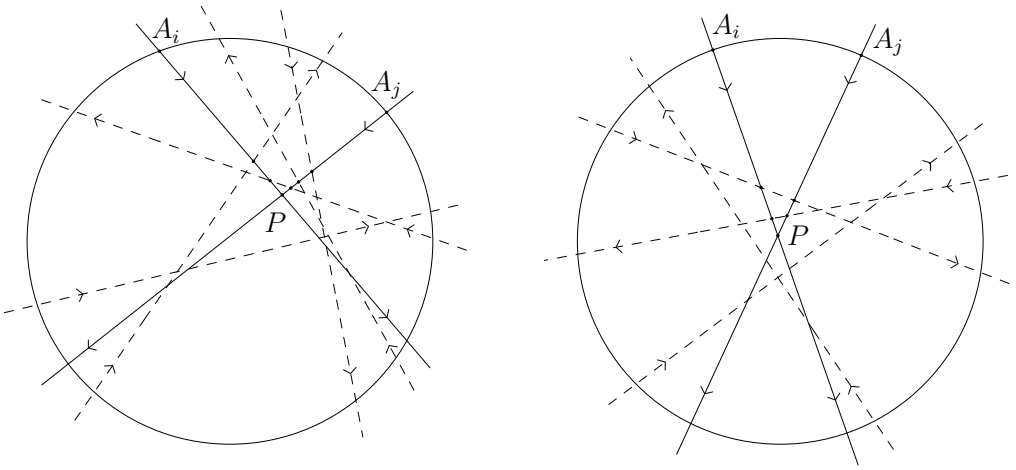
C7. Let $n \geqslant 2$ be an integer. In the plane, there are n segments given in such a way that any two segments have an intersection point in the interior, and no three segments intersect at a single point. Jeff places a snail at one of the endpoints of each of the segments and claps his hands $n-1$ times. Each time when he claps his hands, all the snails move along their own segments and stay at the next intersection points until the next clap. Since there are $n-1$ intersection points on each segment, all snails will reach the furthest intersection points from their starting points after $n-1$ claps.

(a) Prove that if n is odd then Jeff can always place the snails so that no two of them ever occupy the same intersection point.

(b) Prove that if n is even then there must be a moment when some two snails occupy the same intersection point no matter how Jeff places the snails.

Solution. We consider a big disk which contains all the segments. We extend each segment to a line l_i so that each of them cuts the disk at two distinct points A_i, B_i.

(a) For odd n, we travel along the circumference of the disk and mark each of the points A_i or B_i 'in' and 'out' alternately. Since every pair of lines intersect in the disk, there are exactly $n-1$ points between A_i and B_i for any fixed $1 \leqslant i \leqslant n$. As n is odd, this means one of A_i and B_i is marked 'in' and the other is marked 'out'. Then Jeff can put a snail on the endpoint of each segment which is closer to the 'in' side of the corresponding line. We claim that the snails on l_i and l_j do not meet for any pairs i, j, hence proving part (a).

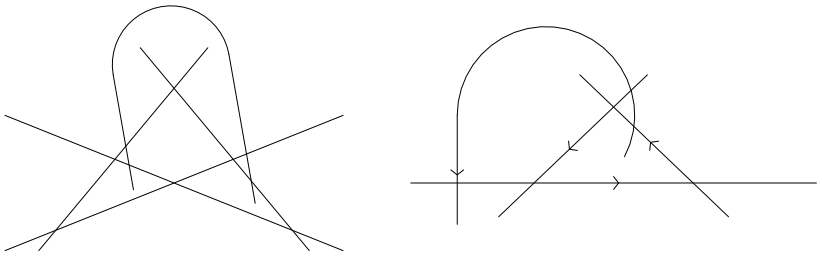


Without loss of generality, we may assume the snails start at A_i and A_j respectively. Let l_i intersect l_j at P. Note that there is an odd number of points between arc A_iA_j. Each of these points belongs to a line l_k. Such a line l_k must intersect exactly one of

the segments A_iP and A_jP, making an odd number of intersections. For the other lines, they may intersect both segments A_iP and A_jP, or meet none of them. Therefore, the total number of intersection points on segments A_iP and A_jP (not counting P) is odd. However, if the snails arrive at P at the same time, then there should be the same number of intersections on A_iP and A_jP, which gives an even number of intersections. This is a contradiction so the snails do not meet each other.

(b) For even n, we consider any way that Jeff places the snails and mark each of the points A_i or B_i 'in' and 'out' according to the directions travelled by the snails. In this case there must be two neighbouring points A_i and A_j both of which are marked 'in'. Let P be the intersection of the segments A_iB_i and A_jB_j. Then any other segment meeting one of the segments A_iP and A_jP must also meet the other one, and so the number of intersections on A_iP and A_jP are the same. This shows the snails starting from A_i and A_j will meet at P.

Comment. The conclusions do not hold for pseudosegments, as can be seen from the following examples.

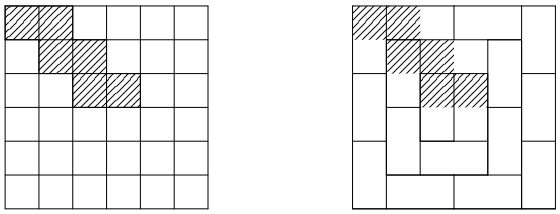


C8. Let n be a positive integer. Determine the smallest positive integer k with the following property: it is possible to mark k cells on a $2n \times 2n$ board so that there exists a unique partition of the board into 1×2 and 2×1 dominoes, none of which contains two marked cells.

Answer. $2n$.

Solution. We first construct an example of marking $2n$ cells satisfying the requirement. Label the rows and columns $1, 2, \ldots, 2n$ and label the cell in the i-th row and the j-th column (i, j).

For $i = 1, 2, \ldots, n$, we mark the cells (i, i) and $(i, i + 1)$. We claim that the required partition exists and is unique. The two diagonals of the board divide the board into four regions. Note that the domino covering cell $(1, 1)$ must be vertical. This in turn shows that each domino covering $(2, 2), (3, 3), \ldots, (n, n)$ is vertical. By induction, the dominoes in the left region are all vertical. By rotational symmetry, the dominoes in the bottom region are horizontal, and so on. This shows that the partition exists and is unique.



It remains to show that this value of k is the smallest possible. Assume that only $k < 2n$ cells are marked, and there exists a partition P satisfying the requirement. It suffices to show there exists another desirable partition distinct from P. Let d be the main diagonal of the board.

Construct the following graph with edges of two colours. Its vertices are the cells of the board. Connect two vertices with a red edge if they belong to the same domino of P. Connect two vertices with a blue edge if their reflections in d are connected by a red edge. It is possible that two vertices are connected by edges of both colours. Clearly, each vertex has both red and blue degrees equal to 1. Thus the graph splits into cycles where the colours of edges in each cycle alternate (a cycle may have length 2).

Consider any cell c lying on the diagonal d. Its two edges are symmetrical with respect to d. Thus they connect c to different cells. This shows c belongs to a cycle $C(c)$ of length at least 4. Consider a part of this cycle $c_0, c_1, \ldots, c_m$ where $c_0 = c$ and m is the least positive integer such that c_m lies on d. Clearly, c_m is distinct from c. From the construction, the path symmetrical to this with respect to d also lies in the graph, and so these paths together form $C(c)$. Hence, $C(c)$ contains exactly two cells from d. Then all $2n$ cells in d belong to n cycles $C_1, C_2, \ldots, C_n$, each has length at least 4.

By the Pigeonhole Principle, there exists a cycle C_i containing at most one of the k marked cells. We modify P as follows. We remove all dominoes containing the vertices of C_i, which

correspond to the red edges of C_i. Then we put the dominoes corresponding to the blue edges of C_i. Since C_i has at least 4 vertices, the resultant partition P' is different from P. Clearly, no domino in P' contains two marked cells as C_i contains at most one marked cell. This shows the partition is not unique and hence k cannot be less than $2n$.

Geometry

G1. In a convex pentagon $ABCDE$, let F be a point on AC such that $\angle FBC = 90°$. Suppose triangles ABF, ACD and ADE are similar isosceles triangles with

$$\angle FAB = \angle FBA = \angle DAC = \angle DCA = \angle EAD = \angle EDA. \tag{1}$$

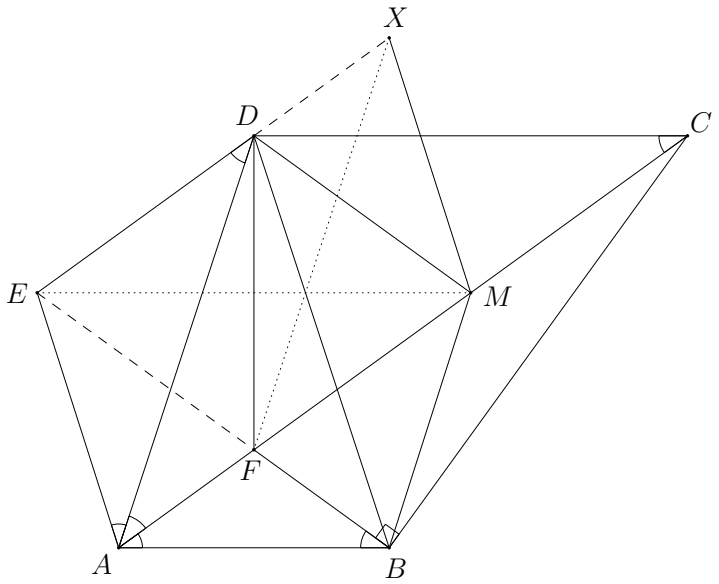
Let M be the midpoint of CF. Point X is chosen such that $AMXE$ is a parallelogram. Show that BD, EM and FX are concurrent.

Solution 1. Denote the common angle in (1) by θ. As $\triangle ABF \sim \triangle ACD$, we have $\frac{AB}{AC} = \frac{AF}{AD}$ so that $\triangle ABC \sim \triangle AFD$. From $EA = ED$, we get

$$\angle AFD = \angle ABC = 90° + \theta = 180° - \frac{1}{2}\angle AED.$$

Hence, F lies on the circle with centre E and radius EA. In particular, $EF = EA = ED$. As $\angle EFA = \angle EAF = 2\theta = \angle BFC$, points B, F, E are collinear.

As $\angle EDA = \angle MAD$, we have $ED // AM$ and hence E, D, X are collinear. As M is the midpoint of CF and $\angle CBF = 90°$, we get $MF = MB$. In the isosceles triangles EFA and MFB, we have $\angle EFA = \angle MFB$ and $AF = BF$. Therefore, they are congruent to each other. Then we have $BM = AE = XM$ and $BE = BF + FE = AF + FM = AM = EX$. This shows $\triangle EMB \cong \triangle EMX$. As F and D lie on EB and EX respectively and $EF = ED$, we know that lines BD and XF are symmetric with respect to EM. It follows that the three lines are concurrent.



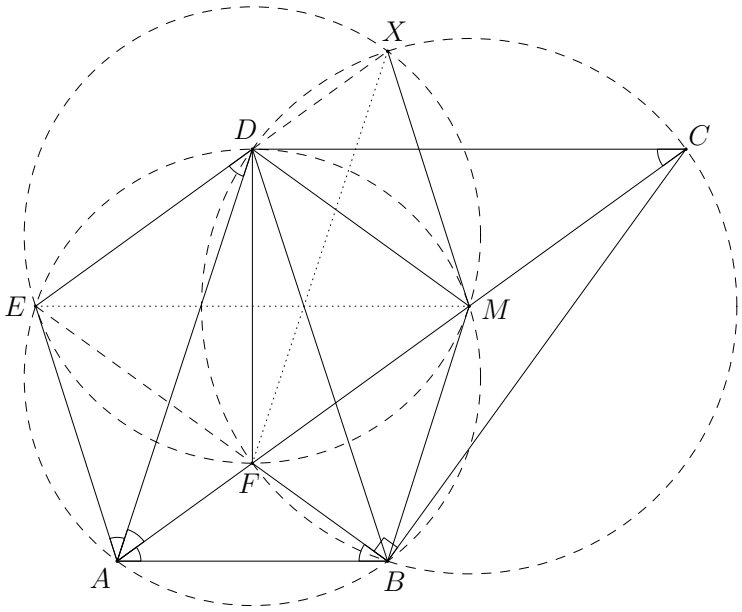
Solution 2. From $\angle CAD = \angle EDA$, we have $AC//ED$. Together with $AC//EX$, we know that E, D, X are collinear. Denote the common angle in (1) by θ. From $\triangle ABF \sim \triangle ACD$, we get $\frac{AB}{AC} = \frac{AF}{AD}$ so that $\triangle ABC \sim \triangle AFD$. This yields $\angle AFD = \angle ABC = 90^\circ + \theta$ and hence $\angle FDC = 90^\circ$, implying that $BCDF$ is cyclic. Let Γ_1 be its circumcircle.

Next, from $\triangle ABF \sim \triangle ADE$, we have $\frac{AB}{AD} = \frac{AF}{AE}$ so that $\triangle ABD \sim \triangle AFE$. Therefore,

$$\angle AFE = \angle ABD = \theta + \angle FBD = \theta + \angle FCD = 2\theta = 180^\circ - \angle BFA.$$

This implies B, F, E are collinear. Note that F is the incentre of triangle DAB. Point E lies on the internal angle bisector of $\angle DBA$ and lies on the perpendicular bisector of AD. It follows that E lies on the circumcircle Γ_2 of triangle ABD, and $EA = EF = ED$.

Also, since CF is a diameter of Γ_1 and M is the midpoint of CF, M is the centre of Γ_1 and hence $\angle AMD = 2\theta = \angle ABD$. This shows M lies on Γ_2. Next, $\angle MDX = \angle MAE = \angle DXM$ since $AMXE$ is a parallelogram. Hence $MD = MX$ and X lies on Γ_1.



We now have two ways to complete the solution.

• **Method 1.** From $EF = EA = XM$ and $EX//FM$, $EFMX$ is an isosceles trapezoid and is cyclic. Denote its circumcircle by Γ_3. Since BD, EM, FX are the three radical axes of $\Gamma_1, \Gamma_2, \Gamma_3$, they must be concurrent.

• **Method 2.** As $\angle DMF = 2\theta = \angle BFM$, we have $DM//EB$. Also,

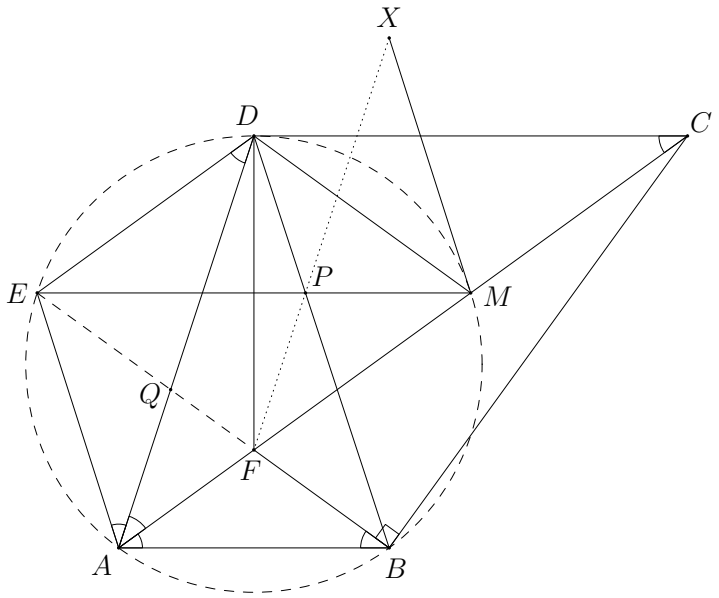
$$\angle BFD + \angle XBF = \angle BFC + \angle CFD + 90^\circ - \angle CBX = 2\theta + (90^\circ - \theta) + 90^\circ - \theta = 180^\circ$$

implies $DF//XB$. These show the corresponding sides of triangles DMF and BEX are parallel. By Desargues' Theorem, the two triangles are perspective and hence DB, ME, FX meet at a point.

Comment. In Solution 2, both the Radical Axis Theorem and Desargues' Theorem could imply DB, ME, FX are parallel. However, this is impossible as can be seen from the configuration. For example, it is obvious that DB and ME meet each other.

Solution 3. Let the common angle in (1) be θ. From $\triangle ABF \sim \triangle ACD$, we have $\frac{AB}{AC} = \frac{AF}{AD}$ so that $\triangle ABC \sim \triangle AFD$. Then $\angle ADF = \angle ACB = 90^\circ - 2\theta = 90^\circ - \angle BAD$ and hence $DF \perp AB$. As $FA = FB$, this implies $\triangle DAB$ is isosceles with $DA = DB$. Then F is the incentre of $\triangle DAB$.

Next, from $\angle AED = 180^\circ - 2\theta = 180^\circ - \angle DBA$, points A, B, D, E are concyclic. Since we also have $EA = ED$, this shows E, F, B are collinear and $EA = EF = ED$.



Note that C lies on the internal angle bisector of $\angle BAD$ and lies on the external angle bisector of $\angle DBA$. It follows that it is the A-excentre of triangle DAB. As M is the midpoint of CF, M lies on the circumcircle of triangle DAB and it is the centre of the circle passing through D, F, B, C. By symmetry, $DEFM$ is a rhombus. Then the midpoints of AX, EM and DF coincide, and it follows that $DAFX$ is a parallelogram.

Let P be the intersection of BD and EM, and Q be the intersection of AD and BE. From $\angle BAC = \angle DCA$, we know that DC, AB, EM are parallel. Thus we have $\frac{DP}{PB} = \frac{CM}{MA}$. This is further equal to $\frac{AE}{BE}$ since $CM = DM = DE = AE$ and $MA = BE$. From $\triangle AEQ \sim \triangle BEA$, we find that

$$\frac{DP}{PB} = \frac{AE}{BE} = \frac{AQ}{BA} = \frac{QF}{FB}$$

by the Angle Bisector Theorem. This implies $QD // FP$ and hence F, P, X are collinear, as desired.

G2. Let ABC be a triangle with circumcircle Γ and incentre I. Let M be the midpoint of side BC. Denote by D the foot of perpendicular from I to side BC. The line through I perpendicular to AI meets sides AB and AC at F and E respectively. Suppose the circumcircle of triangle AEF intersects Γ at a point X other than A. Prove that lines XD and AM meet on Γ.

Solution 1. Let AM meet Γ again at Y and XY meet BC at D'. It suffices to show $D' = D$. We shall apply the following fact.

• **Claim.** For any cyclic quadrilateral $PQRS$ whose diagonals meet at T, we have

$$\frac{QT}{TS} = \frac{PQ \cdot QR}{PS \cdot SR}.$$

Proof. We use $[W_1W_2W_3]$ to denote the area of $W_1W_2W_3$. Then

$$\frac{QT}{TS} = \frac{[PQR]}{[PSR]} = \frac{\frac{1}{2}PQ \cdot QR \sin \angle PQR}{\frac{1}{2}PS \cdot SR \sin \angle PSR} = \frac{PQ \cdot QR}{PS \cdot SR}.$$

□

Applying the Claim to $ABYC$ and $XBYC$ respectively, we have $1 = \frac{BM}{MC} = \frac{AB \cdot BY}{AC \cdot CY}$ and $\frac{BD'}{D'C} = \frac{XB \cdot BY}{XC \cdot CY}$. These combine to give

$$\frac{BD'}{CD'} = \frac{XB}{XC} \cdot \frac{BY}{CY} = \frac{XB}{XC} \cdot \frac{AC}{AB}. \tag{1}$$

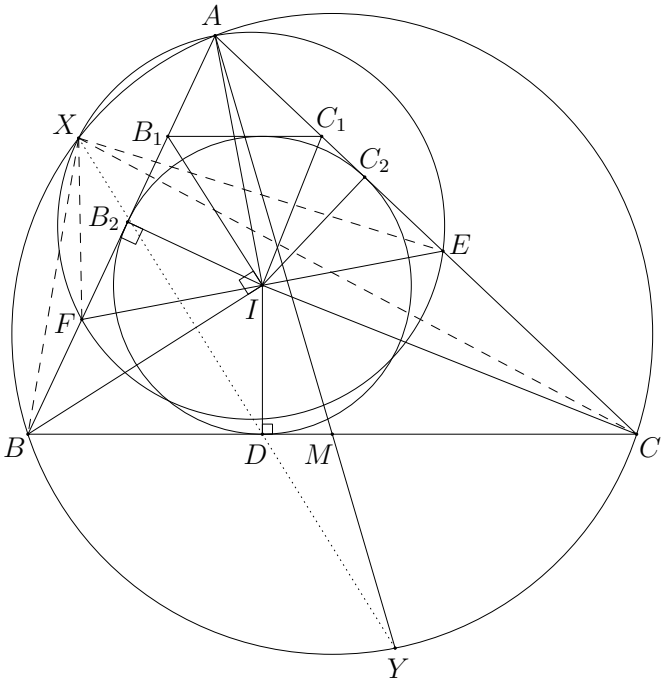
Next, we use directed angles to find that $\measuredangle XBF = \measuredangle XBA = \measuredangle XCA = \measuredangle XCE$ and $\measuredangle XFB = \measuredangle XFA = \measuredangle XEA = \measuredangle XEC$. This shows triangles XBF and XCE are directly similar. In particular, we have

$$\frac{XB}{XC} = \frac{BF}{CE}. \tag{2}$$

In the following, we give two ways to continue the proof.

• **Method 1.** Here is a geometrical method. As $\angle FIB = \angle AIB - 90^\circ = \frac{1}{2}\angle ACB = \angle ICB$ and $\angle FBI = \angle IBC$, the triangles FBI and IBC are similar. Analogously, triangles EIC and IBC are also similar. Hence, we get

$$\frac{FB}{IB} = \frac{BI}{BC} \quad \text{and} \quad \frac{EC}{IC} = \frac{IC}{BC}. \tag{3}$$



Next, construct a line parallel to BC and tangent to the incircle. Suppose it meets sides AB and AC at B_1 and C_1 respectively. Let the incircle touch AB and AC at B_2 and C_2 respectively. By homothety, the line B_1I is parallel to the external angle bisector of $\angle ABC$, and hence $\angle B_1IB = 90^\circ$. Since $\angle BB_2I = 90^\circ$, we get $BB_2 \cdot BB_1 = BI^2$, and similarly $CC_2 \cdot CC_1 = CI^2$. Hence,

$$\frac{BI^2}{CI^2} = \frac{BB_2 \cdot BB_1}{CC_2 \cdot CC_1} = \frac{BB_1}{CC_1} \cdot \frac{BD}{CD} = \frac{AB}{AC} \cdot \frac{BD}{CD}. \tag{4}$$

Combining (1), (2), (3) and (4), we conclude

$$\frac{BD'}{CD'} = \frac{XB}{XC} \cdot \frac{AC}{AB} = \frac{BF}{CE} \cdot \frac{AC}{AB} = \frac{BI^2}{CI^2} \cdot \frac{AC}{AB} = \frac{BD}{CD}$$

so that $D' = D$. The result then follows.

• **Method 2.** We continue the proof of Solution 1 using trigonometry. Let $\beta = \frac{1}{2}\angle ABC$ and $\gamma = \frac{1}{2}\angle ACB$. Observe that $\angle FIB = \angle AIB - 90^\circ = \gamma$. Hence, $\frac{BF}{FI} = \frac{\sin\angle FIB}{\sin\angle IBF} = \frac{\sin\gamma}{\sin\beta}$. Similarly, $\frac{CE}{EI} = \frac{\sin\beta}{\sin\gamma}$. As $FI = EI$, we get

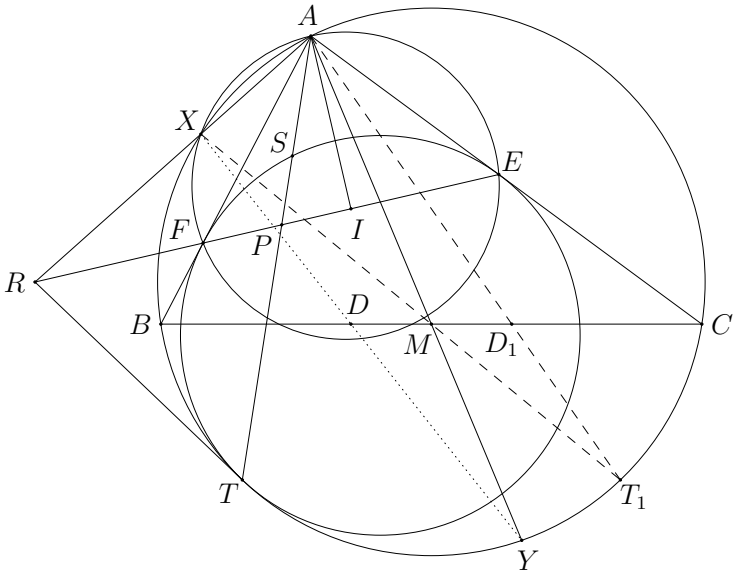
$$\frac{BF}{CE} = \frac{BF}{FI} \cdot \frac{EI}{CE} = \left(\frac{\sin\gamma}{\sin\beta}\right)^2. \tag{5}$$

Together with (1) and (2), we find that

$$\frac{BD'}{CD'} = \frac{AC}{AB} \cdot \left(\frac{\sin\gamma}{\sin\beta}\right)^2 = \frac{\sin 2\beta}{\sin 2\gamma} \cdot \left(\frac{\sin\gamma}{\sin\beta}\right)^2 = \frac{\tan\gamma}{\tan\beta} = \frac{ID/CD}{ID/BD} = \frac{BD}{CD}.$$

This shows $D' = D$ and the result follows.

Solution 2. Let ω_A be the A-mixtilinear incircle of triangle ABC. From the properties of mixtilinear incircles, ω_A touches sides AB and AC at F and E respectively. Suppose ω_A is tangent to Γ at T. Let AM meet Γ again at Y, and let D_1, T_1 be the reflections of D and T with respect to the perpendicular bisector of BC respectively. It is well-known that $\angle BAT = \angle D_1AC$ so that A, D_1, T_1 are collinear.



We then show that X, M, T_1 are collinear. Let R be the radical centre of ω_A, Γ and the circumcircle of triangle AEF. Then R lies on AX, EF and the tangent at T to Γ. Let AT meet ω_A again at S and meet EF at P. Obviously, $SFTE$ is a harmonic quadrilateral. Projecting from T, the pencil $(R, P; F, E)$ is harmonic. We further project the pencil onto Γ from A, so that $XBTC$ is a harmonic quadrilateral. As $TT_1 // BC$, the projection from T_1 onto BC maps T to a point at infinity, and hence maps X to the midpoint of BC, which is M. This shows X, M, T_1 are collinear.

We have two ways to finish the proof.

• **Method 1.** Note that both AY and XT_1 are chords of Γ passing through the midpoint M of the chord BC. By the Butterfly Theorem, XY and AT_1 cut BC at a pair of symmetric points with respect to M, and hence X, D, Y are collinear. The proof is thus complete.

• **Method 2.** Here, we finish the proof without using the Butterfly Theorem. As DTT_1D_1 is an isosceles trapezoid, we have

$$\measuredangle YTD = \measuredangle YTT_1 + \measuredangle T_1TD = \measuredangle YAT_1 + \measuredangle AD_1D = \measuredangle YMD$$

so that D, T, Y, M are concyclic. As X, M, T_1 are collinear, we have

$$\measuredangle AYD = \measuredangle MTD = \measuredangle D_1T_1M = \measuredangle AT_1X = \measuredangle AYX.$$

This shows X, D, Y are collinear.

G3. Let $B = (-1, 0)$ and $C = (1, 0)$ be fixed points on the coordinate plane. A nonempty, bounded subset S of the plane is said to be *nice* if

(i) there is a point T in S such that for every point Q in S, the segment TQ lies entirely in S; and

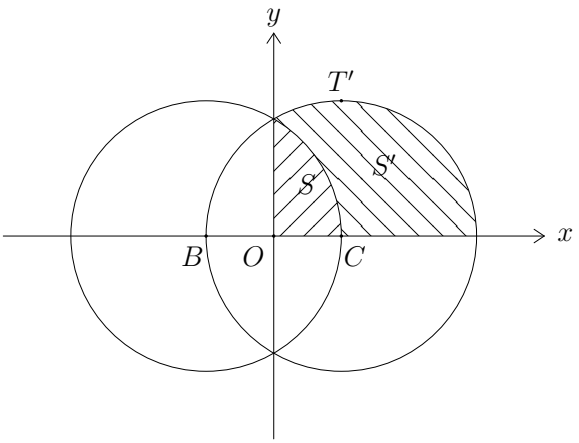
(ii) for any triangle $P_1P_2P_3$, there exists a unique point A in S and a permutation σ of the indices $\{1, 2, 3\}$ for which triangles ABC and $P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$ are similar.

Prove that there exist two distinct nice subsets S and S' of the set $\{(x, y) : x \geqslant 0, y \geqslant 0\}$ such that if $A \in S$ and $A' \in S'$ are the unique choices of points in (ii), then the product $BA \cdot BA'$ is a constant independent of the triangle $P_1P_2P_3$.

Solution. If in the similarity of $\triangle ABC$ and $\triangle P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$, BC corresponds to the longest side of $\triangle P_1P_2P_3$, then we have $BC \geqslant AB \geqslant AC$. The condition $BC \geqslant AB$ is equivalent to $(x + 1)^2 + y^2 \leqslant 4$, while $AB \geqslant AC$ is trivially satisfied for any point in the first quadrant. Then we first define

$$S = \{(x, y) : (x + 1)^2 + y^2 \leqslant 4, x \geqslant 0, y \geqslant 0\}.$$

Note that S is the intersection of a disk and the first quadrant, so it is bounded and convex, and we can choose any $T \in S$ to meet condition (i). For any point A in S, the relation $BC \geqslant AB \geqslant AC$ always holds. Therefore, the point A in (ii) is uniquely determined, while its existence is guaranteed by the above construction.



Next, if in the similarity of $\triangle A'BC$ and $\triangle P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$, BC corresponds to the second longest side of $\triangle P_1P_2P_3$, then we have $A'B \geqslant BC \geqslant A'C$. The two inequalities are equivalent to $(x + 1)^2 + y^2 \geqslant 4$ and $(x - 1)^2 + y^2 \leqslant 4$ respectively. Then we define

$$S' = \{(x, y) : (x + 1)^2 + y^2 \geqslant 4, (x - 1)^2 + y^2 \leqslant 4, x \geqslant 0, y \geqslant 0\}.$$

The boundedness condition is satisfied while (ii) can be argued as in the previous case. For (i), note that S' contains points inside the disk $(x-1)^2+y^2 \leqslant 4$ and outside the disk $(x+1)^2+y^2 \geqslant 4$. This shows we can take $T'=(1,2)$ in (i), which is the topmost point of the circle $(x-1)^2+y^2=4$.

It remains to check that the product $BA \cdot BA'$ is a constant. Suppose we are given a triangle $P_1P_2P_3$ with $P_1P_2 \geqslant P_2P_3 \geqslant P_3P_1$. By the similarity, we have

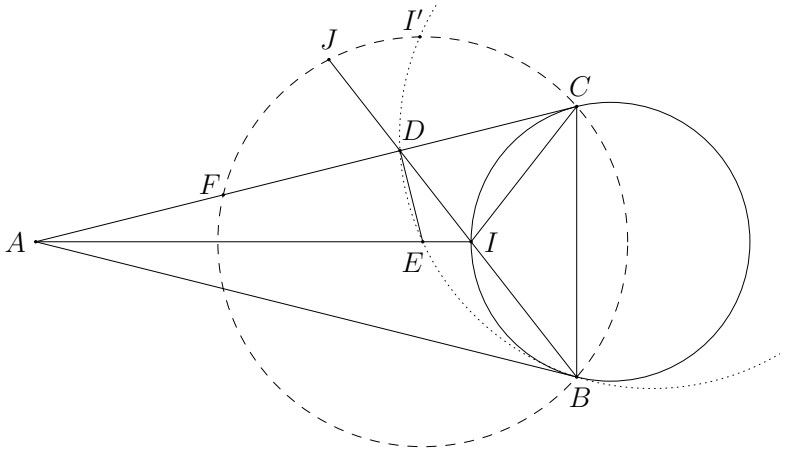
$$BA = BC \cdot \frac{P_2P_3}{P_1P_2} \quad \text{and} \quad BA' = BC \cdot \frac{P_1P_2}{P_2P_3}.$$

Thus $BA \cdot BA' = BC^2 = 4$, which is certainly independent of the triangle $P_1P_2P_3$.

Comment. The original version of this problem includes the condition that the interiors of S and S' are disjoint. We remove this condition since it is hard to define the interior of a point set rigorously.

G4. Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D, and the line through D perpendicular to AC meets AI at E. Prove that the reflection of I in AC lies on the circumcircle of triangle BDE.

Solution 1.



Let Γ be the circle with centre E passing through B and C. Since $ED \perp AC$, the point F symmetric to C with respect to D lies on Γ. From $\angle DCI = \angle ICB = \angle CBI$, the line DC is a tangent to the circumcircle of triangle IBC. Let J be the symmetric point of I with respect to D. Using directed lengths, from

$$DC \cdot DF = -DC^2 = -DI \cdot DB = DJ \cdot DB,$$

the point J also lies on Γ. Let I' be the reflection of I in AC. Since IJ and CF bisect each other, $CJFI$ is a parallelogram. From $\angle FI'C = \angle CIF = \angle FJC$, we find that I' lies on Γ. This gives $EI' = EB$.

Note that AC is the internal angle bisector of $\angle BDI'$. This shows DE is the external angle bisector of $\angle BDI'$ as $DE \perp AC$. Together with $EI' = EB$, it is well-known that E lies on the circumcircle of triangle BDI'.

Solution 2. Let I' be the reflection of I in AC and let S be the intersection of $I'C$ and AI. Using directed angles, we let $\theta = \measuredangle ACI = \measuredangle ICB = \measuredangle CBI$. We have

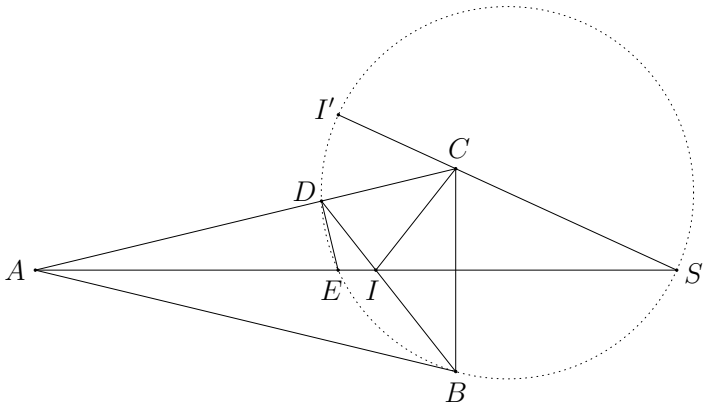
$$\measuredangle I'SE = \measuredangle I'CA + \measuredangle CAI = \theta + \left(\frac{\pi}{2} + 2\theta\right) = 3\theta + \frac{\pi}{2}$$

and

$$\measuredangle I'DE = \measuredangle I'DC + \frac{\pi}{2} = \measuredangle CDI + \frac{\pi}{2} = \measuredangle DCB + \measuredangle CBD + \frac{\pi}{2} = 3\theta + \frac{\pi}{2}.$$

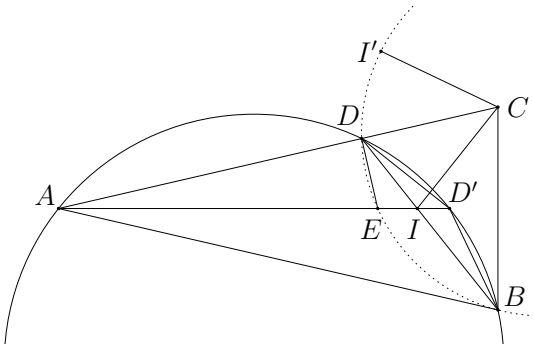
This shows I', D, E, S are concyclic.

Next, we find $\measuredangle I'SB = 2\measuredangle I'SE = 6\theta$ and $\measuredangle I'DB = 2\measuredangle CDI = 6\theta$. Therefore, I', D, B, S are concyclic so that I', D, E, B, S lie on the same circle. The result then follows.



Comment. The point S constructed in Solution 2 may lie on the same side as A of BC. Also, since S lies on the circumcircle of the non-degenerate triangle BDE, we implicitly know that S is not an ideal point. Indeed, one can verify that $I'C$ and AI are parallel if and only if triangle ABC is equilateral.

Solution 3. Let I' be the reflection of I in AC, and let D' be the second intersection of AI and the circumcircle of triangle ABD. Since AD' bisects $\angle BAD$, point D' is the midpoint of the arc BD and $DD' = BD' = CD'$. Obviously, A, E, D' lie on AI in this order.



We find that $\angle ED'D = \angle AD'D = \angle ABD = \angle IBC = \angle ICB$. Next, since D' is the circumcentre of triangle BCD, we have $\angle EDD' = 90^\circ - \angle D'DC = \angle CBD = \angle IBC$. The two relations show that triangles $ED'D$ and ICB are similar. Therefore, we have

$$\frac{BC}{CI'} = \frac{BC}{CI} = \frac{DD'}{D'E} = \frac{BD'}{D'E}.$$

Also, we get

$$\angle BCI' = \angle BCA + \angle ACI' = \angle BCA + \angle ICA = \angle BCA + \angle DBC = \angle BDA = \angle BD'E.$$

These show triangles BCI' and $BD'E$ are similar, and hence triangles BCD' and $BI'E$ are similar. As BCD' is isosceles, we obtain $BE = I'E$.

As DE is the external angle bisector of $\angle BDI'$ and $EI' = EB$, we know that E lies on the circumcircle of triangle BDI'.

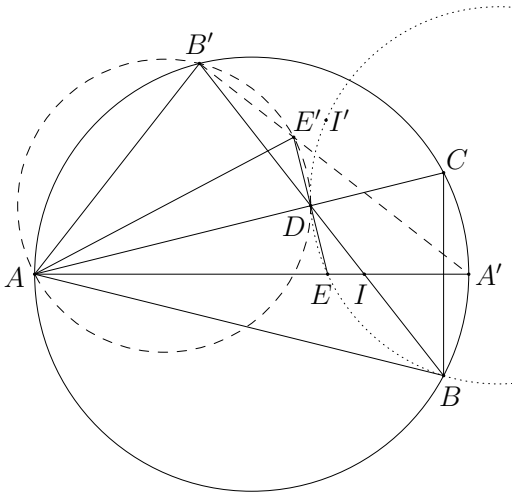
Solution 4. Let AI and BI meet the circumcircle of triangle ABC again at A' and B' respectively, and let E' be the reflection of E in AC. From

$$\begin{aligned}\angle B'AE' &= \angle B'AD - \angle E'AD = \frac{\angle ABC}{2} - \frac{\angle BAC}{2} = 90^\circ - \angle BAC - \frac{\angle ABC}{2} \\ &= 90^\circ - \angle B'DA = \angle B'DE',\end{aligned}$$

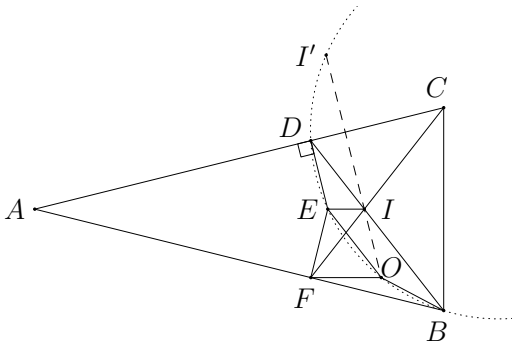
points B', A, D, E' are concyclic. Then

$$\angle DB'E' = \angle DAE' = \frac{\angle BAC}{2} = \angle BAA' = \angle DB'A'$$

and hence B', E', A' are collinear. It is well-known that $A'B'$ is the perpendicular bisector of CI, so that $CE' = IE'$. Let I' be the reflection of I in AC. This implies $BE = CE = I'E$. As DE is the external angle bisector of $\angle BDI'$ and $EI' = EB$, we know that E lies on the circumcircle of triangle BDI'.



Solution 5. Let F be the intersection of CI and AB. Clearly, F and D are symmetric with respect to AI. Let O be the circumcentre of triangle BIF, and let I' be the reflection of I in AC.



From $\angle BFO = 90^\circ - \angle FIB = \frac{1}{2}\angle BAC = \angle BAI$, we get $EI//FO$. Also, from the relation $\angle OIB = 90^\circ - \angle BFI = 90^\circ - \angle CDI = \angle I'ID$, we know that O, I, I' are collinear.

Note that $ED//OI$ since both are perpendicular to AC. Then $\angle FEI = \angle DEI = \angle OIE$. Together with $EI//FO$, the quadrilateral $EFOI$ is an isosceles trapezoid. Therefore, we find that $\angle DIE = \angle FIE = \angle OEI$ so $OE//ID$. Then $DEOI$ is a parallelogram. Hence, we have $DI' = DI = EO$, which shows $DEOI'$ is an isosceles trapezoid. In addition, $ED = OI = OB$ and $OE//BD$ imply $EOBD$ is another isosceles trapezoid. In particular, both $DEOI'$ and $EOBD$ are cyclic. This shows B, D, E, I' are concyclic.

Solution 6. Let I' be the reflection of I in AC. Denote by T and M the projections from I to sides AB and BC respectively. Since BI is the perpendicular bisector of TM, we have

$$DT = DM. \tag{1}$$

Since $\angle ADE = \angle ATI = 90^\circ$ and $\angle DAE = \angle TAI$, we have $\triangle ADE \sim \triangle ATI$. This shows $\frac{AD}{AE} = \frac{AT}{AI} = \frac{AT}{AI'}$. Together with $\angle DAT = 2\angle DAE = \angle EAI'$, this yields $\triangle DAT \sim \triangle EAI'$. In particular, we have

$$\frac{DT}{EI'} = \frac{AT}{AI'} = \frac{AT}{AI}. \tag{2}$$

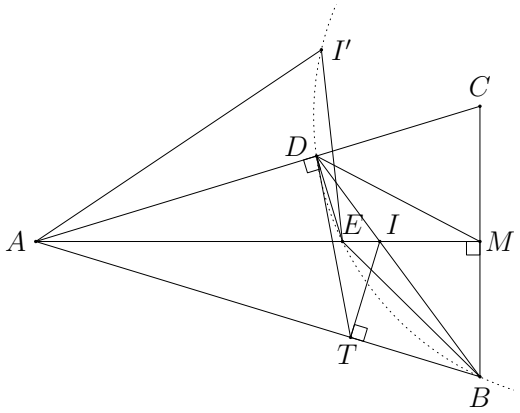
Obviously, the right-angled triangles AMB and ATI are similar. Then we get

$$\frac{AM}{AB} = \frac{AT}{AI}. \tag{3}$$

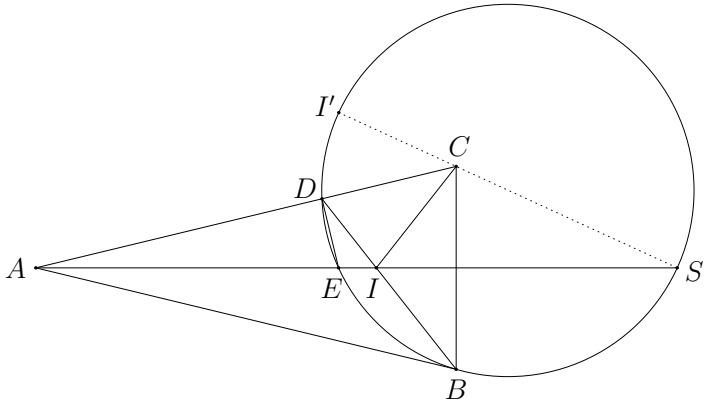
Next, from $\triangle AMB \sim \triangle ATI \sim \triangle ADE$, we have $\frac{AM}{AB} = \frac{AD}{AE}$ so that $\triangle ADM \sim \triangle AEB$. It follows that

$$\frac{DM}{EB} = \frac{AM}{AB}. \tag{4}$$

Combining (1), (2), (3) and (4), we get $EB = EI'$. As DE is the external angle bisector of $\angle BDI'$, we know that E lies on the circumcircle of triangle BDI'.



Comment. A stronger version of this problem is to ask the contestants to prove the reflection of I in AC lies on the circumcircle of triangle BDE if and only if $AB = AC$. Some of the above solutions can be modified to prove the converse statement to the original problem. For example, we borrow some ideas from Solution 2 to establish the converse as follows.



Let I' be the reflection of I in AC and suppose B, E, D, I' lie on a circle Γ. Let AI meet Γ again at S. As DE is the external angle bisector of $\angle BDI'$, we have $EB = EI'$. Using directed angles, we get

$$\measuredangle CI'S = \measuredangle CI'D + \measuredangle DI'S = \measuredangle DIC + \measuredangle DES = \measuredangle DIC + \measuredangle DEA = \measuredangle DIC + \measuredangle DCB = 0.$$

This means I', C, S are collinear. From this we get $\measuredangle BSE = \measuredangle ESI' = \measuredangle ESC$ and hence AS bisects both $\angle BAC$ and $\angle BSC$. Clearly, S and A are distinct points. It follows that $\triangle BAS \cong \triangle CAS$ and thus $AB = AC$.

As in some of the above solutions, an obvious way to prove the stronger version is to establish the following equivalence: $BE = EI'$ if and only if $AB = AC$. In addition to the ideas used in those solutions, one can use trigonometry to express the lengths of BE and EI' in terms of the side lengths of triangle ABC and to establish the equivalence.

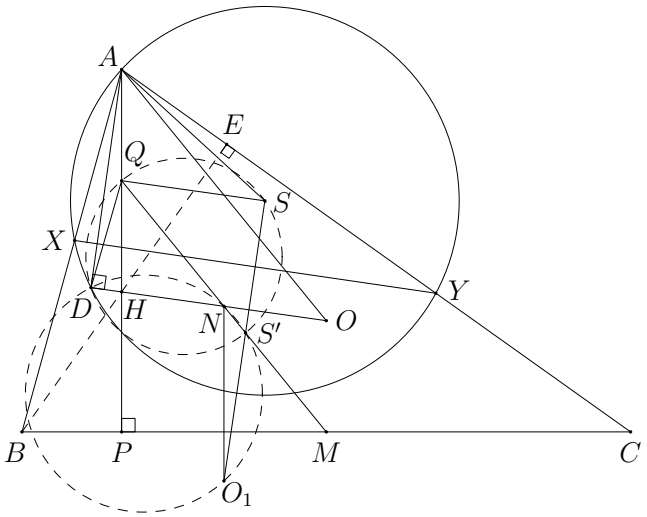
G5. Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC. A circle ω with centre S passes through A and D, and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC, and let M be the midpoint of BC. Prove that the circumcentre of triangle XSY is equidistant from P and M.

Solution 1. Denote the orthocentre and circumcentre of triangle ABC by H and O respectively. Let Q be the midpoint of AH and N be the nine-point centre of triangle ABC. It is known that Q lies on the nine-point circle of triangle ABC, N is the midpoint of QM and that QM is parallel to AO.

Let the perpendicular from S to XY meet line QM at S'. Let E be the foot of altitude from B to side AC. Since Q and S lie on the perpendicular bisector of AD, using directed angles, we have

$$\begin{aligned}\measuredangle SDQ = \measuredangle QAS = \measuredangle XAS - \measuredangle XAQ &= \left(\frac{\pi}{2} - \measuredangle AYX\right) - \measuredangle BAP = \measuredangle CBA - \measuredangle AYX \\ &= (\measuredangle CBA - \measuredangle ACB) - \measuredangle BCA - \measuredangle AYX = \measuredangle PEM - (\measuredangle BCA + \measuredangle AYX) \\ &= \measuredangle PQM - \angle(BC, XY) = \frac{\pi}{2} - \angle(S'Q, BC) - \angle(BC, XY) = \measuredangle SS'Q.\end{aligned}$$

This shows D, S', S, Q are concyclic.



Let the perpendicular from N to BC intersect line SS' at O_1. (Note that the two lines coincide when S is the midpoint of AO, in which case the result is true since the circumcentre of triangle XSY must lie on this line.) It suffices to show that O_1 is the circumcentre of triangle XSY since N lies on the perpendicular bisector of PM. From

$$\measuredangle DS'O_1 = \measuredangle DQS = \measuredangle SQA = \angle(SQ, QA) = \angle(OD, O_1N) = \measuredangle DNO_1$$

since $SQ//OD$ and $QA//O_1N$, we know that D, O_1, S', N are concyclic. Therefore, we get

$$\measuredangle SDS' = \measuredangle SQS' = \angle(SQ, QS') = \angle(ND, NS') = \measuredangle DNS',$$

so that SD is a tangent to the circle through D, O_1, S', N. Then we have

$$SS' \cdot SO_1 = SD^2 = SX^2. \tag{1}$$

Next, we show that S and S' are symmetric with respect to XY. By the Sine Law, we have

$$\frac{SS'}{\sin\angle SQS'} = \frac{SQ}{\sin\angle SS'Q} = \frac{SQ}{\sin\angle SDQ} = \frac{SQ}{\sin\angle SAQ} = \frac{SA}{\sin\angle SQA}.$$

It follows that

$$SS' = SA \cdot \frac{\sin\angle SQS'}{\sin\angle SQA} = SA \cdot \frac{\sin\angle HOA}{\sin\angle OHA} = SA \cdot \frac{AH}{AO} = SA \cdot 2\cos A,$$

which is twice the distance from S to XY. Note that S and C lie on the same side of the perpendicular bisector of PM if and only if $\angle SAC < \angle OAC$ if and only if $\angle YXA > \angle CBA$. This shows S and O_1 lie on different sides of XY. As S' lies on ray SO_1, it follows that S and S' cannot lie on the same side of XY. Therefore, S and S' are symmetric with respect to XY.

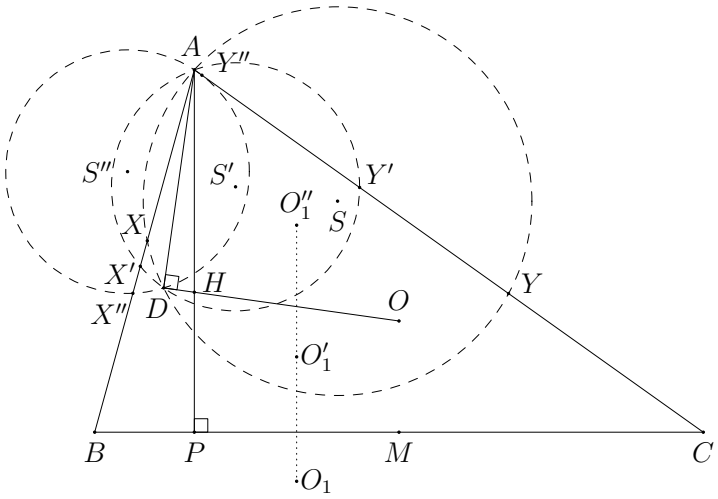
Let d be the diameter of the circumcircle of triangle XSY. As SS' is twice the distance from S to XY and $SX = SY$, we have $SS' = 2\frac{SX^2}{d}$. It follows from (1) that $d = 2SO_1$. As SO_1 is the perpendicular bisector of XY, point O_1 is the circumcentre of triangle XSY.

Solution 2. Denote the orthocentre and circumcentre of triangle ABC by H and O respectively. Let O_1 be the circumcentre of triangle XSY. Consider two other possible positions of S. We name them S' and S'' and define the analogous points $X', Y', O_1', X'', Y''O_1''$ accordingly. Note that S, S', S'' lie on the perpendicular bisector of AD.

As XX' and YY' meet at A and the circumcircles of triangles AXY and $AX'Y'$ meet at D, there is a spiral similarity with centre D mapping XY to $X'Y'$. We find that

$$\measuredangle SXY = \frac{\pi}{2} - \measuredangle YAX = \frac{\pi}{2} - \measuredangle Y'AX' = \measuredangle S'X'Y'$$

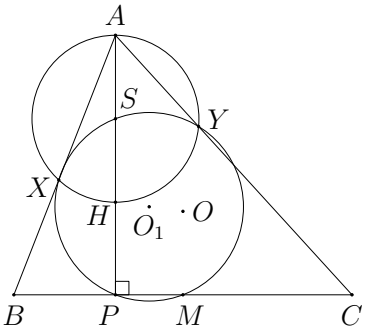
and similarly $\measuredangle SYX = \measuredangle S'Y'X'$. This shows triangles SXY and $S'X'Y'$ are directly similar. Then the spiral similarity with centre D takes points S, X, Y, O_1 to S', X', Y', O_1'. Similarly, there is a spiral similarity with centre D mapping S, X, Y, O_1 to S'', X'', Y'', O_1''. From these, we see that there is a spiral similarity taking the corresponding points S, S', S'' to points O_1, O_1', O_1''. In particular, O_1, O_1', O_1'' are collinear.

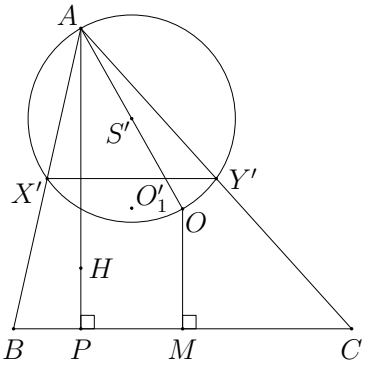


It now suffices to show that O_1 lies on the perpendicular bisector of PM for two special cases.

Firstly, we take S to be the midpoint of AH. Then X and Y are the feet of altitudes from C and B respectively. It is well-known that the circumcircle of triangle XSY is the nine-point circle of triangle ABC. Then O_1 is the nine-point centre and $O_1P = O_1M$. Indeed, P and M also lies on the nine-point circle.

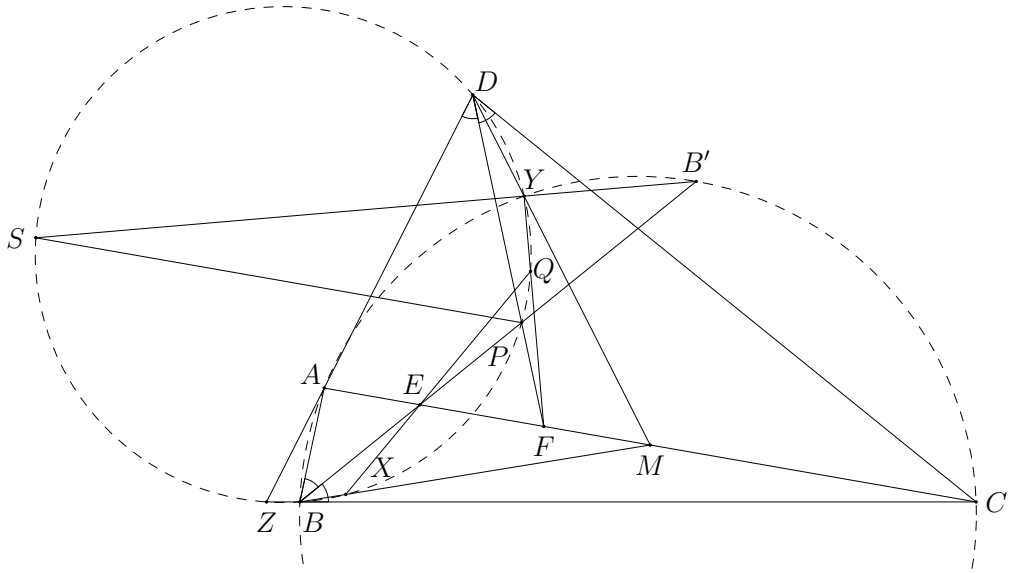
Secondly, we take S' to be the midpoint of AO. Then X' and Y' are the midpoints of AB and AC respectively. Then $X'Y'//BC$. Clearly, S' lies on the perpendicular bisector of PM. This shows the perpendicular bisectors of $X'Y'$ and PM coincide. Hence, we must have $O'_1P = O'_1M$.





G6. Let $ABCD$ be a convex quadrilateral with $\angle ABC = \angle ADC < 90°$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at E and F respectively, and meet each other at point P. Let M be the midpoint of AC and let ω be the circumcircle of triangle BPD. Segments BM and DM intersect ω again at X and Y respectively. Denote by Q the intersection point of lines XE and YF. Prove that $PQ \perp AC$.

Solution 1.



Let ω_1 be the circumcircle of triangle ABC. We first prove that Y lies on ω_1. Let Y' be the point on ray MD such that $MY' \cdot MD = MA^2$. Then triangles MAY' and MDA are oppositely similar. Since $MC^2 = MA^2 = MY' \cdot MD$, triangles MCY' and MDC are also oppositely similar. Therefore, using directed angles, we have

$$\measuredangle AY'C = \measuredangle AY'M + \measuredangle MY'C = \measuredangle MAD + \measuredangle DCM = \measuredangle CDA = \measuredangle ABC$$

so that Y' lies on ω_1.

Let Z be the intersection point of lines BC and AD. Since $\measuredangle PDZ = \measuredangle PBC = \measuredangle PBZ$, point Z lies on ω. In addition, from $\measuredangle Y'BZ = \measuredangle Y'BC = \measuredangle Y'AC = \measuredangle Y'AM = \measuredangle Y'DZ$, we also know that Y' lies on ω. Note that $\angle ADC$ is acute implies $MA \neq MD$ so $MY' \neq MD$. Therefore, Y' is the second intersection of DM and ω. Then $Y' = Y$ and hence Y lies on ω_1.

Next, by the Angle Bisector Theorem and the similar triangles, we have

$$\frac{FA}{FC} = \frac{AD}{CD} = \frac{AD}{AM} \cdot \frac{CM}{CD} = \frac{YA}{YM} \cdot \frac{YM}{YC} = \frac{YA}{YC}.$$

Hence, FY is the internal angle bisector of $\angle AYC$.

Let B' be the second intersection of the internal angle bisector of $\angle CBA$ and ω_1. Then B' is the midpoint of arc AC not containing B. Therefore, YB' is the external angle bisector of $\angle AYC$, so that $B'Y \perp FY$.

Denote by l the line through P parallel to AC. Suppose l meets line $B'Y$ at S. From

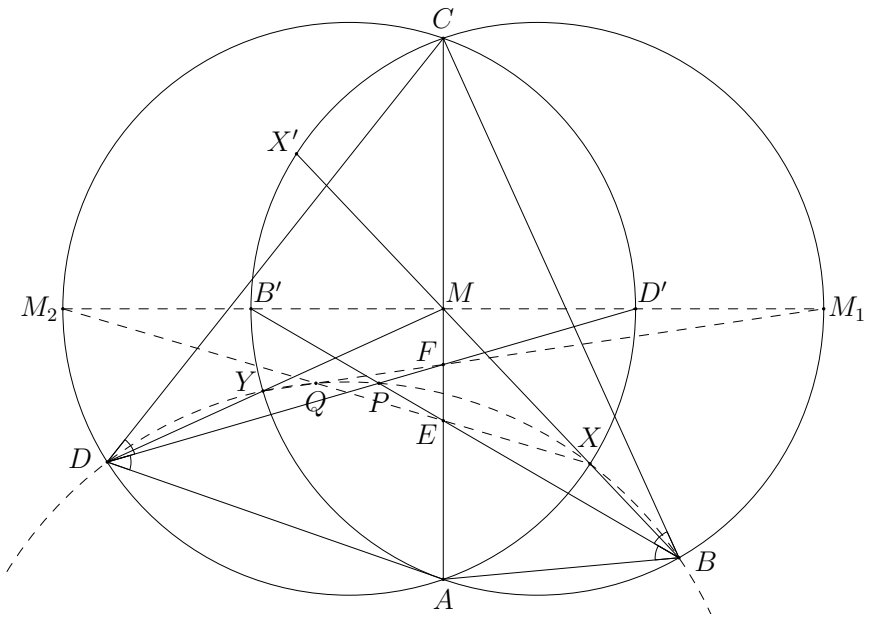
$$\begin{aligned}\measuredangle PSY &= \angle(AC, B'Y) = \measuredangle ACY + \measuredangle CYB' = \measuredangle ACY + \measuredangle CAB' = \measuredangle ACY + \measuredangle B'CA \\ &= \measuredangle B'CY = \measuredangle B'BY = \measuredangle PBY,\end{aligned}$$

the point S lies on ω. Similarly, the line through X perpendicular to XE also passes through the second intersection of l and ω, which is the point S. From $QY \perp YS$ and $QX \perp XS$, point Q lies on ω and QS is a diameter of ω. Therefore, $PQ \perp PS$ so that $PQ \perp AC$.

Solution 2. Denote by ω_1 and ω_2 the circumcircles of triangles ABC and ADC respectively. Since $\angle ABC = \angle ADC$, we know that ω_1 and ω_2 are symmetric with respect to the midpoint M of AC.

Firstly, we show that X lies on ω_2. Let X_1 be the second intersection of ray MB and ω_2 and X' be its symmetric point with respect to M. Then X' lies on ω_1 and $X'AX_1C$ is a parallelogram. Hence, we have

$$\begin{aligned}\measuredangle DX_1B &= \measuredangle DX_1A + \measuredangle AX_1B = \measuredangle DCA + \measuredangle AX_1X' = \measuredangle DCA + \measuredangle CX'X_1 \\ &= \measuredangle DCA + \measuredangle CAB = \angle(CD, AB).\end{aligned}$$



Also, we have

$$\measuredangle DPB = \measuredangle PDC + \angle(CD, AB) + \measuredangle ABP = \angle(CD, AB).$$

These yield $\measuredangle DX_1B = \measuredangle DPB$ and hence X_1 lies on ω. It follows that $X_1 = X$ and X lies on ω_2. Similarly, Y lies on ω_1.

Next, we prove that Q lies on ω. Suppose the perpendicular bisector of AC meet ω_1 at B' and M_1 and meet ω_2 at D' and M_2, so that B, M_1 and D' lie on the same side of AC. Note that B' lies on the angle bisector of $\angle ABC$ and similarly D' lies on DP.

If we denote the area of $W_1W_2W_3$ by $[W_1W_2W_3]$, then

$$\frac{BA \cdot X'A}{BC \cdot X'C} = \frac{\frac{1}{2}BA \cdot X'A \sin \angle BAX'}{\frac{1}{2}BC \cdot X'C \sin \angle BCX'} = \frac{[BAX']}{[BCX']} = \frac{MA}{MC} = 1.$$

As BE is the angle bisector of $\angle ABC$, we have

$$\frac{EA}{EC} = \frac{BA}{BC} = \frac{X'C}{X'A} = \frac{XA}{XC}.$$

Therefore, XE is the angle bisector of $\angle AXC$, so that M_2 lies on the line joining X, E, Q. Analogously, M_1, F, Q, Y are collinear. Thus,

$$\begin{aligned}\measuredangle XQY &= \measuredangle M_2QM_1 = \measuredangle QM_2M_1 + \measuredangle M_2M_1Q = \measuredangle XM_2D' + \measuredangle B'M_1Y \\ &= \measuredangle XDD' + \measuredangle B'BY = \measuredangle XDP + \measuredangle PBY = \measuredangle XBP + \measuredangle PBY = \measuredangle XBY,\end{aligned}$$

which implies Q lies on ω.

Finally, as M_1 and M_2 are symmetric with respect to M, the quadrilateral $X'M_2XM_1$ is a parallelogram. Consequently,

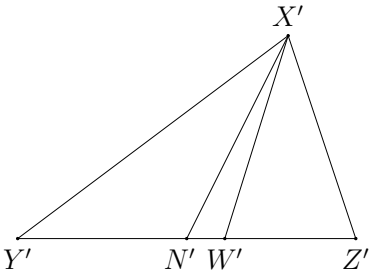
$$\measuredangle XQP = \measuredangle XBP = \measuredangle X'BB' = \measuredangle X'M_1B' = \measuredangle XM_2M_1.$$

This shows $QP // M_2M_1$. As $M_2M_1 \perp AC$, we get $QP \perp AC$.

Solution 3. We first state two results which will be needed in our proof.

• **Claim 1.** In $\triangle X'Y'Z'$ with $X'Y' \neq X'Z'$, let N' be the midpoint of $Y'Z'$ and W' be the foot of internal angle bisector from X'. Then $\tan^2 \measuredangle W'X'Z' = \tan \measuredangle N'X'W' \tan \measuredangle Z'W'X'$.

Proof.



Without loss of generality, assume $X'Y' > X'Z'$. Then W' lies between N' and Z'. The signs of both sides agree so it suffices to establish the relation for ordinary angles. Let $\angle W'X'Z' = \alpha$, $\angle N'X'W' = \beta$ and $\angle Z'W'X' = \gamma$. We have

$$\frac{\sin(\gamma - \alpha)}{\sin(\alpha - \beta)} = \frac{N'X'}{N'Y'} = \frac{N'X'}{N'Z'} = \frac{\sin(\gamma + \alpha)}{\sin(\alpha + \beta)}.$$

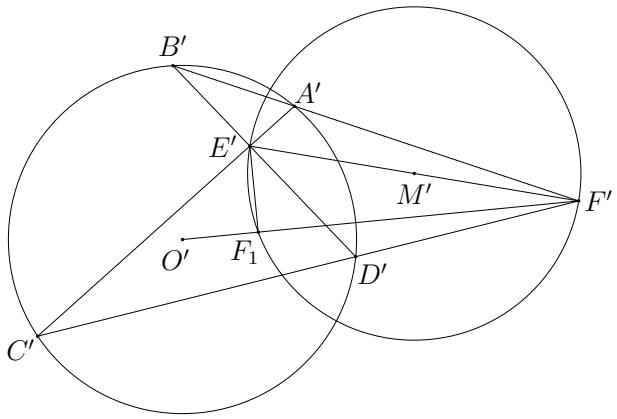
This implies

$$\frac{\tan\gamma-\tan\alpha}{\tan\gamma+\tan\alpha}=\frac{\sin\gamma\cos\alpha-\cos\gamma\sin\alpha}{\sin\gamma\cos\alpha+\cos\gamma\sin\alpha}=\frac{\sin\alpha\cos\beta-\cos\alpha\sin\beta}{\sin\alpha\cos\beta+\cos\alpha\sin\beta}=\frac{\tan\alpha-\tan\beta}{\tan\alpha+\tan\beta}.$$

Expanding and simplifying, we get the desired result $\tan^2\alpha=\tan\beta\tan\gamma$. □

• **Claim 2.** Let $A'B'C'D'$ be a quadrilateral inscribed in circle Γ. Let diagonals $A'C'$ and $B'D'$ meet at E', and F' be the intersection of lines $A'B'$ and $C'D'$. Let M' be the midpoint of $E'F'$. Then the power of M' with respect to Γ is equal to $(M'E')^2$.

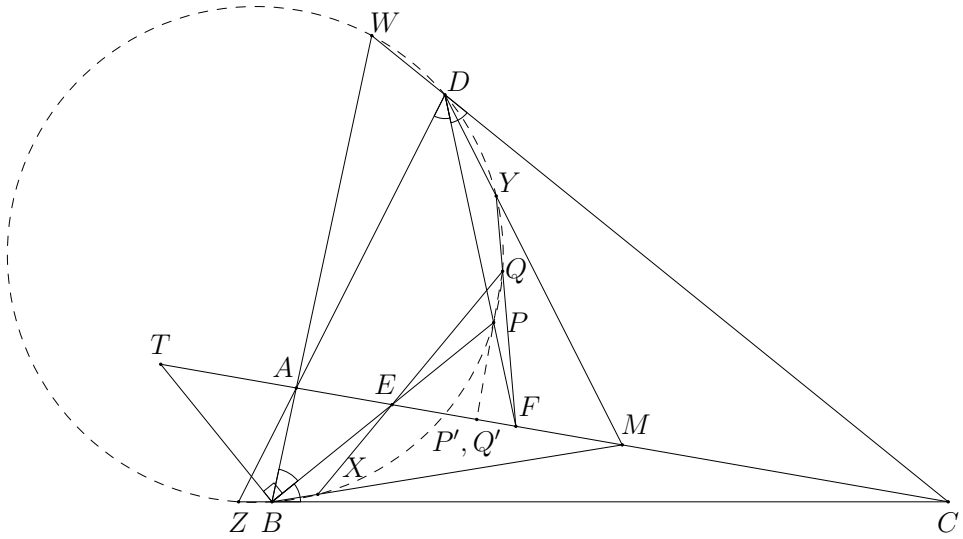
Proof.



Let O' be the centre of Γ and let Γ' be the circle with centre M' passing through E'. Let F_1 be the inversion image of F' with respect to Γ. It is well-known that E' lies on the polar of F' with respect to Γ. This shows $E'F_1\perp O'F'$ and hence F_1 lies on Γ'. It follows that the inversion image of Γ' with respect to Γ is Γ' itself. This shows Γ' is orthogonal to Γ, and thus the power of M' with respect to Γ is the square of radius of Γ', which is $(M'E')^2$. □

We return to the main problem. Let Z be the intersection of lines AD and BC, and W be the intersection of lines AB and CD. Since $\measuredangle PDZ=\measuredangle PBC=\measuredangle PBZ$, point Z lies on ω. Similarly, W lies on ω. Applying Claim 2 to the cyclic quadrilateral $ZBDW$, we know that the power of M with respect to ω is MA^2. Hence, $MX\cdot MB=MA^2$.

Suppose the line through B perpendicular to BE meets line AC at T. Then BE and BT are the angle bisectors of $\angle CBA$. This shows $(T,E;A,C)$ is harmonic. Thus, we have $ME\cdot MT=MA^2=MX\cdot MB$. It follows that E,T,B,X are concyclic.



The result is trivial for the special case $AD = CD$ since P, Q lie on the perpendicular bisector of AC in that case. Similarly, the case $AB = CB$ is trivial. It remains to consider the general cases where we can apply Claim 1 in the latter part of the proof.

Let the projections from P and Q to AC be P' and Q' respectively. Then $PQ \perp AC$ if and only if $P' = Q'$ if and only if $\frac{EP'}{FP'} = \frac{EQ'}{FQ'}$ in terms of directed lengths. Note that

$$\frac{EP'}{FP'} = \frac{\tan \measuredangle EFP}{\tan \measuredangle FEP} = \frac{\tan \measuredangle AFD}{\tan \measuredangle AEB}.$$

Next, we have $\frac{EQ'}{FQ'} = \frac{\tan \measuredangle EFQ}{\tan \measuredangle FEQ}$ where $\measuredangle FEQ = \measuredangle TEX = \measuredangle TBX = \frac{\pi}{2} + \measuredangle EBM$ and by symmetry $\measuredangle EFQ = \frac{\pi}{2} + \measuredangle FDM$. Combining all these, it suffices to show

$$\frac{\tan \measuredangle AFD}{\tan \measuredangle AEB} = \frac{\tan \measuredangle MBE}{\tan \measuredangle MDF}.$$

We now apply Claim 1 twice to get

$$\tan \measuredangle AFD \tan \measuredangle MDF = \tan^2 \measuredangle FDC = \tan^2 \measuredangle EBA = \tan \measuredangle MBE \tan \measuredangle AEB.$$

The result then follows.

G7. Let I be the incentre of a non-equilateral triangle ABC, I_A be the A-excentre, I'_A be the reflection of I_A in BC, and l_A be the reflection of line AI'_A in AI. Define points I_B, I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B.

(a) Prove that P lies on line OI where O is the circumcentre of triangle ABC.

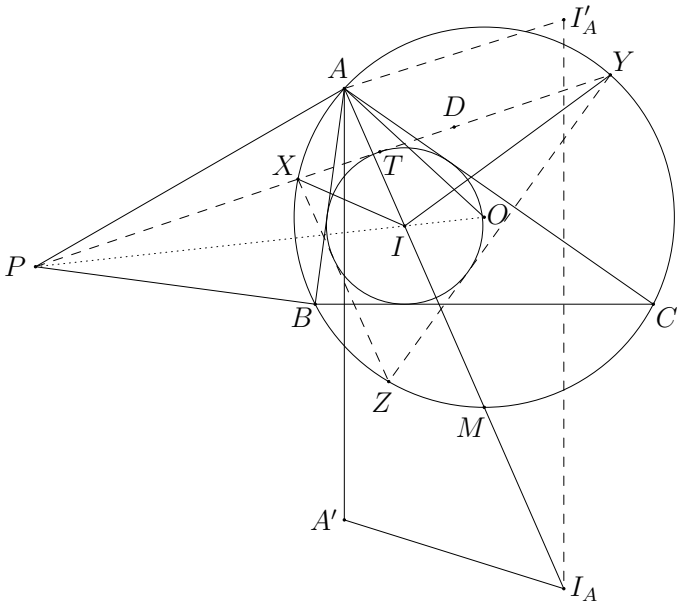
(b) Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y. Show that $\angle XIY = 120°$.

Solution 1.

(a) Let A' be the reflection of A in BC and let M be the second intersection of line AI and the circumcircle Γ of triangle ABC. As triangles ABA' and AOC are isosceles with $\angle ABA' = 2\angle ABC = \angle AOC$, they are similar to each other. Also, triangles ABI_A and AIC are similar. Therefore we have

$$\frac{AA'}{AI_A} = \frac{AA'}{AB} \cdot \frac{AB}{AI_A} = \frac{AC}{AO} \cdot \frac{AI}{AC} = \frac{AI}{AO}.$$

Together with $\angle A'AI_A = \angle IAO$, we find that triangles $AA'I_A$ and AIO are similar.



Denote by P' the intersection of line AP and line OI. Using directed angles, we have

$$\begin{aligned}\measuredangle MAP' &= \measuredangle I'_A AI_A = \measuredangle I'_A AA' - \measuredangle I_A AA' = \measuredangle AA'I_A - \angle(AM, OM) \\ &= \measuredangle AIO - \measuredangle AMO = \measuredangle MOP'.\end{aligned}$$

This shows M, O, A, P' are concyclic.

Denote by R and r the circumradius and inradius of triangle ABC. Then

$$IP' = \frac{IA \cdot IM}{IO} = \frac{IO^2 - R^2}{IO}$$

is independent of A. Hence, BP also meets line OI at the same point P' so that $P' = P$, and P lies on OI.

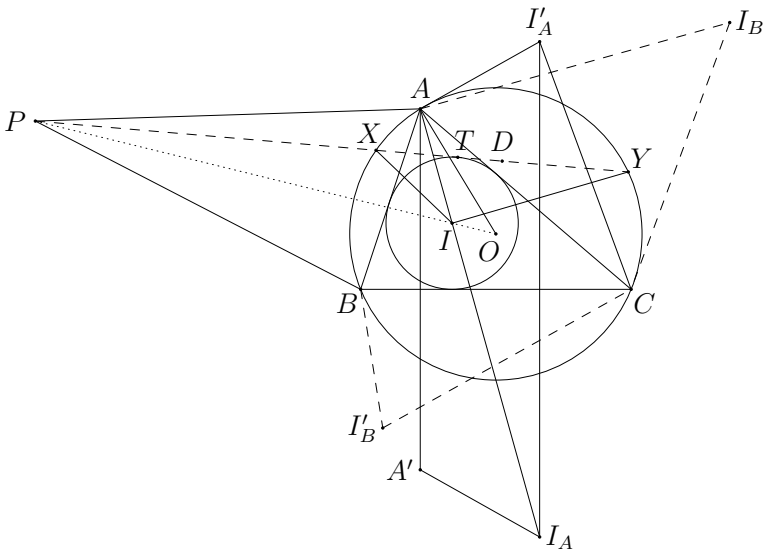
(b) By Poncelet's Porism, the other tangents to the incircle of triangle ABC from X and Y meet at a point Z on Γ. Let T be the touching point of the incircle to XY, and let D be the midpoint of XY. We have

$$\begin{aligned} OD &= IT \cdot \frac{OP}{IP} = r\left(1 + \frac{OI}{IP}\right) = r\left(1 + \frac{OI^2}{OI \cdot IP}\right) = r\left(1 + \frac{R^2 - 2Rr}{R^2 - IO^2}\right) \\ &= r\left(1 + \frac{R^2 - 2Rr}{2Rr}\right) = \frac{R}{2} = \frac{OX}{2}. \end{aligned}$$

This shows $\angle XZY = 60^\circ$ and hence $\angle XIY = 120^\circ$.

Solution 2.

(a) Note that triangles AI_BC and I_ABC are similar since their corresponding interior angles are equal. Therefore, the four triangles AI'_BC, AI_BC, I_ABC and I'_ABC are all similar. From $\triangle AI'_BC \sim \triangle I'_ABC$, we get $\triangle AI'_AC \sim \triangle I'_BBC$. From $\measuredangle ABP = \measuredangle I'_BBC = \measuredangle AI'_AC$ and $\measuredangle BAP = \measuredangle I'_AAC$, the triangles ABP and AI'_AC are directly similar.



Consider the inversion with centre A and radius $\sqrt{AB \cdot AC}$ followed by the reflection in AI. Then B and C are mapped to each other, and I and I_A are mapped to each other.

From the similar triangles obtained, we have $AP \cdot AI'_A = AB \cdot AC$ so that P is mapped to I'_A under the transformation. In addition, line AO is mapped to the altitude from A, and hence O is mapped to the reflection of A in BC, which we call point A'. Note that $AA'I_AI'_A$ is an isosceles trapezoid, which shows it is inscribed in a circle. The preimage of this circle is a straight line, meaning that O, I, P are collinear.

(b) Denote by R and r the circumradius and inradius of triangle ABC. Note that by the above transformation, we have $\triangle APO \sim \triangle AA'I'_A$ and $\triangle AA'I_A \sim \triangle AIO$. Therefore, we find that

$$PO = A'I'_A \cdot \frac{AO}{AI'_A} = AI_A \cdot \frac{AO}{A'I_A} = \frac{AI_A}{A'I_A} \cdot AO = \frac{AO}{IO} \cdot AO.$$

This shows $PO \cdot IO = R^2$, and it follows that P and I are mapped to each other under the inversion with respect to the circumcircle Γ of triangle ABC. Then $PX \cdot PY$, which is the power of P with respect to Γ, equals $PI \cdot PO$. This yields X, I, O, Y are concyclic.

Let T be the touching point of the incircle to XY, and let D be the midpoint of XY. Then

$$OD = IT \cdot \frac{PO}{PI} = r \cdot \frac{PO}{PO - IO} = r \cdot \frac{R^2}{R^2 - IO^2} = r \cdot \frac{R^2}{2Rr} = \frac{R}{2}.$$

This shows $\angle DOX = 60^\circ$ and hence $\angle XIY = \angle XOY = 120^\circ$.

Comment. A simplification of this problem is to ask part (a) only. Note that the question in part (b) implicitly requires P to lie on OI, or otherwise the angle is not uniquely determined as we can find another tangent from P to the incircle.

G8. Let A_1, B_1 and C_1 be points on sides BC, CA and AB of an acute triangle ABC respectively, such that AA_1, BB_1 and CC_1 are the internal angle bisectors of triangle ABC. Let I be the incentre of triangle ABC, and H be the orthocentre of triangle $A_1B_1C_1$. Show that

$$AH + BH + CH \geqslant AI + BI + CI.$$

Solution. Without loss of generality, assume $\alpha = \angle BAC \leqslant \beta = \angle CBA \leqslant \gamma = \angle ACB$. Denote by a, b, c the lengths of BC, CA, AB respectively. We first show that triangle $A_1B_1C_1$ is acute.

Choose points D and E on side BC such that $B_1D // AB$ and B_1E is the internal angle bisector of $\angle BB_1C$. As $\angle B_1DB = 180^\circ - \beta$ is obtuse, we have $BB_1 > B_1D$. Thus,

$$\frac{BE}{EC} = \frac{BB_1}{B_1C} > \frac{DB_1}{B_1C} = \frac{BA}{AC} = \frac{BA_1}{A_1C}.$$

Therefore, $BE > BA_1$ and $\frac{1}{2}\angle BB_1C = \angle BB_1E > \angle BB_1A_1$. Similarly, $\frac{1}{2}\angle BB_1A > \angle BB_1C_1$. It follows that

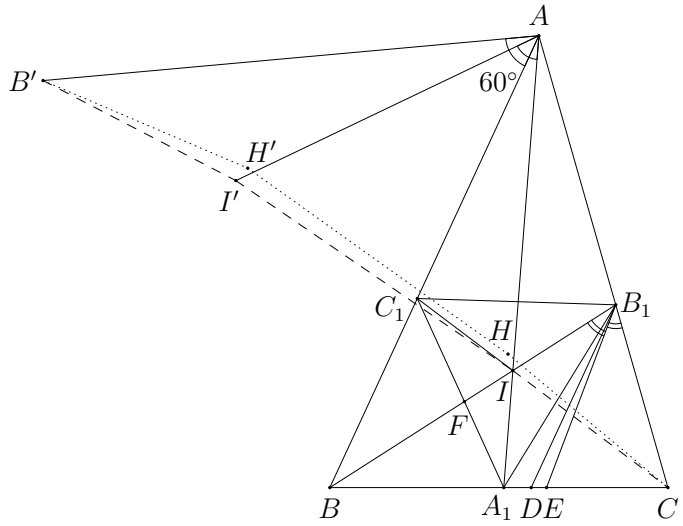
$$\angle A_1B_1C_1 = \angle BB_1A_1 + \angle BB_1C_1 < \frac{1}{2}(\angle BB_1C + \angle BB_1A) = 90^\circ$$

is acute. By symmetry, triangle $A_1B_1C_1$ is acute.

Let BB_1 meet A_1C_1 at F. From $\alpha \leqslant \gamma$, we get $a \leqslant c$, which implies

$$BA_1 = \frac{ca}{b+c} \leqslant \frac{ac}{a+b} = BC_1$$

and hence $\angle BC_1A_1 \leqslant \angle BA_1C_1$. As BF is the internal angle bisector of $\angle A_1BC_1$, this shows $\angle B_1FC_1 = \angle BFA_1 \leqslant 90^\circ$. Hence, H lies on the same side of BB_1 as C_1. This shows H lies inside triangle BB_1C_1. Similarly, from $\alpha \leqslant \beta$ and $\beta \leqslant \gamma$, we know that H lies inside triangles CC_1B_1 and AA_1C_1.



As $\alpha \leqslant \beta \leqslant \gamma$, we have $\alpha \leqslant 60^\circ \leqslant \gamma$. Then $\angle BIC \leqslant 120^\circ \leqslant \angle AIB$. Firstly, suppose $\angle AIC \geqslant 120^\circ$.

Rotate points B, I, H through 60° about A to B', I', H' so that B' and C lie on different sides of AB. Since triangle $AI'I$ is equilateral, we have

$$AI + BI + CI = I'I + B'I' + IC = B'I' + I'I + IC. \tag{1}$$

Similarly,

$$AH + BH + CH = H'H + B'H' + HC = B'H' + H'H + HC. \tag{2}$$

As $\angle AII' = \angle AI'I = 60^\circ$, $\angle AI'B' = \angle AIB \geqslant 120^\circ$ and $\angle AIC \geqslant 120^\circ$, the quadrilateral $B'I'IC$ is convex and lies on the same side of $B'C$ as A.

Next, since H lies inside triangle ACC_1, H lies outside $B'I'IC$. Also, H lying inside triangle ABI implies H' lies inside triangle $AB'I'$. This shows H' lies outside $B'I'IC$ and hence the convex quadrilateral $B'I'IC$ is contained inside the quadrilateral $B'H'HC$. It follows that the perimeter of $B'I'IC$ cannot exceed the perimeter of $B'H'HC$. From (1) and (2), we conclude that

$$AH + BH + CH \geqslant AI + BI + CI.$$

For the case $\angle AIC < 120^\circ$, we can rotate B, I, H through 60° about C to B', I', H' so that B' and A lie on different sides of BC. The proof is analogous to the previous case and we still get the desired inequality.

Number Theory

N1. For any positive integer k, denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer $n \geqslant 2016$, the integer $P(n)$ is positive and

$$S(P(n)) = P(S(n)). \tag{1}$$

Answer.

- $P(x) = c$ where $1 \leqslant c \leqslant 9$ is an integer; or
- $P(x) = x$.

Solution 1. We consider three cases according to the degree of P.

• **Case 1.** $P(x)$ is a constant polynomial.

Let $P(x) = c$ where c is an integer constant. Then (1) becomes $S(c) = c$. This holds if and only if $1 \leqslant c \leqslant 9$.

• **Case 2.** $\deg P = 1$.

We have the following observation. For any positive integers m, n, we have

$$S(m+n) \leqslant S(m) + S(n), \tag{2}$$

and equality holds if and only if there is no carry in the addition $m + n$.

Let $P(x) = ax + b$ for some integers a, b where $a \neq 0$. As $P(n)$ is positive for large n, we must have $a \geqslant 1$. The condition (1) becomes $S(an+b) = aS(n) + b$ for all $n \geqslant 2016$. Setting $n = 2025$ and $n = 2020$ respectively, we get

$$S(2025a + b) - S(2020a + b) = (aS(2025) + b) - (aS(2020) + b) = 5a.$$

On the other hand, (2) implies

$$S(2025a + b) = S((2020a + b) + 5a) \leqslant S(2020a + b) + S(5a).$$

These give $5a \leqslant S(5a)$. As $a \geqslant 1$, this holds only when $a = 1$, in which case (1) reduces to $S(n + b) = S(n) + b$ for all $n \geqslant 2016$. Then we find that

$$S(n + 1 + b) - S(n + b) = (S(n + 1) + b) - (S(n) + b) = S(n + 1) - S(n). \tag{3}$$

If $b > 0$, we choose n such that $n + 1 + b = 10^k$ for some sufficiently large k. Note that all the digits of $n + b$ are 9's, so that the left-hand side of (3) equals $1 - 9k$. As n is a positive integer less than $10^k - 1$, we have $S(n) < 9k$. Therefore, the right-hand side of (3) is at least $1 - (9k - 1) = 2 - 9k$, which is a contradiction.

The case $b < 0$ can be handled similarly by considering $n + 1$ to be a large power of 10. Therefore, we conclude that $P(x) = x$, in which case (1) is trivially satisfied.

• **Case 3.** $\deg P \geqslant 2$.

Suppose the leading term of P is $a_d n^d$ where $a_d \neq 0$. Clearly, we have $a_d > 0$. Consider $n = 10^k - 1$ in (1). We get $S(P(n)) = P(9k)$. Note that $P(n)$ grows asymptotically as fast as n^d, so $S(P(n))$ grows asymptotically as no faster than a constant multiple of k. On the other hand, $P(9k)$ grows asymptotically as fast as k^d. This shows the two sides of the last equation cannot be equal for sufficiently large k since $d \geqslant 2$.

Therefore, we conclude that $P(x) = c$ where $1 \leqslant c \leqslant 9$ is an integer, or $P(x) = x$.

Solution 2. Let $P(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_0$. Clearly $a_d > 0$. There exists an integer $m \geqslant 1$ such that $|a_i| < 10^m$ for all $0 \leqslant i \leqslant d$. Consider $n = 9 \times 10^k$ for a sufficiently large integer k in (1). If there exists an index $0 \leqslant i \leqslant d-1$ such that $a_i < 0$, then all digits of $P(n)$ in positions from 10^{ik+m+1} to $10^{(i+1)k-1}$ are all 9's. Hence, we have $S(P(n)) \geqslant 9(k-m-1)$. On the other hand, $P(S(n)) = P(9)$ is a fixed constant. Therefore, (1) cannot hold for large k. This shows $a_i \geqslant 0$ for all $0 \leqslant i \leqslant d-1$.

Hence, $P(n)$ is an integer formed by the nonnegative integers $a_d \times 9^d, a_{d-1} \times 9^{d-1}, \ldots, a_0$ by inserting some zeros in between. This yields

$$S(P(n)) = S(a_d \times 9^d) + S(a_{d-1} \times 9^{d-1}) + \cdots + S(a_0).$$

Combining with (1), we have

$$S(a_d \times 9^d) + S(a_{d-1} \times 9^{d-1}) + \cdots + S(a_0) = P(9) = a_d \times 9^d + a_{d-1} \times 9^{d-1} + \cdots + a_0.$$

As $S(m) \leqslant m$ for any positive integer m, with equality when $1 \leqslant m \leqslant 9$, this forces each $a_i \times 9^i$ to be a positive integer between 1 and 9. In particular, this shows $a_i = 0$ for $i \geqslant 2$ and hence $d \leqslant 1$. Also, we have $a_1 \leqslant 1$ and $a_0 \leqslant 9$. If $a_1 = 1$ and $1 \leqslant a_0 \leqslant 9$, we take $n = 10^k + (10 - a_0)$ for sufficiently large k in (1). This yields a contradiction since

$$S(P(n)) = S(10^k + 10) = 2 \neq 11 = P(11 - a_0) = P(S(n)).$$

The zero polynomial is also rejected since $P(n)$ is positive for large n. The remaining candidates are $P(x) = x$ or $P(x) = a_0$ where $1 \leqslant a_0 \leqslant 9$, all of which satisfy (1), and hence are the only solutions.

N2. Let $\tau(n)$ be the number of positive divisors of n. Let $\tau_1(n)$ be the number of positive divisors of n which have remainders 1 when divided by 3. Find all possible integral values of the fraction $\frac{\tau(10n)}{\tau_1(10n)}$.

Answer. All composite numbers together with 2.

Solution. In this solution, we always use p_i to denote primes congruent to 1 mod 3, and use q_j to denote primes congruent to 2 mod 3. When we express a positive integer m using its prime factorization, we also include the special case $m = 1$ by allowing the exponents to be zeros. We first compute $\tau_1(m)$ for a positive integer m.

• **Claim.** Let $m = 3^x p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s} q_1^{b_1} q_2^{b_2} \cdots q_t^{b_t}$ be the prime factorization of m. Then

$$\tau_1(m) = \prod_{i=1}^{s} (a_i + 1) \left\lceil \frac{1}{2} \prod_{j=1}^{t} (b_j + 1) \right\rceil. \tag{1}$$

Proof. To choose a divisor of m congruent to 1 mod 3, it cannot have the prime divisor 3, while there is no restriction on choosing prime factors congruent to 1 mod 3. Also, we have to choose an even number of prime factors (counted with multiplicity) congruent to 2 mod 3.

If $\prod_{j=1}^{t} (b_j + 1)$ is even, then we may assume without loss of generality $b_1 + 1$ is even. We can choose the prime factors $q_2, q_3, \ldots, q_t$ freely in $\prod_{j=2}^{t} (b_j + 1)$ ways. Then the parity of the number of q_1 is uniquely determined, and hence there are $\frac{1}{2}(b_1 + 1)$ ways to choose the exponent of q_1. Hence (1) is verified in this case.

If $\prod_{j=1}^{t} (b_j + 1)$ is odd, we use induction on t to count the number of choices. When $t = 1$, there are $\lceil \frac{b_1+1}{2} \rceil$ choices for which the exponent is even and $\lfloor \frac{b_1+1}{2} \rfloor$ choices for which the exponent is odd. For the inductive step, we find that there are

$$\left\lceil \frac{1}{2} \prod_{j=1}^{t-1} (b_j + 1) \right\rceil \cdot \left\lceil \frac{b_t + 1}{2} \right\rceil + \left\lfloor \frac{1}{2} \prod_{j=1}^{t-1} (b_j + 1) \right\rfloor \cdot \left\lfloor \frac{b_t + 1}{2} \right\rfloor = \left\lceil \frac{1}{2} \prod_{j=1}^{t} (b_j + 1) \right\rceil$$

choices with an even number of prime factors and hence $\lfloor \frac{1}{2} \prod_{j=1}^{t} (b_j + 1) \rfloor$ choices with an odd number of prime factors. Hence (1) is also true in this case. □

Let $n = 3^x 2^y 5^z p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s} q_1^{b_1} q_2^{b_2} \cdots q_t^{b_t}$. Using the well-known formula for computing the divisor function, we get

$$\tau(10n) = (x + 1)(y + 2)(z + 2) \prod_{i=1}^{s} (a_i + 1) \prod_{j=1}^{t} (b_j + 1). \tag{2}$$

By the Claim, we have

$$\tau_1(10n) = \prod_{i=1}^{s} (a_i + 1) \left\lceil \frac{1}{2}(y + 2)(z + 2) \prod_{j=1}^{t} (b_j + 1) \right\rceil. \tag{3}$$

If $c = (y+2)(z+2)\prod_{j=1}^{t}(b_j+1)$ is even, then (2) and (3) imply

$$\frac{\tau(10n)}{\tau_1(10n)} = 2(x+1).$$

In this case $\frac{\tau(10n)}{\tau_1(10n)}$ can be any even positive integer as x runs through all nonnegative integers.

If c is odd, which means y, z are odd and each b_j is even, then (2) and (3) imply

$$\frac{\tau(10n)}{\tau_1(10n)} = \frac{2(x+1)c}{c+1}. \tag{4}$$

For this to be an integer, we need $c+1$ divides $2(x+1)$ since c and $c+1$ are relatively prime. Let $2(x+1) = k(c+1)$. Then (4) reduces to

$$\frac{\tau(10n)}{\tau_1(10n)} = kc = k(y+2)(z+2)\prod_{j=1}^{t}(b_j+1). \tag{5}$$

Noting that y, z are odd, the integers $y+2$ and $z+2$ are at least 3. This shows the integer in this case must be composite. On the other hand, for any odd composite number ab with $a, b \geqslant 3$, we may simply take $n = 3^{\frac{ab-1}{2}} \cdot 2^{a-2} \cdot 5^{b-2}$ so that $\frac{\tau(10n)}{\tau_1(10n)} = ab$ from (5).

We conclude that the fraction can be any even integer or any odd composite number. Equivalently, it can be 2 or any composite number.

N3. Define $P(n) = n^2 + n + 1$. For any positive integers a and b, the set

$$\{P(a), P(a+1), P(a+2), \ldots, P(a+b)\}$$

is said to be *fragrant* if none of its elements is relatively prime to the product of the other elements. Determine the smallest size of a fragrant set.

Answer. 6.

Solution. We have the following observations.

(i) $(P(n), P(n+1)) = 1$ for any n.

We have $(P(n), P(n+1)) = (n^2+n+1, n^2+3n+3) = (n^2+n+1, 2n+2)$. Noting that n^2+n+1 is odd and $(n^2+n+1, n+1) = (1, n+1) = 1$, the claim follows.

(ii) $(P(n), P(n+2)) = 1$ for $n \not\equiv 2 \pmod 7$ and $(P(n), P(n+2)) = 7$ for $n \equiv 2 \pmod 7$.

From $(2n+7)P(n) - (2n-1)P(n+2) = 14$ and the fact that $P(n)$ is odd, $(P(n), P(n+2))$ must be a divisor of 7. The claim follows by checking $n \equiv 0, 1, \ldots, 6 \pmod 7$ directly.

(iii) $(P(n), P(n+3)) = 1$ for $n \not\equiv 1 \pmod 3$ and $3 | (P(n), P(n+3))$ for $n \equiv 1 \pmod 3$.

From $(n+5)P(n) - (n-1)P(n+3) = 18$ and the fact that $P(n)$ is odd, $(P(n), P(n+3))$ must be a divisor of 9. The claim follows by checking $n \equiv 0, 1, 2 \pmod 3$ directly.

Suppose there exists a fragrant set with at most 5 elements. We may assume it contains exactly 5 elements $P(a), P(a+1), \ldots, P(a+4)$ since the following argument also works with fewer elements. Consider $P(a+2)$. From (i), it is relatively prime to $P(a+1)$ and $P(a+3)$. Without loss of generality, assume $(P(a), P(a+2)) > 1$. From (ii), we have $a \equiv 2 \pmod 7$. The same observation implies $(P(a+1), P(a+3)) = 1$. In order that the set is fragrant, $(P(a), P(a+3))$ and $(P(a+1), P(a+4))$ must both be greater than 1. From (iii), this holds only when both a and $a+1$ are congruent to 1 mod 3, which is a contradiction.

It now suffices to construct a fragrant set of size 6. By the Chinese Remainder Theorem, we can take a positive integer a such that

$$a \equiv 7 \pmod{19}, \quad a+1 \equiv 2 \pmod 7, \quad a+2 \equiv 1 \pmod 3.$$

For example, we may take $a = 197$. From (ii), both $P(a+1)$ and $P(a+3)$ are divisible by 7. From (iii), both $P(a+2)$ and $P(a+5)$ are divisible by 3. One also checks from $19 | P(7) = 57$ and $19 | P(11) = 133$ that $P(a)$ and $P(a+4)$ are divisible by 19. Therefore, the set $\{P(a), P(a+1), \ldots, P(a+5)\}$ is fragrant.

Therefore, the smallest size of a fragrant set is 6.

Comment. "Fragrant Harbour" is the English translation of "Hong Kong".

A stronger version of this problem is to show that there exists a fragrant set of size k for any $k \geqslant 6$. We present a proof here.

For each even positive integer m which is not divisible by 3, since $m^2 + 3 \equiv 3 \pmod 4$, we can find a prime $p_m \equiv 3 \pmod 4$ such that $p_m | m^2 + 3$. Clearly, $p_m > 3$.

If $b = 2t \geqslant 6$, we choose a such that $3|2(a+t)+1$ and $p_m|2(a+t)+1$ for each $1 \leqslant m \leqslant b$ with $m \equiv 2, 4 \pmod 6$. For $0 \leqslant r \leqslant t$ and $3|r$, we have $a + t \pm r \equiv 1 \pmod 3$ so that $3|P(a+t\pm r)$. For $0 \leqslant r \leqslant t$ and $(r, 3) = 1$, we have

$$4P(a+t\pm r) \equiv (-1 \pm 2r)^2 + 2(-1 \pm 2r) + 4 = 4r^2 + 3 \equiv 0 \pmod{p_{2r}}.$$

Hence, $\{P(a), P(a+1), \ldots, P(a+b)\}$ is fragrant.

If $b = 2t + 1 \geqslant 7$ (the case $b = 5$ has been done in the original problem), we choose a such that $3|2(a+t)+1$ and $p_m|2(a+t)+1$ for $1 \leqslant m \leqslant b$ with $m \equiv 2, 4 \pmod 6$, and that $a + b \equiv 9 \pmod{13}$. Note that a exists by the Chinese Remainder Theorem since $p_m \neq 13$ for all m. The even case shows that $\{P(a), P(a+1), \ldots, P(a+b-1)\}$ is fragrant. Also, one checks from $13|P(9) = 91$ and $13|P(3) = 13$ that $P(a+b)$ and $P(a+b-6)$ are divisible by 13. The proof is thus complete.

N4. Let n, m, k and l be positive integers with $n \neq 1$ such that $n^k + mn^l + 1$ divides $n^{k+l} - 1$. Prove that

- $m = 1$ and $l = 2k$; or
- $l|k$ and $m = \frac{n^{k-l}-1}{n^l-1}$.

Solution 1. It is given that

$$n^k + mn^l + 1 | n^{k+l} - 1. \tag{1}$$

This implies

$$n^k + mn^l + 1 | (n^{k+l} - 1) + (n^k + mn^l + 1) = n^{k+l} + n^k + mn^l. \tag{2}$$

We have two cases to discuss.

- **Case 1.** $l \geqslant k$.
  Since $(n^k + mn^l + 1, n) = 1$, (2) yields

$$n^k + mn^l + 1 | n^l + mn^{l-k} + 1.$$

In particular, we get $n^k + mn^l + 1 \leqslant n^l + mn^{l-k} + 1$. As $n \geqslant 2$ and $k \geqslant 1$, $(m-1)n^l$ is at least $2(m-1)n^{l-k}$. It follows that the inequality cannot hold when $m \geqslant 2$. For $m = 1$, the above divisibility becomes

$$n^k + n^l + 1 | n^l + n^{l-k} + 1.$$

Note that $n^l + n^{l-k} + 1 < n^l + n^l + 1 < 2(n^k + n^l + 1)$. Thus we must have $n^l + n^{l-k} + 1 = n^k + n^l + 1$ so that $l = 2k$, which gives the first result.

- **Case 2.** $l < k$.
  This time (2) yields

$$n^k + mn^l + 1 | n^k + n^{k-l} + m.$$

In particular, we get $n^k + mn^l + 1 \leqslant n^k + n^{k-l} + m$, which implies

$$m \leqslant \frac{n^{k-l} - 1}{n^l - 1}. \tag{3}$$

On the other hand, from (1) we may let $n^{k+l} - 1 = (n^k + mn^l + 1)t$ for some positive integer t. Obviously, t is less than n^l, which means $t \leqslant n^l - 1$ as it is an integer. Then we have $n^{k+l} - 1 \leqslant (n^k + mn^l + 1)(n^l - 1)$, which is the same as

$$m \geqslant \frac{n^{k-l} - 1}{n^l - 1}. \tag{4}$$

Equations (3) and (4) combine to give $m = \frac{n^{k-l}-1}{n^l-1}$. As this is an integer, we have $l|k - l$. This means $l|k$ and it corresponds to the second result.

Solution 2. As in Solution 1, we begin with equation (2).

- **Case 1.** $l \geqslant k$.
  Then (2) yields
$$n^k + mn^l + 1 | n^l + mn^{l-k} + 1.$$

Since $2(n^k+mn^l+1) > 2mn^l+1 > n^l+mn^{l-k}+1$, it follows that $n^k+mn^l+1 = n^l+mn^{l-k}+1$, that is,
$$m(n^l - n^{l-k}) = n^l - n^k.$$

If $m \geqslant 2$, then $m(n^l - n^{l-k}) \geqslant 2n^l - 2n^{l-k} \geqslant 2n^l - n^l > n^l - n^k$ gives a contradiction. Hence $m = 1$ and $l - k = k$, which means $m = 1$ and $l = 2k$.

- **Case 2.** $l < k$.
  Then (2) yields
$$n^k + mn^l + 1 | n^k + n^{k-l} + m.$$

Since $2(n^k+mn^l+1) > 2n^k+m > n^k+n^{k-l}+m$, it follows that $n^k+mn^l+1 = n^k+n^{k-l}+m$. This gives $m = \frac{n^{k-l}-1}{n^l-1}$. Note that $n^l - 1 | n^{k-l} - 1$ implies $l|k - l$ and hence $l|k$. The proof is thus complete.

Comment. Another version of this problem is as follows: let n, m, k and l be positive integers with $n \neq 1$ such that k and l do not divide each other. Show that $n^k + mn^l + 1$ does not divide $n^{k+l} - 1$.

N5. Let a be a positive integer which is not a square number. Denote by A the set of all positive integers k such that

$$k = \frac{x^2 - a}{x^2 - y^2} \tag{1}$$

for some integers x and y with $x > \sqrt{a}$. Denote by B the set of all positive integers k such that (1) is satisfied for some integers x and y with $0 \leqslant x < \sqrt{a}$. Prove that $A = B$.

Solution 1. We first prove the following preliminary result.

• **Claim.** For fixed k, let x, y be integers satisfying (1). Then the numbers x_1, y_1 defined by

$$x_1 = \frac{1}{2}\left(x - y + \frac{(x-y)^2 - 4a}{x+y}\right), \quad y_1 = \frac{1}{2}\left(x - y - \frac{(x-y)^2 - 4a}{x+y}\right)$$

are integers and satisfy (1) (with x, y replaced by x_1, y_1 respectively).

Proof. Since $x_1 + y_1 = x - y$ and

$$x_1 = \frac{x^2 - xy - 2a}{x+y} = -x + \frac{2(x^2 - a)}{x+y} = -x + 2k(x-y),$$

both x_1 and y_1 are integers. Let $u = x + y$ and $v = x - y$. The relation (1) can be rewritten as

$$u^2 - (4k-2)uv + (v^2 - 4a) = 0.$$

By Vieta's Theorem, the number $z = \frac{v^2 - 4a}{u}$ satisfies

$$v^2 - (4k-2)vz + (z^2 - 4a) = 0.$$

Since x_1 and y_1 are defined so that $v = x_1 + y_1$ and $z = x_1 - y_1$, we can reverse the process and verify (1) for x_1, y_1. □

We first show that $B \subset A$. Take any $k \in B$ so that (1) is satisfied for some integers x, y with $0 \leqslant x < \sqrt{a}$. Clearly, $y \neq 0$ and we may assume y is positive. Since a is not a square, we have $k > 1$. Hence, we get $0 \leqslant x < y < \sqrt{a}$. Define

$$x_1 = \frac{1}{2}\left|x - y + \frac{(x-y)^2 - 4a}{x+y}\right|, \quad y_1 = \frac{1}{2}\left(x - y - \frac{(x-y)^2 - 4a}{x+y}\right).$$

By the Claim, x_1, y_1 are integers satisfying (1). Also, we have

$$x_1 \geqslant -\frac{1}{2}\left(x - y + \frac{(x-y)^2 - 4a}{x+y}\right) = \frac{2a + x(y-x)}{x+y} \geqslant \frac{2a}{x+y} > \sqrt{a}.$$

This implies $k \in A$ and hence $B \subset A$.

Next, we shall show that $A \subset B$. Take any $k \in A$ so that (1) is satisfied for some integers x, y with $x > \sqrt{a}$. Again, we may assume y is positive. Among all such representations of k, we choose the one with smallest $x + y$. Define

$$x_1 = \frac{1}{2}\left|x - y + \frac{(x-y)^2 - 4a}{x+y}\right|, \quad y_1 = \frac{1}{2}\left(x - y - \frac{(x-y)^2 - 4a}{x+y}\right).$$

By the Claim, x_1, y_1 are integers satisfying (1). Since $k > 1$, we get $x > y > \sqrt{a}$. Therefore, we have $y_1 > \frac{4a}{x+y} > 0$ and $\frac{4a}{x+y} < x + y$. It follows that

$$x_1 + y_1 \leqslant \max\left\{x - y, \frac{4a - (x-y)^2}{x+y}\right\} < x + y.$$

If $x_1 > \sqrt{a}$, we get a contradiction due to the minimality of $x + y$. Therefore, we must have $0 \leqslant x_1 < \sqrt{a}$, which means $k \in B$ so that $A \subset B$.

The two subset relations combine to give $A = B$.

Solution 2. The relation (1) is equivalent to

$$ky^2 - (k-1)x^2 = a. \tag{2}$$

Motivated by Pell's Equation, we prove the following, which is essentially the same as the Claim in Solution 1.

• **Claim.** If (x_0, y_0) is a solution to (2), then $((2k-1)x_0 \pm 2ky_0, (2k-1)y_0 \pm 2(k-1)x_0)$ is also a solution to (2).

Proof. We check directly that

$$\begin{aligned} &k((2k-1)y_0 \pm 2(k-1)x_0)^2 - (k-1)((2k-1)x_0 \pm 2ky_0)^2 \\ &= (k(2k-1)^2 - (k-1)(2k)^2)y_0^2 + (k(2(k-1))^2 - (k-1)(2k-1)^2)x_0^2 \\ &= ky_0^2 - (k-1)x_0^2 = a. \end{aligned}$$

□

If (2) is satisfied for some $0 \leqslant x < \sqrt{a}$ and nonnegative integer y, then clearly (1) implies $y > x$. Also, we have $k > 1$ since a is not a square number. By the Claim, consider another solution to (2) defined by

$$x_1 = (2k-1)x + 2ky, \quad y_1 = (2k-1)y + 2(k-1)x.$$

It satisfies $x_1 \geqslant (2k-1)x + 2k(x+1) = (4k-1)x + 2k > x$. Then we can replace the old solution by a new one which has a larger value in x. After a finite number of replacements, we must get a solution with $x > \sqrt{a}$. This shows $B \subset A$.

If (2) is satisfied for some $x > \sqrt{a}$ and nonnegative integer y, by the Claim we consider another solution to (2) defined by

$$x_1 = |(2k-1)x - 2ky|, \quad y_1 = (2k-1)y - 2(k-1)x.$$

From (2), we get $\sqrt{k}y > \sqrt{k-1}x$. This implies $ky > \sqrt{k(k-1)}x > (k-1)x$ and hence $(2k-1)x - 2ky < x$. On the other hand, the relation (1) implies $x > y$. Then it is clear that $(2k-1)x - 2ky > -x$. These combine to give $x_1 < x$, which means we have found a solution to (2) with x having a smaller absolute value. After a finite number of steps, we shall obtain a solution with $0 \leqslant x < \sqrt{a}$. This shows $A \subset B$.

The desired result follows from $B \subset A$ and $A \subset B$.

Solution 3. It suffices to show $A \cup B$ is a subset of $A \cap B$. We take any $k \in A \cup B$, which means there exist integers x, y satisfying (1). Since a is not a square, it follows that $k \neq 1$. As in Solution 2, the result follows readily once we have proved the existence of a solution (x_1, y_1) to (1) with $|x_1| > |x|$, and, in case of $x > \sqrt{a}$, another solution (x_2, y_2) with $|x_2| < |x|$.

Without loss of generality, assume $x, y \geqslant 0$. Let $u = x + y$ and $v = x - y$. Then $u \geqslant v$ and (1) becomes

$$k = \frac{(u+v)^2 - 4a}{4uv}. \tag{3}$$

This is the same as

$$v^2 + (2u - 4ku)v + u^2 - 4a = 0.$$

Let $v_1 = 4ku - 2u - v$. Then $u + v_1 = 4ku - u - v \geqslant 8u - u - v > u + v$. By Vieta's Theorem, v_1 satisfies

$$v_1^2 + (2u - 4ku)v_1 + u^2 - 4a = 0.$$

This gives $k = \frac{(u+v_1)^2 - 4a}{4uv_1}$. As k is an integer, $u + v_1$ must be even. Therefore, $x_1 = \frac{u+v_1}{2}$ and $y_1 = \frac{v_1 - u}{2}$ are integers. By reversing the process, we can see that (x_1, y_1) is a solution to (1), with $x_1 = \frac{u+v_1}{2} > \frac{u+v}{2} = x \geqslant 0$. This completes the first half of the proof.

Suppose $x > \sqrt{a}$. Then $u + v > 2\sqrt{a}$ and (3) can be rewritten as

$$u^2 + (2v - 4kv)u + v^2 - 4a = 0.$$

Let $u_2 = 4kv - 2v - u$. By Vieta's Theorem, we have $uu_2 = v^2 - 4a$ and

$$u_2^2 + (2v - 4kv)u_2 + v^2 - 4a = 0. \tag{4}$$

By $u > 0$, $u + v > 2\sqrt{a}$ and (3), we have $v > 0$. If $u_2 \geqslant 0$, then $vu_2 \leqslant uu_2 = v^2 - 4a < v^2$. This shows $u_2 < v \leqslant u$ and $0 < u_2 + v < u + v$. If $u_2 < 0$, then $(u_2 + v) + (u + v) = 4kv > 0$ and $u_2 + v < u + v$ imply $|u_2 + v| < u + v$. In any case, since $u_2 + v$ is even from (4), we can define $x_2 = \frac{u_2+v}{2}$ and $y_2 = \frac{u_2-v}{2}$ so that (1) is satisfied with $|x_2| < x$, as desired. The proof is thus complete.

N6. Denote by $\mathbb{N}$ the set of all positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all positive integers m and n, the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

Answer. $f(n) = n^2$ for any $n \in \mathbb{N}$.

Solution. It is given that

$$f(m) + f(n) - mn | mf(m) + nf(n). \tag{1}$$

Taking $m = n = 1$ in (1), we have $2f(1) - 1 | 2f(1)$. Then $2f(1) - 1 | 2f(1) - (2f(1) - 1) = 1$ and hence $f(1) = 1$.

Let $p \geqslant 7$ be a prime. Taking $m = p$ and $n = 1$ in (1), we have $f(p) - p + 1 | pf(p) + 1$ and hence

$$f(p) - p + 1 | pf(p) + 1 - p(f(p) - p + 1) = p^2 - p + 1.$$

If $f(p) - p + 1 = p^2 - p + 1$, then $f(p) = p^2$. If $f(p) - p + 1 \neq p^2 - p + 1$, as $p^2 - p + 1$ is an odd positive integer, we have $p^2 - p + 1 \geqslant 3(f(p) - p + 1)$, that is,

$$f(p) \leqslant \frac{1}{3}(p^2 + 2p - 2). \tag{2}$$

Taking $m = n = p$ in (1), we have $2f(p) - p^2 | 2pf(p)$. This implies

$$2f(p) - p^2 | 2pf(p) - p(2f(p) - p^2) = p^3.$$

By (2) and $f(p) \geqslant 1$, we get

$$-p^2 < 2f(p) - p^2 \leqslant \frac{2}{3}(p^2 + 2p - 2) - p^2 < -p$$

since $p \geqslant 7$. This contradicts the fact that $2f(p) - p^2$ is a factor of p^3. Thus we have proved that $f(p) = p^2$ for all primes $p \geqslant 7$.

Let n be a fixed positive integer. Choose a sufficiently large prime p. Consider $m = p$ in (1). We obtain

$$f(p) + f(n) - pn | pf(p) + nf(n) - n(f(p) + f(n) - pn) = pf(p) - nf(n) + pn^2.$$

As $f(p) = p^2$, this implies $p^2 - pn + f(n) | p(p^2 - pn + n^2)$. As p is sufficiently large and n is fixed, p cannot divide $f(n)$, and so $(p, p^2 - pn + f(n)) = 1$. It follows that $p^2 - pn + f(n) | p^2 - pn + n^2$ and hence

$$p^2 - pn + f(n) | (p^2 - pn + n^2) - (p^2 - pn + f(n)) = n^2 - f(n).$$

Note that $n^2 - f(n)$ is fixed while $p^2 - pn + f(n)$ is chosen to be sufficiently large. Therefore, we must have $n^2 - f(n) = 0$ so that $f(n) = n^2$ for any positive integer n.

Finally, we check that when $f(n) = n^2$ for any positive integer n, then

$$f(m) + f(n) - mn = m^2 + n^2 - mn$$

and

$$mf(m) + nf(n) = m^3 + n^3 = (m + n)(m^2 + n^2 - mn).$$

The latter expression is divisible by the former for any positive integers m, n. This shows $f(n) = n^2$ is the only solution.

N7. Let n be an odd positive integer. In the Cartesian plane, a cyclic polygon P with area S is chosen. All its vertices have integral coordinates, and the squares of its side lengths are all divisible by n. Prove that $2S$ is an integer divisible by n.

Solution. Let $P = A_1A_2 \dots A_k$ and let $A_{k+i} = A_i$ for $i \geqslant 1$. By the Shoelace Formula, the area of any convex polygon with integral coordinates is half an integer. Therefore, $2S$ is an integer. We shall prove by induction on $k \geqslant 3$ that $2S$ is divisible by n. Clearly, it suffices to consider $n = p^t$ where p is an odd prime and $t \geqslant 1$.

For the base case $k = 3$, let the side lengths of P be $\sqrt{na}, \sqrt{nb}, \sqrt{nc}$ where a, b, c are positive integers. By Heron's Formula,

$$16S^2 = n^2(2ab + 2bc + 2ca - a^2 - b^2 - c^2).$$

This shows $16S^2$ is divisible by n^2. Since n is odd, $2S$ is divisible by n.

Assume $k \geqslant 4$. If the square of length of one of the diagonals is divisible by n, then that diagonal divides P into two smaller polygons, to which the induction hypothesis applies. Hence we may assume that none of the squares of diagonal lengths is divisible by n. As usual, we denote by $\nu_p(r)$ the exponent of p in the prime decomposition of r. We claim the following.

• **Claim.** $\nu_p(A_1A_m^2) > \nu_p(A_1A_{m+1}^2)$ for $2 \leqslant m \leqslant k-1$.

Proof. The case $m = 2$ is obvious since $\nu_p(A_1A_2^2) \geqslant p^t > \nu_p(A_1A_3^2)$ by the condition and the above assumption.

Suppose $\nu_p(A_1A_2^2) > \nu_p(A_1A_3^2) > \cdots > \nu_p(A_1A_m^2)$ where $3 \leqslant m \leqslant k-1$. For the induction step, we apply Ptolemy's Theorem to the cyclic quadrilateral $A_1A_{m-1}A_mA_{m+1}$ to get

$$A_1A_{m+1} \times A_{m-1}A_m + A_1A_{m-1} \times A_mA_{m+1} = A_1A_m \times A_{m-1}A_{m+1},$$

which can be rewritten as

$$\begin{aligned} A_1A_{m+1}^2 \times A_{m-1}A_m^2 = {} & A_1A_{m-1}^2 \times A_mA_{m+1}^2 + A_1A_m^2 \times A_{m-1}A_{m+1}^2 \\ & - 2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}. \end{aligned} \tag{1}$$

From this, $2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}$ is an integer. We consider the component of p of each term in (1). By the inductive hypothesis, we have $\nu_p(A_1A_{m-1}^2) > \nu_p(A_1A_m^2)$. Also, we have $\nu_p(A_mA_{m+1}^2) \geqslant p^t > \nu_p(A_{m-1}A_{m+1}^2)$. These give

$$\nu_p(A_1A_{m-1}^2 \times A_mA_{m+1}^2) > \nu_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2). \tag{2}$$

Next, we have $\nu_p(4A_1A_{m-1}^2 \times A_mA_{m+1}^2 \times A_1A_m^2 \times A_{m-1}A_{m+1}^2) = \nu_p(A_1A_{m-1}^2 \times A_mA_{m+1}^2) + \nu_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2) > 2\nu_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2)$ from (2). This implies

$$\nu_p(2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}) > \nu_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2). \tag{3}$$

Combining (1), (2) and (3), we conclude that

$$\nu_p(A_1A_{m+1}^2 \times A_{m-1}A_m^2) = \nu_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2).$$

By $\nu_p(A_{m-1}A_m^2) \geqslant p^t > \nu_p(A_{m-1}A_{m+1}^2)$, we get $\nu_p(A_1A_{m+1}^2) < \nu_p(A_1A_m^2)$. The Claim follows by induction. □

From the Claim, we get a chain of inequalities

$$p^t > \nu_p(A_1A_3^2) > \nu_p(A_1A_4^2) > \cdots > \nu_p(A_1A_k^2) \geqslant p^t,$$

which yields a contradiction. Therefore, we can show by induction that $2S$ is divisible by n.

Comment. The condition that P is cyclic is crucial. As a counterexample, consider the rhombus with vertices $(0,3),(4,0),(0,-3),(-4,0)$. Each of its squares of side lengths is divisible by 5, while $2S = 48$ is not.

The proposer also gives a proof for the case n is even. One just needs an extra technical step for the case $p = 2$.

N8. Find all polynomials $P(x)$ of odd degree d and with integer coefficients satisfying the following property: for each positive integer n, there exist n positive integers $x_1, x_2, \ldots, x_n$ such that $\frac{1}{2} < \frac{P(x_i)}{P(x_j)} < 2$ and $\frac{P(x_i)}{P(x_j)}$ is the d-th power of a rational number for every pair of indices i and j with $1 \leqslant i, j \leqslant n$.

Answer. $P(x) = a(rx+s)^d$ where a, r, s are integers with $a \neq 0$, $r \geqslant 1$ and $(r, s) = 1$.

Solution. Let $P(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Consider the substitution $y = da_d x + a_{d-1}$. By defining $Q(y) = P(x)$, we find that Q is a polynomial with rational coefficients without the term y^{d-1}. Let $Q(y) = b_d y^d + b_{d-2}y^{d-2} + b_{d-3}y^{d-3} + \cdots + b_0$ and $B = \max_{0 \leqslant i \leqslant d} \{|b_i|\}$ (where $b_{d-1} = 0$).

The condition shows that for each $n \geqslant 1$, there exist integers $y_1, y_2, \ldots, y_n$ such that $\frac{1}{2} < \frac{Q(y_i)}{Q(y_j)} < 2$ and $\frac{Q(y_i)}{Q(y_j)}$ is the d-th power of a rational number for $1 \leqslant i, j \leqslant n$. Since n can be arbitrarily large, we may assume all x_i's and hence y_i's are integers larger than some absolute constant in the following.

By Dirichlet's Theorem, since d is odd, we can find a sufficiently large prime p such that $p \equiv 2 \pmod{d}$. In particular, we have $(p-1, d) = 1$. For this fixed p, we choose n to be sufficiently large. Then by the Pigeonhole Principle, there must be $d+1$ of $y_1, y_2, \ldots, y_n$ which are congruent mod p. Without loss of generality, assume $y_i \equiv y_j \pmod{p}$ for $1 \leqslant i, j \leqslant d+1$. We shall establish the following.

• **Claim.** $\frac{Q(y_i)}{Q(y_1)} = \frac{y_i^d}{y_1^d}$ for $2 \leqslant i \leqslant d+1$.

Proof. Let $\frac{Q(y_i)}{Q(y_1)} = \frac{l^d}{m^d}$ where $(l, m) = 1$ and $l, m > 0$. This can be rewritten in the expanded form

$$b_d(m^d y_i^d - l^d y_1^d) = -\sum_{j=0}^{d-2} b_j(m^d y_i^j - l^d y_1^j). \tag{1}$$

Let c be the common denominator of Q, so that $cQ(k)$ is an integer for any integer k. Note that c depends only on P and so we may assume $(p, c) = 1$. Then $y_1 \equiv y_i \pmod{p}$ implies $cQ(y_1) \equiv cQ(y_i) \pmod{p}$.

• **Case 1.** $p | cQ(y_1)$.

In this case, there is a cancellation of p in the numerator and denominator of $\frac{cQ(y_i)}{cQ(y_1)}$, so that $m^d \leqslant p^{-1}|cQ(y_1)|$. Noting $|Q(y_1)| < 2By_1^d$ as y_1 is large, we get

$$m \leqslant p^{-\frac{1}{d}}(2cB)^{\frac{1}{d}} y_1. \tag{2}$$

For large y_1 and y_i, the relation $\frac{1}{2} < \frac{Q(y_i)}{Q(y_1)} < 2$ implies

$$\frac{1}{3} < \frac{y_i^d}{y_1^d} < 3. \tag{3}$$

We also have

$$\frac{1}{2} < \frac{l^d}{m^d} < 2. \tag{4}$$

Now, the left-hand side of (1) is

$$b_d(my_i - ly_1)(m^{d-1}y_i^{d-1} + m^{d-2}y_i^{d-2}ly_1 + \cdots + l^{d-1}y_1^{d-1}).$$

Suppose on the contrary that $my_i - ly_1 \neq 0$. Then the absolute value of the above expression is at least $|b_d|m^{d-1}y_i^{d-1}$. On the other hand, the absolute value of the right-hand side of (1) is at most

$$\begin{aligned}\sum_{j=0}^{d-2} B(m^d y_i^j + l^d y_1^j) &\leqslant (d-1)B(m^d y_i^{d-2} + l^d y_1^{d-2}) \\ &\leqslant (d-1)B(7m^d y_i^{d-2}) \\ &\leqslant 7(d-1)B(p^{-\frac{1}{d}}(2cB)^{\frac{1}{d}}y_1)m^{d-1}y_i^{d-2} \\ &\leqslant 21(d-1)Bp^{-\frac{1}{d}}(2cB)^{\frac{1}{d}}m^{d-1}y_i^{d-1}\end{aligned}$$

by using successively (3), (4), (2) and again (3). This shows

$$|b_d|m^{d-1}y_i^{d-1} \leqslant 21(d-1)Bp^{-\frac{1}{d}}(2cB)^{\frac{1}{d}}m^{d-1}y_i^{d-1},$$

which is a contradiction for large p as b_d, B, c, d depend only on the polynomial P. Therefore, we have $my_i - ly_1 = 0$ in this case.

• **Case 2.** $(p, cQ(y_1)) = 1$.

From $cQ(y_1) \equiv cQ(y_i) \pmod p$, we have $l^d \equiv m^d \pmod p$. Since $(p-1, d) = 1$, we use Fermat Little Theorem to conclude $l \equiv m \pmod p$. Then $p|my_i - ly_1$. Suppose on the contrary that $my_i - ly_1 \neq 0$. Then the left-hand side of (1) has absolute value at least $|b_d|pm^{d-1}y_i^{d-1}$. Similar to Case 1, the right-hand side of (1) has absolute value at most

$$21(d-1)B(2cB)^{\frac{1}{d}}m^{d-1}y_i^{d-1},$$

which must be smaller than $|b_d|pm^{d-1}y_i^{d-1}$ for large p. Again this yields a contradiction and hence $my_i - ly_1 = 0$.

In both cases, we find that $\frac{Q(y_i)}{Q(y_1)} = \frac{l^d}{m^d} = \frac{y_i^d}{y_1^d}$. □

From the Claim, the polynomial $Q(y_1)y^d - y_1^d Q(y)$ has roots $y = y_1, y_2, \ldots, y_{d+1}$. Since its degree is at most d, this must be the zero polynomial. Hence, $Q(y) = b_d y^d$. This implies $P(x) = a_d(x + \frac{a_{d-1}}{da_d})^d$. Let $\frac{a_{d-1}}{da_d} = \frac{s}{r}$ with integers r, s where $r \geqslant 1$ and $(r, s) = 1$. Since P has integer coefficients, we need $r^d|a_d$. Let $a_d = r^d a$. Then $P(x) = a(rx + s)^d$. It is obvious that such a polynomial satisfies the conditions.

Comment. In the proof, the use of prime and Dirichlet's Theorem can be avoided. One can easily show that each $P(x_i)$ can be expressed in the form uv_i^d where u, v_i are integers and u cannot be divisible by the d-th power of a prime (note that u depends only on P). By fixing a large integer q and by choosing a large n, we can apply the Pigeonhole Principle and assume

$x_1 \equiv x_2 \equiv \cdots \equiv x_{d+1} \pmod{q}$ and $v_1 \equiv v_2 \equiv \cdots \equiv v_{d+1} \pmod{q}$. Then the remaining proof is similar to Case 2 of the Solution.

Alternatively, we give another modification of the proof as follows.

We take a sufficiently large n and consider the corresponding positive integers $y_1, y_2, \ldots, y_n$. For each $2 \leqslant i \leqslant n$, let $\frac{Q(y_i)}{Q(y_1)} = \frac{l_i^d}{m_i^d}$.

As in Case 1, if there are d indices i such that the integers $\frac{c|Q(y_1)|}{m_i^d}$ are bounded below by a constant depending only on P, we can establish the Claim using those y_i's and complete the proof. Similarly, as in Case 2, if there are d indices i such that the integers $|m_i y_i - l_i y_1|$ are bounded below, then the proof goes the same. So it suffices to consider the case where $\frac{c|Q(y_1)|}{m_i^d} \leqslant M$ and $|m_i y_i - l_i y_1| \leqslant N$ for all $2 \leqslant i \leqslant n'$ where M, N are fixed constants and n' is large. Since there are only finitely many choices for m_i and $m_i y_i - l_i y_1$, by the Pigeonhole Principle, we can assume without loss of generality $m_i = m$ and $m_i y_i - l_i y_1 = t$ for $2 \leqslant i \leqslant d+2$. Then

$$\frac{Q(y_i)}{Q(y_1)} = \frac{l_i^d}{m^d} = \frac{(my_i - t)^d}{m^d y_1^d}$$

so that $Q(y_1)(my - t)^d - m^d y_1^d Q(y)$ has roots $y = y_2, y_3, \ldots, y_{d+2}$. Its degree is at most d and hence it is the zero polynomial. Therefore, $Q(y) = \frac{b_d}{m^d}(my - t)^d$. Indeed, Q does not have the term y^{d-1}, which means t should be 0. This gives the corresponding $P(x)$ of the desired form.

The two modifications of the Solution work equally well when the degree d is even.